WellSky® Blood Centers

Installation Guide

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WBCGI2026R1_01

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Revision History

Software Version & Revision Date	Document Number	Content Affected	Description of Change
2026 R1 November 2025	WBCGI2026R1_01	All	Updated the document to reflect the current release version and release date.
2026 R1 November 2025	WBCGI2026R1_01	2.1 Pre-installation Check List	Updated references to the 2026 R1 release throughout the procedure. Step 1: Removed Xeon 5460 (3.16 GHz 1333 FSB), Update GB memory to 32, Updated Disks RAID 1 to 2 x 200 and 2 x 400. Step 4: Updated Oracle WebLogic Server to 14.1.2.0m, Oracle Fusion Middleware Forms & Reports to 14.1.2.0 (64-Bit), removed AVA SE Runtime Environment 8 Version 261.
2026 R1 November 2025	WBCGI2026R1_01	2.4 Training Area Preparation	Updated trainee's PC minimum requirements:16 GB RAM memory, Microsoft Windows 11, removed Internet Explorer Version 11 and JRE 8 Update 261, added Microsoft Edge and Google Chrome.
2026 R1 November 2025	WBCGI2026R1_01	3.1 Oracle Linux Installation Procedure	Replaced Oracle Linux 8.4 Installation instructions with Redhat Enterprise Linux 9_0 Installation instructions.
2026 R1 November 2025	WBCGI2026R1_01	3.2 Tmp.conf Reconfiguration	Updated entries in tmp.conf in Step 4. See yellow highlight.
2026 R1 November 2025	WBCGI2026R1_01	4.1 Pre-Oracle Installation Procedure	Updated references to the 2026 R1 release in content and screenshots throughout the procedure.
2026 R1 November 2025	WBCGI2026R1_01	4.2 Download Oracle Database	Added new steps 13 - 17.
2026 R1 November 2025	WBCGI2026R1_01	4.3 Oracle Installation	Updated to reflect Software installation to Oracle Database (RDBMS) Software 19.3.0.0 64-Bit Standard Edition 2 and apply the 19.28.0.0 patch.

Software Version & Revision Date	Document Number	Content Affected	Description of Change
2026 R1 November 2025	WBCGI2026R1_01	4.4 Oracle Database Creation	Updated content and screenshot in steps 7,8 and 9.
2026 R1 November 2025	WBCGI2026R1_01	4.5 Oracle Database Upgrade	Removed this section from Install Guide.
2026 R1 November 2025	WBCGI2026R1_01	4.6 APEX Software Installation	Removed this section from Install Guide.
2026 R1 November 2025	WBCGI2026R1_01	5.1 Database Configuration Procedure	Updated references to the 2026 R1 release and replaced "bcprd" with "prddb" throughout the procedure.
2026 R1 November 2025	WBCGI2026R1_01	6.1 Download Fusion Middleware	Updated references to Oracle Fusion Middleware to 14c (Oracle Forms 14.1.2.0.0) in content and screenshots.
2026 R1 November 2025	WBCGI2026R1_01	6.2 Oracle WebLogic Server Installation	Updated references to Oracle Fusion Middleware to 14c (Oracle Forms 14.1.2.0.0) in content and screenshots.
2026 R1 November 2025	WBCGI2026R1_01	6.3 Oracle Forms and Reports Installation	Updated references to Oracle Fusion Middleware to 14c (Oracle Forms 14.1.2.0.0) in content and screenshots.
2026 R1 November 2025	WBCGI2026R1_01	6.4 Repository Creation Utility	Updated references to Oracle Fusion Middleware to 14c (Oracle Forms 14.1.2.0.0) in content and screenshots.
2026 R1 November 2025	WBCGI2026R1_01	6.5 FMW Configuration Wizard	Updated references to Oracle Fusion Middleware to 14c (Oracle Forms 14.1.2.0.0) in content and screenshots.
2026 R1 November 2025	WBCGI2026R1_01	6.6 Application Server Configuration	Updated references to Oracle Fusion Middleware to 14c (Oracle Forms 14.1.2.0.0) in content and screenshots.

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Software Version & Revision Date	Document Number	Content Affected	Description of Change
2026 R1 November 2025	WBCGI2026R1_01	7.2.1 Client Setup	Updated references to the 2026 R1 release throughout the procedure. Removed Java Client Setup section. Removed Blood Centers Forms Stand Alone Application Setup and Launching section.
2026 R1 November 2025	WBCGI2026R1_01	7.2.2 Installing the Open JDK 17 (Optional)	Removed this section from Install Guide.
2026 R1 November 2025	WBCGI2026R1_01	8.1 Overview	Changed Oracle 12c to 19c and Linux 8.4+ to Linux 9.6
2026 R1 November 2025	WBCGI2026R1_01	8.2 Upgrading Blood Centers	Changed 2024R1 to 2026R1 and BC2019 R4 to 2022 R1 in the first and third bullets.
2026 R1 November 2025	WBCGI2026R1_01	8.2.1 Upgrade Procedure in BC2026R1 Database Server	Changed 2024R1 to 2026R1 and "prddb" to "bcprd" throughout the procedure. Inserted new Steps 1 - 4. In Step 6, changed BC2019 R4 to 2022 R1, removed SQL>@UPG_UPD_BC2022R1.sql from liost of upgrade scripts and updated release numbers in the not immediately following the bullets.
2026 R1 November 2025	WBCGI2026R1_01	9.1 ASP.NET Installation and Configuration Procedure	Updated Step 3 screenshots.
2026 R1 November 2025	WBCGI2026R1_01	A.5 Blood Centers Application Links	Removed A.5.1 Oracle-Based Blood Centers from guide. In the renumbered A.5.1 .NET-Based Blood Centers Applications, changed bc2019 to bc2026. Reorganizd and updated links Under OnSite Application and Fastrack Appilication.
2026 R1 November 2025	WBCGI2026R1_01	A.7 Oracle Wallet Setup and Configuration	New Appendix section.
2026 R1 November 2025	WBCGI2026R1_01	A.8 OKTA Setup and Configuration	New Appendix section.

Contents

Chapter 1 - Introduction 1
1.1 Overview 2
1.2 Wireless Configuration Overview 2
1.2.1 Wireless Description 2
1.2.2 Computers 10
1.2.3 Wireless and Cellular Quality of Service 11
1.2.4 Security Measures 12
1.2.5 Electromagnetic Compatibility 13
1.2.6 Other Important Information Regarding Use of Wireless Devices 15
1.3 Document Organization 19
Chapter 2 - Pre-Installation 21
2.1 Pre-installation Check List 22
2.2 Supported Equipment 23
2.2.1 Barcode Printers 23
2.2.2 Barcode Scanners 23
2.2.3 Scales 23
2.3 DB Server Installation Preparation 24
2.4 Training Area Preparation 25
Chapter 3 - Oracle Linux Installation 27
3.1 Oracle Linux Installation Procedure 28
3.2 Tmp.conf Reconfiguration 35
Chapter 4 - Oracle DBMS Installation and Database Creation 37
4.1 Pre-Oracle Installation Procedure 38
4.2 Download Oracle Database 44
4.3 Oracle Installation 45
4.4 Oracle Database Creation 56
Chapter 5 - Blood Centers Database Configuration 71
5.1 Database Configuration Procedure 72
Chapter 6 - Oracle Fusion Middleware Installation and Configuration 75

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6.1 Download Fusion Middleware 76
6.2 Oracle WebLogic Server Installation 78
6.3 Oracle Forms and Reports Installation 86
6.4 Repository Creation Utility 95
6.5 FMW Configuration Wizard 106
6.6 Application Server Configuration 126
6.7 Set Up Cron Jobs 137
6.7.1 Required Cron Jobs 137
6.7.2 Optional Cron Jobs 138
6.8 Automatically Printing Reports 141
6.8.1 Overview 141
6.8.2 Adding Report Names to the UPD_REP_STATUS.sql File 141
6.9 Linux NFS Setup Between the Oracle Application and the Database Server 144
Chapter 7 - Post Donor Application and Database Checklist 147
7.1 Update the Donor Database Tables with Donor Application Server Configuration 148
7.1.1 Configuration 148
7.2 Blood Centers Client Setup 150
7.2.1 Client Setup 150
7.2.2 Installing the WebCam 151
7.2.3 Setting Up the Internal Camera for On-Site 155
7.3 HemaTrax-UNITY Printing 158
7.3.1 Install HemaTrax-UNITY 158
7.3.2 Configuring the HemaTrax Client Utility 164
7.3.3 Configuring Label Printers in Blood Centers 168
7.3.4 HemaTrax Print Server 169
7.4 Installation Cleanup 171
Chapter 8 - Upgrading Blood Centers 173
8.1 Overview 174
8.2 Upgrading Blood Centers 175
8.2.1 Upgrade Procedure in BC2026R1 Database Server 175
Chapter 9 - ASP.NET Installation and Configuration 177

9.1 ASP.NET Installation and Configuration Procedure 178
9.2 Deploy Blood Centers On-Site Application 180
9.2.1 Gallon Donor Image File Location 181
9.2.2 Multi-Language Files 182
9.3 Deploy Blood Centers FastTrak Application 184
9.3.1 Multi-Language Files 186
9.4 Deploy Blood Centers Donor Card Application 188
9.5 HTTPS Setup for Blood Centers Applications 189
9.6 External Proxy Setup for FastTrak Application 191
9.7 Deploy Blood Centers Donor Survey Application 192
9.8 Postfix Setup Instructions 193
Appendix A - Setup Guides 195
A.1 Blood Centers 2026 R1 SSL Setup Guide 196
A.1.1 Part I: Create a Java Keystore which contains an SSL Certificate 196
A.1.2 Part II: Configure the WebLogic Server (WLS_Forms) for SSL 200
A.1.3 Part III: Configure the Admin Server Trust Keystore 212
A.2 LT_APPLICATION_CONFIG Setup Guide 214
A.3 CAC Setup and Configuration 216
A.3.1 Setup and Configuration 216
A.4 Blood Centers Application Links 219
A.4.1 .NET-Based Blood Centers Applications 219
A.5 AGMGR Setup Instructions 220
A.5.1 Set Up the Oracle Gateway Software 220
A.5.2 Install and Configure the AGMGR Schema 228
A.5.3 Setting Up the Zebra Label Printer 230
A.6 Oracle Wallet Setup and Configuration 237
A.6.1 Oracle Wallet Setup in Database Server Environment 237
A.6.2 Oracle Wallet Setup in BC Forms and Reports Application Server 242
A.6.3 Oracle Wallet Setup in BC Onsite Application Server 243
A.6.4 Oracle Wallet files modification and updates 243
A.7 OKTA Setup and Configuration 245

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Glossary 249

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1

Chapter 1 - Introduction

1.1 Overview 2
1.2 Wireless Configuration Overview 2
1.2.1 Wireless Description 2
1.2.2 Computers 10
1.2.3 Wireless and Cellular Quality of Service 11
1.2.4 Security Measures 12
1.2.5 Electromagnetic Compatibility 13
1.2.6 Other Important Information Regarding Use of Wireless Devices 15
1.3 Document Organization 19

1.1 Overview

Congratulations on your purchase of the Blood Centers software application for the management of your blood center operation. The WellSky Team looks forward to working with you.

1.2 Wireless Configuration Overview

Wireless networks offer the flexibility of setting up a network infrastructure without the need for cables, etc. to connect the computers. Blood Centers is designed and was tested to perform on either a traditionally wired network or a wireless network with the same response, data accuracy and integrity. The following should be considered when using Blood Centers on a wireless network:

1.2.1 Wireless Description

Donor was tested from RF wireless connections to an appropriately configured and secured wireless network (WLAN) within a blood collection facility, simulated drive location, and mobile collection bus.

WellSky has evaluated the performance of Blood Centers on three distinct wireless network configurations.

- Wireless Local Area Network (WLAN) using 802.11n network topology.
- Wireless Local Area Network (WLAN) using 802.11g network topology.
- Connected to a server WAN link using cellular network air cards.



NOTE – WellSky has not qualified and does not recommend using a WLAN configuration together with cellular network air cards at the same drive.

Wireless Local Area Network (WLAN) using 802.11n and 802.11g Network Topology

WellSky used a WLAN utilizing 802.11n and 802.11g network topology. In this configuration, all computers were connected to a LAN using 802.11n and 802.11g wireless technology. Most of the testing performed was done on a wireless network consisting of a master laptop with two client laptops representing a typical setup for a blood drive. Additionally, a wireless network consisting of a master laptop and six client laptops was tested to verify that Blood Centers would operate correctly on a wireless network with a maximum expected number of laptops.

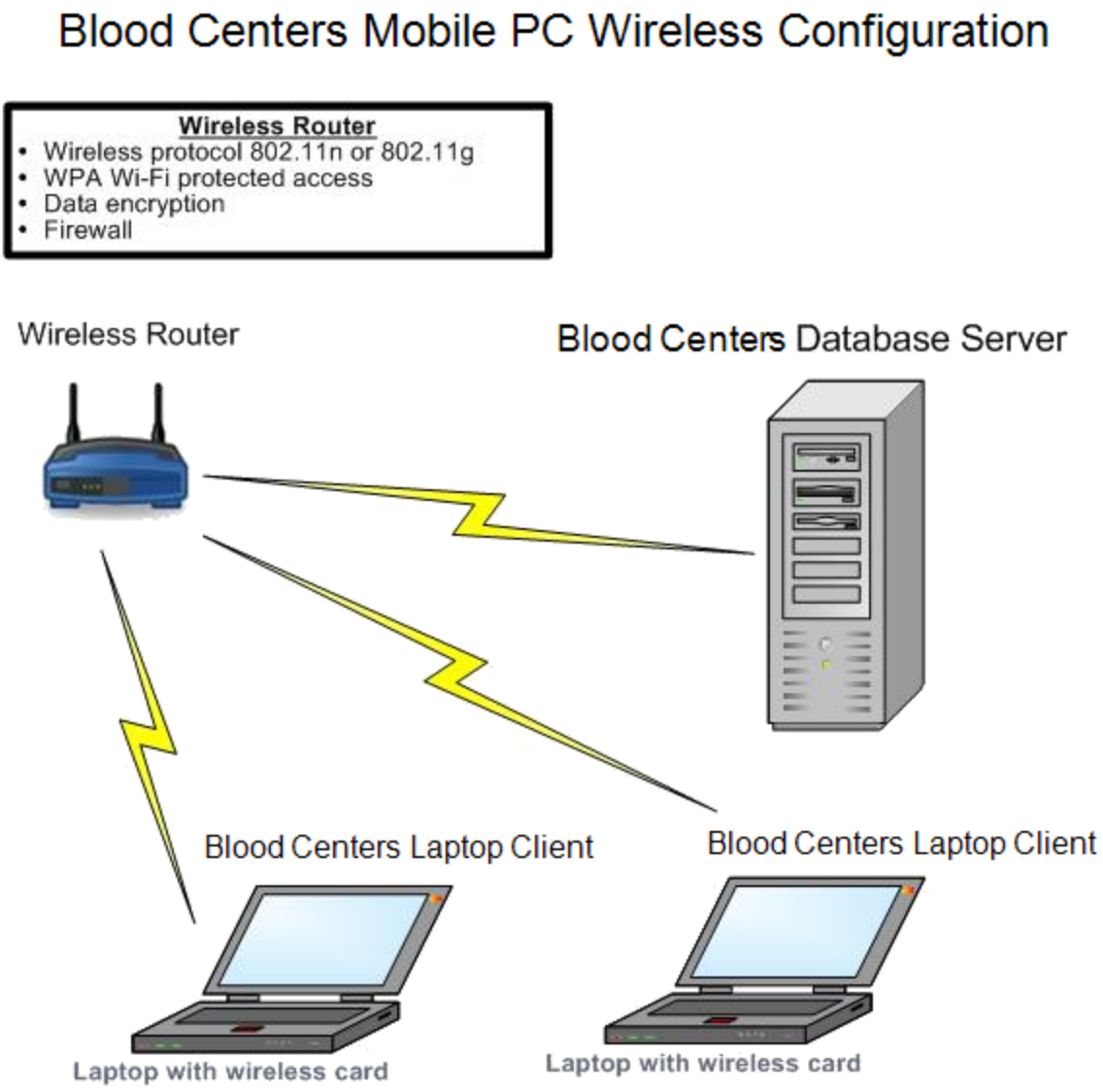
Summary of WLAN Essential Operating Requirements

- Use only 802.11n or 802.11g protocols.
- Do not use WLAN and cellular network cards at the same drive.
- Select an operating location and position network components to avoid sources of electromagnetic interference.
- Position all stations within 75 feet of the wireless router.

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- Maintain a separation distance of three feet between active network components.
- Apply WLAN network security according to recommendations.
- Do not operate a WLAN station above two percent packet loss.
- Do not exceed one master laptop and six client stations one a single WLAN network.
- Follow all of your equipment manufacturer's guidelines.

Example 802.11 Network Configuration



Wireless 802.11n and 802.11g Standard

The 802.11 family includes over-the-air modulation techniques that use the same basic protocol. WellSky has qualified 802.11n and 802.11g wireless networks for Blood Centers. Because of this choice of frequency band, 802.11n and 802.11g equipment can suffer interference from microwave ovens, cordless telephones, and Bluetooth devices.

NOTE – In the following table, MIMO stands for Multiple Input/Multiple Output Streams.

802.11 Protocol	Freq. (GHz)	Bandwidth (MHz)	Data Rate per Stream (MBPS)	Allowable MIMO*	Approx. Indoor Range (Ft)	Approx. Outdoor Range (Ft)	Modulation
g	2.4	20	6, 9, 12, 18, 24, 36, 48, 54	1	125	460	OFDM DSSS
n	2.4/5	20	7.2, 14.4, 21.7, 28.9, 43.3, 57.8, 65, 72.2	4	230	820	OFDM
		40	15, 30, 45, 60, 90, 120, 135, 150				

Wireless 802.11n and 802.11g Range of Operation

Blood Centers is a Blood Establishment Computer Software (BECS) software product that communicates between computers in a computer network. Blood Centers is designed to operate on commercial off-the-shelf (COTS) computer equipment and peripherals. WellSky has tested the Blood Centers software product in a wireless network environment, using industry standard 802.11g and 802.11n wireless networks. It is the responsibility of the customer to follow your equipment manufacturer's specific guidelines regarding separation distances, EMC considerations, etc., and to validate the equipment as suitable for use with Blood Centers in the customer's operating environment.

Examples of manufacturer's requirements for equipment used in testing by WellSky are listed below. The equipment utilized during the testing performed on Blood Centers was labeled on the data plates on the equipment or in the literature accompanying the hardware as FCC compliant to the standards for commercial off-the-shelf (COTS) electronic equipment.

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Table 1-1: COTS 802.11 Equipment Specifications

Equipment	Separation Distance from Manufacturer Documentation
Cisco 1800 Wireless G Router	A minimum 20 cm (7.9 in.) co-located separation distance from other FCC approved indoor/outdoor antennas used with the access point. The co-located 2.4-GHz and 5-GHz antennas support a minimum separation distance of 10 cm (3.9 in.) and are compliant with the applicable FCC RF exposure limit when transmitting simultaneously.
Netgear WNR2000 Wireless N Router	This equipment should be installed and operated with a minimum distance of 20 cm between the radiator and your body. The antennas used for this transmitter must be installed to provide a separation distance of at least 20 cm from all persons and must not be co-located or operating in conjunction with any other antenna or transmitter.
Intel 5100 Wireless network card	In order to comply with FCC RF exposure limits, low gain integrated antennas should be located at a minimum distance of 20 cm (8 inches) or more from the body of all persons. High-gain, wall-mount, or mast-mount antennas are designed to be professionally installed and should be located at a minimum distance of 30 cm (12 inches) or more from the body of all persons. Please contact your professional installer, VAR, or antenna manufacturer for proper installation requirements. To comply with the FCC and ANSI C95.1 RF exposure limits, it is recommended for the adapter installed in a desktop or portable computer, that the antenna for this device be installed so as to provide a separation distance of at least 20 cm (8 inches) from all persons. It is recommended that the user limit exposure time if the antenna is positioned closer than 20 cm (8 inches). Intel® PRO/Wireless LAN products are not designed for use with high-gain directional antennas. Use of such antennas with these products in a manner other than as described in the previous section titled, “Antenna Use” is illegal.

In addition to your manufacturer’s recommendations regarding equipment setup, WellSky recommends the following derived from our own system testing and field experience:

- Do not plan to use wireless communication near known sources of electromagnetic interference, including but not limited to:
    a. Heavy machinery with large motors
    b. Large compressors (e.g. industrial refrigerators)
    c. X-ray machines or any other type of radiology equipment
    d. Microwave emitters such as microwave ovens and other microwave devices
    e. Electric generators
    f. Rooms with walls that block radio signals such as basements and reinforced walls
    g. Rooms that do not allow enough room to properly deploy the network equipment for the blood drive
    h. 2.4 MHz cordless telephones that use wireless network frequencies
    i. Radar equipment
    j. Locations outside the service area of your broadband provider
    k. Known dead zones

- Locate your wireless router centrally to your wireless network and elevated for best results.
- Maintain a separation distance of three feet between operational units for best results.
- Keep operational units within 75 feet of the wireless router for best results.
- Do not allow phones or other hand-held electronic devices to be laid on top of or next to routers or workstations.
- Advise staff and visitors to turn off their personal electronic equipment while in the donation area.
- The WLAN should include a maximum of one master laptop and six client laptops or six laptops connected to your center server through the WLAN. Configurations exceeding this number must be validated prior to use to ensure that they operate properly.

Wireless Cellular WAN Connection

Client laptops located at a blood drive are connected directly to the central server located in the Blood Center through a WAN link using cellular network air cards. This wireless functionality can only be used where a cellular signal can be obtained at the mobile drive. Connections are made through a VPN to access the main database in the central server to verify donor eligibility and to update donation information.

Wireless Cellular Specifications

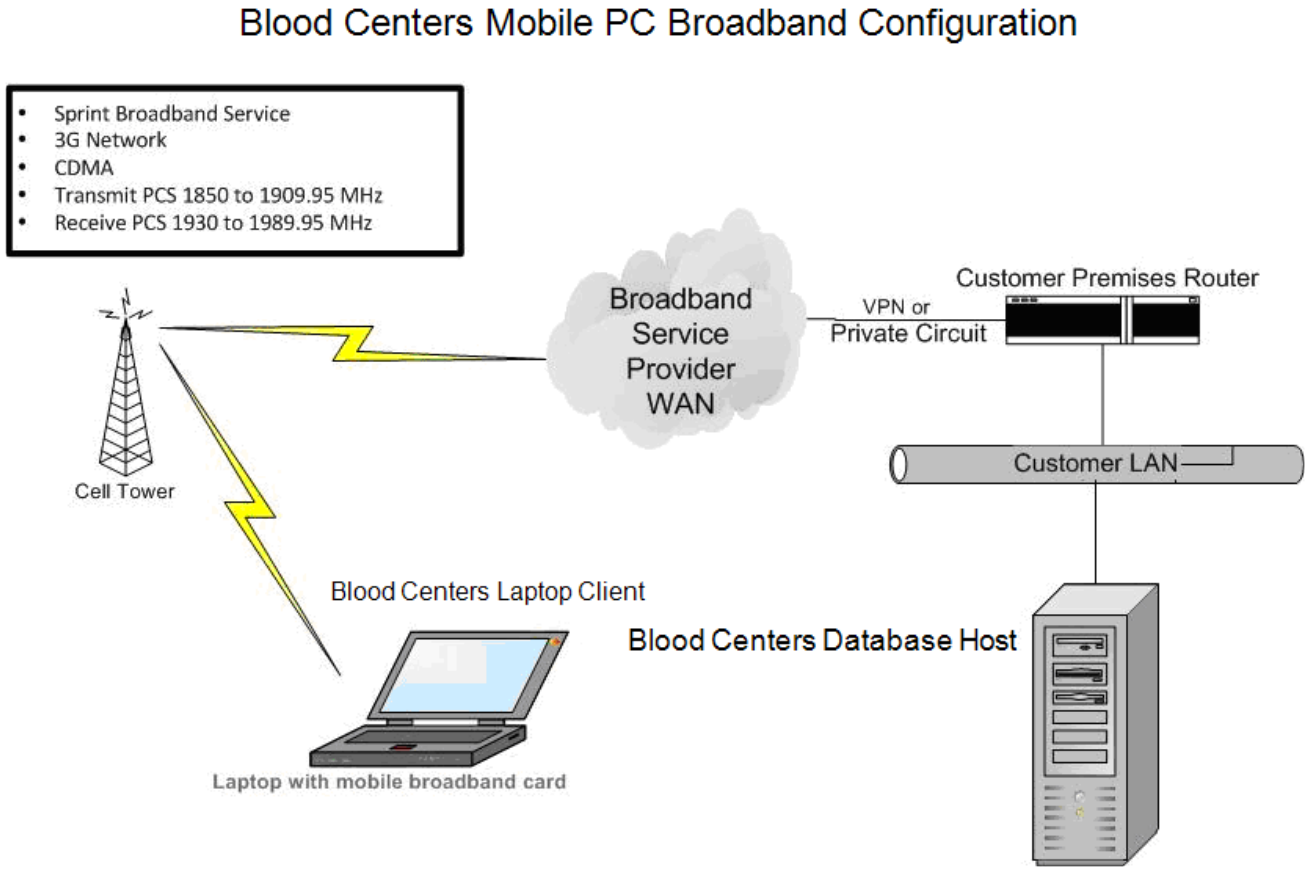
WellSky has tested cellular broadband communication using the following specifications for commercial off-the-shelf (COTS) equipment.

Table 1-2: COTS Cellular Equipment Specifications

Device	RF Frequency	Modulation	Data Rate/ Data Flow
Sprint U301 USB Air card	CDMA 800/1900 MHz	CDMA	Up to 2 MBPS (3G)
Sprint Mobile Broadband Card by Sierra Wireless Air Card 597E	CDMA 824 to 849 MHz	CDMA	Up to 2 MBPS (3G)

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Example Cellular Network Configuration



Wireless Cellular Range of Operation

Blood Centers is a Blood Establishment Computer Software (BECS) software product that communicates between computers in a computer network. Blood Centers is designed to operate on commercial off-the-shelf (COTS) computer equipment and peripherals. WellSky has tested the Blood Centers software product with cellular connectivity to the Internet using 3G and 4G cellular technology. It is the responsibility of the customer to follow your manufacturer's specific guidelines regarding separation distances, EMC considerations, etc., and to validate the equipment as suitable for use with Blood Centers in the customer's operating environment.

Examples of manufacturer's requirements for equipment used in testing by WellSky are listed below. The equipment utilized during the testing performed on Blood Centers was labeled on the data plates on the equipment or in the literature accompanying the hardware as FCC compliant to the standards for off-the-shelf electronic equipment.

Table 1-3: Example COTS Wireless Cellular Equipment Specifications

Equipment	Separation Distance from Manufacturer Documentation
Sprint U301 USB Aircard	This modem must be 5 mm (0.2 inches) or more from the user and bystanders during operation to satisfy FCC (regulatory) RF exposure requirements. It is approved for use with normal-size notebook computers only (typically with 12-inch or larger display screens). For notebook computers with a display screen smaller than 12 inches or if the orientation of the USB slot does not allow for at least 5 mm between the modem and the user or bystanders, use the provided extension cable to ensure a safe distance between the modem antenna and users or bystanders. To meet FCC requirements, use only the supplied USB extension cable; other cables should not be used.
Sprint Mobile Broadband Card by Sierra Wireless AirCard 597E	At least 20 cm separation distance between the antenna and the user's body must be maintained at all times.

In addition to your manufacturer's recommendations regarding equipment setup, WellSky recommends the following derived from our own system testing and field experience when using cellular equipment:

- Do not plan to use wireless communication near known sources of electromagnetic interference, including but not limited to:
    a. Heavy machinery with large motors
    b. Large compressors (e.g. industrial refrigerators)
    c. X-ray machines or any other type of radiology equipment
    d. Microwave emitters such as microwave ovens and other microwave devices
    e. Electric generators
    f. Rooms with walls that block radio signals such as basements and reinforced walls
    g. Rooms that do not allow enough room to properly deploy the network equipment for the blood drive
    h. 2.4 MHz cordless telephones that use wireless network frequencies
    i. Radar equipment
    j. Locations outside the service area of your broadband provider
    k. Known dead zones
- Maintain a separation distance of three feet between operational units for best results.
- Do not allow phones or other hand-held electronic devices to be laid on top of or next to the cellular equipped workstation.
- Advise staff and visitors to turn off their personal electronic equipment while in the donation area.

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Electromagnetic Compatibility

Customers should be aware that electronic equipment, including computers and peripherals, has the potential for causing interference with other electronic equipment present in the environment where the equipment operates. Although no specific interference was noted in the testing performed by WellSky on Blood Centers, customers must be aware of the potential for interference as part of the installation and validation process.

Recognizing Electromagnetic Interference

Blood Centers wireless communication was validated to a standard of less than 2% packet loss on a wireless network. You can easily verify your wireless setup to the same standard.

1. From any client station on your wireless network, ping the laptop server on the network. If ping reports packet loss of more than two percent, then that station does not meet the standard to continue operation on the wireless network.
2. Watch for these other warning signs of packet loss.
    - Response from the laptop server is slow
    - The client hangs while communicating with the laptop server
    - System reports wireless connection lost

Procedures for Dealing with Lost or Problem Wireless Connections

If you lose or are unable to establish a connection to your wireless network or become aware that your wireless network is interfering with other equipment in the area, consider the following mitigation and strategies as appropriate for your situation and in accordance with your facility's Standard Operating Procedures (SOPs).

1. Reboot the workstation to reestablish your connection.
2. Use another workstation.
3. Relocate or change the orientation of the workstation.
4. Increase the distance from other nearby electronic components.
5. Move closer to the wireless router.
6. Change your power source.
7. Reorient your wireless antenna.
8. Connect to the host directly using a wired connection instead of a wireless connection.
9. Reset your wireless router if multiple stations are affected.
10. If no wireless network can be established, then continue drive operations on the laptop server alone if practical to do so.
11. Invoke the manual screening and draw procedures for your facility.
12. Fall back to backup copies of the screening database on the remote workstation or other options provided by your facility's configuration and its SOPs.

13. Discontinue the drive if that is your facility's SOP or appropriate for the situation (e.g. drive is almost over).
14. Seek expert advice from the manufacturer of the equipment and other wireless networking experts.

1.2.2 Computers

The following describes the equipment and configurations that were tested for wireless connection and transmission of donor data and information at the specified Quality of Service:

Servers

Mobile Donor Laptop

- Lenovo T510
- Dual Core I5 with Hyper-threading Technology @2.67 GHz - 64-bit
- 4GB RAM Memory
- 250GB Hard Disk
- 10/100/1000 Ethernet Network Interface
- Intel 5100 wireless network card

Clients

- Lenovo T500
- 2.66 GHz Intel Core 2 Duo CPU
- 4 GB Ram memory
- 150 GB hard disk
- Windows 7, 32-bit or 64-bit
- 10/100/1000 Ethernet Network Interface
- Intel 5100 wireless network card

Network Devices

- Cisco 1811W, Security router with dual 10/100 WAN ports, 8-port 10/100BASE-T switch, V.92 analog modem backup, Cisco IOS Advanced IP Services, 32 MB Flash, 128 MB DRAM, and integrated FCC-compliant 802.11n and 802.11g wireless capability
- Sprint U301 USB Device 3G/4G Mobile Broadband, Sprint CDMA frequency of 800 MHz/1.9 GHz and Sprint WiMAX frequency band 2.5 GHz
- Netgear WNR2000
- Sierra Wireless Compass 597

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1.2.3 Wireless and Cellular Quality of Service

Wireless

- Maintain the network according to IEEE 802.11n or 802.11g standards with encryption.
- The wireless access security measures outlined below should be applied,
- The wireless devices should be at least three feet and no more than 75 feet from the wireless access point to maintain connectivity.
- The error rate of data transfer must be no more than 2% packet loss.
- During user testing at WellSky, typical devices that could coexist with the wireless network during a drive were tested. These included a tube sealer, cell phone, laptop computer, printer, microwave oven, and apheresis equipment. The system was also tested in a mobile bus while operating. No interference with the wireless network was detected. Users should be aware that other electronic devices, especially those with wireless capabilities could potentially interfere with wireless data transmission.
  - Personal electronics including cell phones, portable phones, music players, Bluetooth devices should be kept at least six inches from laptops that are being used during a blood drive. As a general policy, recommend that blood drive personnel instruct donors to power-off any electronic devices in their possession while in the blood drive area.
  - Common lunch or break-room equipment including microwave ovens, refrigerators and coffee makers should be kept at least six feet from laptops that are being used during a blood drive
  - Blood processing equipment commonly found in the blood donation sites including aphaeresis equipment and heat sealing devices should be kept at least three feet from laptops being used during a blood drive
- An alternative backup system such as a wired network or manual paper records should be available at the donation site to be used if the wireless network is not working properly.

Cellular

- The cellular network should be maintained on a 3G network
- The wireless access security measures outlined below should be applied
- The wireless devices should be at least 3 feet and no more than 75 feet from the wireless access point to maintain connectivity
- The error rate of data transfer must be no more than 2% packet loss
- Typical devices that we believe could co-exist with the wireless network at the blood center or during a remote blood drive include tube sealer, cell phone/PDA, laptop computers, printers, microwave ovens and aphaeresis equipment. Therefore, we tested the Blood Centers software operating on a wireless network against each of the identified devices. The system was also tested on a mobile bus while the bus was in operation.
  - Personal electronics including cell phones, portable phones, music players, Bluetooth devices should be kept at least six inches from laptops that are being used during a blood drive. As a general policy, recommend that blood drive personnel instruct donors to power-off any electronic devices in their possession while in the blood drive area.

- Common lunch or break-room equipment including microwave ovens, refrigerators and coffee makers should be kept at least six feet from laptops that are being used during a blood drive
- Blood processing equipment commonly found in the blood donation sites including aphaeresis equipment and heat sealing devices should be kept at least three feet from laptops being used during a blood drive

1.2.4 Security Measures

These security measures are recommended for all wireless access point devices when using the Blood Centers application. These security measures are general guidelines; refer to the owner's manual of the wireless access point to implement specific settings.

Wireless

- The wireless access point administration will be configured with a password. The password must be changed from the default provided by the manufacturer. The recommended minimum password length is eight characters. The password should include lower case alphabetic, upper case alphabetic, numeric and special characters. At least one of each type is required.
- The default wireless Service Set Identifier (SSID) or network name should be changed. The name should have a length of at least eight characters and require both alphabetic and numeric characters. Additionally, broadcast of the SSID should be disabled.
- The wireless devices (routers and network cards) should be secured using encryption technology. It is recommended to use the WPA2 authentication level. The encryption should be AES (Advanced Encryption Standard). The WPA2 shared key should be a minimum of fifteen characters. These should include lower case alphabetic, upper case alphabetic, numeric and special characters. At least one of each type should be required.
- To prevent unauthorized computers from communicating with the wireless access points, it is recommended to use the Media Access Control (MAC) filtering. MAC filtering allows only authorized equipment to connect to the wireless access point. This prevents unauthorized equipment from connecting to a wireless network

Cellular

Connections made over the cellular network must be made over a virtual private network (VPN). This will ensure that the traffic will be encrypted while being transmitted over the public Internet.

Risks and Mitigation

a. Risk of failure of the wireless network is that individual client stations cannot be used for screening on the drive. WellSky recommends that this risk be covered by client operating procedures since failure of an individual laptop is always an attendant risk of any remote blood drive and must be planned for regardless of whether wireless communication is utilized or not. In the case of a drive that plans to use wireless communication, additional mitigations are available since multiple laptops provide redundancy in the case of a single laptop or wireless network failure including the option of using the database server itself to continue screening when no wireless network can be established.

b. Risk of failure of the wireless connection is that the remote workstation cannot connect to the central database host for direct interaction with the system. WellSky recommends that this risk must also be covered by client operating since failure of an individual laptop is always an attendant risk of any remote blood drive

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and must be planned for regardless of whether wireless communication is utilized or not. Mitigations for a failed remote workstation (or its wireless connection) range from full manual procedures, deployment of backup workstations, call-in screening procedures, up to cancelling the drive. Failure of wireless connection alone allows the additional option of falling back to screening and registration from a backup local copy of the database on the workstation if the facility so chooses to configure their remote workstations.

1.2.5 Electromagnetic Compatibility

Users should be aware that electronic equipment, including computers and peripherals, has the potential for causing interference with other electronic equipment present in the environment where the equipment operates. Although no specific interference was noted in the testing performed by WellSky on Blood Centers, users should be aware of the potential for interference as part of the installation and validation process.

When setting up your equipment for a blood drive where you plan to use wireless communication consider the following guidelines.

Selecting a Location

"Location, location, location." It is true in real estate and for wireless networks. Do not plan to use wireless communication near known sources of electromagnetic interference, including but not limited to the following:

- Heavy machinery with large motors
- Large compressors (e.g. industrial refrigerators)
- X-ray machines or any other type of radiology equipment
- Microwave emitters such as microwave ovens and other microwave devices
- Electric generators
- Rooms with walls that block radio signals such as basements and reinforced walls
- Rooms that do not allow enough room to properly deploy the network equipment for the blood drive
- 2.4 MHz cordless telephones that use wireless network frequencies
- Radar equipment
- Locations outside the service area of your broadband provider
- Known dead zones

Setup

Observe the following guidelines in setting up your wireless network:

a. Position equipment in accordance with WellSky guidelines and the specific recommendations of your commercial off-the-shelf wireless network equipment manufacturer.

b. WellSky recommended guidelines include:
    i. Locate your wireless router centrally to your wireless network and elevated for best results.
    ii. Maintain a separation distance of three feet between operational units for best results.

iii. Do not overextend your wireless network. Keep operational units within 75 feet of the wireless router for best results.

iv. Do not allow phones or other hand-held electronic devices to be laid on top of or next to routers or workstations.

v. Advise staff and visitors to turn off their personal electronic equipment while in the donation area

Recognizing Electromagnetic Interference

Blood Centers wireless communication was validated to a standard of less than 2% packet loss on a wireless network. You can easily verify your wireless setup to the same standard, as follows:

a. From any client station on your wireless network, ping the laptop server on the network. If ping reports packet loss of more than two percent, then that station does not meet the standard to continue operation on the wireless network.

b. Watch for these other warning signs of packet loss.
   i. Response from the laptop server is slow.
   ii. The client hangs while communicating with the laptop server.
   iii. System reports wireless connection lost

Mitigation Strategies

If you lose or are unable to establish a connection to your wireless network or become aware that your wireless network is interfering with other equipment in the area, consider the following mitigation and strategies as appropriate for your situation and in accordance with your facility's Standard Operating Procedures (SOPs).

a. Reboot the workstation to reestablish your connection.

b. Use another workstation.

c. Relocate or change the orientation of the workstation.

d. Increase the distance from other nearby electronic components.

e. Move closer to the wireless router.

f. Change your power source.

g. Reorient your wireless antenna.

h. Connect to the host directly using a wired connection instead of a wireless connection.

i. Reset your wireless router if multiple stations are affected.

j. If no wireless network can be established, then continue drive operations on the laptop server alone if practical to do so.

k. Invoke the manual screening and draw procedures for your facility.

l. Fall back to backup copies of the screening database on the remote workstation or other options provided by your facility's configuration and its SOPs.

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m. Discontinue the drive if that is your facility's SOP or appropriate for the situation (e.g. drive is almost over).

n. Seek expert advice from the manufacturer of the equipment and other wireless networking experts.

1.2.6 Other Important Information Regarding Use of Wireless Devices

During WellSky's internal and external validation testing of Blood Centers several wireless devices were used to evaluate the product. Wireless devices are required by the FCC to be tested for the potential for interference with other devices that are located in the same general area as the wireless device. Additionally, devices are tested for the potential effect on persons that are using the devices or are in the general vicinity of these devices when they are being used. This information is usually provided in the labeling shipped with the device or may be available on the equipment supplier's website. When setting up Blood Centers for use in a wireless network, you should be aware of such information and to utilize the any cautions and warnings contained therein in the planning for your network. Some examples of the information provided by the manufacturer of the devices used by WellSky during testing are summarized in the following:

Device	Separation Distance from Manufacturer Documentation
Cisco 1800 Wireless G Router	A minimum 20 cm (7.9 in.) co-located separation distance from other FCC approved indoor/outdoor antennas used with the access point. The co-located 2.4-GHz and 5-GHz antennas support a minimum separation distance of 10 cm (3.9 in.) and are compliant with the applicable FCC RF exposure limit when transmitting simultaneously.
Netgear WNR2000 Wireless N Router	This equipment should be installed and operated with a minimum distance of 20 centimeters between the radiator and your body. The antennas used for this transmitter must be installed to provide a separation distance of at least 20 cm from all persons and must not be co-located or operating in conjunction with any other antenna or transmitter.
Intel 5100 Wireless network card	In order to comply with FCC RF exposure limits, low gain integrated antennas should be located at a minimum distance of 20 cm (8 inches) or more from the body of all persons. High-gain, wall-mount, or mast-mount antennas are designed to be professionally installed and should be located at a minimum distance of 30 cm (12 inches) or more from the body of all persons. Please contact your professional installer, VAR, or antenna manufacturer for proper installation requirements. To comply with the FCC and ANSI C95.1 RF exposure limits, it is recommended for the adapter installed in a desktop or portable computer, that the antenna for this device be installed so as to provide a separation distance of at least 20 cm (8 inches) from all persons. It is recommended that the user limit exposure time if the antenna is positioned closer than 20 cm (8 inches). Intel® PRO/Wireless LAN products are not designed for use with high-gain directional antennas. Use of such antennas with these products in a manner other than as described in the previous section titled, "Antenna Use" is illegal.

Device	Separation Distance from Manufacturer Documentation
Sprint U301 USB Aircard	This modem must be 5 mm (0.2 inches) or more from the user and bystanders during operation to satisfy FCC (regulatory) RF exposure requirements. It is approved for use with normal-size notebook computers only (typically with 12-inch or larger display screens). For notebook computers with a display screen smaller than 12 inches or if the orientation of the USB slot does not allow for at least 5 mm between the modem and the user or bystanders, use the provided extension cable to ensure a safe distance between the modem antenna and users or bystanders. To meet FCC requirements, use only the supplied USB extension cable; other cables should not be used.
Sprint Mobile Broadband Card by Sierra WirelessAirCard 597E	At least 20 cm separation distance between the antenna and the user's body must be maintained at all times.

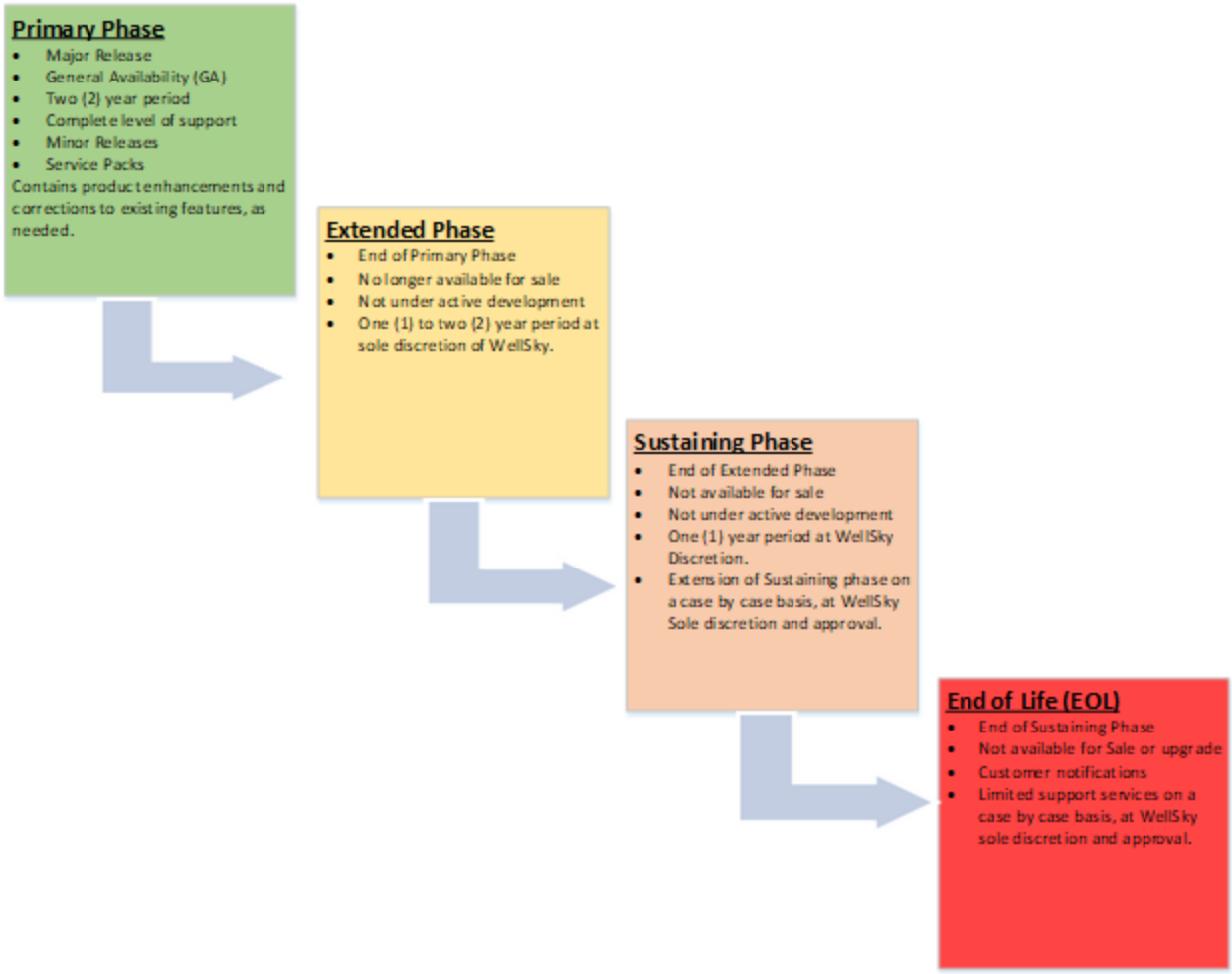
Solution Life Cycle

The WellSky® Solution Life Cycle describes the sequence of phases our solutions go through from General Availability (GA) through maturity and decline. The value of identifying the phases and applying timeframes, i.e. actual dates to each phase, allows you to plan your upgrades and migrations without any gaps in support.

To help realize the maximum value from WellSky's applications, customers should install the latest version of your WellSky solution. The latest versions allow customer(s) to benefit from the latest functionality and feature enhancements, are designed to handle modern workloads, and provide access to our most complete technical support and engineering assistance and resources.

WellSky strives to make each version easier to upgrade, and we also have a professional services team to assist with upgrade plans as well. WellSky understands that business constraints may delay customers from upgrading, and that certain customers may need to have the flexibility either to upgrade to new versions or decide to remain on an older version for a while longer. WellSky, at its discretion, may provide customers with additional support offerings that can extend certain limited features of maintenance/support coverage until the customer is able to upgrade.

Milestone	Estimated Schedules
Major Release	Every two (2) years, at a minimum
Minor Release	At least one during the life cycle
Service Packs	One (1) – two (2) per life cycle, according to need during the life cycle


Primary Phase
Major Release
General Availability (GA)
Two (2) year period
Complete level of support
Minor Releases
Service Packs
Contains product enhancements and corrections to existing features, as needed.
Extended Phase
End of Primary Phase
No longer available for sale
Not under active development
One (1) to two (2) year period at sole discretion of WellSky.
Sustaining Phase
End of Extended Phase
Not available for sale
Not under active development
One (1) year period at WellSky Discretion.
Extension of Sustaining phase on a case by case basis, at WellSky Sole discretion and approval.
End of Life (EOL)
End of Sustaining Phase
Not available for Sale or upgrade
Customer notifications
Limited support services on a case by case basis, at WellSky sole discretion and approval.

1.3 Document Organization

This document is organized into the following chapters:

1. Introduction
2. Pre-Installation
3. Oracle Linux Installation
4. Oracle DBMS Installation and Database Creation
5. Blood Centers Database Configuration
6. Oracle Fusion Middleware Installation and Configuration
7. Post Blood Centers Application and Database Checklist
8. Upgrading Blood Centers
9. FastTrak Installation and Configuration

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2

Chapter 2 - Pre-Installation

2.1 Pre-installation Check List 22
2.2 Supported Equipment 23
2.2.1 Barcode Printers 23
2.2.2 Barcode Scanners 23
2.2.3 Scales 23
2.3 DB Server Installation Preparation 24
2.4 Training Area Preparation 25

2.1 Pre-installation Check List

1. Confirm that the server meets minimum hardware requirements.
    - Xeon 5460 (3.16 GHz 1333 FSB)
    - 32 GB Memory
    - Integrated RAID Controller
    - 2 x 200 GB (minimum) Disks RAID 1
    - 2 x 400 GB (minimum) Disks RAID 1
    - 2 x Gigabit Ethernet BaseT LAN Adapter
    - Network Backup System or 1 x 20/40 DDS-4
2. Confirm Ethernet connectivity to Training Area, Lab and remote, as follows:

    Ports 21, 22, 23, 80, 81, 88, 98, 443, 1085, 1156, 1381, 1521, 1550, 3128, 4444 (if using SSL), 5000, 5001, 5010, 5011, 5500, 5800-5809, 5901-5904, 7001, 8000, 8010, 8080, 8090, 8383, 8888, 9001, 9002, 9003, 9090, 9100, and 10000
3. Verify that the following software CD/DVDs are on-hand and available:
    - Oracle Database 19.3.0.0.0 (64-Bit)
    - Oracle WebLogic Server 14.1.2.0 Generic (64-Bit)
    - Oracle Fusion Middleware Forms & Reports 14.1.2.0 (64-Bit)
    - Adobe Acrobat Reader DC - can be downloaded from the Internet

Refer to Section 4.2 for instructions on downloading the Oracle Database and Fusion Middleware.

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2.2 Supported Equipment

The following equipment is supported by Blood Centers.

2.2.1 Barcode Printers

- Digi-Trax Zebra BB-ZT410E
- Digi-Trax Zebra BB-ZM400+
- Digi-Trax Zebra Z-4M BBE+
- Digi-Trax Zebra 4000 BB

2.2.2 Barcode Scanners

2D/3D Scanners

- Symbol DS6708-SR20007ZZR
- Zebra DS8108-HC40008VZWW

1D Scanners

- Symbol LS4004-i049
- Symbol LS2208-SR20371R (No Codabar concatenation)
- Symbol LS2208-SR20401R
- Symbol LS4208 (No ISBT concatenation)
- Welch Allyn IT3800
- Welch Allyn ST5700

2.2.3 Scales

- Sartorius TE6100
- Sartorius BL6
- Sartorius PT6
- AND EK 6000G
- Mettler Toledo Viper PM BM 6

2.3 DB Server Installation Preparation

1. Verify that the appropriate UPS power has been installed and has been charged for at least 24 hours.
2. Verify Warranty Cards have been collected.
3. Verify space for the new equipment has been identified and meets the following requirements:
    - The space is protected from environmental hazards including heat, cold, moisture, dust, static, and fire.
    - The space is secure.
    - The server is rack-mounted.
4. Verify that the following network requirements are met:
    - Network cables have been run to the location of the new equipment.
    - Network cables are CAT-5, 100 base TX, with RJ-45 connectors.
    - Available network ports have been identified and configured for the new equipment.
    - The network has sufficient available capacity to support the planned Blood Centers servers and users. (The network should have capacity and speed to routinely transmit up to 1 megabyte form to each Blood Centers user without delaying the user's workflow.)
5. If not using Network Backup, verify that sufficient media has been purchased for system backups.
6. Verify that DDS-3 or DDS-4 DAT tapes are available for initial backups.
7. Procure host names, DNS names, IP addresses, and network submasks for configuring DB server, as follows:
    - 3 IP addresses for DB server (NIC#1, NIC#2, PPP)
    - Blood Centers application servers
    - remote access servers
    - network submask
    - network gateway
    - primary and secondary DNS servers
    - primary and secondary WINS servers
    - DNS domain name
8. Obtain a list of required DNS names and IP addresses for the following:
    - label printers
    - network printers
9. Obtain a list of Blood Centers application users.

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2.4 Training Area Preparation

An adequate location for training must be available. It must have enough space to accommodate the following:

- The Trainer
- The Trainer's laptop or PC
- The trainees
- The trainees' laptops or PCs
- Sufficient network connections
- Network printers (or available)

Each trainee's PC must meet the following minimum requirements:

- Lenovo T60 (or equivalent)
- 2.5 GHz dual core processor
- 16 GB RAM memory
- 150 GB hard disk
- 10/100 Ethernet Network Interface
- Battery (if laptop)
- Microsoft Windows 11
- Microsoft Edge
- Google Chrome with latest update
- Adobe Acrobat Reader DC

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3

Chapter 3 - Oracle Linux Installation

3.1 Oracle Linux Installation Procedure 28
3.2 Tmp.conf Reconfiguration 35

3.1 Oracle Linux Installation Procedure

Perform the following procedure to install the Linux operating system.

NOTE – This Guide has Red Hat Linux 9 installation. There is no difference between Oracle Linux 9 and Red Hat Linux in terms of installation or qualification. Oracle Linux 9.5 64 bit is qualified version for BC 2026.

NOTE – If you are installing Blood Centers on two servers (one for the database and one for the application), this procedure must be completed on both servers.

NOTE – If this procedure has already been performed on the server, it does not need to be performed a second time.

NOTE – Chapters 3-7 cover a new Blood Centers installation. If you are upgrading an existing Blood Centers installation, proceed to Chapter 8.

NOTE – On the 2 x 200 GB (minimum) portion of RAID 1, you will be installing the operating system, Oracle, and the Blood Centers applications.
On the 2 x 400 GB (minimum) portion of RAID 1, you will be installing the Oracle databases and temporary Hot and Cold Backups.

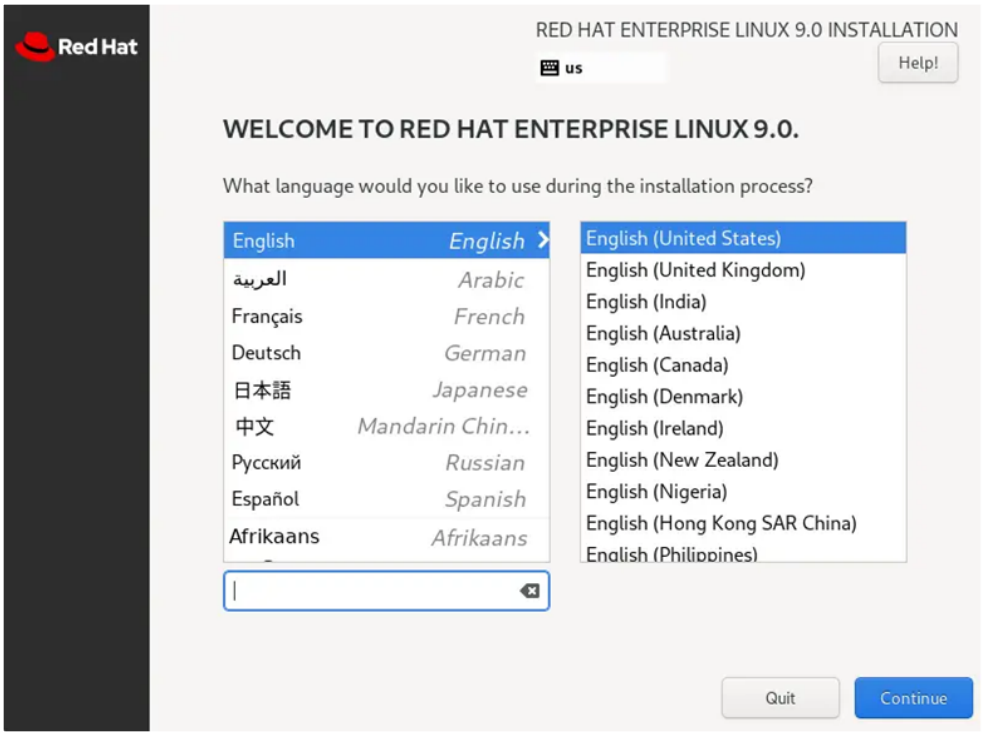
NOTE – If you only have RAID 5, configure four drives, leaving one drive as a spare.

1. From the Oracle Linux 9.0 Install menu, select Install Oracle Linux 9.0, as shown in the following figure.

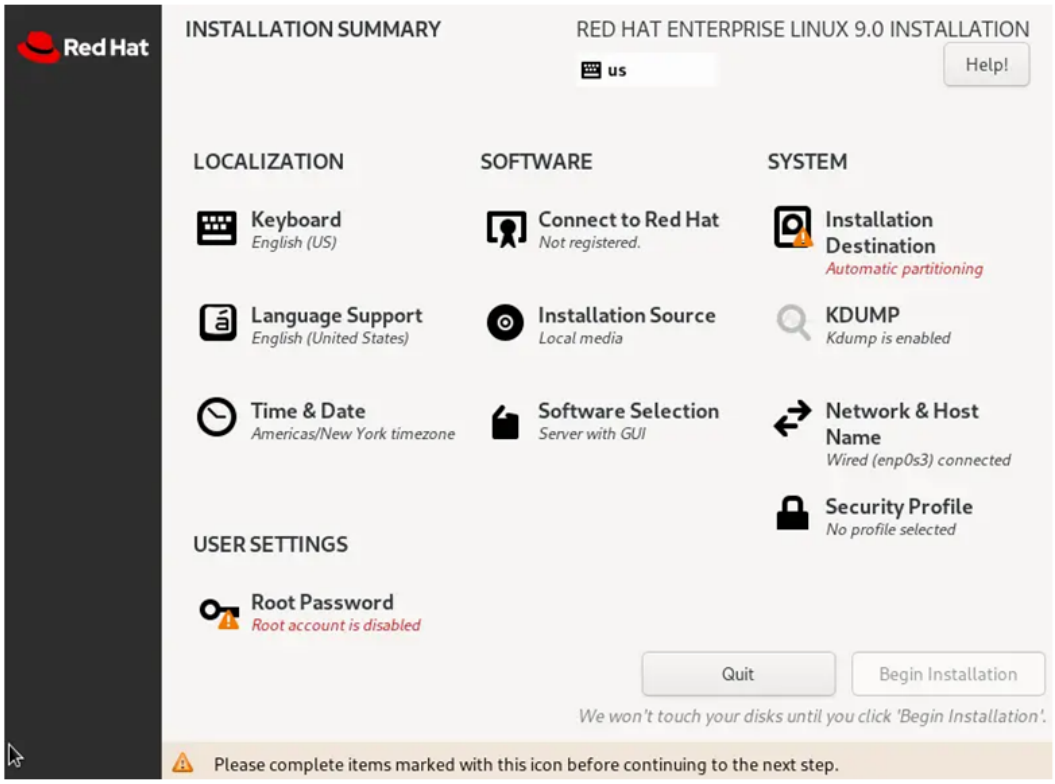


2. Choose a Language for RHEL 9 Installation. Select your preferred language which will be used during OS installation, as shown in the following figure.

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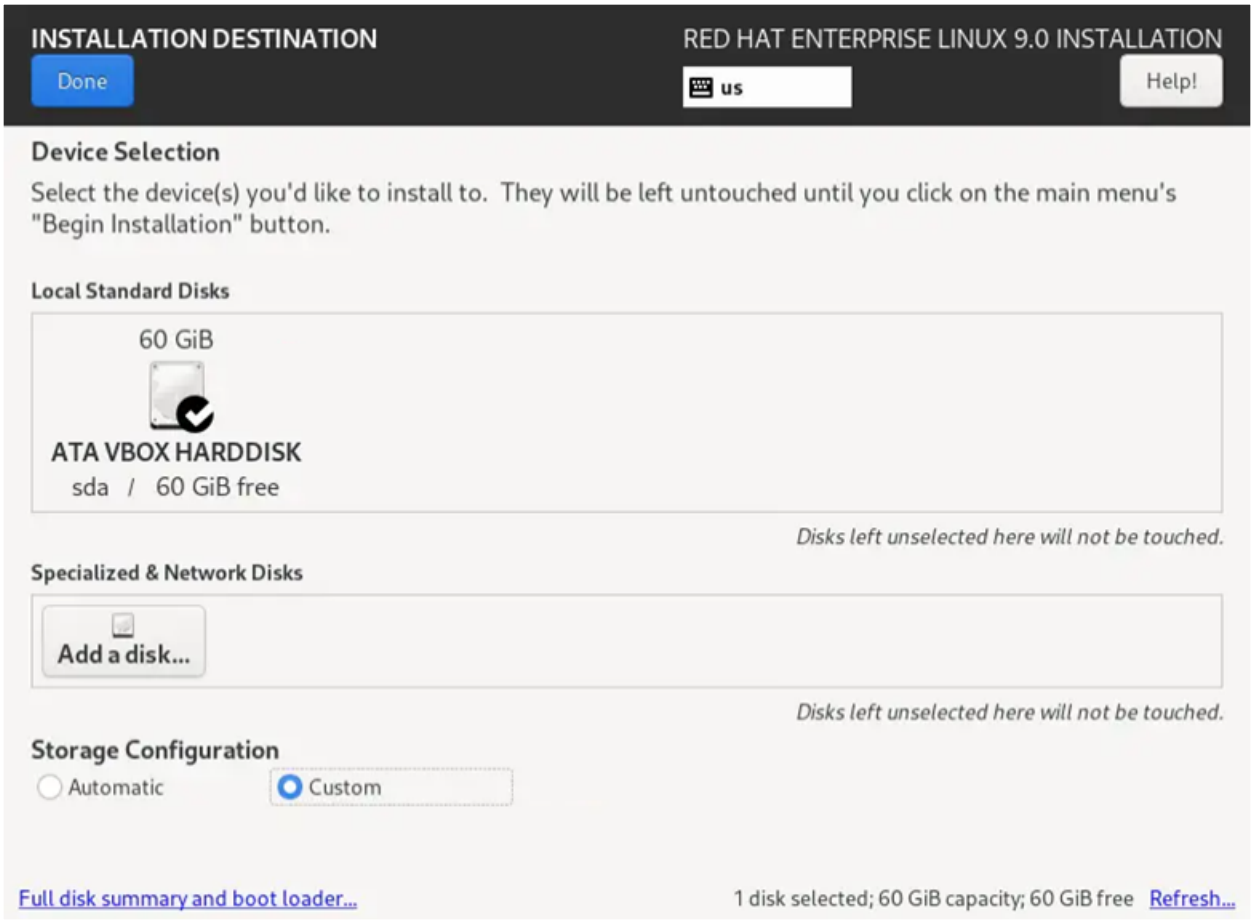


3. Click **Continue**
4. On the Installation Summary screen click **Installation Destination**.



Important! As it is mentioned clearly, you must first complete mandatory items before initiating the installation.

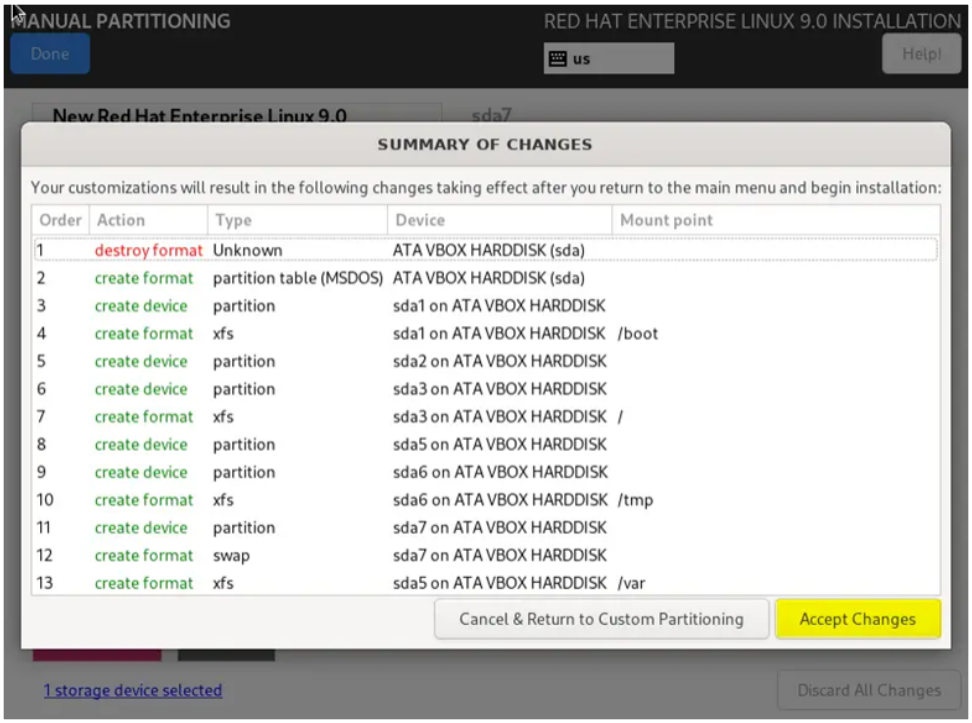
5. On the Manual Partitioning screen, click the **+** icon to Configure partitions based on allocated disk size.



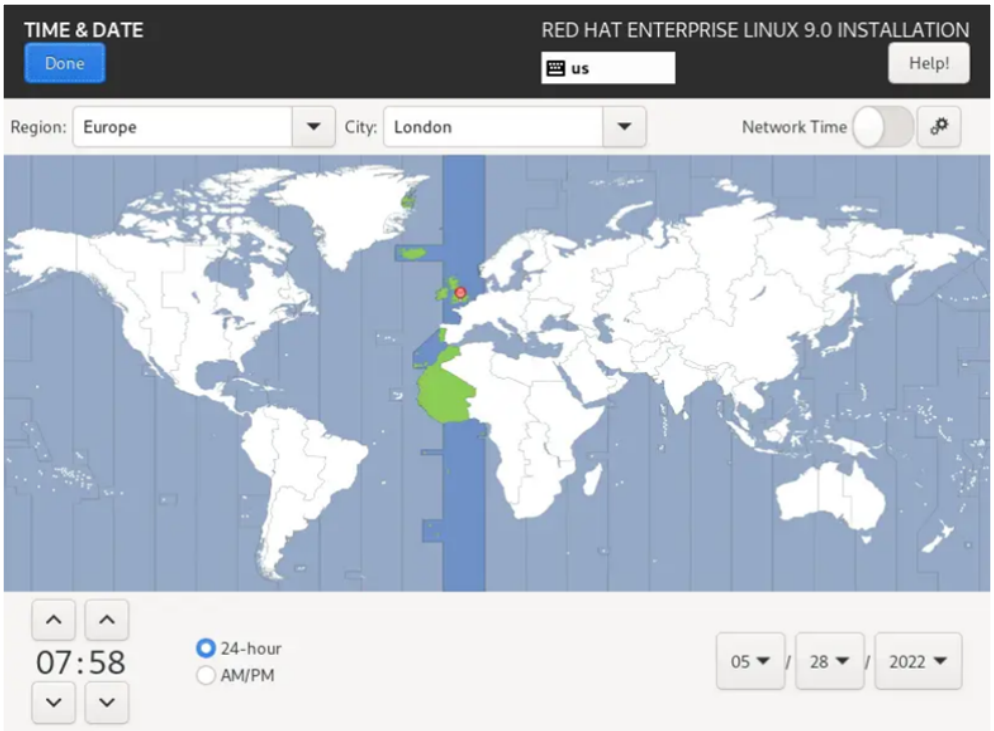
6. Under *Storage Configuration*, choose the **Custom** option then click **Done**.
7. On the Manual Partitioning screen, click the **+** icon to add the below partitions.
    - /boot – 5GB
    - /root – 100 GB
    - /home – 50GB
    - /var – 25 GB
    - /usr – 25 GB
    - Swap – 16 GB
    - /apps – 100 GB
    - /oradata – 280 GB

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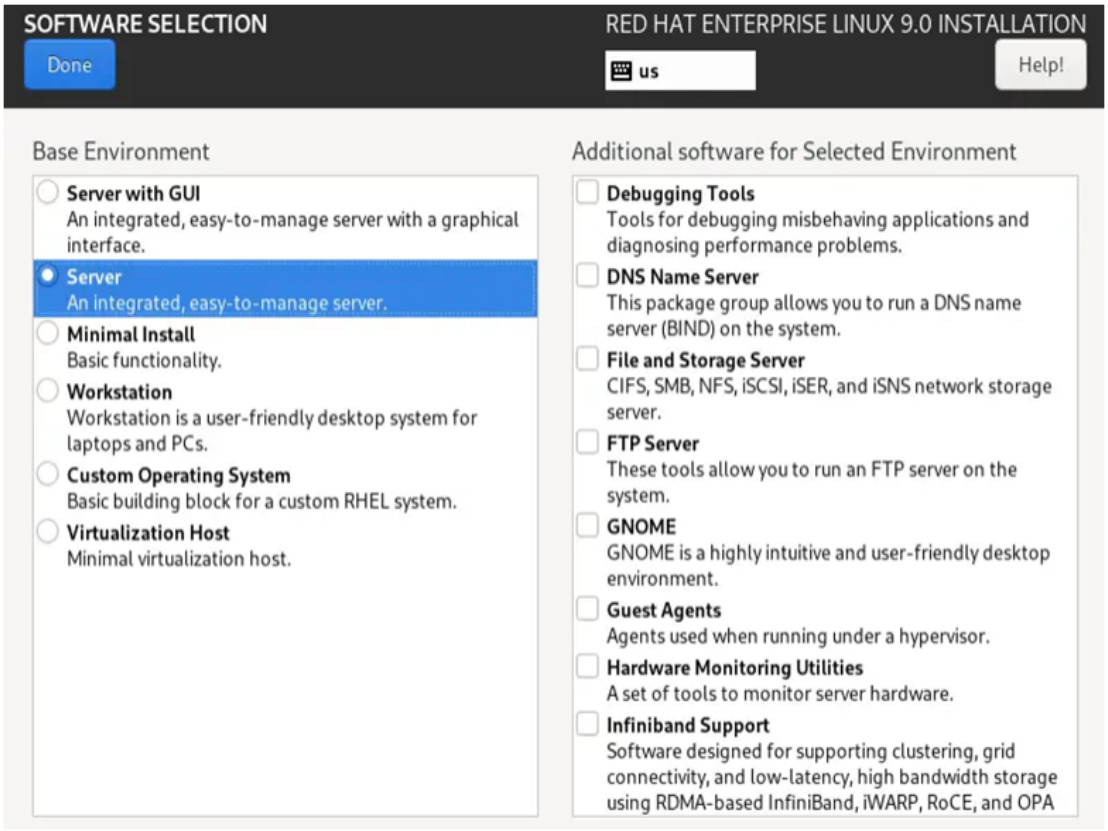
8. Click **Accept Changes**. This will take you to the Installation Summary screen.
9. The Summary of Changes screen displays, as shown in the following figure.



10. To change the Time-Zone, click **Time & Date**, choose the location as per your setup, then click **Done**.

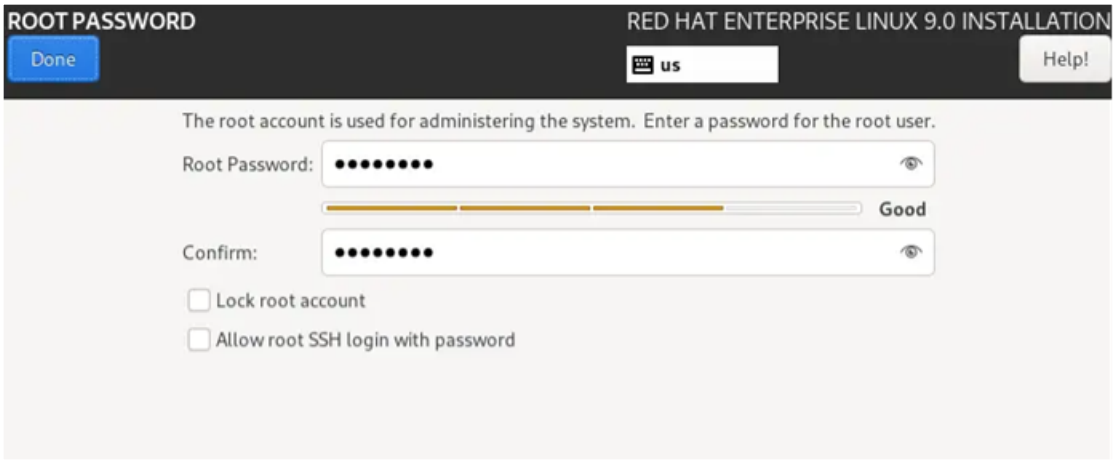


11. On the Software Selection screen, for each option in the *Base Environment* section, select the appropriate options in the *Additional software for Selected Environment* section, as shown in the following figure.



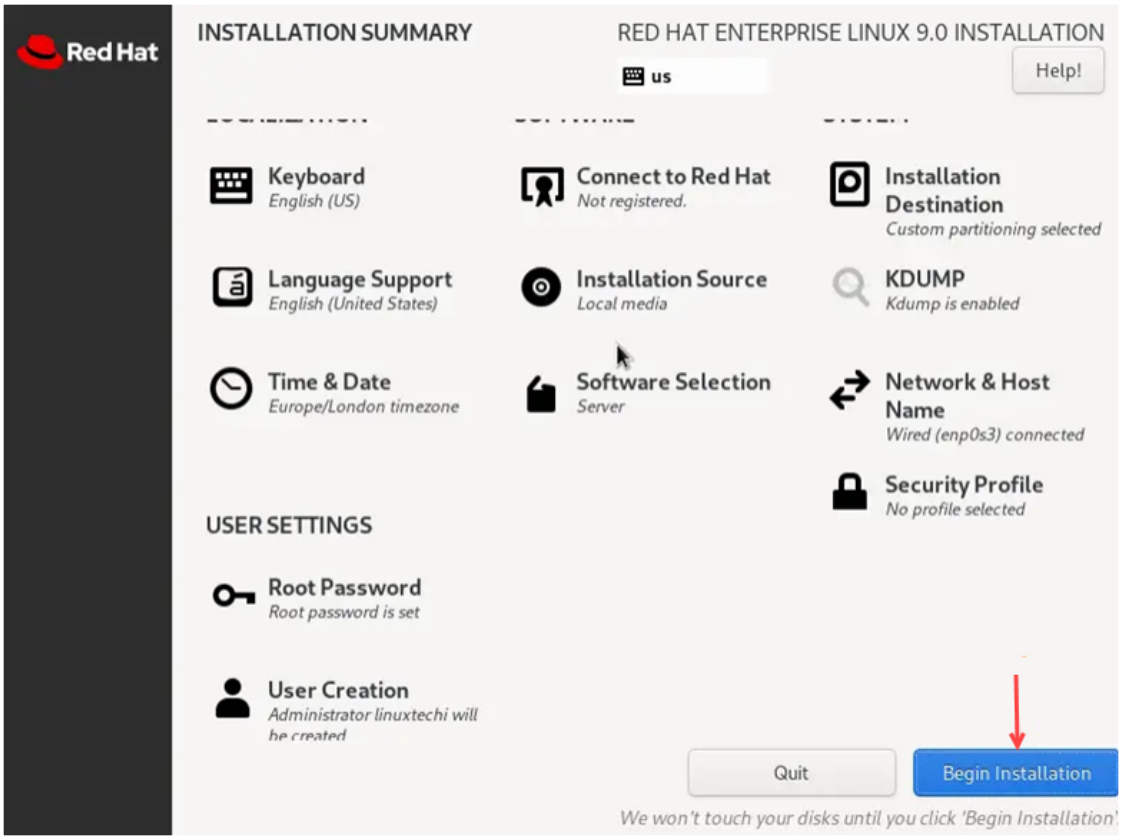
12. In *Base Environment*, select **Server with GUI**. In *Additional Software for Selected Environment*, select the **Graphical Administration** tools.

13. Click **Root Password** to set the password for the root user then click **Done**.

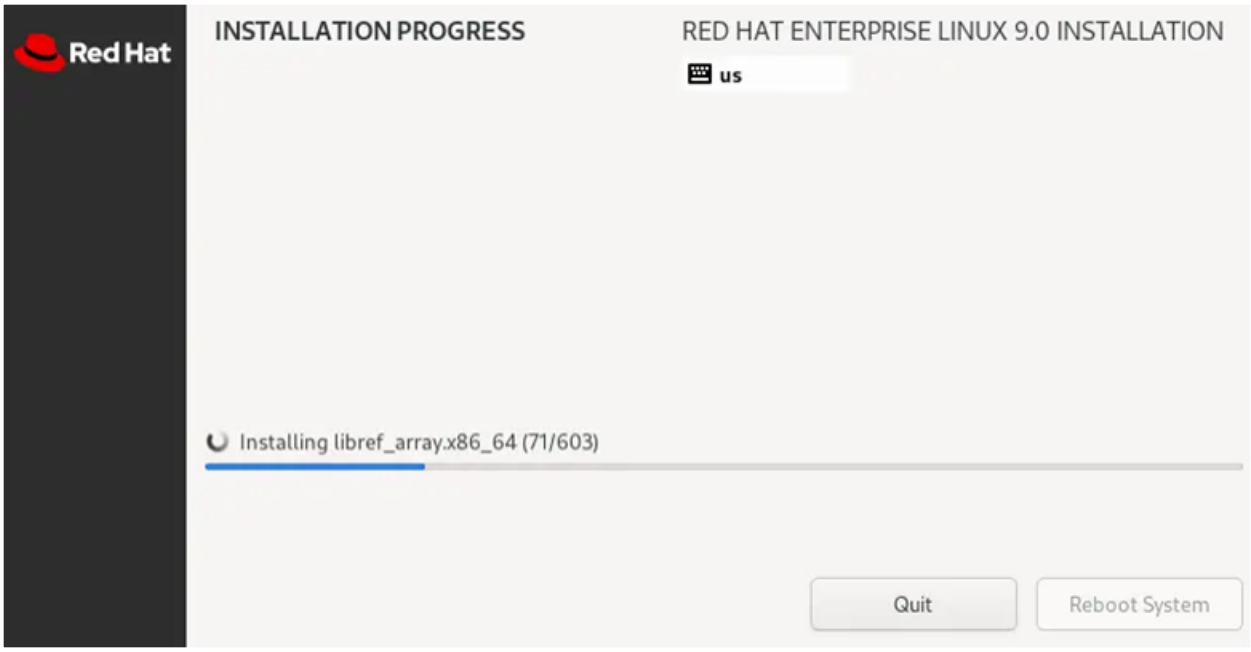


14. On the Installation Summary screen, click **Begin Installation**.

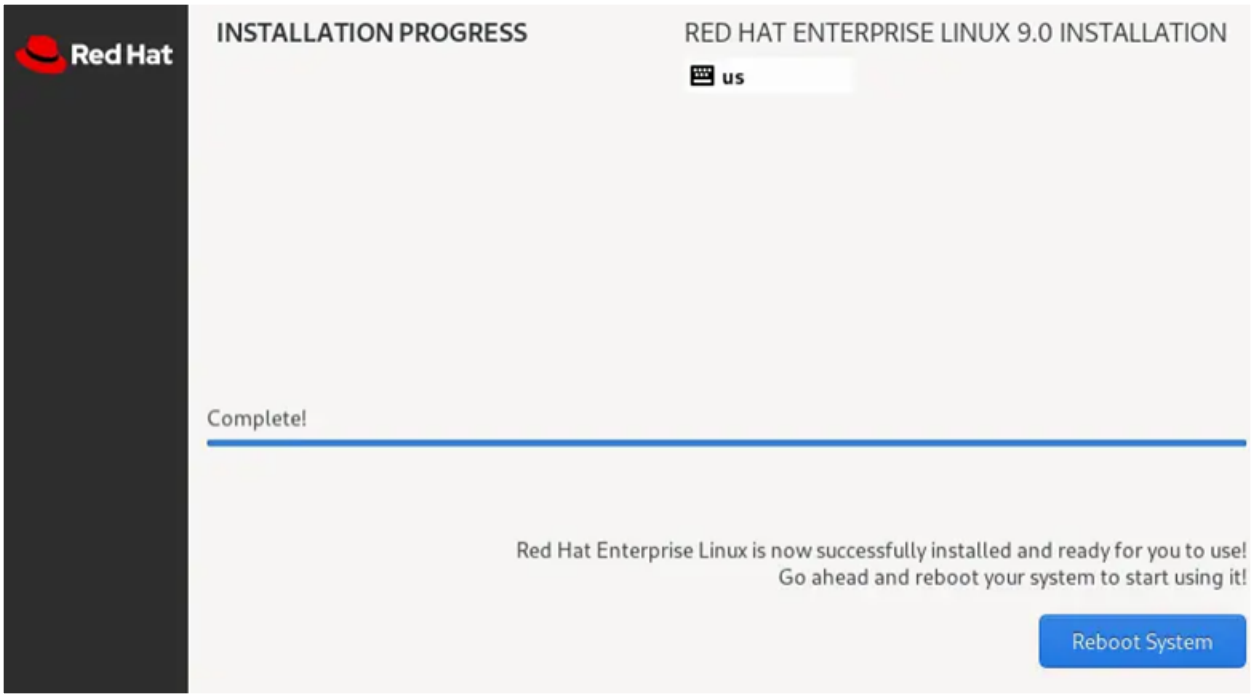
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15. At this step, RHEL 9 installation has started and is in progress.



16. Once the installation is completed, the installer will instruct you to reboot the system.



17. Click **Reboot System** to restart the system.

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3.2 Tmp.conf Reconfiguration

NOTE – If this procedure has already been performed on the server, it need not be performed a second time. Proceed to the next section.

It is necessary to prevent the Linux OS from deleting **/tmp/** directories and files. The following procedure will perform this function.

NOTE – If you are installing Blood Centers on two servers (one for the database and one for the application), this procedure must be completed on both servers.

1. Right-click on the desktop and select **Open Terminal**.
2. Change directories by entering the following command:
   `#cd /usr/lib/tmpfiles.d`
3. Open the file for editing by entering this command:
   `#vi tmp.conf`
4. Edit entries in tmp.conf to read as follows.

   NOTE – Changes are highlighted in yellow.

```
# Exclude namespace mountpoints created with PrivateTmp=yes
x /tmp/systemd-private-%b-*
X /tmp/systemd-private-%b-*/tmp
x /var/tmp/systemd-private-%b-*
X /var/tmp/systemd-private-%b-*/tmp
x /tmp/prddb_util
X /tmp/prddb_util
x /tmp/qasdb_util
X /tmp/qasdb_util
x /tmp/tstdb_util
X /tmp/tstdb_util
x /tmp/lblprt.log
X /tmp/lblprt.log
x /tmp/webutil/log
X /tmp/webutil/log
```

5. Save and close the file.
6. Close the terminal window.

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4

Chapter 4 - Oracle DBMS Installation and Database Creation

4.1 Pre-Oracle Installation Procedure 38
4.2 Download Oracle Database 44
4.3 Oracle Installation 45
4.4 Oracle Database Creation 56

4.1 Pre-Oracle Installation Procedure

1. Log on as **root**.
2. Right-click on the desktop, and select **Open Terminal** from the menu.
3. Update the **/etc/hostname** file with the Fully Qualified Domain Name (FQDN), in the following format:
   `bc2026r1.wellsky.com`
4. Remove or comment out the lines of /etc/hosts default values for localdomain4 (ipv4) and localdomain6 (ipv6).
5. Update the /etc/hosts file with the exact /etc/hostname file value and the IP address of the server using the following format:
   ```
   #127.0.0.1 localhost localhost.localdomain localhost4 localhost4.localdomain4
   #::1 localhost localhost.localdomain localhost6 localhost6.localdomain6
   ```

127.0.0.1	**localhost.localdomain**	localhost
172.31.10.230	**bc2026.wellsky.com**	bc2026

6. Insert the Blood Centers software CD in the CD-ROM drive.
7. Create the **oradata** directory by entering the following command:
   `#mkdir /oradata`
8. Change to the **/oradata** directory by entering the following command:
   `#cd /oradata`
9. Copy **bc2026r1.tar.gz** from the Software CD to the **/oradata** directory.
10. Extract the **bc2026r1.tar.gz** file by entering the following command:
    `#tar -xvzf bc2026r1.tar.gz`
11. Disable the firewall by entering the following commands.
    ```
    #service firewalld stop
    #systemctl disable firewalld
    ```

    NOTE – You can enable the firewall once the server is up and running and you have opened the specific ports.

12. In the terminal window, change the directory to the root folder by entering the following command:
    `#cd /oradata/bc2026/BC_SERVER_INSTALL/root`
13. Execute the Pre_BC2026_Users_Setup.sh to create the required users (oracle, donsys, and sbcsys) and its directories, and to change the kernel parameters for Oracle by entering the following command:
    `#sh Pre_BC2026_Users_Setup.sh`

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Example:

```
[root@bc2026 root]# pwd
/oradata/bc2026/BC_SERVER_INSTALL/root
[root@bc2026 root]# sh Pre_BC2026_Users_Setup.sh
```

14. The system will prompt you to enter the passwords for the oracle, donsys, and sbcsys users. Retype the password for each user, as pictured below, to confirm.

```
Updated Server Kernel Parameters and oracle owner limits for Blood Center
Enter Password for oracle user
Changing password for user oracle.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
Enter Password for DONOR System user donsys
Changing password for user donsys.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
Enter Password for LAB System user sbcsys
Changing password for user sbcsys.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
```

15. Execute the following command as root user. This script will install the required Linux packages for the Oracle software, including tiger vnc.

```
# sh Setup_Linux9_PKG_For_Oracle.sh
```

Example:

```
[root@bc2026 root]# pwd
/oradata/bc2026/BC_SERVER_INSTALL/root
[root@bc2026 root]# sh Setup_Linux9_PKG_For_Oracle.sh
```

It will take some time to update all of the packages. When the update is complete, you should see messages similar to the following example:

```
Installed:
  libXp-1.0.3-3.el8.x86_64                motif-2.3.4-16.el8.x86_64

Complete!
Last metadata expiration check: 2:37:49 ago on Wed 05 Jan 2022 05:47:43 PM GMT.
Package make-1:4.2.1-10.el8.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
Last metadata expiration check: 2:37:52 ago on Wed 05 Jan 2022 05:47:43 PM GMT.
Package sysstat-11.7.3-6.0.1.el8.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

16. Run the RPM_Check.sh script. It will list the packages installed using the script in step 15.

```
[root@bc2026 root]# pwd
/oradata/bc2026/BC_SERVER_INSTALL/root
[root@bc2026 root]# sh RPM_Check.sh


Compare the Below package list with required packges in the install guide



tigervnc-license-1.14.1-8.el9_6_noarch.rpm
tigervnc-icons-1.14.1-8.el9_6_noarch.rpm
tigervnc-1.14.1-8.el9_6_x86_64.rpm
tigervnc-selinux-1.14.1-8.el9_6_noarch.rpm
tigervnc-server-minimal-1.14.1-8.el9_6_x86_64.rpm
tigervnc-server-1.14.1-8.el9_6_x86_64.rpm
tigervnc-server-module-1.14.1-8.el9_6_x86_64.rpm
oracle-database-preinstall-19c-1.0-1.el9_x86_64.rpm
binutils-gold-2.35.2-63.0.1.el9_x86_64.rpm
binutils-2.35.2-63.0.1.el9_x86_64.rpm
elfutils-libelf-0.192-6.el9_6_x86_64.rpm
gcc-11.5.0-5.0.1.el9_5_x86_64.rpm
glibc-gconv-extra-2.34-168.0.1.el9_6.23_x86_64.rpm
glibc-langpack-en-2.34-168.0.1.el9_6.23_x86_64.rpm
glibc-common-2.34-168.0.1.el9_6.23_x86_64.rpm
glibc-2.34-168.0.1.el9_6.23_x86_64.rpm
glibc-all-langpacks-2.34-168.0.1.el9_6.23_x86_64.rpm
glibc-headers-2.34-168.0.1.el9_6.23_x86_64.rpm
glibc-devel-2.34-168.0.1.el9_6.23_x86_64.rpm
glibc-common-2.34-168.0.1.el9_6.23_x86_64.rpm
ksh-1.0.6-7.0.1.el9_6.1_x86_64.rpm
libaio-0.3.111-13.el9_x86_64.rpm
libaio-devel-0.3.111-13.el9_x86_64.rpm
libstdc++-11.5.0-5.0.1.el9_5_x86_64.rpm
libstdc++-devel-11.5.0-5.0.1.el9_5_x86_64.rpm
make-4.3-8.el9_x86_64.rpm
sysstat-12.5.4-9.0.2.el9_x86_64.rpm
[root@bc2026 root]#
```

17. Restart the Linux server by running the following command for all of the above updates:
    `# init 6`

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18. Once you have logged in, verify the Linux server name by entering the following command:
    `# hostname`

```
[root@bc2026 ~]# hostname
bc2026.wellsky.com
[root@bc2026 ~]#
```

19. Set up the VNC logins for the Oracle users for installing the Oracle database and application server software in GUI mode.
    a. a. Uncomment the following line in the /etc/gdm/custom.conf file as root user and save it:
    `WaylandEnable=false`
    b. Set the VNC password for the Oracle user:
    `# su - oracle`

    `$ vncpasswd`

    You will be prompted for the VNC password for the Oracle user. Enter it and confirm it.

```
[root@bc2024rl ~]#
[root@bc2024rl ~]# su - oracle
Last login: Wed Jan  5 21:34:03 GMT 2022 on pts/0
[oracle@bc2024rl ~]$
[oracle@bc2024rl ~]$ vncpasswd
Password:
Verify:
Would you like to enter a view-only password (y/n)? n
A view-only password is not used
[oracle@bc2024rl ~]$
```

    c. Append the Oracle user account and Xserver Display for VNC service to /etc/tigervnc/vncserver.users file.

```
[root@bc2024rl ~]# more /etc/tigervnc/vncserver.users
# TigerVNC User assignment
#
# This file assigns users to specific VNC display numbers.
# The syntax is <display>=<username>. E.g.:
#
# :2=andrew
# :3=lisa
:2=oracle
```

d. Append the default desktop and screen resolution in the **/etc/tigervnc/vncserver-config-defaults** file:
```
session=gnome
geometry=1280x1024
```

```
[root@bc2024r1 ~]# more /etc/tigervnc/vncserver-config-defaults
## Default settings for VNC servers started by the vncserver service
#
# Any settings given here will override the builtin defaults, but can
# also be overriden by ~/.vnc/config and vncserver-config-mandatory.
#
# See the following manpages for more details: vncserver(1) Xvnc(1)
#
# Several common settings are shown below. Uncomment and modify to your
# liking.

# securitytypes=vncauth,tlsvnc
# desktop=sandbox
# geometry=2000x1200
# localhost
# alwaysshared

# Default to GNOME session
# Note: change this only when you know what are you doing
session=gnome
geometry=1280x1024
```

e. By default, the VNC server terminates when the last user signs out. You can manually force the VNC server to persist by creating the **/etc/systemd/system/vncserver@.service.d/10-restart.conf** file as the root user, and then adding the following content:
```
[Service]
Restart=always
```

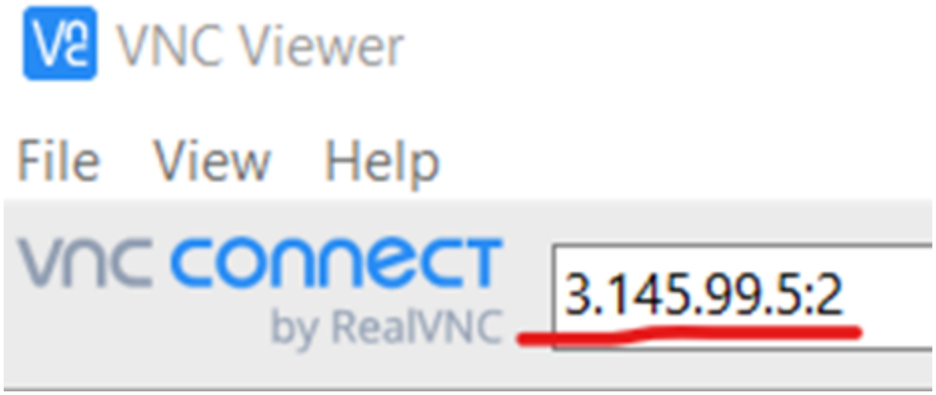
```
[root@bc2024r1 ~]#
[root@bc2024r1 ~]# mkdir -p /etc/systemd/system/vncserver@.service.d
[root@bc2024r1 ~]# vi /etc/systemd/system/vncserver@.service.d/10-restart.conf
[root@bc2024r1 ~]#
[root@bc2024r1 ~]# more /etc/systemd/system/vncserver@.service.d/10-restart.conf
[Service]
Restart=always
```

f. Set up the VNC service by running the following commands:
```
# systemctl daemon-reload
# systemctl enable --now vncserver@:2.service
```

```
[root@bc2024r1 ~]#
[root@bc2024r1 ~]# systemctl daemon-reload
[root@bc2024r1 ~]# systemctl enable --now vncserver@:2.service
Created symlink /etc/systemd/system/multi-user.target.wants/vnc
[root@bc2024r1 ~]#
```


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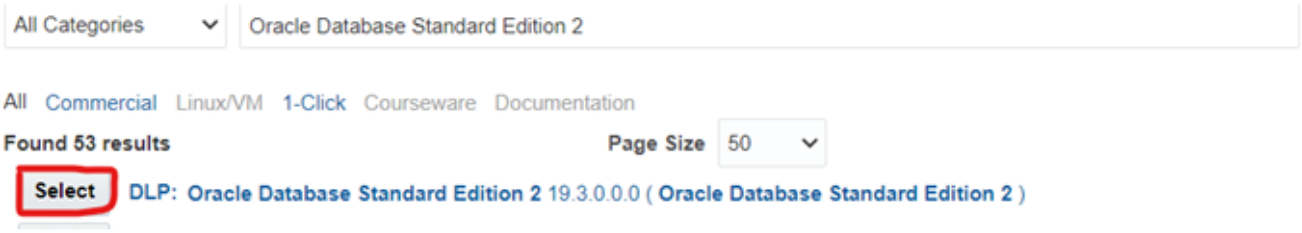
g. Open the VNC client software and run the Linux Server IP with terminal 2, as shown below:





4.2 Download Oracle Database

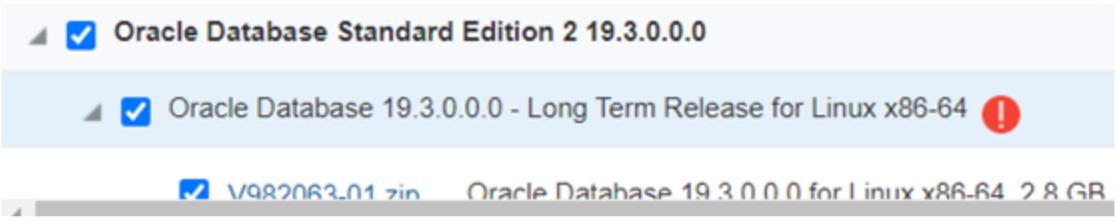
1. Open a Web browser, and navigate to https://edelivery.oracle.com/osdc/faces/Home.jspx.
2. Log in with Oracle Support Credentials.
3. Under *All Categories*, select the **License** option.
4. In the **Search** box, enter `Oracle Database Standard Edition 2`, and click **Search**.
5. Click the **DLP: Oracle Database Standard Edition 2 19.3.0.0.0 (Oracle Database Standard Edition 2)**. It will be added to the cart.



6. Click **Selected Software** on the right side of the screen.
7. On the Selected Software screen, unselect everything.
8. Select the **Oracle Database 19.3.0.0.0 - Long Term Release** option, as shown in the following figure.



9. Under *Platforms/Languages*, select **Linux x86-64** from the drop-down.
10. Click **Continue**.
11. Select the agreement, and click **Download**. The V982063-01.zip - Oracle Database 19.3.0.0.0 file is downloaded, as shown in the following figure.



12. Copy the downloaded software V982063-01.zip to the **/apps/Software** directory as an oracle user.
13. Log in to **support.oracle.com**.
14. Download **Patch 37952354: DATABASE RELEASE UPDATE 19.28.0.0.0 for Linux x86-64.**.
15. 15. Copy the downloaded p37952354_190000_Linux-x86-64.zip to the **/apps/Software/PATCHS** directory.
16. Download **Patch 6880880: OPatch 12.2.0.1.47 for DB 23.0.0.0.0 for Linux x86-64**.
17. Copy the downloaded p6880880_190000_Linux-x86-64.zip to the **/apps/Software** directory.

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4.3 Oracle Installation

NOTE – If this procedure has already been performed on the server, it need not be performed a second time. Proceed to the next section.

NOTE – If you are installing Blood Centers on two servers (one for the database and one for the application), this procedure is completed on the database server.

1. Log on as **oracle** through VNC.
2. Right-click on the desktop and select **Open Terminal**.
3. Change to the /apps/Software directory using the following command:
   $ `cd /apps/Software`

NOTE – Verify the oracle document ID 2982833.1 before starting oracle database installation. The next steps will install the Oracle Database (RDBMS) Software 19.3.0.0 64-Bit Standard Edition 2 and apply the 19.28.0.0 patch. If you want to install the Enterprise Edition or any customized installation, stop here and follow the Oracle Guides, or contact WellSky Technical Support for suggestions.

NOTE – To install Oracle Database 19c on Oracle Linux 9 or Red Hat Linux 9, you must use 19.3 Oracle Database software binaries and then apply the 19.22 release updates (RUs) during the installation process.

4. Run the below environment variables and unzip the Oracle Database (RDBMS) Software 19.3.0.0.
   mkdir -p /apps/product/19.0.0/dbhome_1
   cd /apps/product/19.0.0/dbhome_1
   unzip /apps/Software/V982063-01.zip

```
[oracle@cac2026 ~]$ mkdir -p /apps/product/19.0.0/dbhome_1
cd /apps/product/19.0.0/dbhome_1
unzip /apps/Software/V982063-01.zip
```

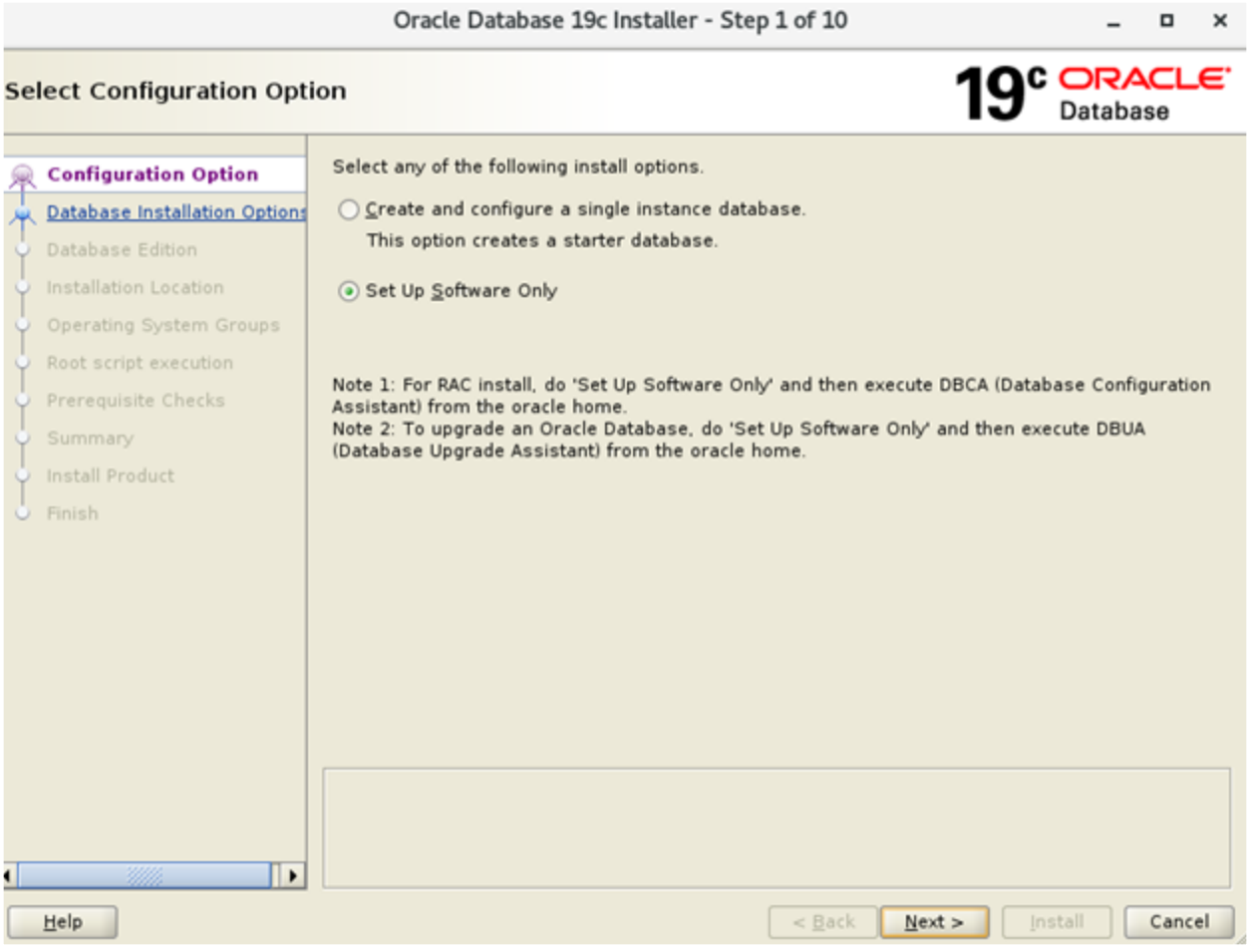
5. Backup the /apps/product/19.0.0/dbhome_1/OPatch directory as OPatch_Bkp.
6. Unzip the /apps/Software/p6880880_190000_Linux-x86-64.zip file. It extracts as OPatch directory in /apps/product/19.0.0/dbhome_1

```
[oracle@cac2026 dbhome_1]$
[oracle@cac2026 dbhome_1]$ mv OPatch OPatch_bkp
[oracle@cac2026 dbhome_1]$ unzip /apps/Software/p6880880_190000_Linux-x86-64.zip
[oracle@cac2026 dbhome_1]$ /apps/product/19.0.0/dbhome_1/OPatch/opatch version
OPatch Version: 12.2.0.1.47

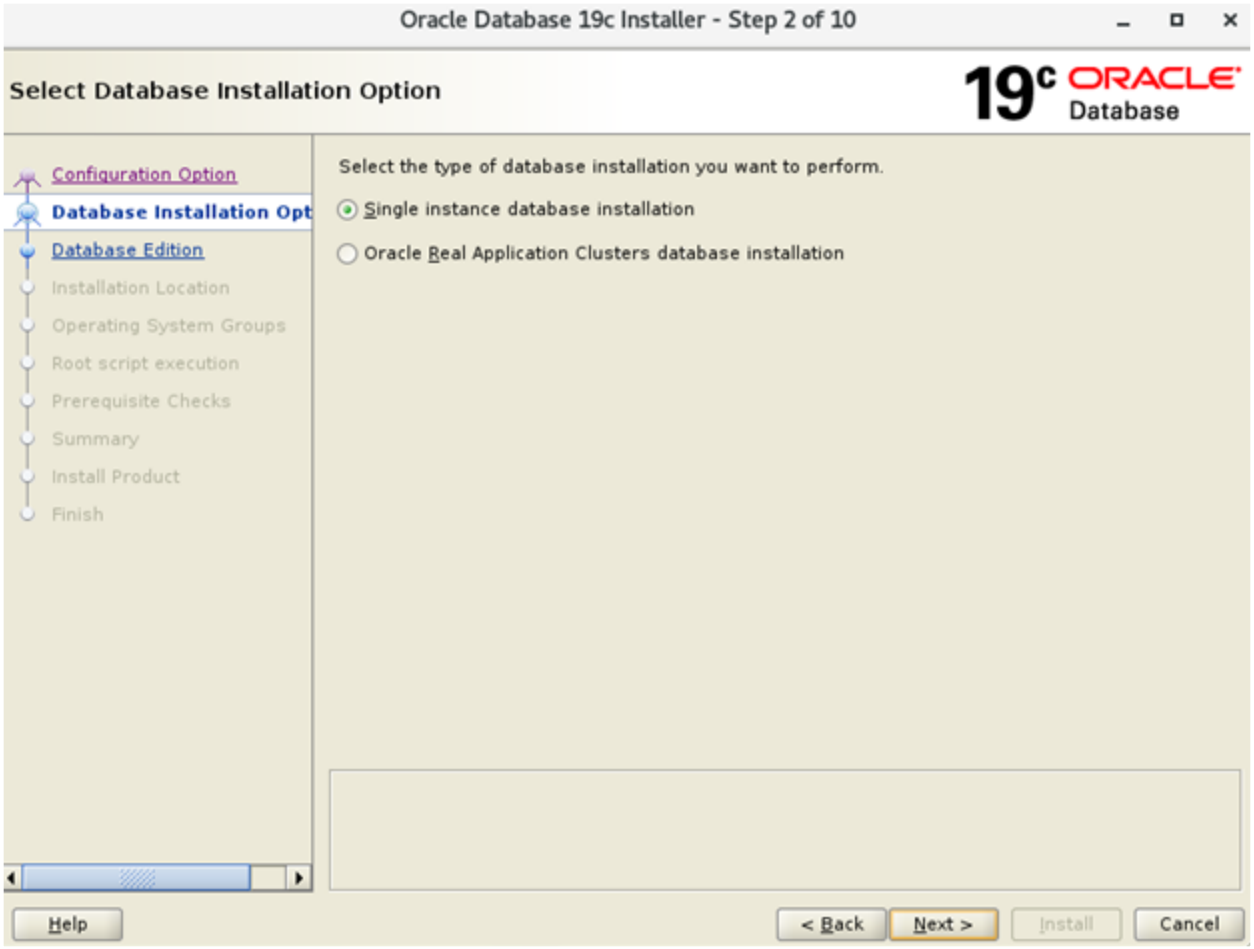
OPatch succeeded.
```

7. Change directory to /apps/Software & Extract /apps/Software/p37952354_190000_Linux-x86-64.zip file under /apps/Software/PATCHS directory. It will extract as /apps/Software/PATCHS/37952354 Directory.
8. Use the command below to install Database Software using Oracle Database 19c Patch 19.28.0.0
   $ `export CV_ASSUME_DISTID=OL8`
   $ `./runInstaller -ignorePrereq -applyRU /apps/Software/PATCHS/37952354/37960098 -applyOneOffs /apps/Software/PATCHS/37952354/37847857`

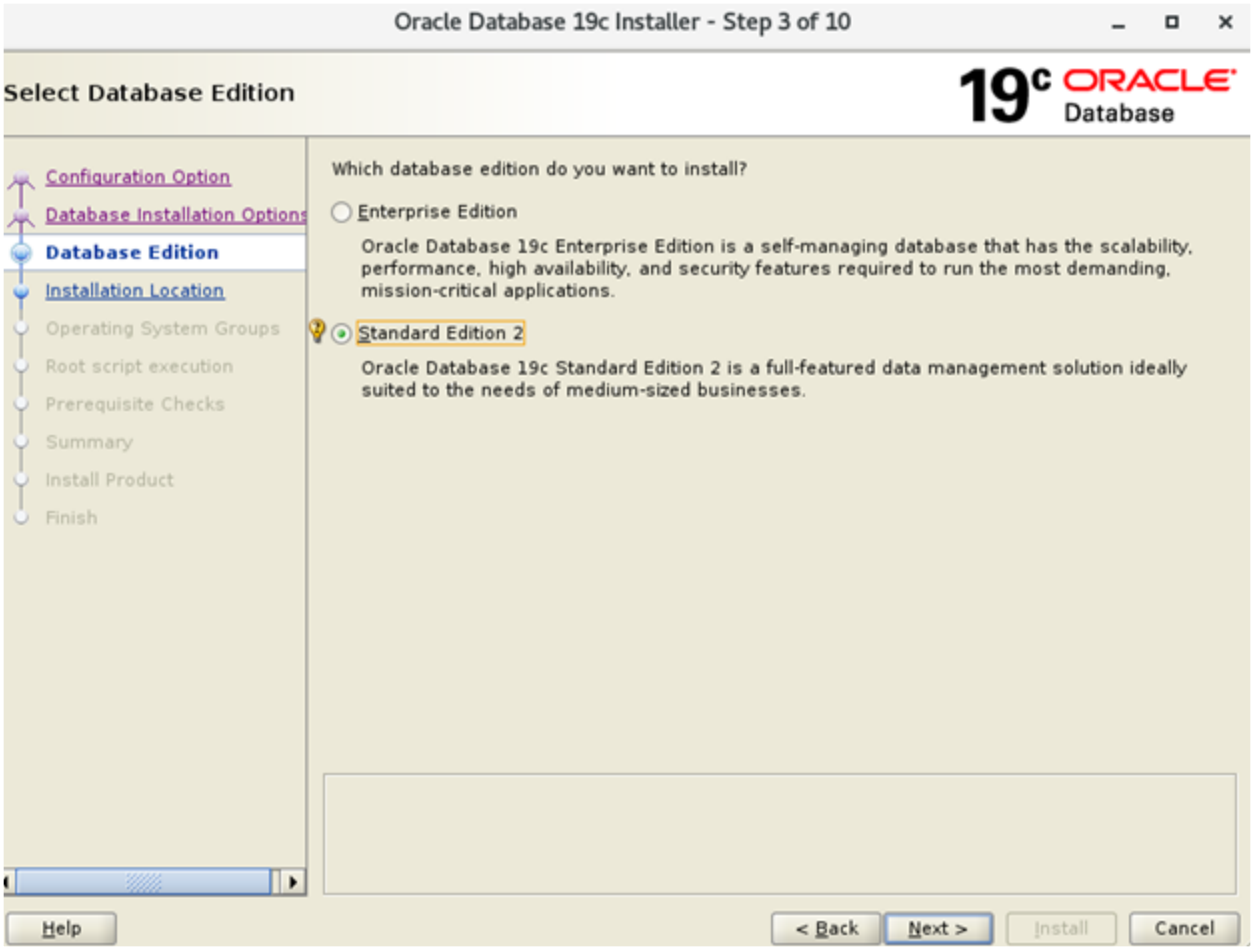
```
[oracle@cac2026 dbhome_1]$
[oracle@cac2026 dbhome_1]$ export CV_ASSUME_DISTID=OL8
[oracle@cac2026 dbhome_1]$ ./runInstaller -ignorePrereq -applyRU /apps/Software/PATCHS/37952354/3796
0098 -applyOneOffs /apps/Software/PATCHS/37952354/37847857
Preparing the home to patch...
Applying the patch /apps/Software/PATCHS/37952354/37960098...
Successfully applied the patch.
Applying the patch /apps/Software/PATCHS/37952354/37847857...
Successfully applied the patch.
The log can be found at: /apps/oraInventory/logs/InstallActions2025-09-15_02-15-45PM/installerPatchA
ctions_2025-09-15_02-15-45PM.log
Launching Oracle Database Setup Wizard...
```



9. Select the **Set Up Software Only** option then click **Next**.

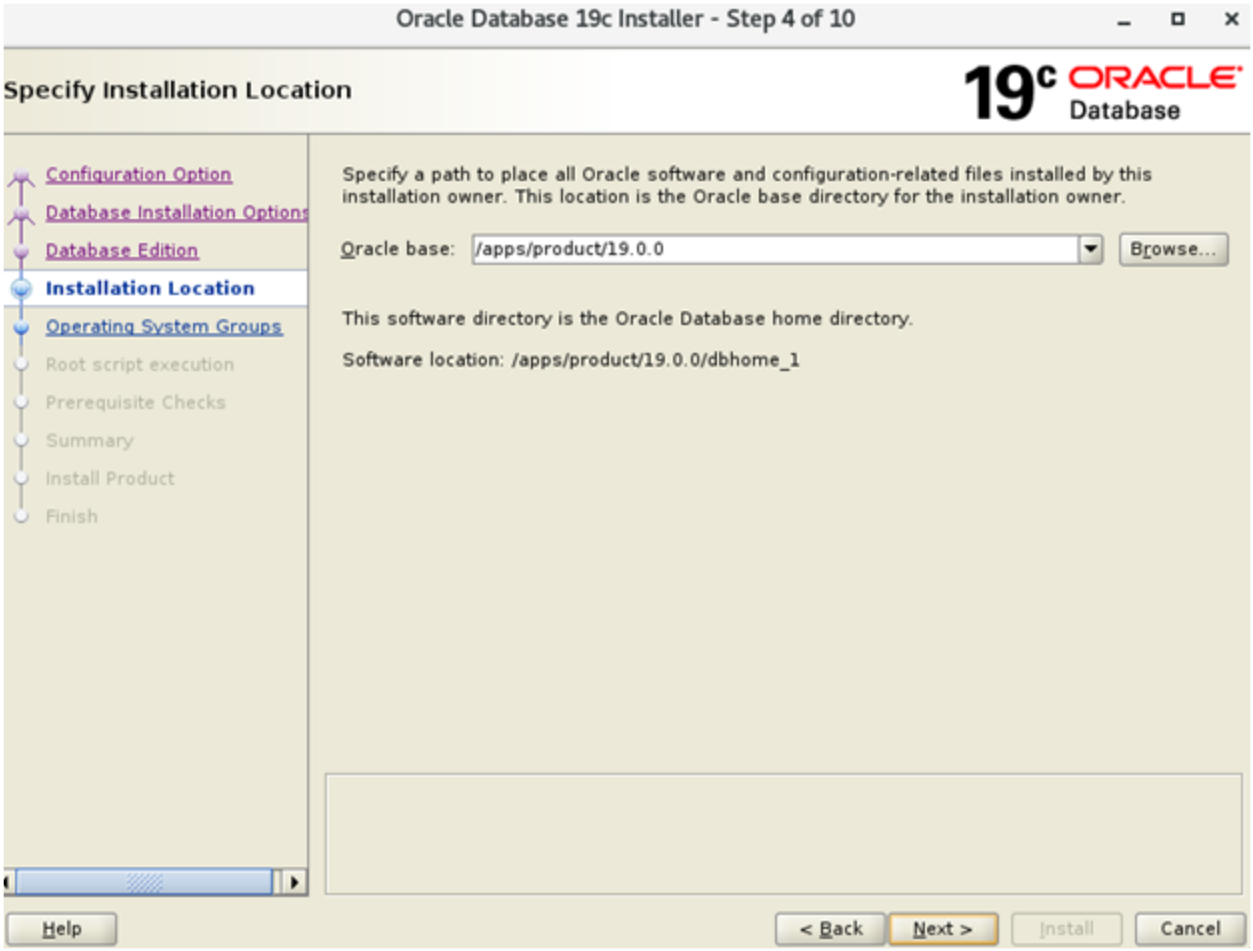


10. Select the **Single instance database installation** option then click **Next**.

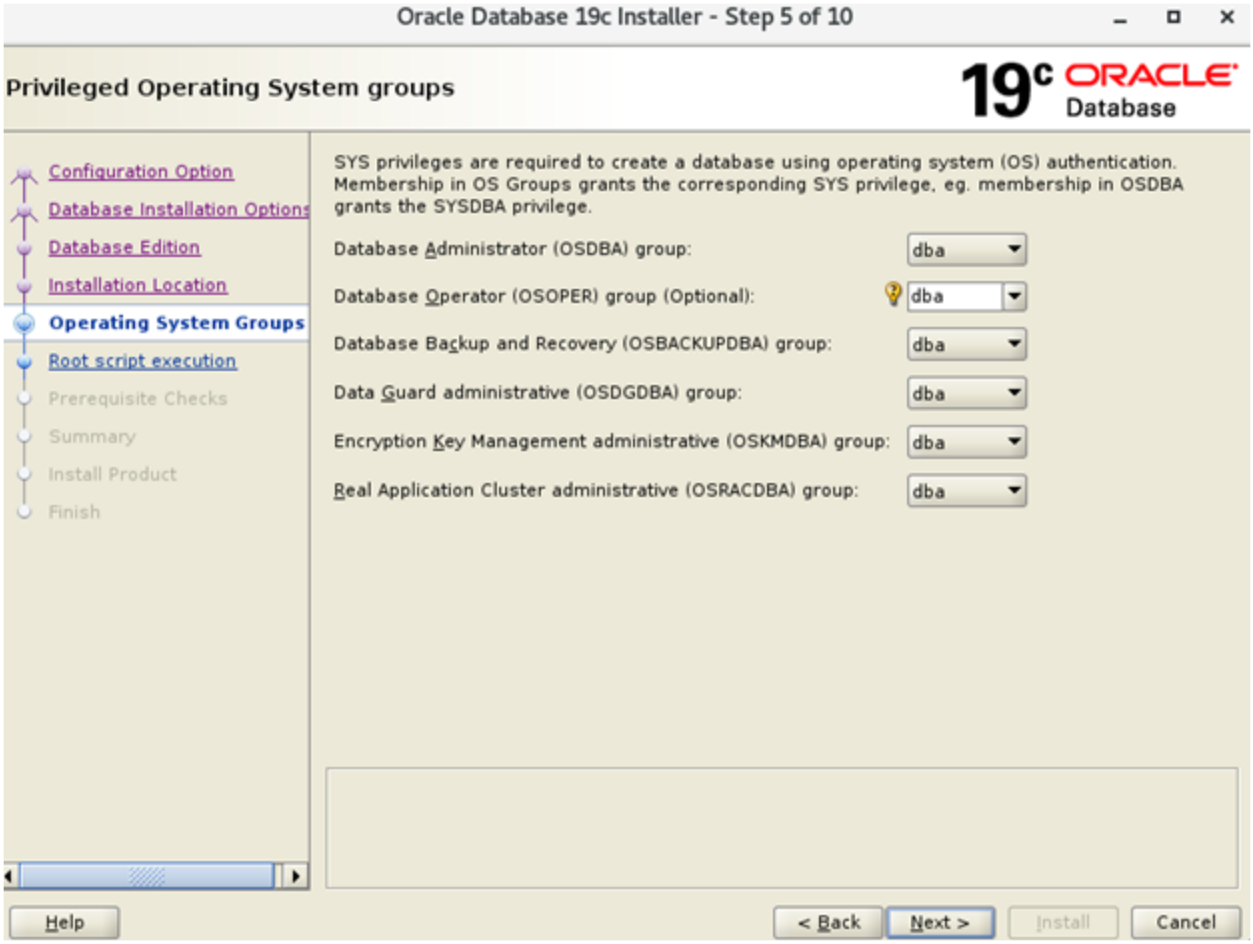


11. Select the Standard Edition 2 option then click Next.

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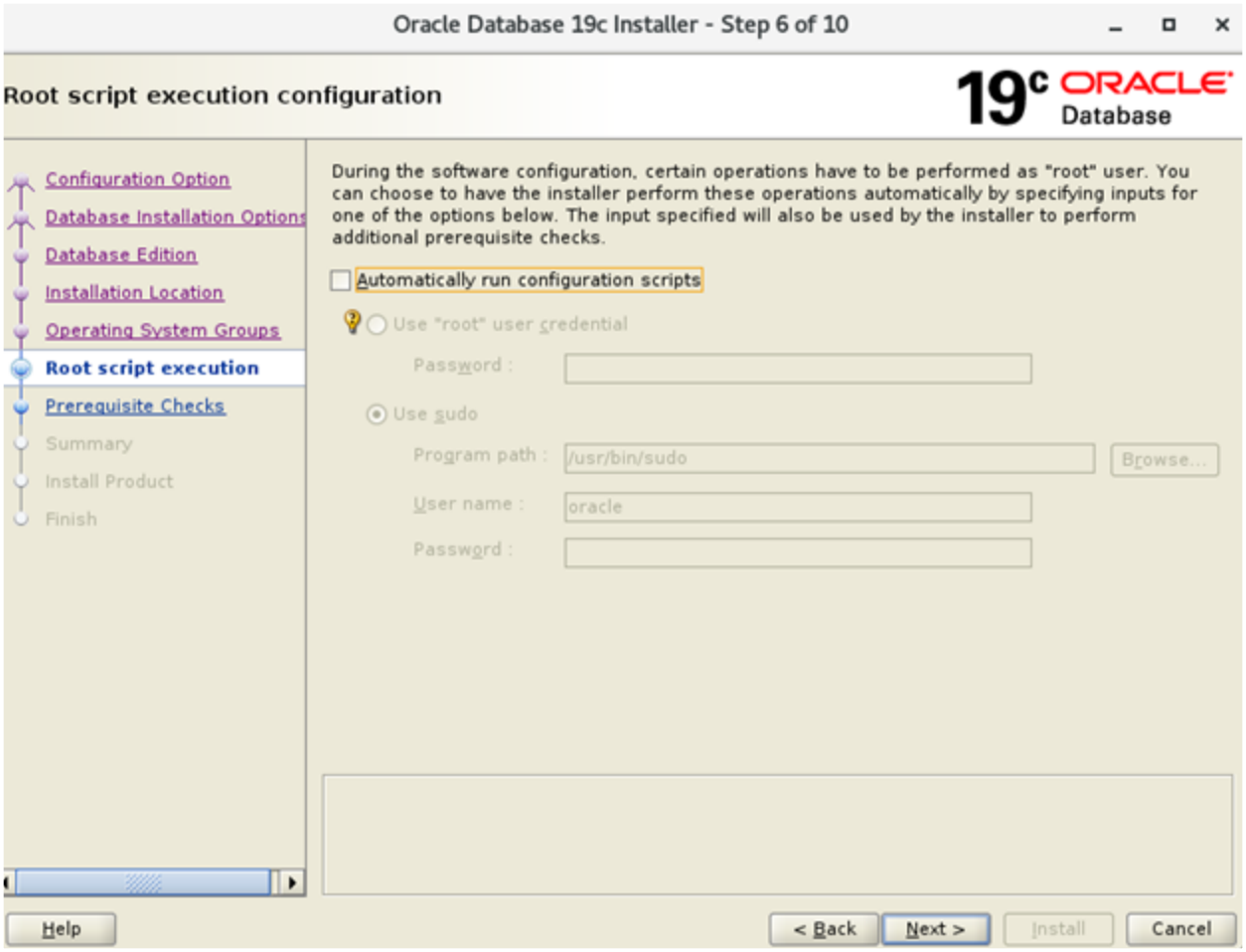


12. Enter **`/apps/product/19.0.0`** for the Oracle base, and it will set ORACLE_HOME as /apps/product/19.0.0/dbhome_1.
13. Click Next.

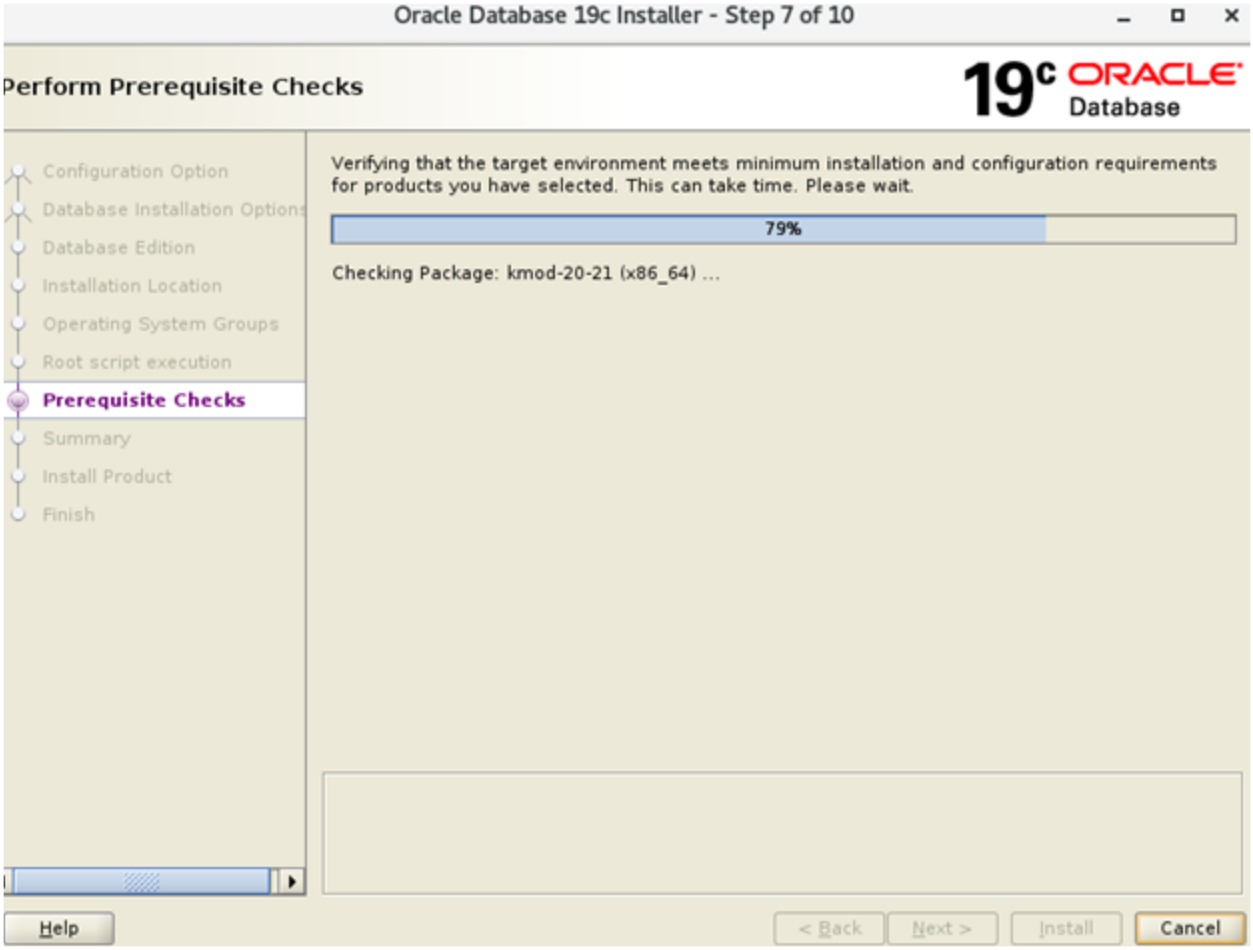


14. Select **dba** from all of the drop-down menus then click **Next**.

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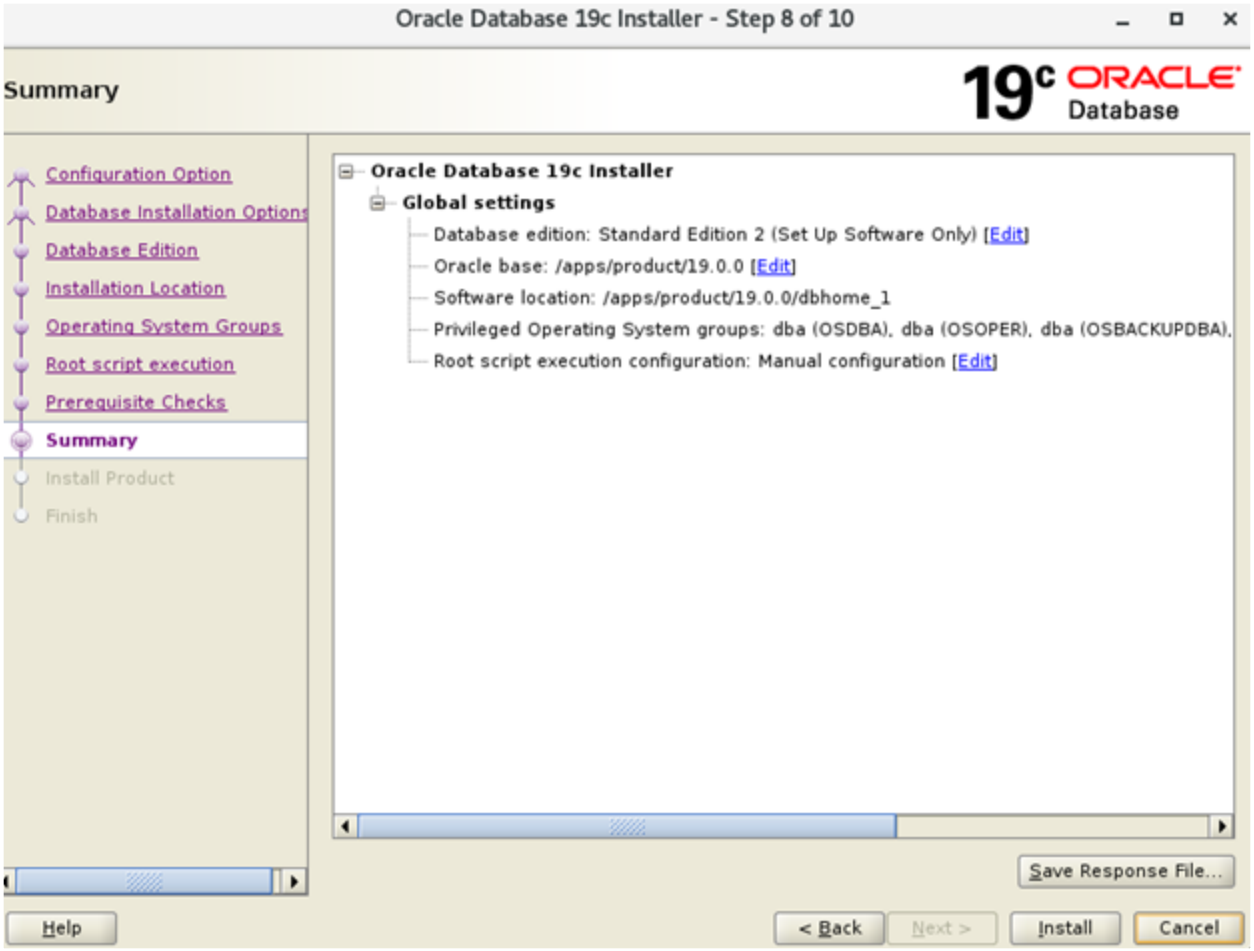


15. Click **Next**.

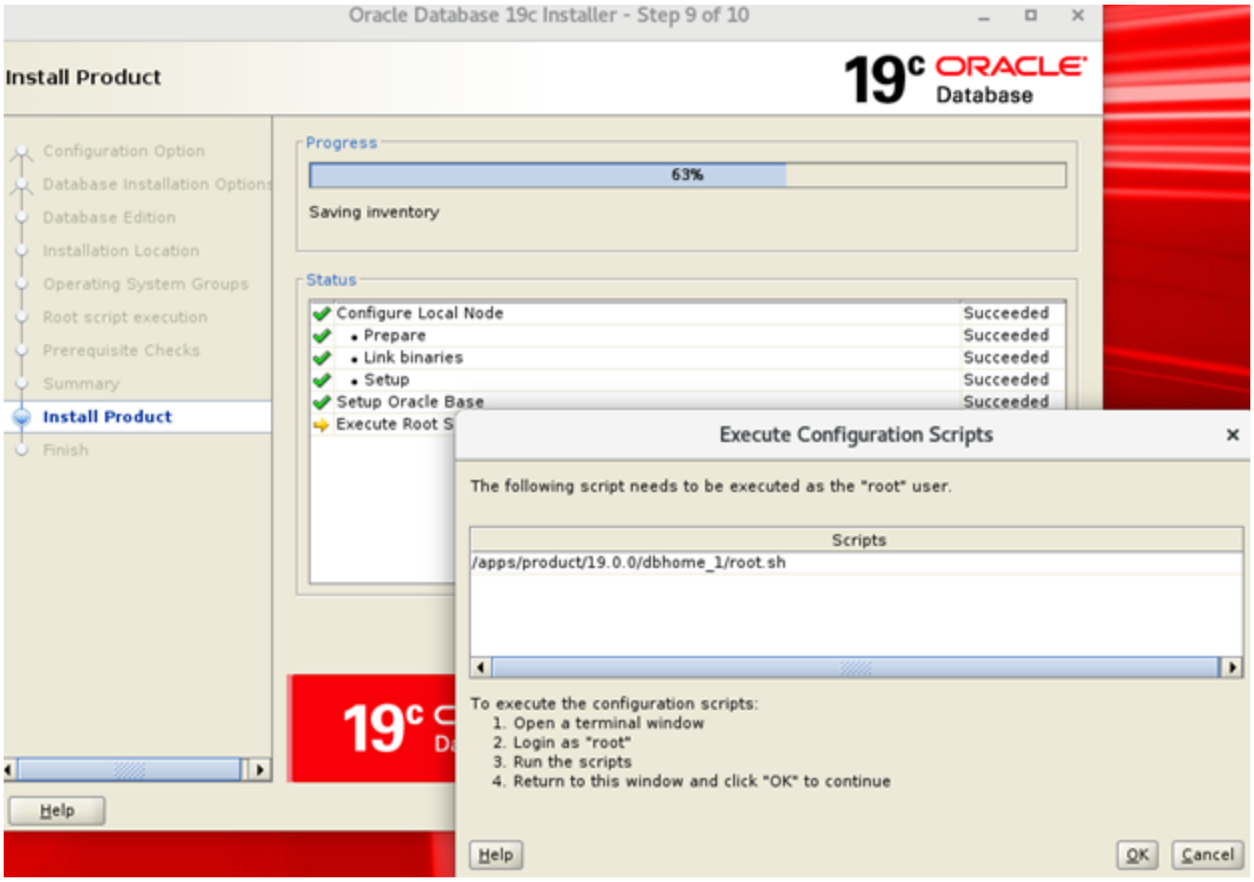


The installer will verify all of the requirements.

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16. Click **Install**.



The installer will prompt for configuration scripts to be executed as root user.

17. Open another terminal as root user, and runn the following script:
`/apps/product/19.0.0/dbhome_1/root.sh`

```
[root@oral9c ~]#
[root@oral9c ~]# sh /apps/product/19.0.0/dbhome_1/root.sh
Performing root user operation.

The following environment variables are set as:
    ORACLE_OWNER= oracle
    ORACLE_HOME=  /apps/product/19.0.0/dbhome_1

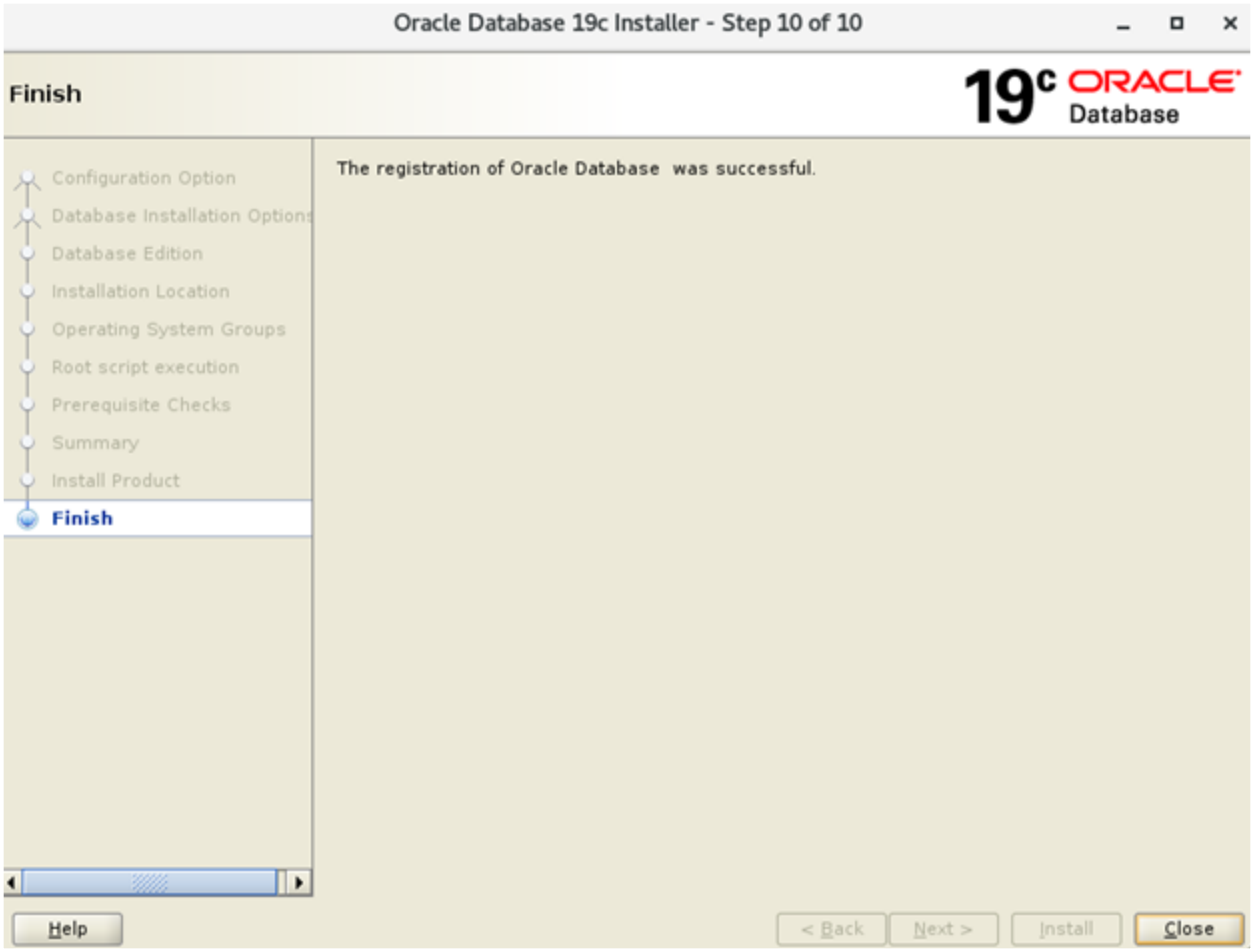
Enter the full pathname of the local bin directory: [/usr/local/bin]:
   Copying dbhome to /usr/local/bin ...
   Copying oraenv to /usr/local/bin ...
   Copying coraenv to /usr/local/bin ...


Creating /etc/oratab file...
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
Oracle Trace File Analyzer (TFA - Standalone Mode) is available at :
    /apps/product/19.0.0/dbhome_1/bin/tfactl

Note :
1. tfactl will use TFA Service if that service is running and user has been granted access
2. tfactl will configure TFA Standalone Mode only if user has no access to TFA Service or TFA is not installed
```

The required environment variables will be set.

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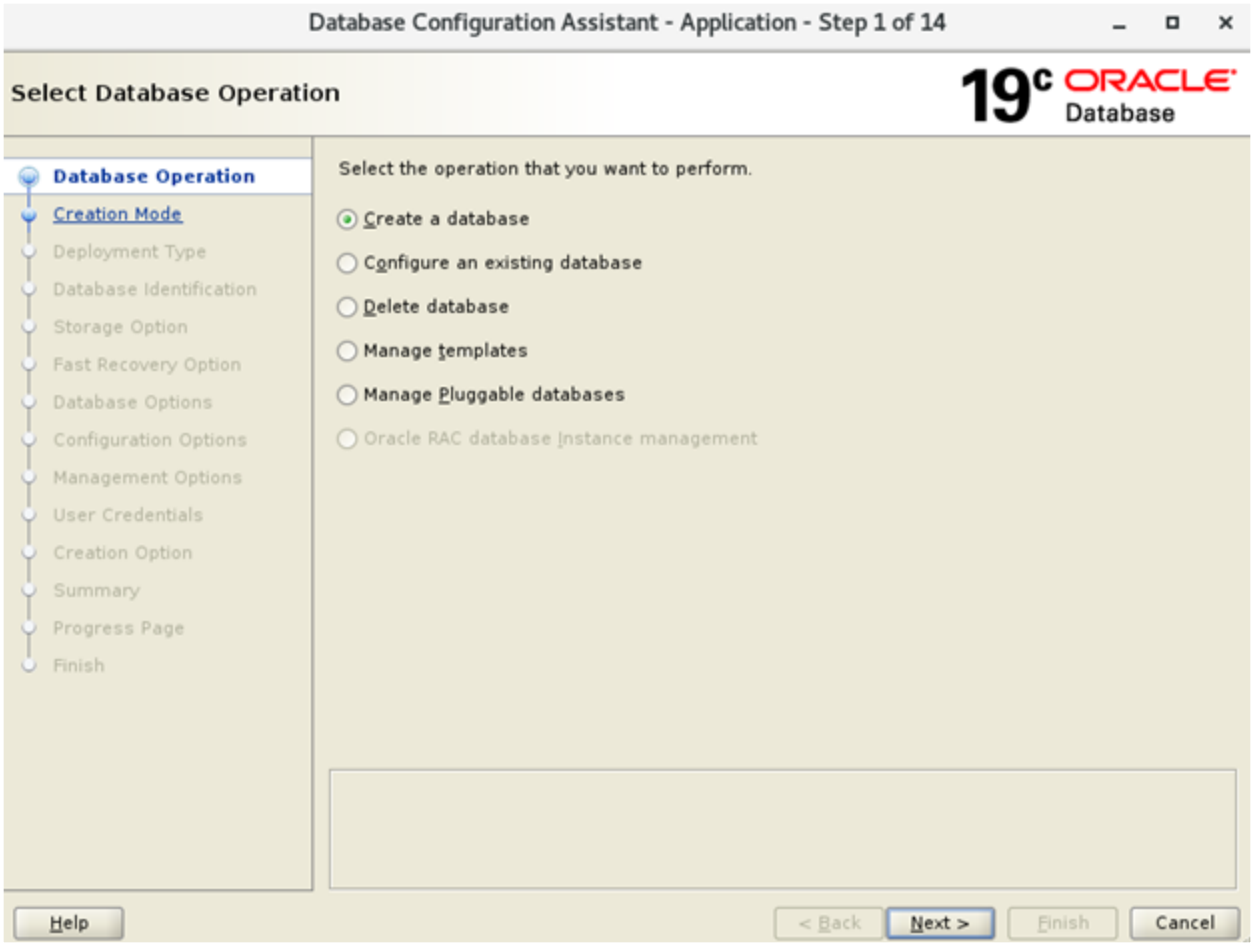
18. Click **Close**.

4.4 Oracle Database Creation

NOTE – If you are installing Blood Centers on two servers (one for the database and one for the application), this procedure is completed on the database server.

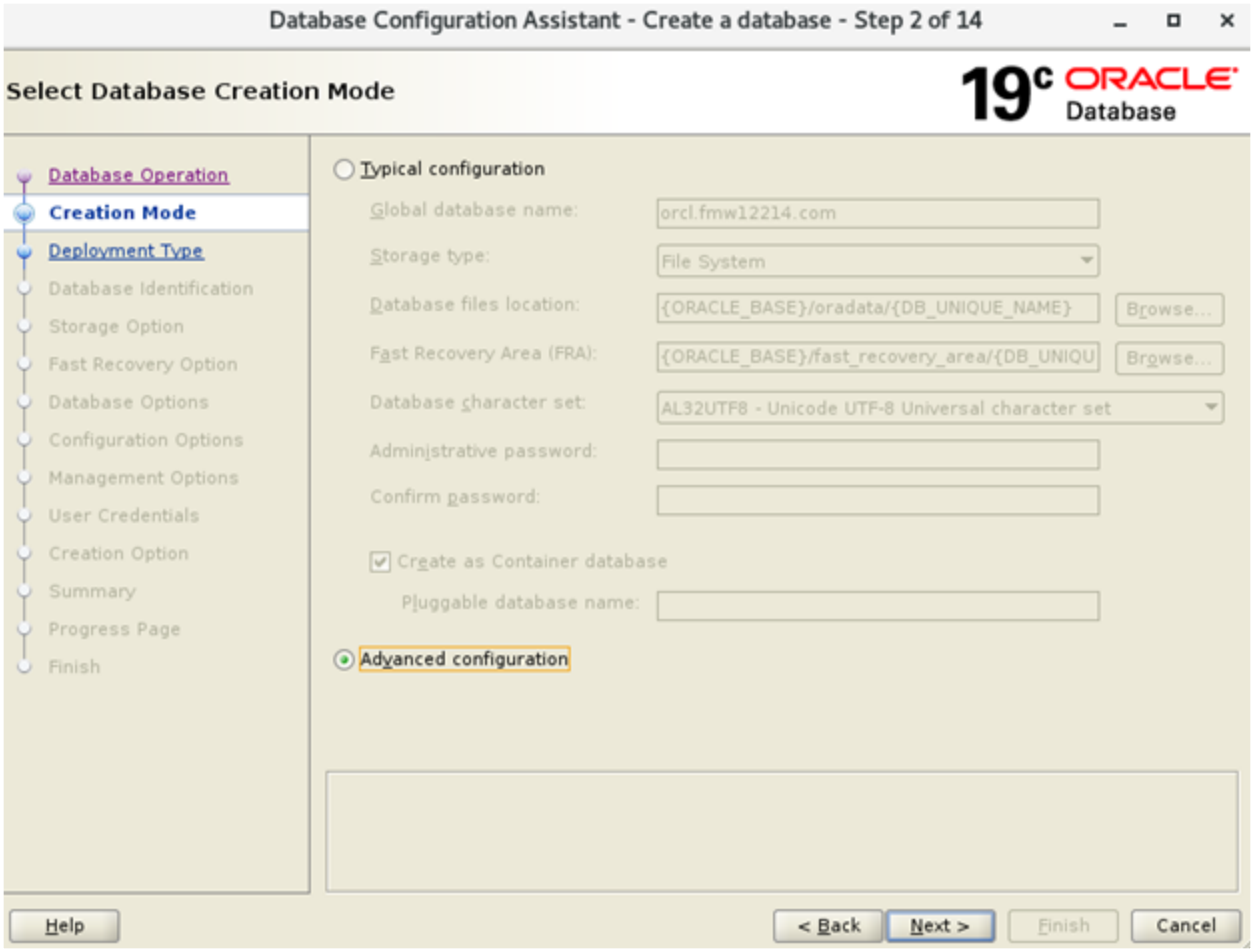
1. Log on as **oracle**.
2. Right-click on the desktop, and select **Open Terminal**.
3. Create the database by entering the following command:
   $ `sh /apps/Software/Create_Database.sh`

   It starts the database listener and opens the Select Database Operation screen, as shown in the following figure.

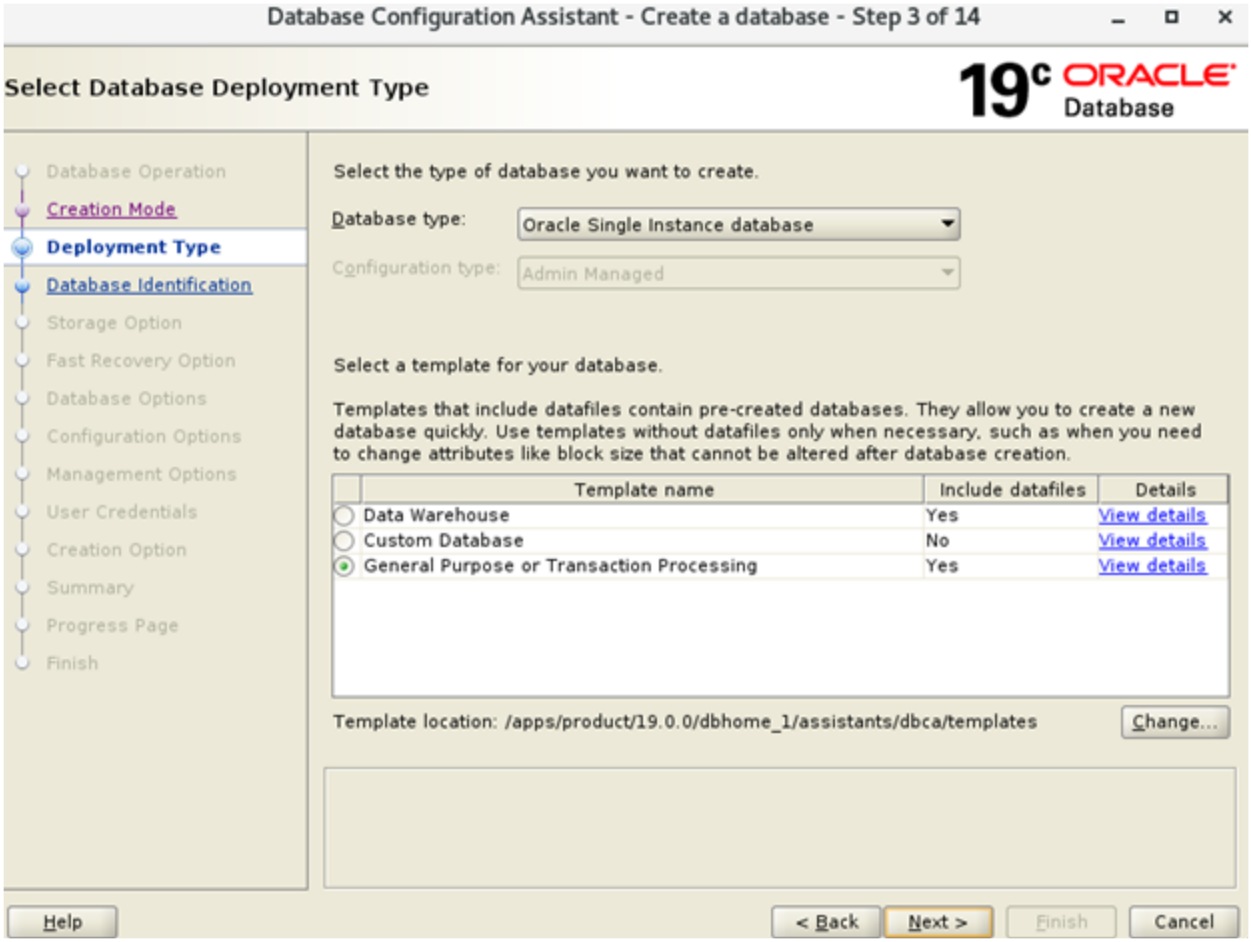


4. Click **Next**. The Select Database Creation Mode screen displays.

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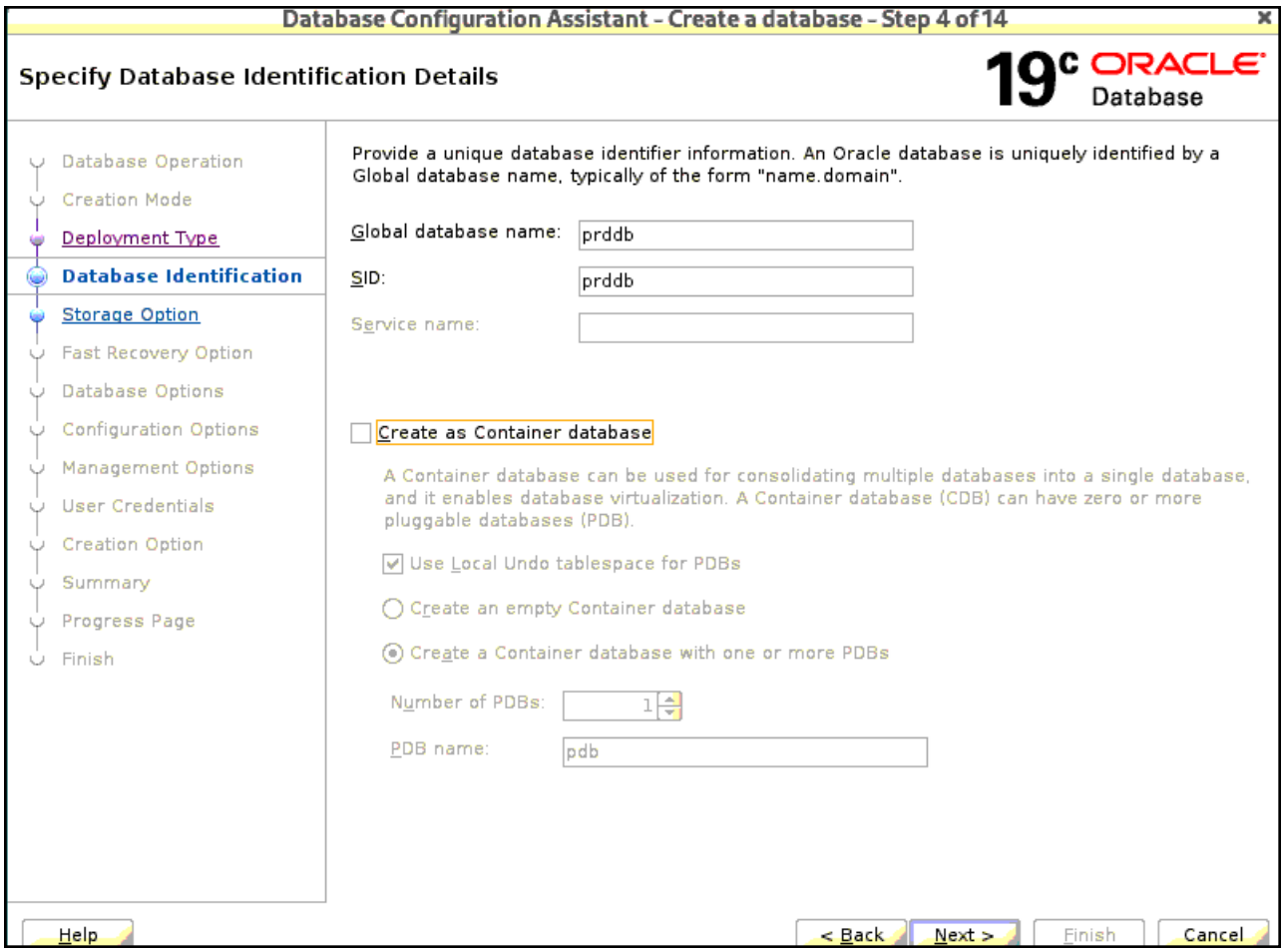


5. Select the **Advanced configuration** option, and click **Next**. The Select Database Deployment Type screen displays, as shown in the following figure.

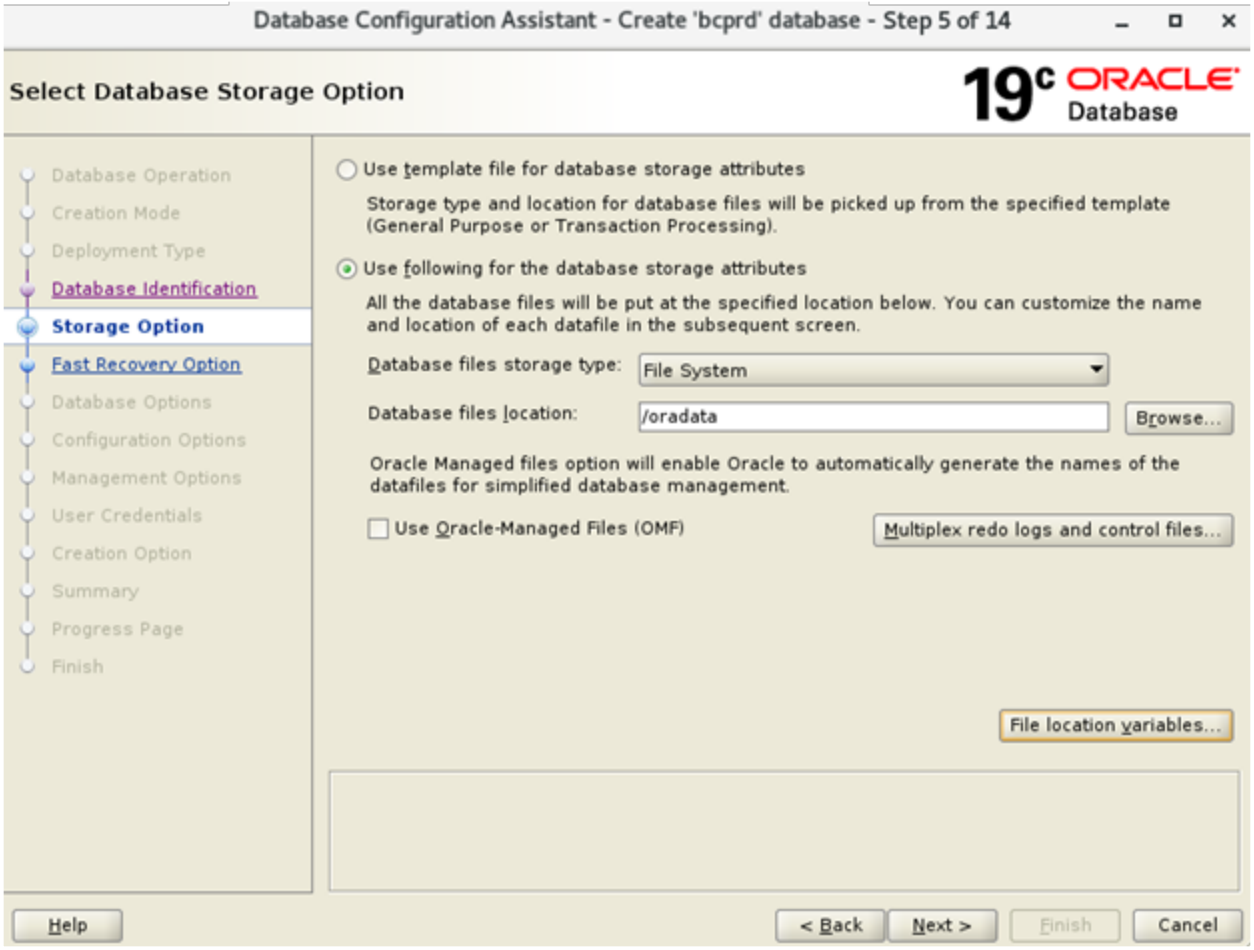


6. Under *Select a template for your database*, make sure the **General Purpose or Transaction Processing** option is selected.
7. Click **Next**. The Specify Database Identification Details screen displays, as shown in the following figure.

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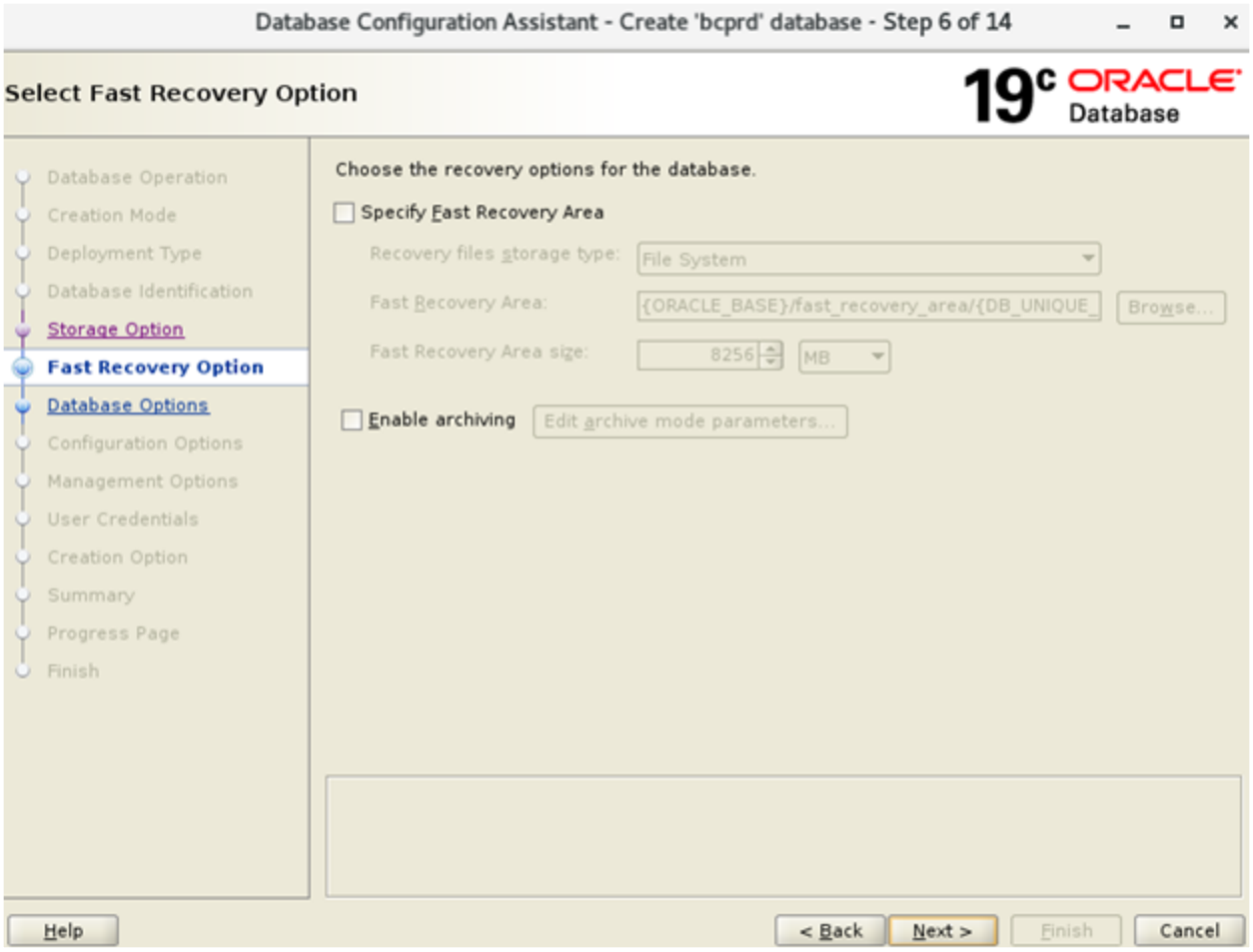


8. Enter the **Global database name** in the space provided. For the production database, enter `prddb`. For the test database, enter `prddb`.
9. Enter the **SID** (e.g. prddb) in the space provided.
10. Click **Next**. The Select Database Storage Option screen displays, as shown in the following figure.

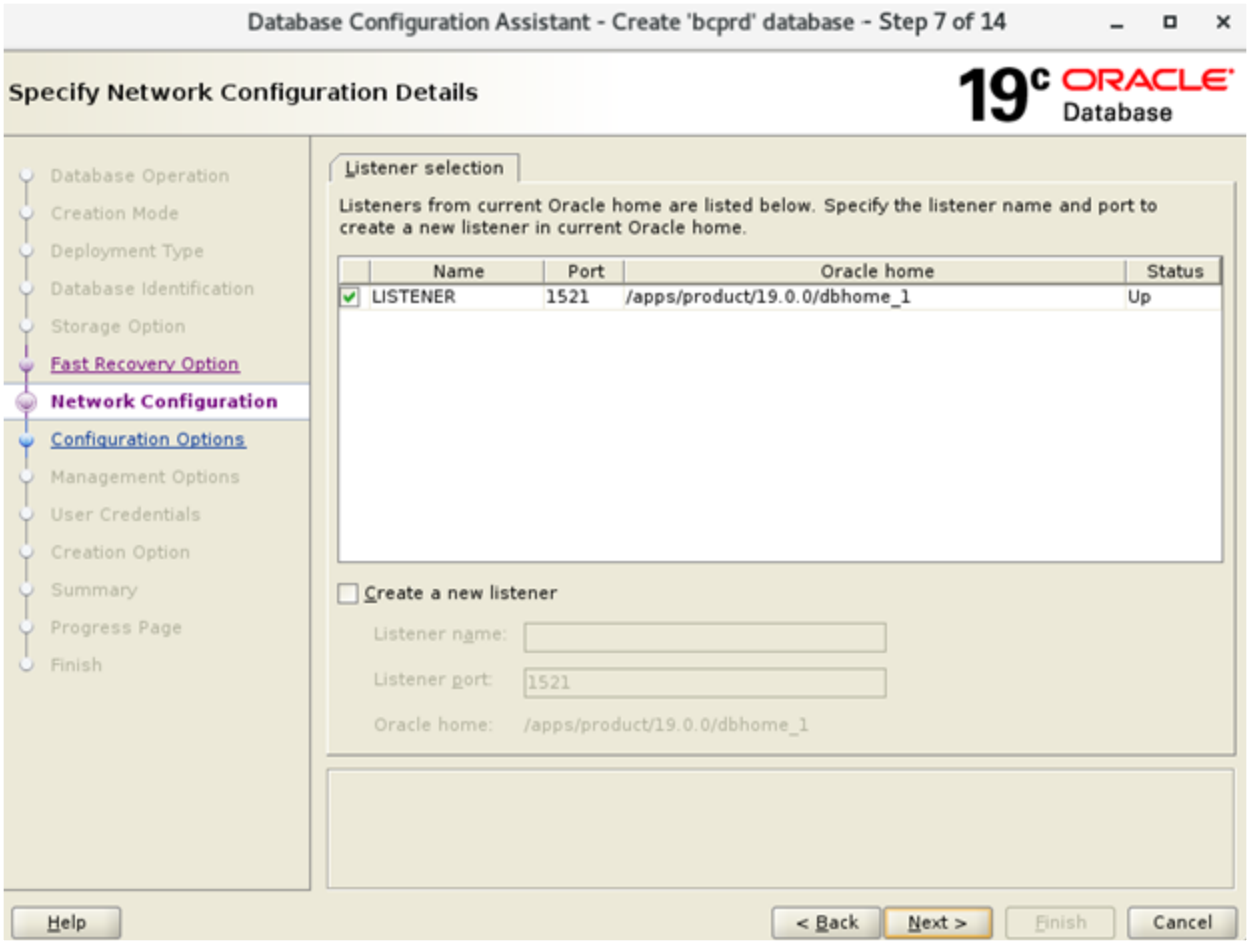


11. Select the **Use following for the database storage attributes** option.
12. Select **File System** from the *Database files storage type* drop-down.
13. In the **Database files location** field, enter the location of the database files (e.g. /oradata).
14. Click **Next**. The Select Fast Recovery Option screen displays, as shown in the following figure.

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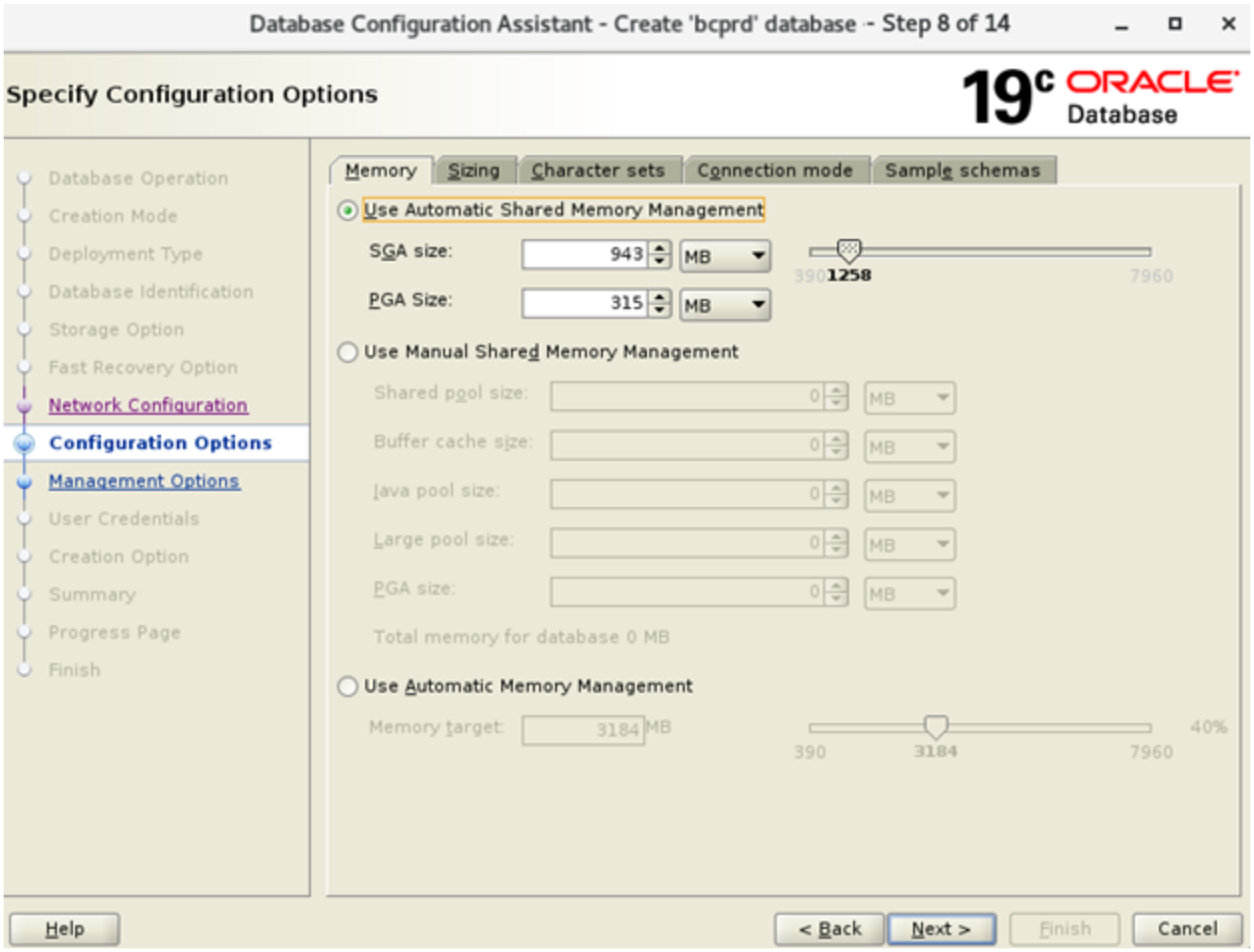


15. Click **Next**. The Specify Network Configuration Details screen displays, as shown in the following figure.



16. Select the existing **LISTENER** option, and click **Next**. The Specify Configuration Options screen displays, as shown in the following figure.

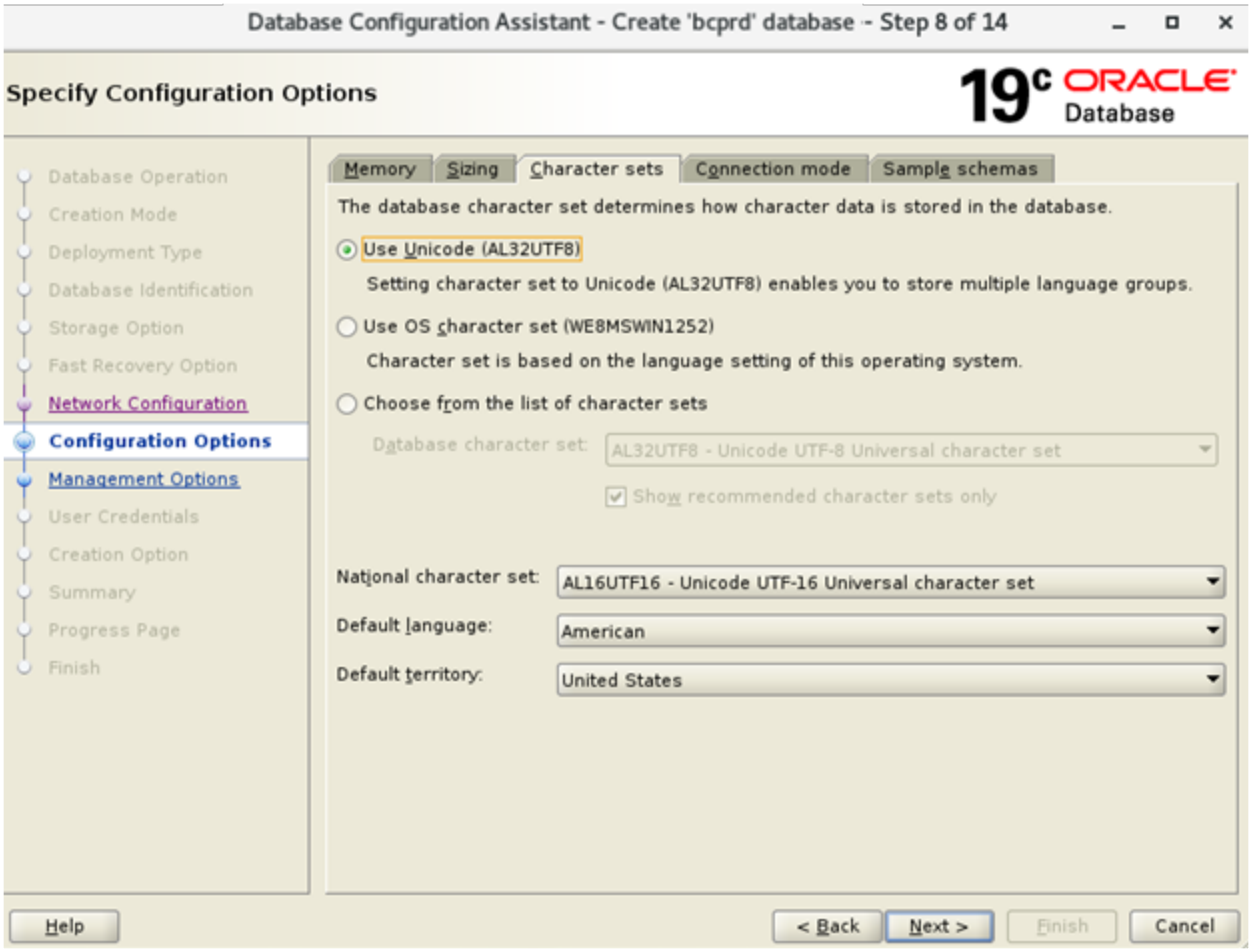
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17. Select the **Use Automatic Shared Memory Management** option.
18. Enter the **SGA size** and **PGA size** by dragging the memory bar.

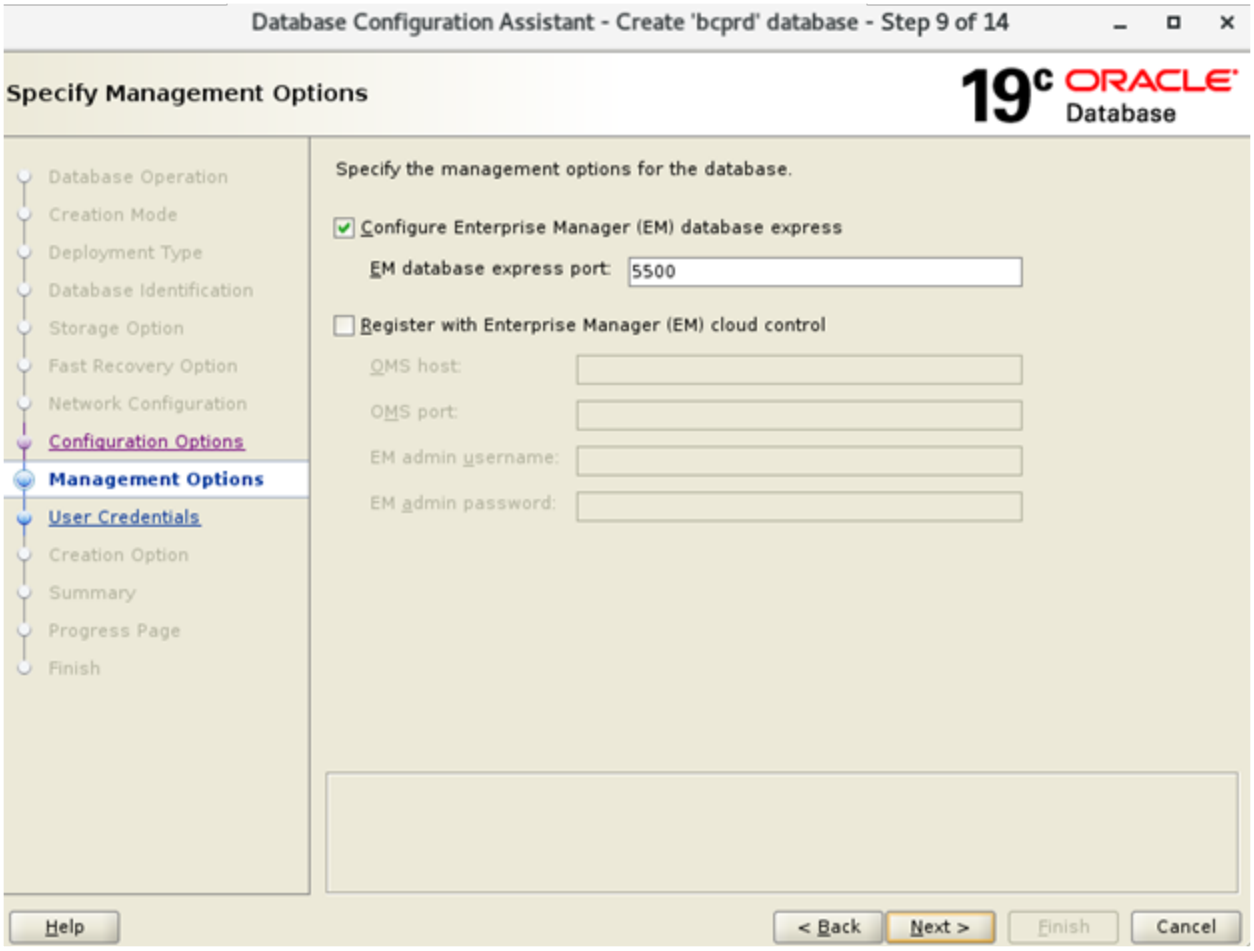
> **NOTE –** Try to keep 20% of OS memory for the database. If needed, these values can be modified later.

19. Click **Next**. The Character Sets tab displays, as shown in the following figure.

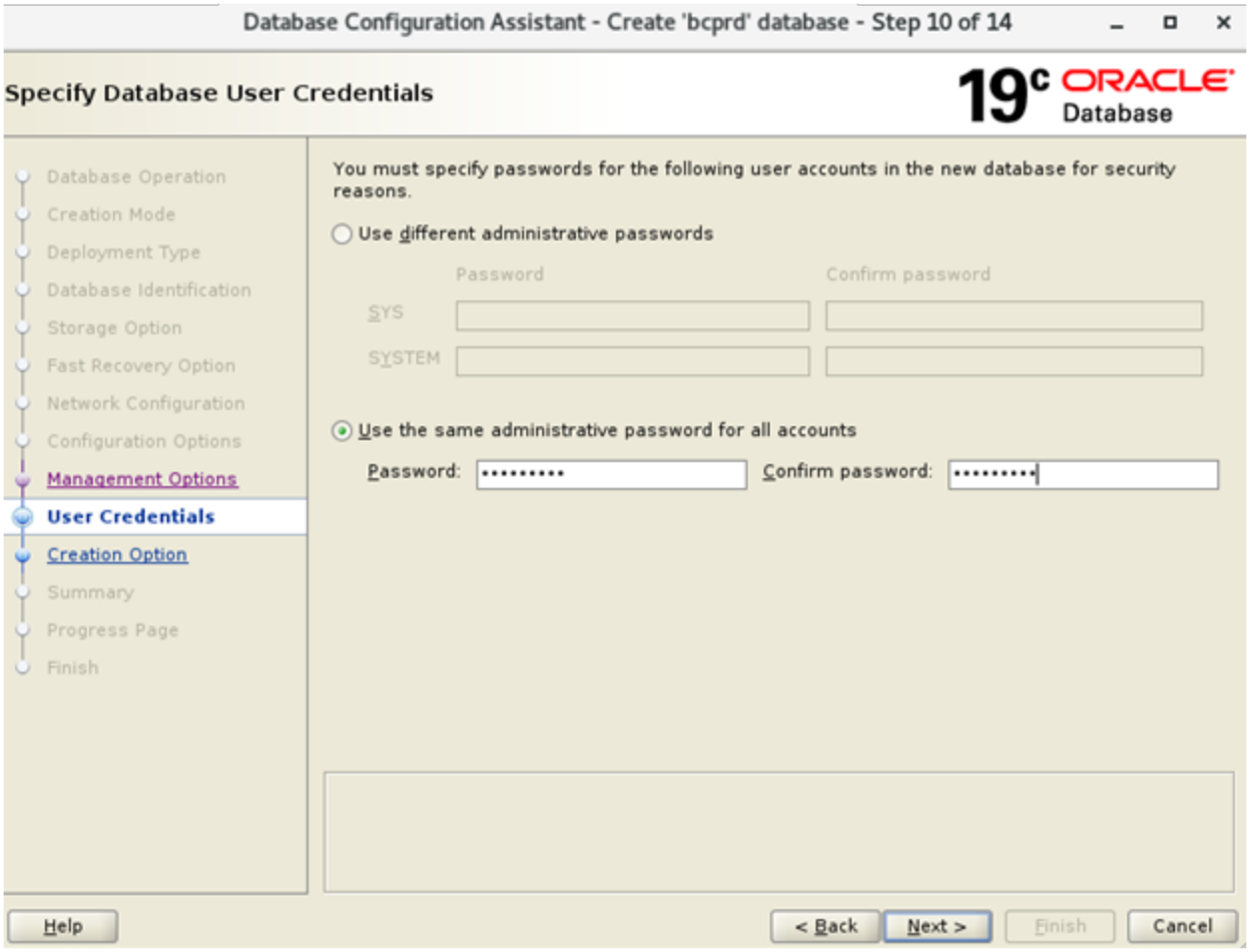


20. Select the **Use Unicode (AL32UTF8)** option, and click **Next**. The Specify Management Options screen displays.

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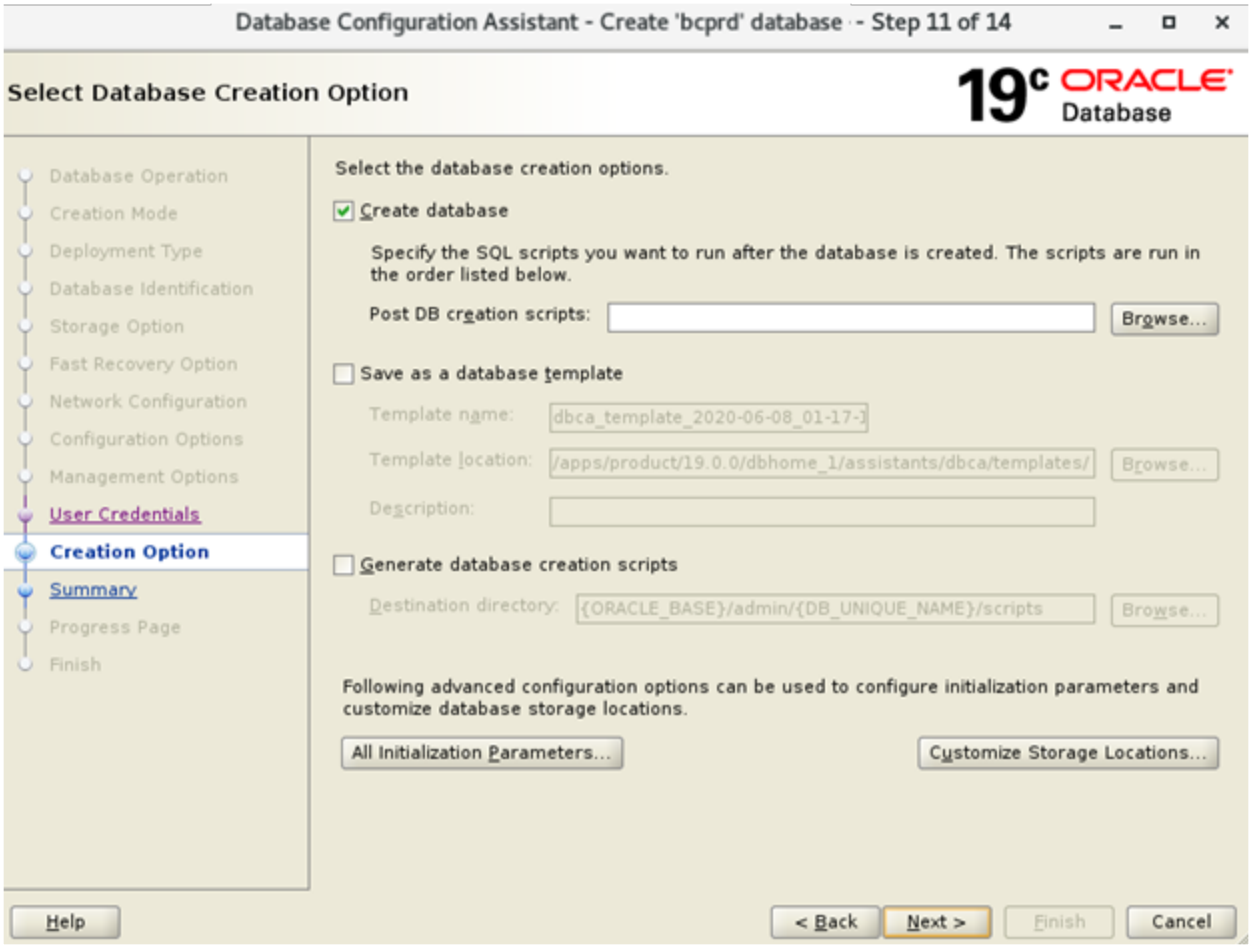


21. Select the **Configure Enterprise Manager (EM) database express** option.
22. Click **Next**. The Specify Database User Credentials screen displays, as shown in the following figure.

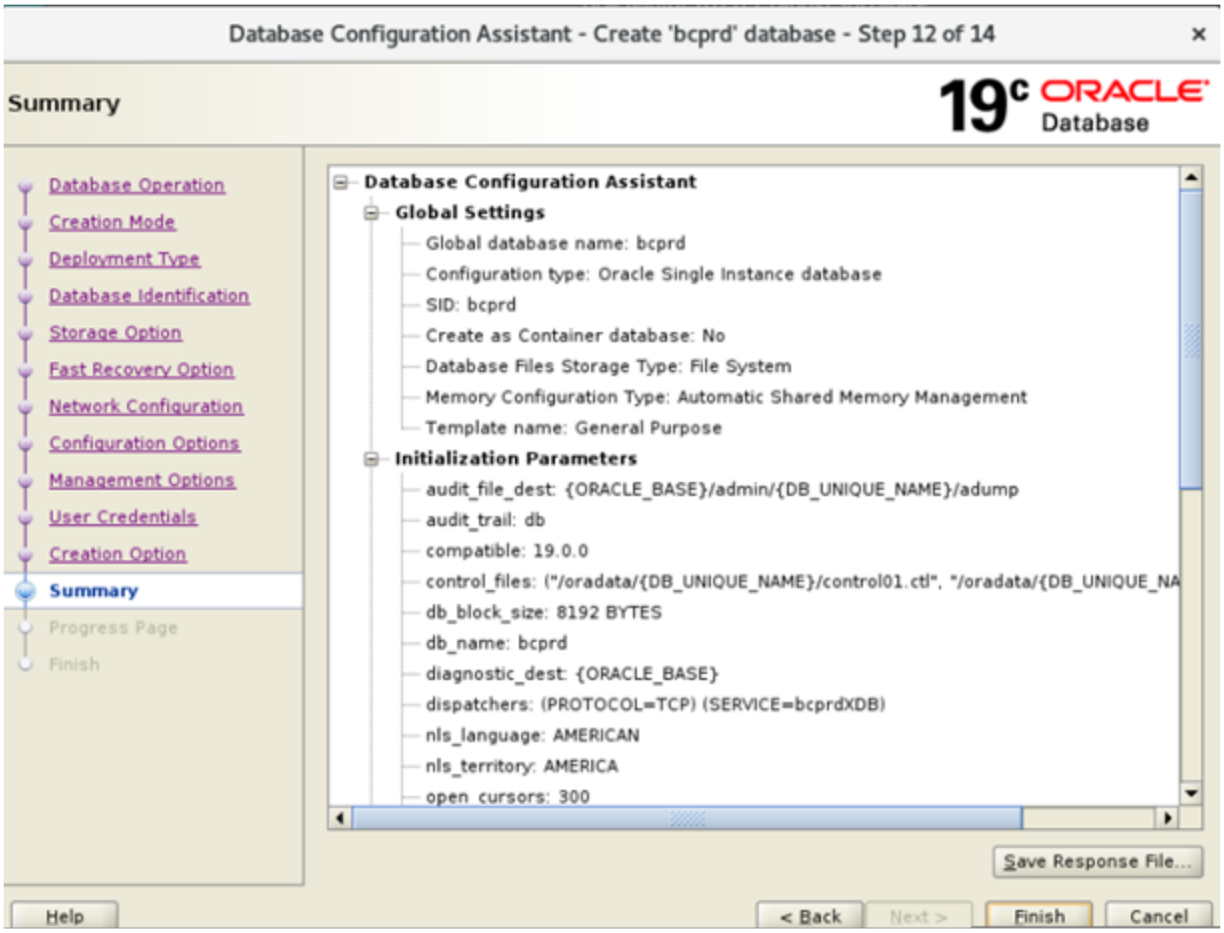


23. Select the **Use the same administrative password for all accounts** option.
24. Enter the password for both SYS and SYSTEM database users in the **Password** box.
25. Enter the password again in the **Confirm password** box.
26. Click **Next**. The Select Database Creation Option screen displays, as shown in the following figure.

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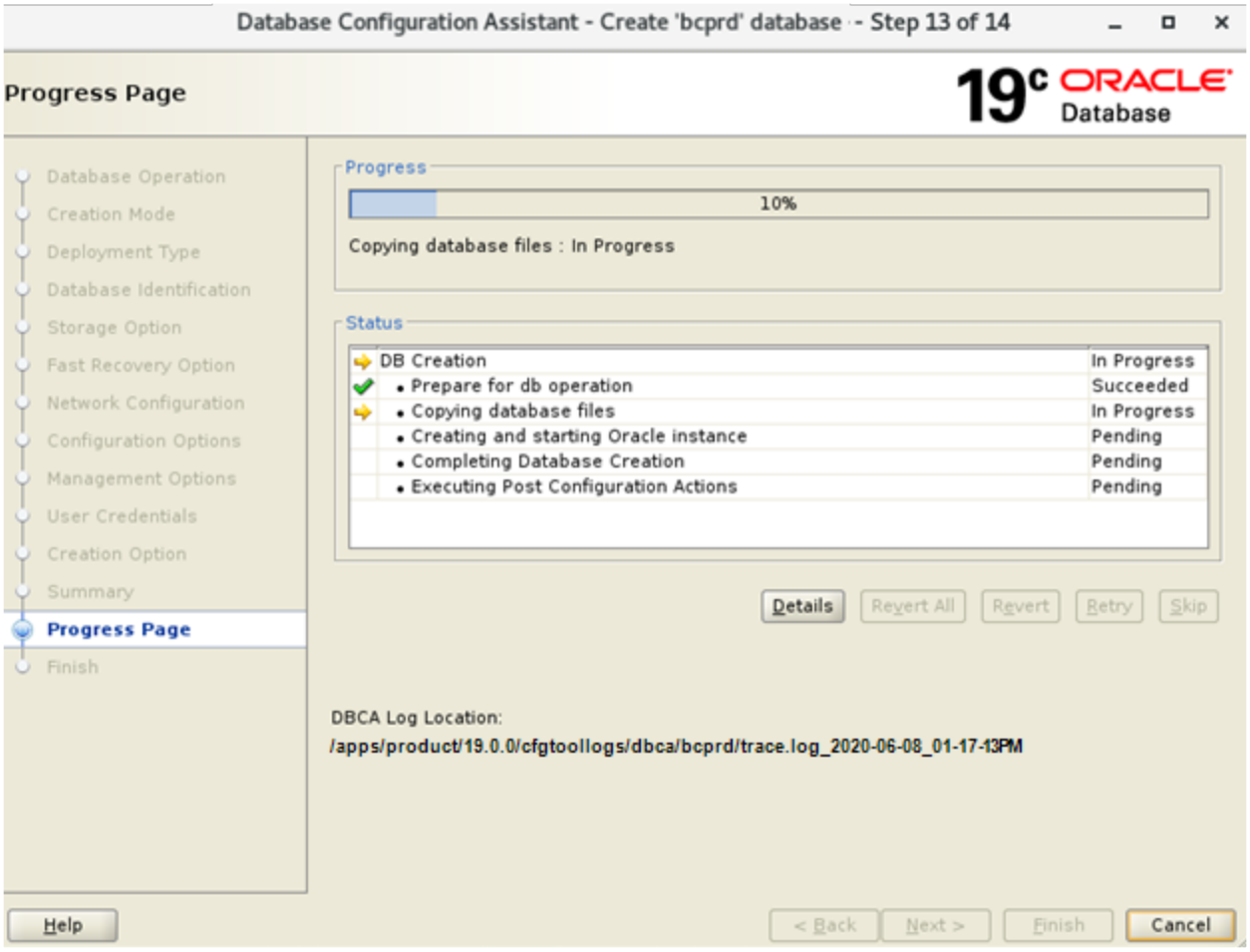


27. Select the **Create Database** option.
28. Click **Next**. The Summary screen displays, as shown in the following figure.

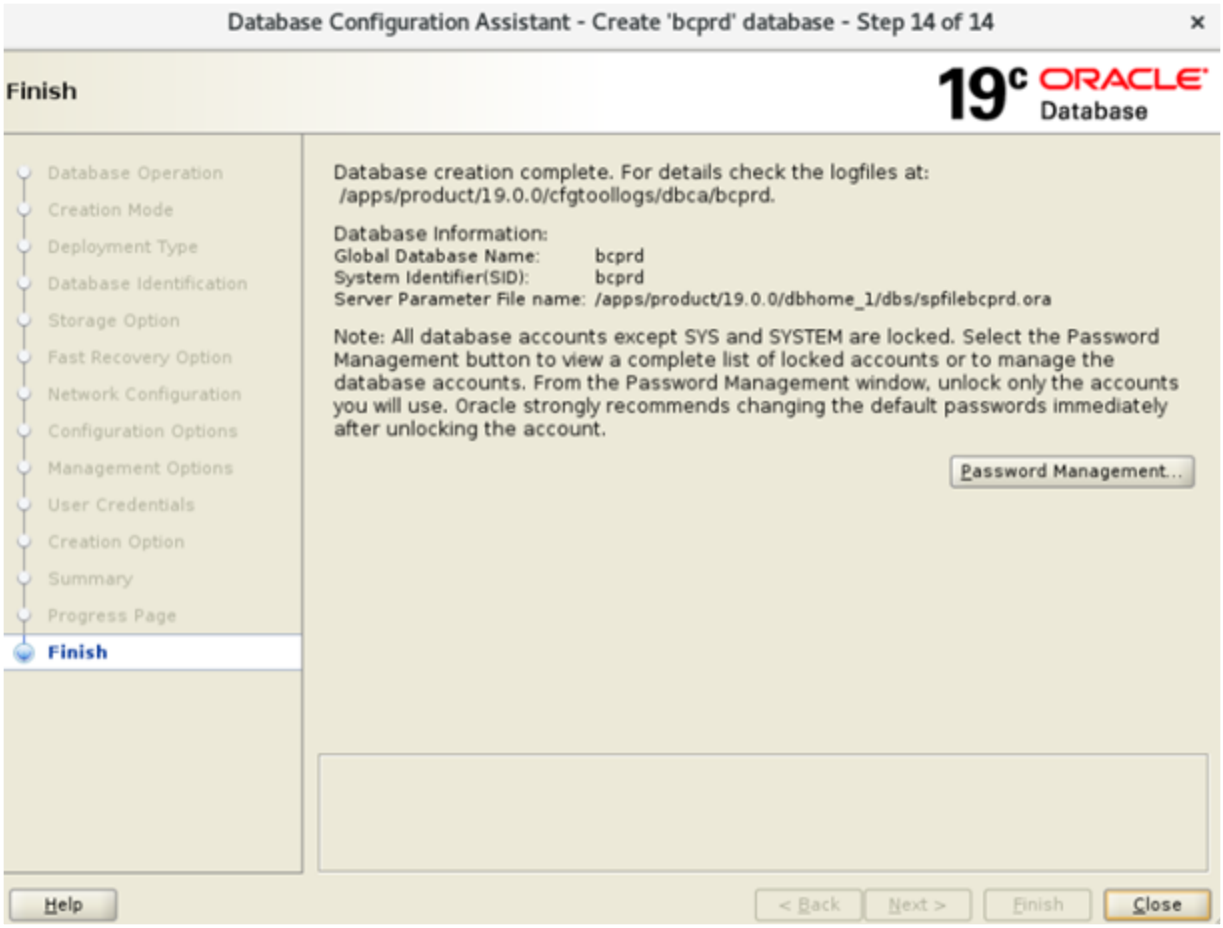


29. Click **Finish**. The Progress Page displays, as shown in the following figure.

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The Progress Page indicates how the database creation is progressing. It will take some time to create the database. When the database is created, the Finish screen displays, as shown in the following figure.



30. Click **Close**.
31. Repeat steps 1-30 for each database to be created.

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5

Chapter 5 - Blood Centers Database Configuration

5.1 Database Configuration Procedure 72

5.1 Database Configuration Procedure

NOTE – If you are installing Blood Centers on two servers (one for the database and one for the application), this procedure is completed on the database server.

1. Log on as **oracle**.
2. Right-click on the desktop and select **Open Terminal**.
3. Change the directory to **/oradata/bc2026/BC_SERVER_INSTALL/bloodc** by entering the following command:
   $ `cd /oradata/bc2026/BC_SERVER_INSTALL/bloodc`
4. Run the oraenv command, and enter the Blood Centers database name when prompted.
   $ `. oraenv`

   When prompted, enter the database name (ex: prddb)

   Example:

```
[oracle@bc2024r1 bloodc]$
[oracle@bc2024r1 bloodc]$ . oraenv
ORACLE_SID = [oracle] ? bcprd
The Oracle base has been set to /apps/product/19.0.0
[oracle@bc2024r1 bloodc]$
```

5. Enter the following command to log in to the database:
   $ `sqlplus "/as sysdba"`
6. Run the **bcdb_config.sql** by entering the following command at the SQL prompt:
   SQL> `@bcdb_config.sql`

   This script will configure the required Blood Centers database tablespace, database users, roles, and required database parameters.

   Example:

```
[oracle@bc2024r1 bloodc]$
[oracle@bc2024r1 bloodc]$ sqlplus "/as sysdba"

SQL*Plus: Release 19.0.0.0.0 - Production on Thu Jan 6 17:36:17 2022
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle.  All rights reserved.


Connected to:
Oracle Database 19c Standard Edition 2 Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> @bcdb_config.sql
```


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7. The script will prompt for **BloodCenter_database_name**. Enter the name of the database in lower case (ex: prddb).
8. The script will prompt for **db_dir_upper_case_name**. Enter the name of the database in upper case (ex. PRDDB).

```
SQL> @bcdb_config.sql
Enter value for bloodcenter_database_name: bcprd
Enter value for db_dir_upper_case_name: BCPRD

System altered.


Session altered.

old   1: Create tablespace bc_data datafile '/oradata/&DB_DIR_NAME/bc_data01.dbf' size 1900M autoextend on next 10M
new   1: Create tablespace bc_data datafile '/oradata/BCPRD/bc_data01.dbf' size 1900M autoextend on next 10M
```

9. If there are any issues, verify the logs in the log directory.
10. Issue the following command to restart the database:
    $ `bcoradbrestart`

    It will prompt for the Blood Centers Database Name. Enter the database name (ex: prddb)

    Example:

```
[oracle@bc2024rl bloodc]$ bcoradbrestart
ORACLE_SID = [bcprd] ? bcprd
```

    After restarting, it will display a message stating the database has restarted.

```
bcprd Database has Restarted
[oracle@bc2024rl bloodc]$
```

11. Enter the following commands to run the **BC2026R1_Synonyms.sql** script:
    $ `. oraenv`

    When prompted, enter the Blood Centers Database Name (ex: prddb).

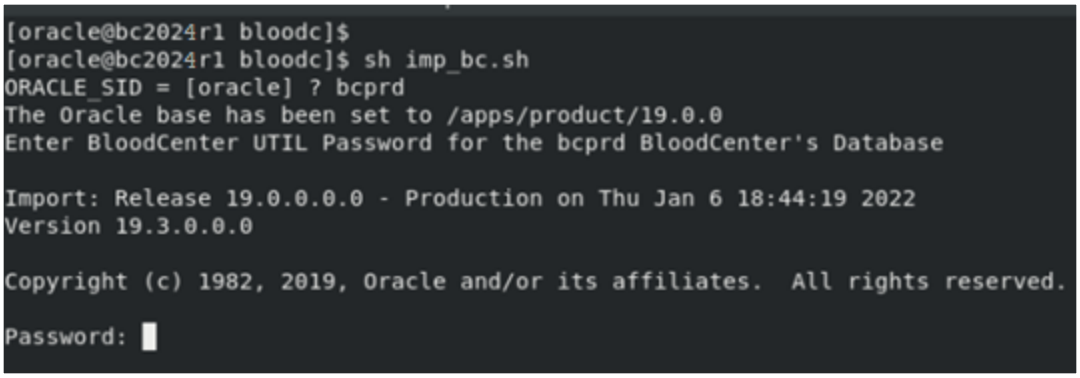
    $ `sqlplus "/as sysdba"`

    At the SQL prompt, enter the following command:
    SQL> `@BC2026R1_Synonyms.sql`
12. Enter the following commands to run the **imp_bc.sh** shell script:
    $ `sh imp_bc.sh`

    When prompted, enter the Blood Centers Database Name (ex: prddb), and then the database Utility password.

NOTE – Directions for changing this password can be found in the Linux Operations Guide.

Example:



NOTE – This will import the required Blood Centers Database schemas.

13. Enter the following commands to run the BC2026R1_Grants.sql script:
    $ `. oraenv`

    When prompted, enter the Blood Centers Database Name (ex: prddb).

    $ `sqlplus "/as sysdba"`

    At the SQL prompt, enter the following command:
    SQL> `@BC2026R1_Grants.sql`
14. Repeat this procedure for each database that needs to be configured.

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6

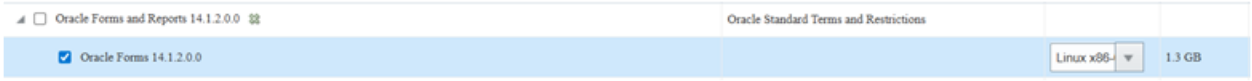
Chapter 6 - Oracle Fusion Middleware Installation and Configuration

6.1 Download Fusion Middleware 76
6.2 Oracle WebLogic Server Installation 78
6.3 Oracle Forms and Reports Installation 86
6.4 Repository Creation Utility 95
6.5 FMW Configuration Wizard 106
6.6 Application Server Configuration 126
6.7 Set Up Cron Jobs 137
6.7.1 Required Cron Jobs 137
6.7.2 Optional Cron Jobs 138
6.8 Automatically Printing Reports 141
6.8.1 Overview 141
6.8.2 Adding Report Names to the UPD_REP_STATUS.sql File 141
6.9 Linux NFS Setup Between the Oracle Application and the Database Server 144

6.1 Download Fusion Middleware

Perform the following procedure to download the Oracle Fusion Middleware software.

1. Go to https://edelivery.oracle.com/osdc/faces/Home.jspx.
2. Log in with Oracle Support credentials.
3. Select **License** in *All Categories*.
4. Enter the Oracle Forms and Reports in the *Search* box.
5. Click **Search**.
6. Click **DLP: Oracle Forms and Reports 14.1.2.0.0 (Oracle Forms and Reports)**.
7. Click **Selected Software** on the right.
8. Unselect everything.
9. Select only **Oracle Forms 14.1.2.0.0**.
10. Select **Linux x86-64** in the *Platforms/Languages* drop-down.

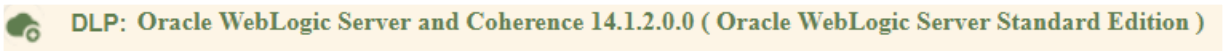


11. Click **Continue**.
12. Select the agreements, and click **Download**.

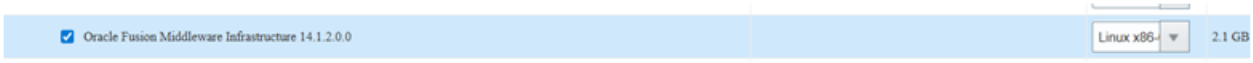


V983392-01_1of2.zip, V983392-01_2of2.zip - Oracle Forms and Reports 14.1.2.0.0 downloads.

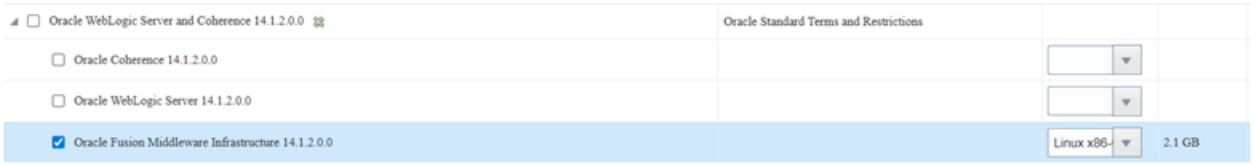
13. Select **License** in *All Categories*.
14. Enter **Oracle WebLogic Server Standard Edition** in the *Search* box.
15. **Select**.



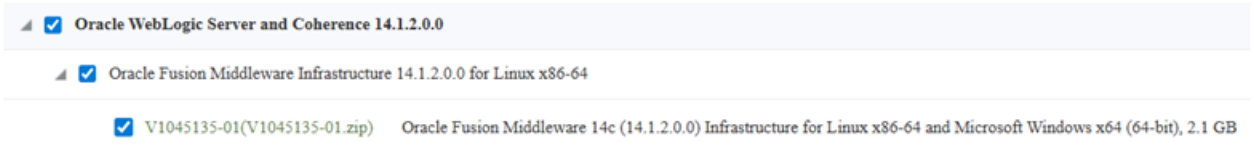
16. DLP: Oracle WebLogic Server and Coherence 14.1.2.0.0 (Oracle WebLogic Server Standard Edition)



17. Click **Select Software**.
18. Unselect the selected software.
19. Select the **Oracle Fusion Middleware 14c Infrastructure 14.1.2.0.0** software.



20. Select **Linux x86-64** in the *Platforms/Languages* drop-down.
21. Click **Continue**.
22. Agree to the license terms, and click **Continue**.
23. Click **Download**. The V1045135-01.zip - Oracle Fusion Middleware 14c (14.1.2.0.0) Infrastructure for Linux x86-64.



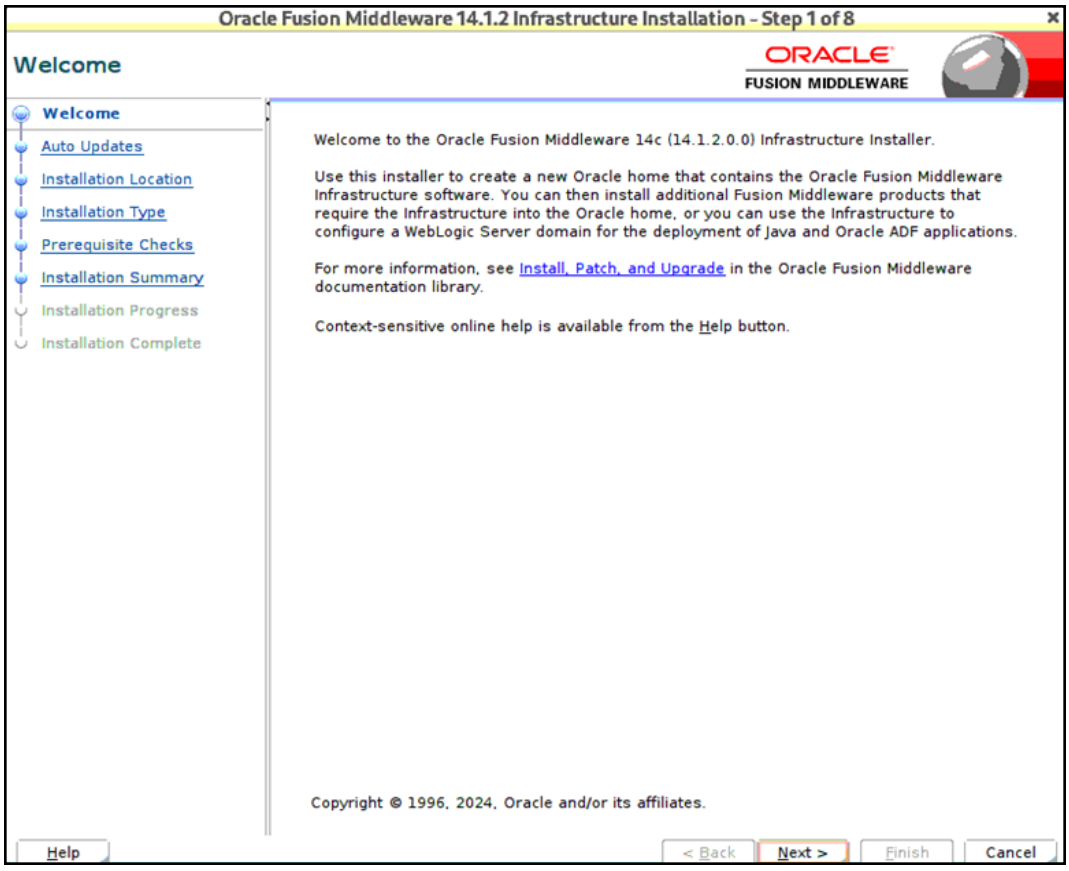
24. Copy the downloaded software in the /apps/Software/FMW directory as an Oracle user.
25. Unzip the downloaded software in the /apps/Software/FMW directory using the following commands:

```
$ cd /apps/Software/FMW
$ unzip V1045135-01.zip
$ unzip V1045121-01.zip
```

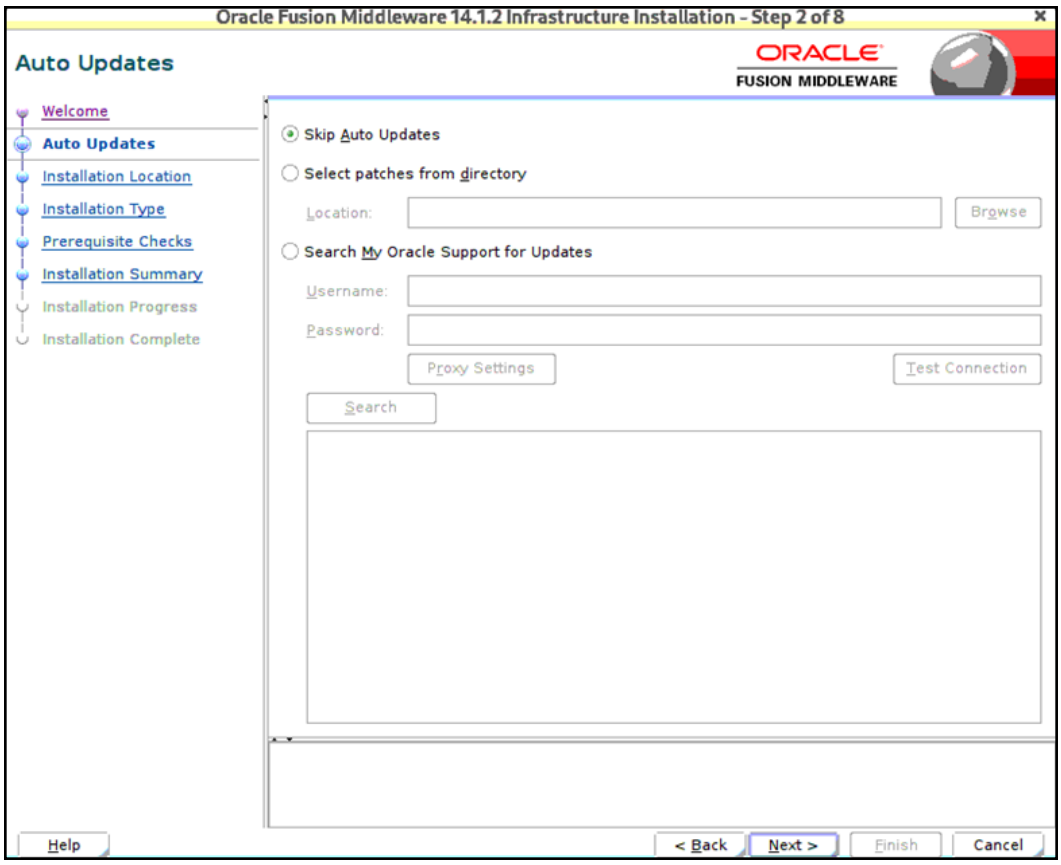
6.2 Oracle WebLogic Server Installation

1. Log on as **oracle** using a GUI terminal like VNC.
2. Right-click on the desktop and select **Open Terminal**.
3. Extract the Oracle Fusion Middleware Application Server Software from the /apps/Software/FMW directory by entering the following commands:
   $ `cd /apps/Software/FMW`
   $ `sh /apps/Software/FMW/Install_WLS_14c.sh`

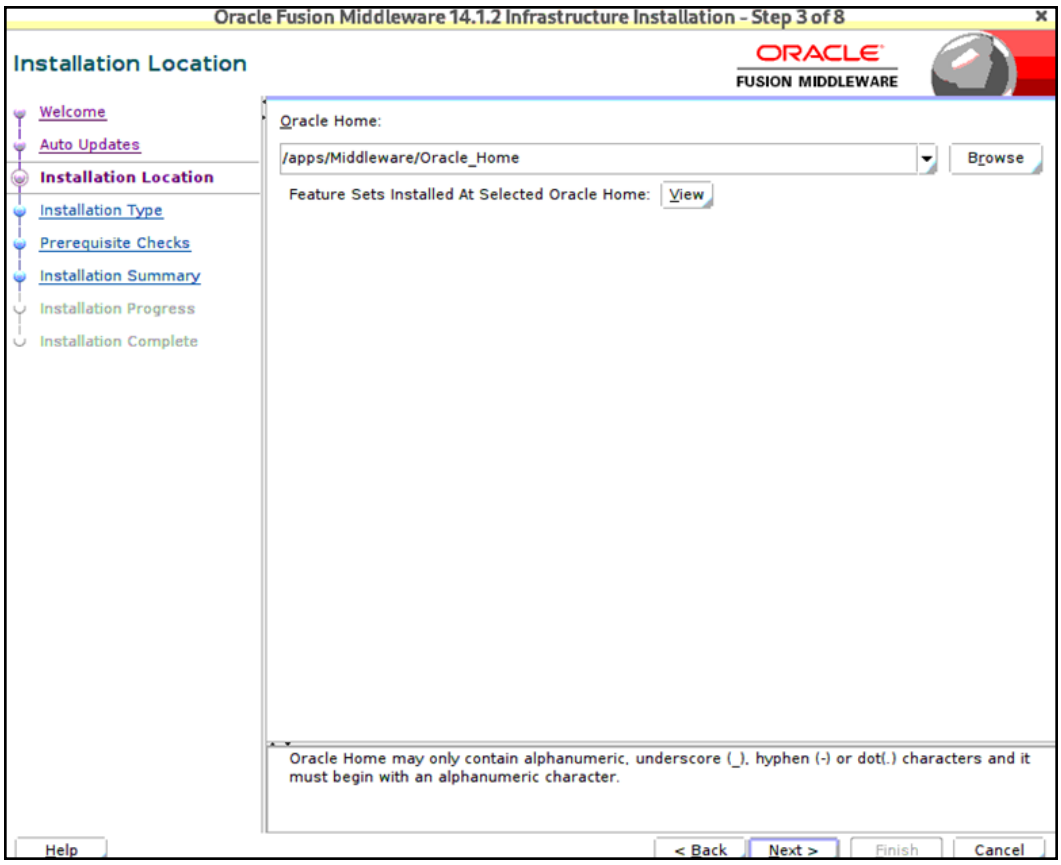
   The Oracle Fusion Middleware 14c Infrastructure Installation Wizard displays, as shown in the following figure.



4. Click **Next**. The Auto Updates screen displays, as shown in the following figure.



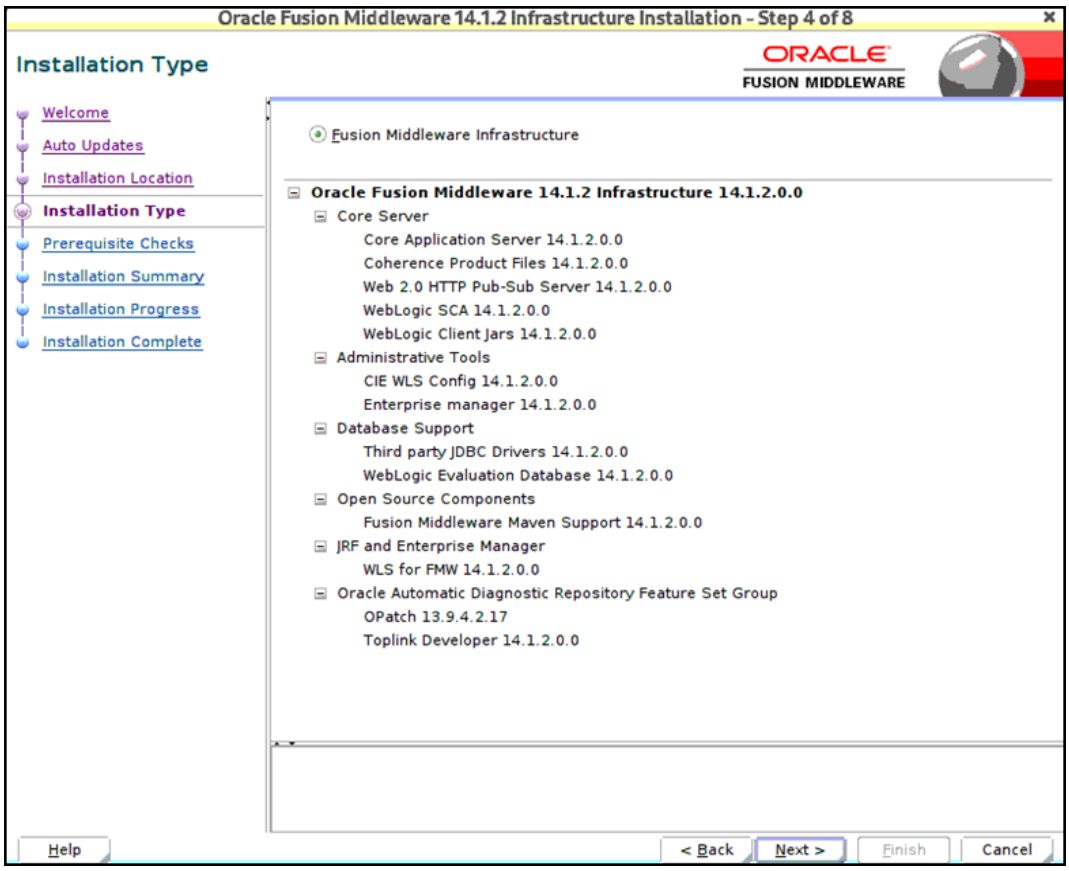
5. Make sure the **Skip Auto Updates** option is selected.
6. Click **Next**. The Installation Location screen displays, as shown in the following figure.



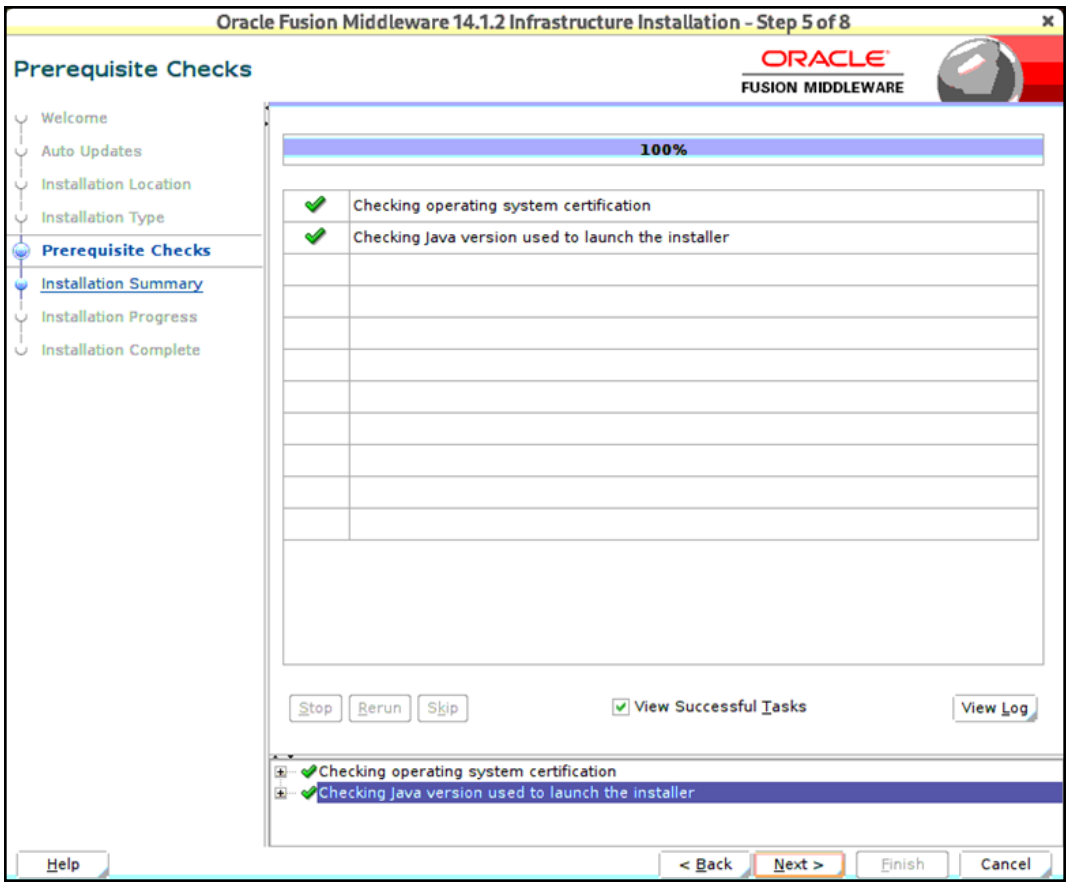
7. Update the file path in the *Oracle Home* box to **/apps/Middleware/Oracle_Home**.

> **NOTE –** The file path is case sensitive.

8. Click **Next**. The Installation Type screen displays, as shown in the following figure.

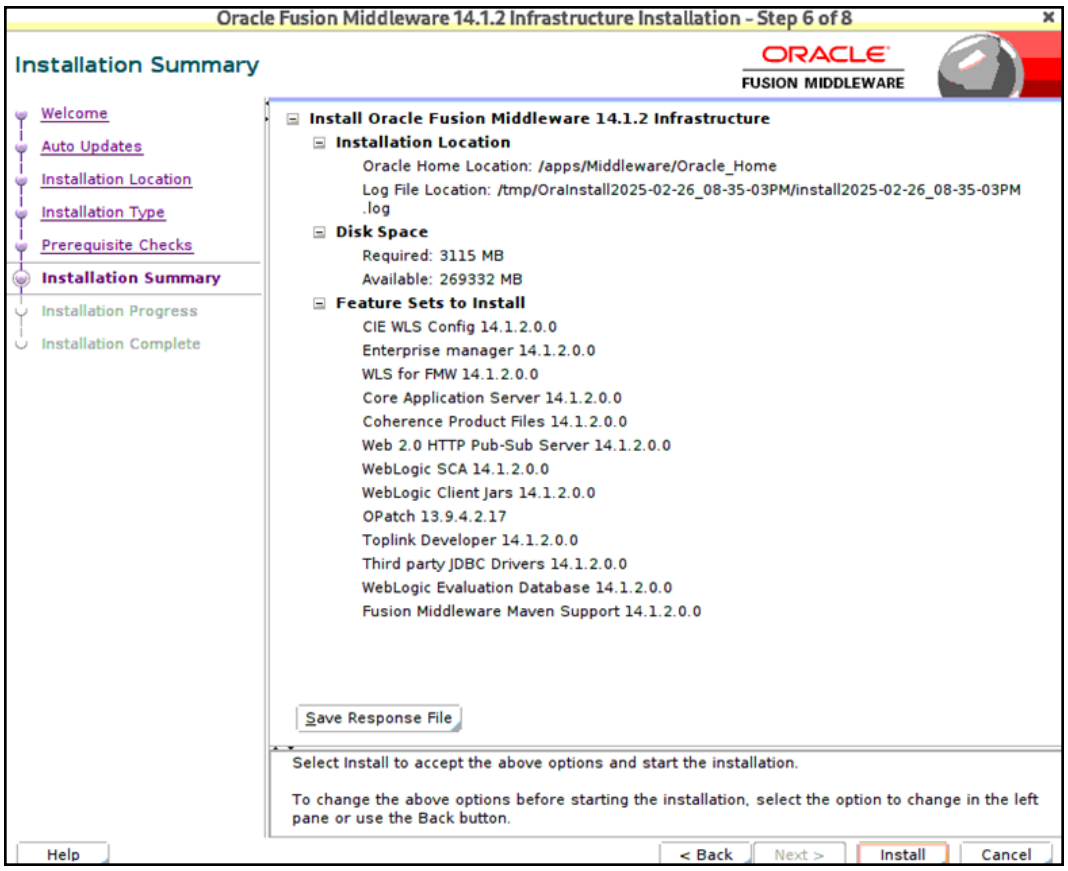


9. Make sure the **Fusion Middleware Infrastructure** option is selected.
10. Click **Next**. The Prerequisite Checks screen displays, as shown in the following figure.

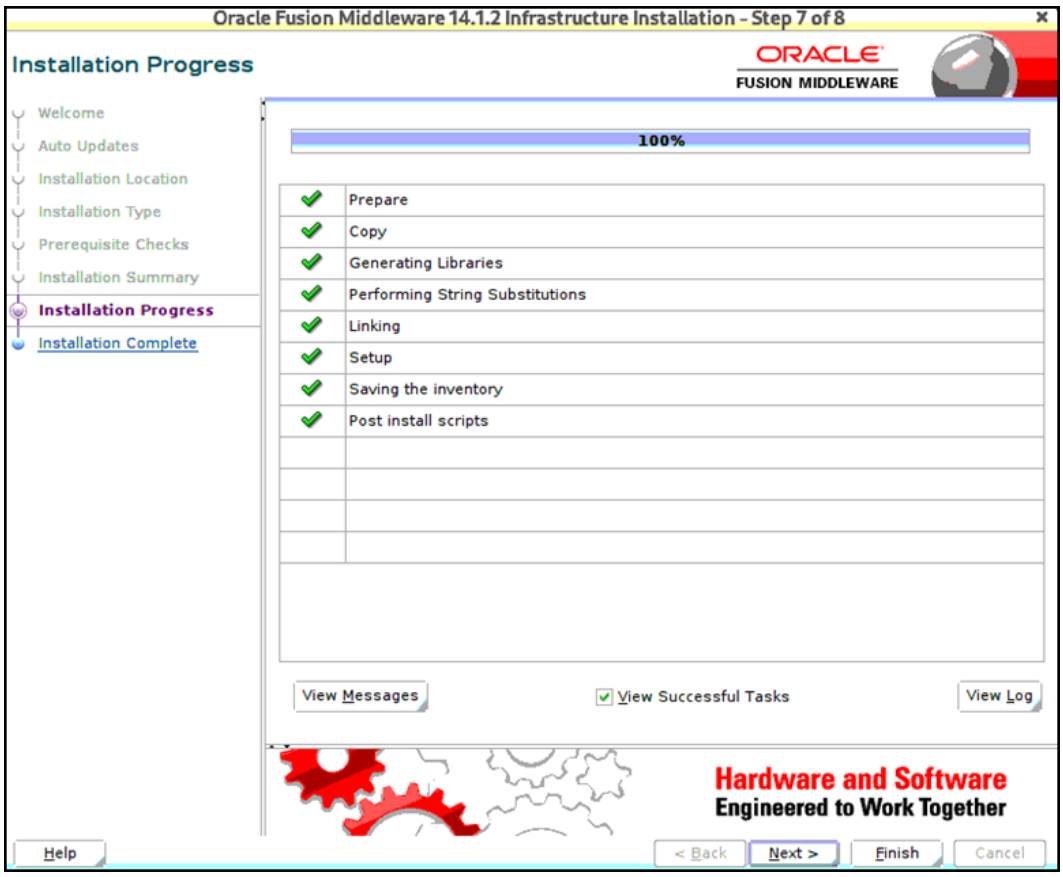


11. When the checks are finished successfully, click **Next**. The Installation Summary screen displays, as shown in the following figure.

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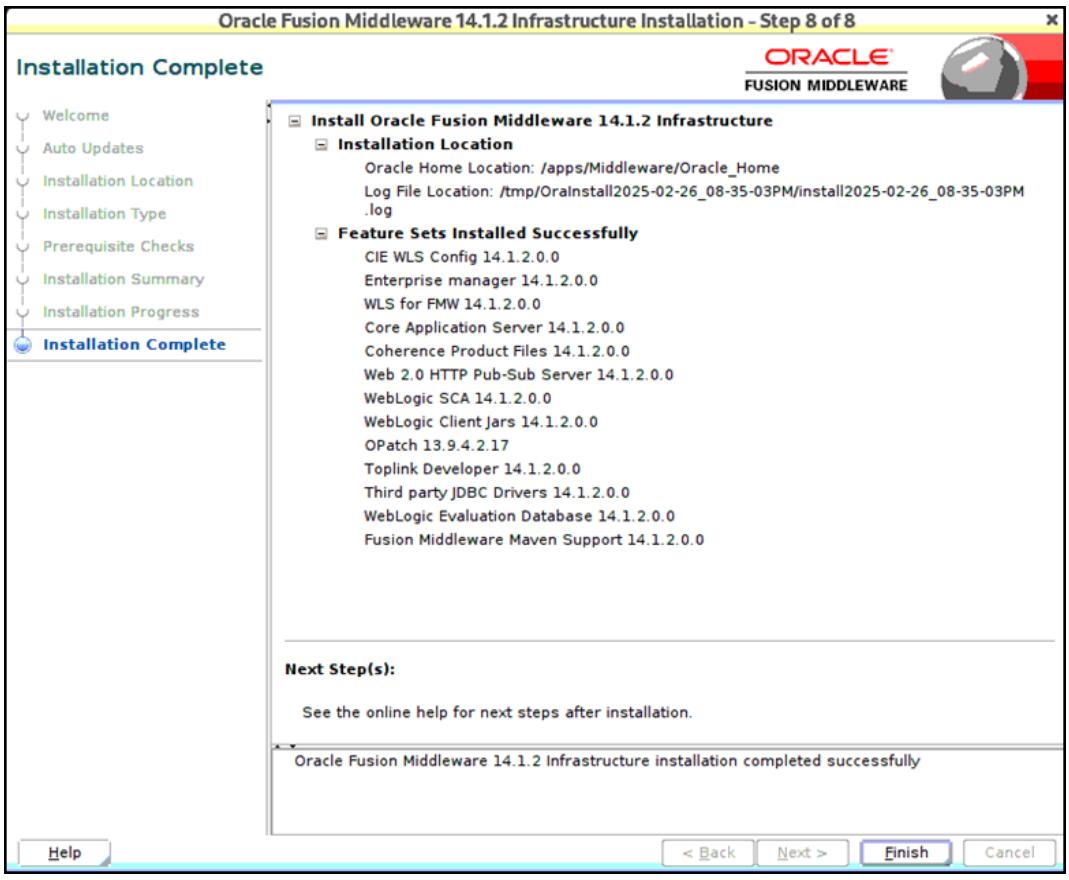


12. Click **Install** to start the installation. The Installation Progress screen displays, as shown in the following figure.



13. When the installation finishes successfully, click **Next**. The Installation Complete screen displays, as shown in the following figure.

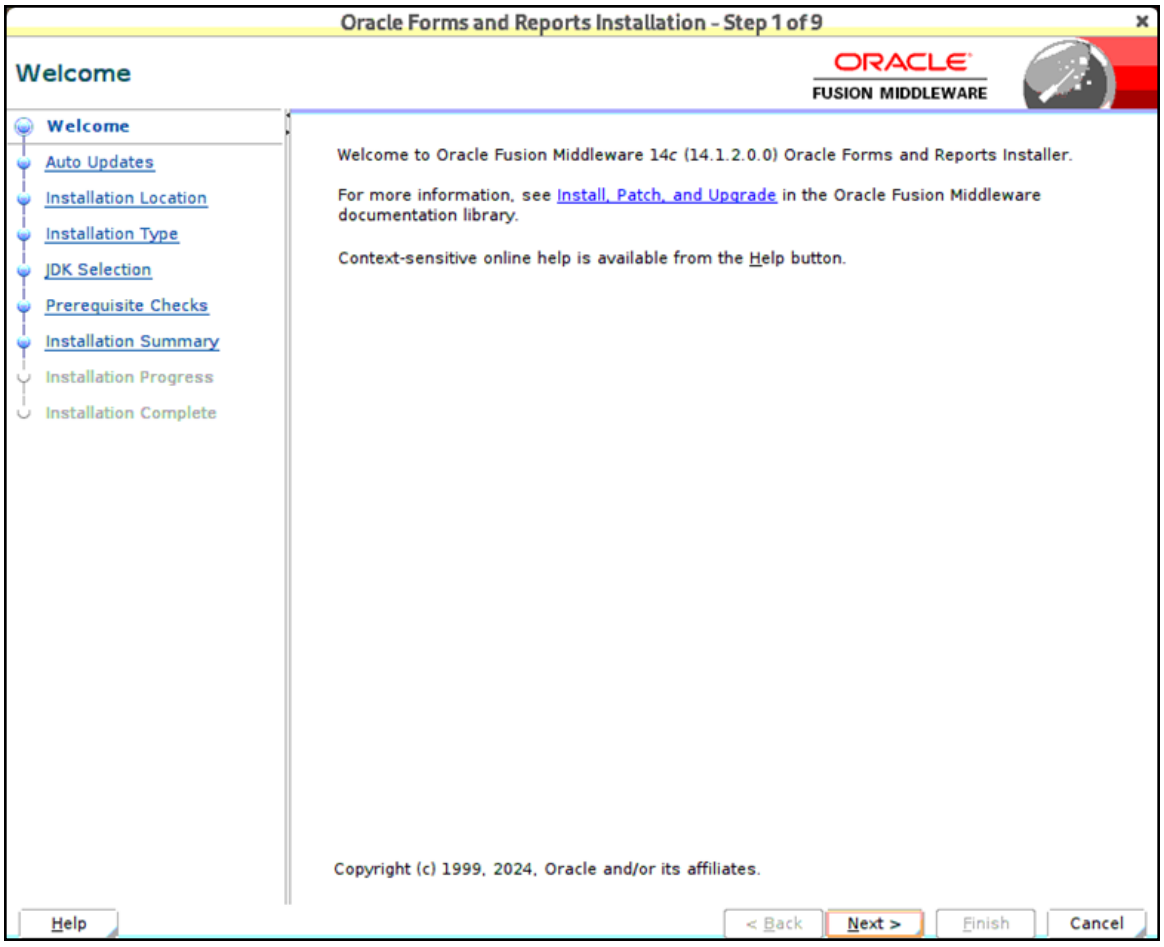
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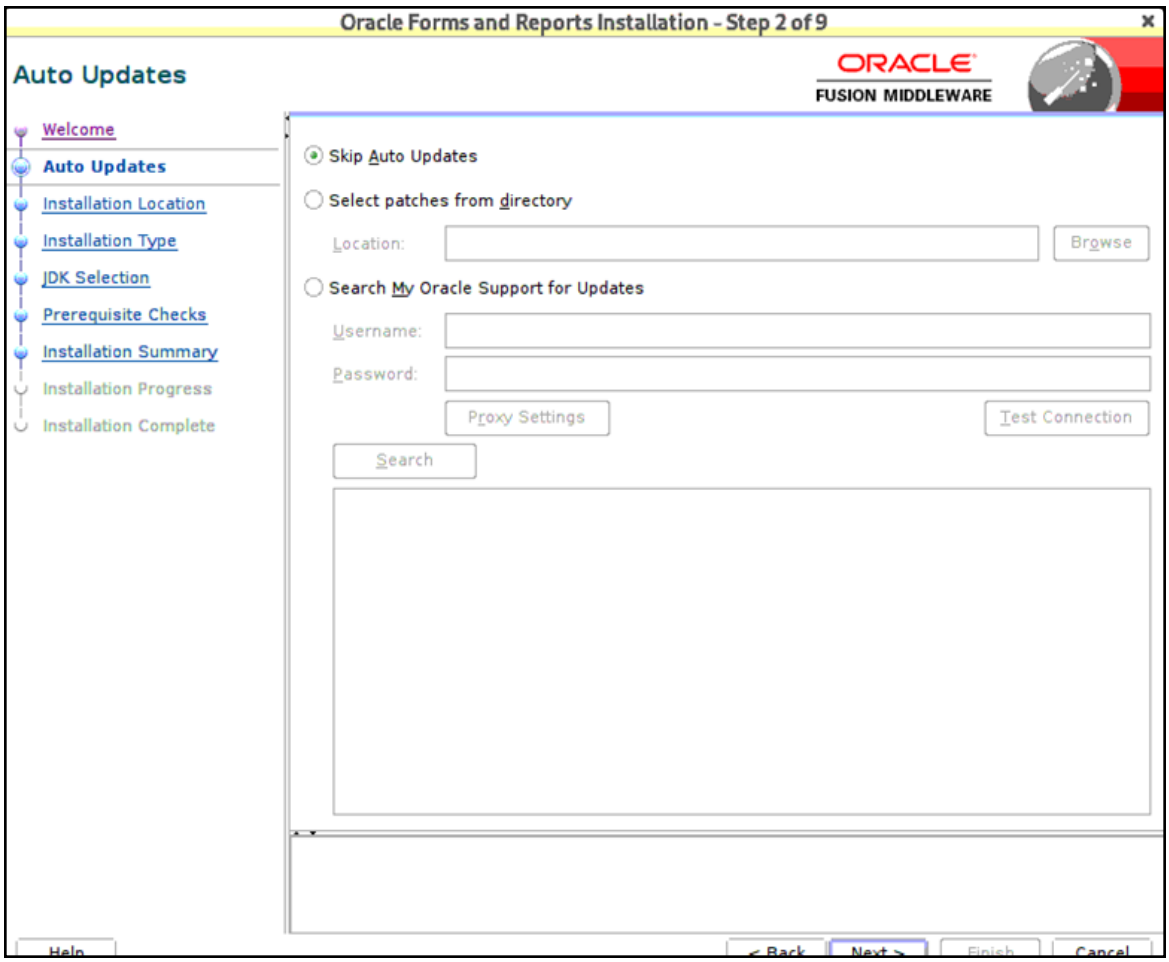
14. Click **Finish**.

6.3 Oracle Forms and Reports Installation

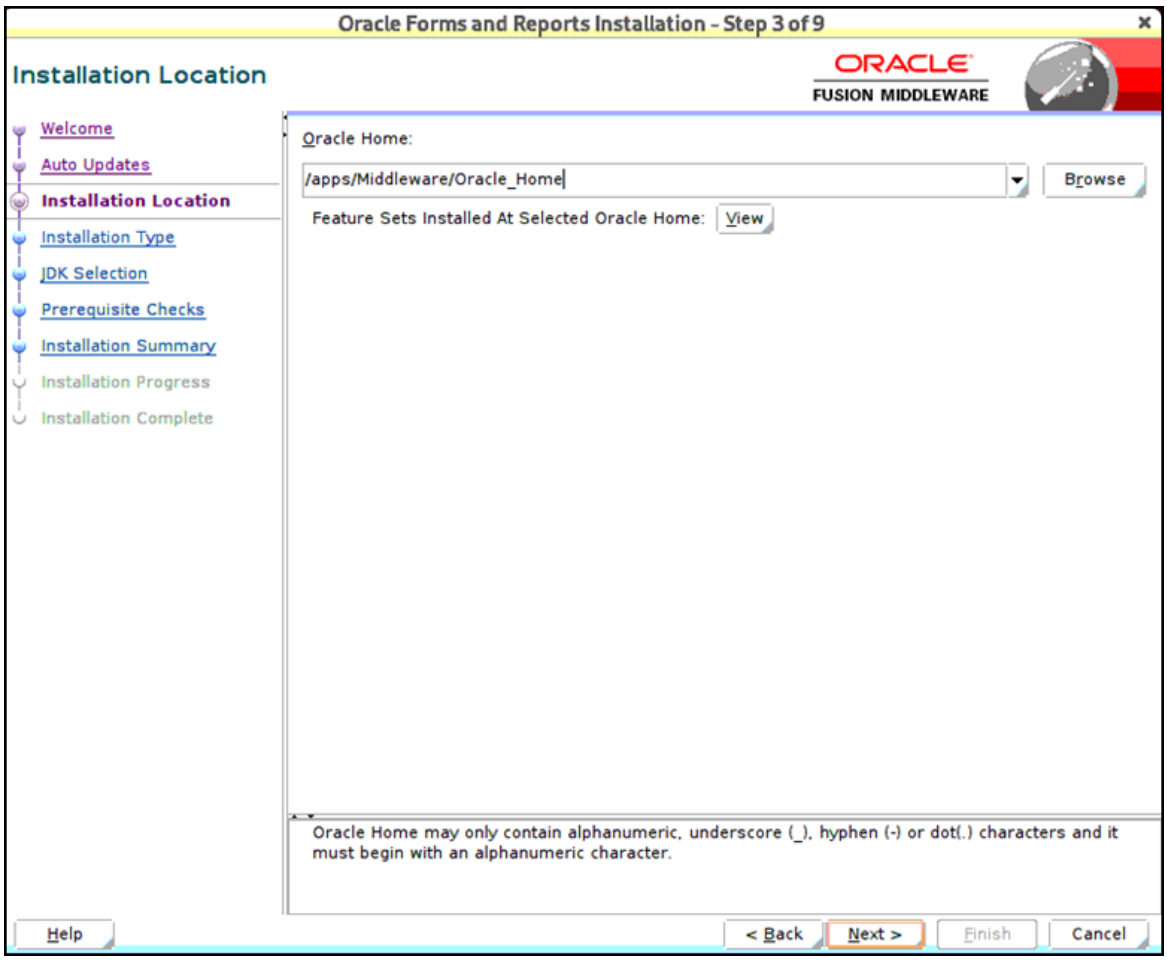
1. Run the following command to install the Fusion Middleware software:
   $ `cd/apps/Software/FMW`
   $ `sh Install_FMW_14c.sh`
   The Oracle Forms and Reports Installer launches, as shown in the following figure.



2. Click **Next**. The Auto Updates screen displays, as shown in the following figure.

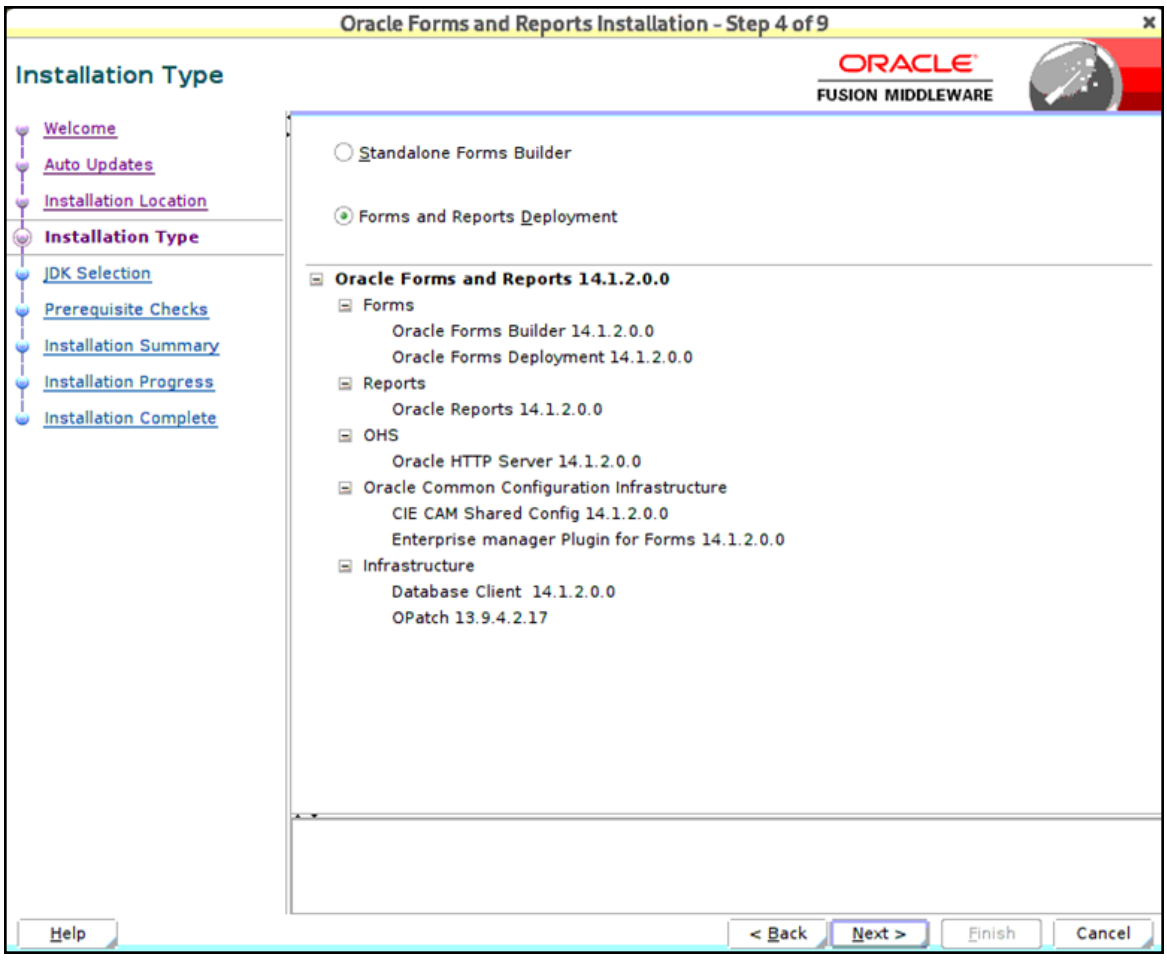


3. Make sure the **Skip Auto Updates** option is selected.
4. Click **Next**. The Installation Location screen displays, as shown in the following figure.

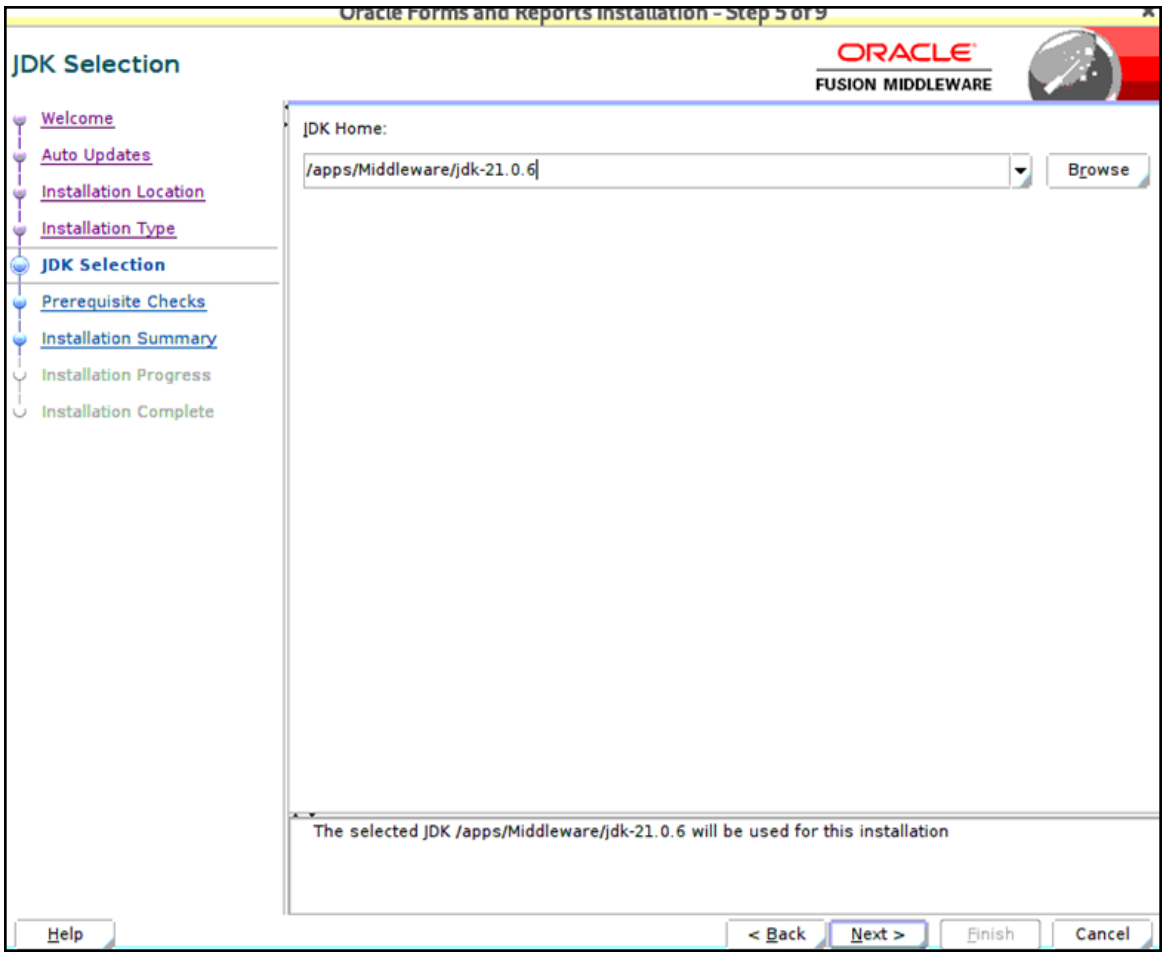


5. Select the **Oracle Home** file path from the drop-down, or click **Browse** to locate it.
6. Click **Next**. The Installation Type screen displays, as shown in the following figure.

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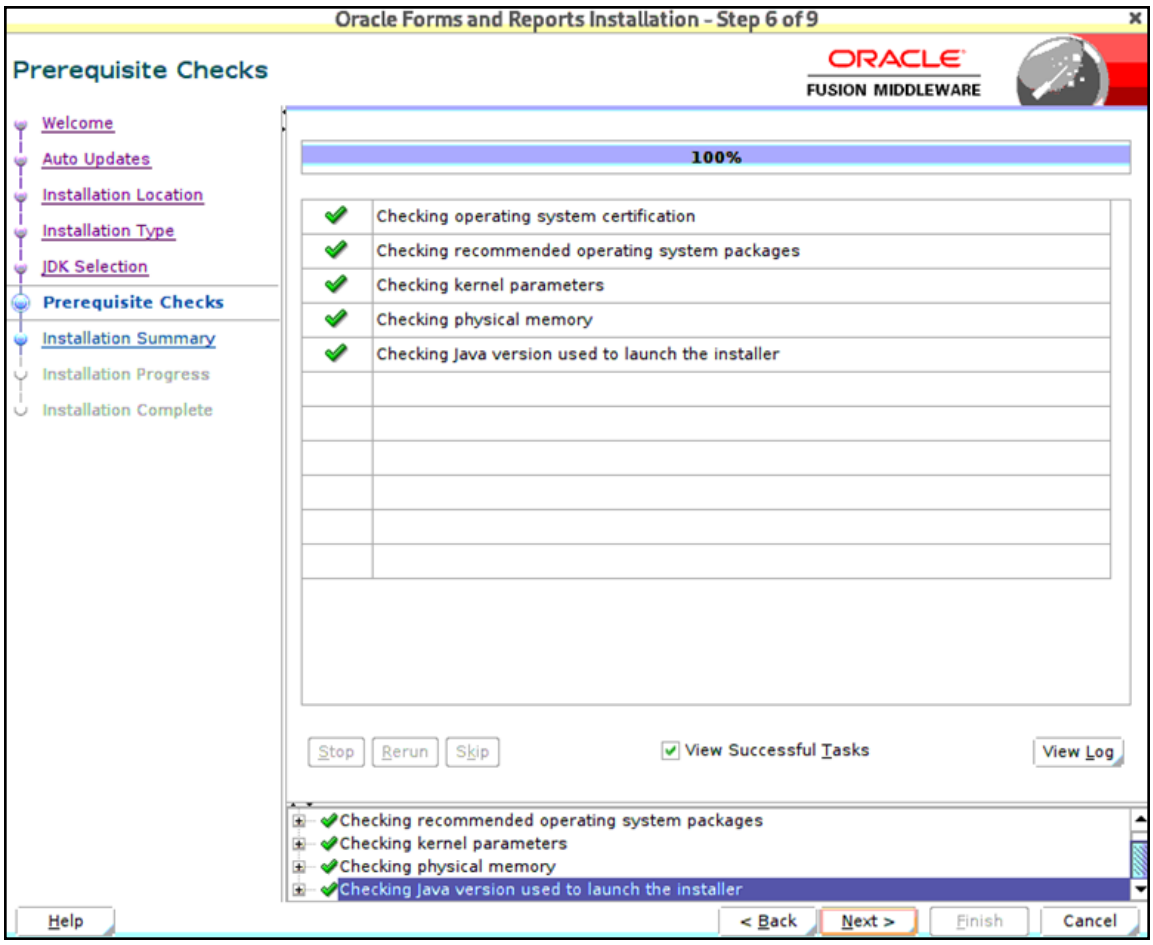


7. Make sure the **Forms and Reports Deployment** option is selected.
8. Click **Next**. The JDK Selection screen displays, as shown in the following figure.

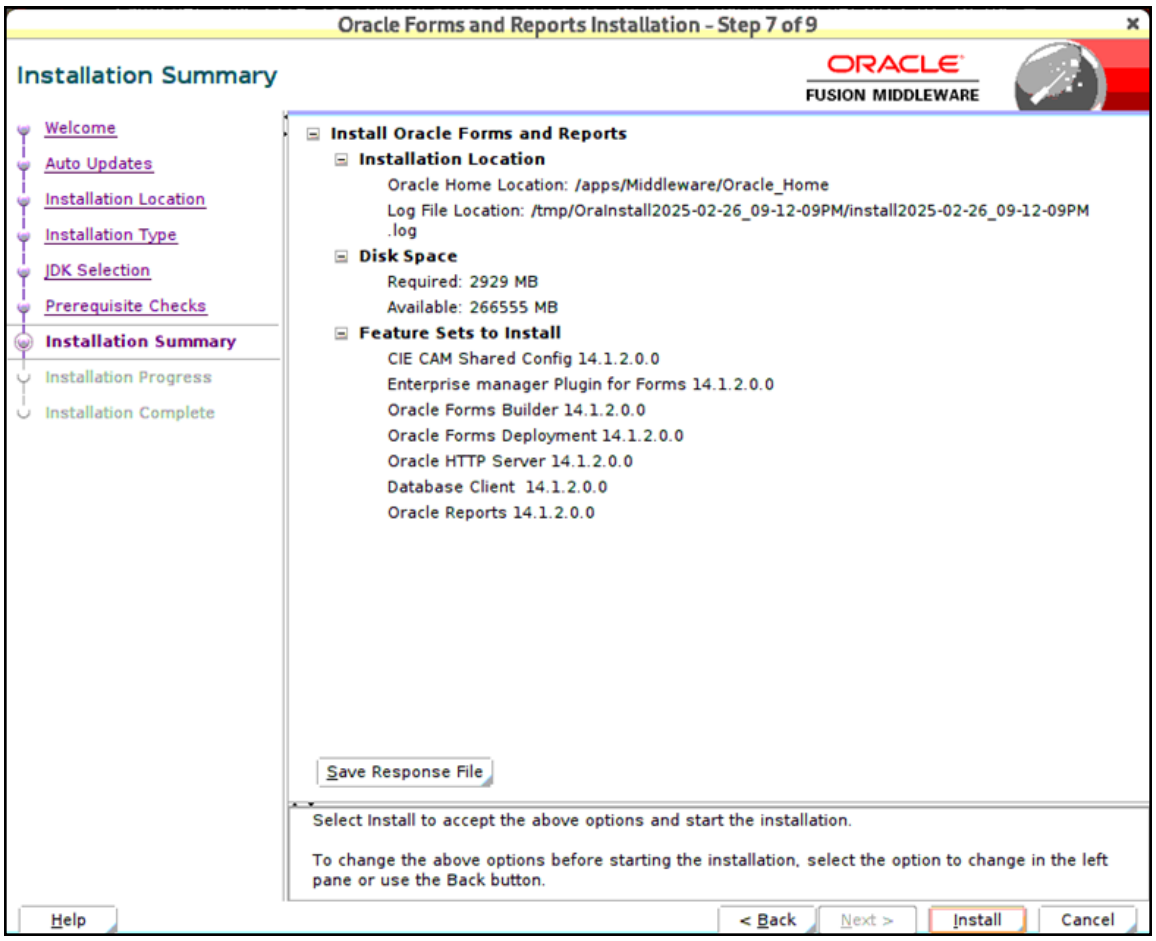


9. Make sure **/apps/Middleware/jdk-21.0.6** is selected in the JDK Home drop-down.
10. Click **Next**. The Prerequisite Checks screen displays, as shown in the following figure.

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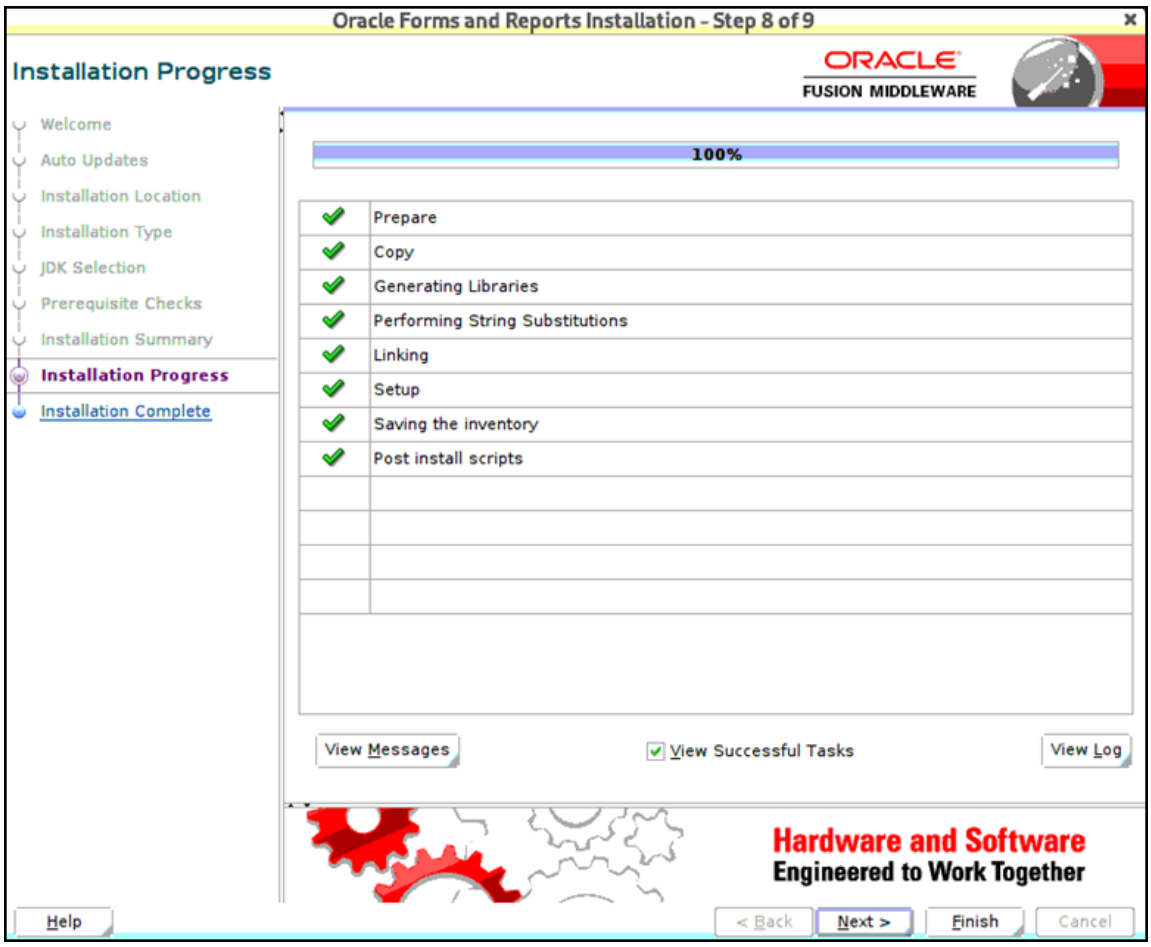


11. When the checks are completed successfully, click **Next**. The Installation Summary screen displays, as shown in the following figure.

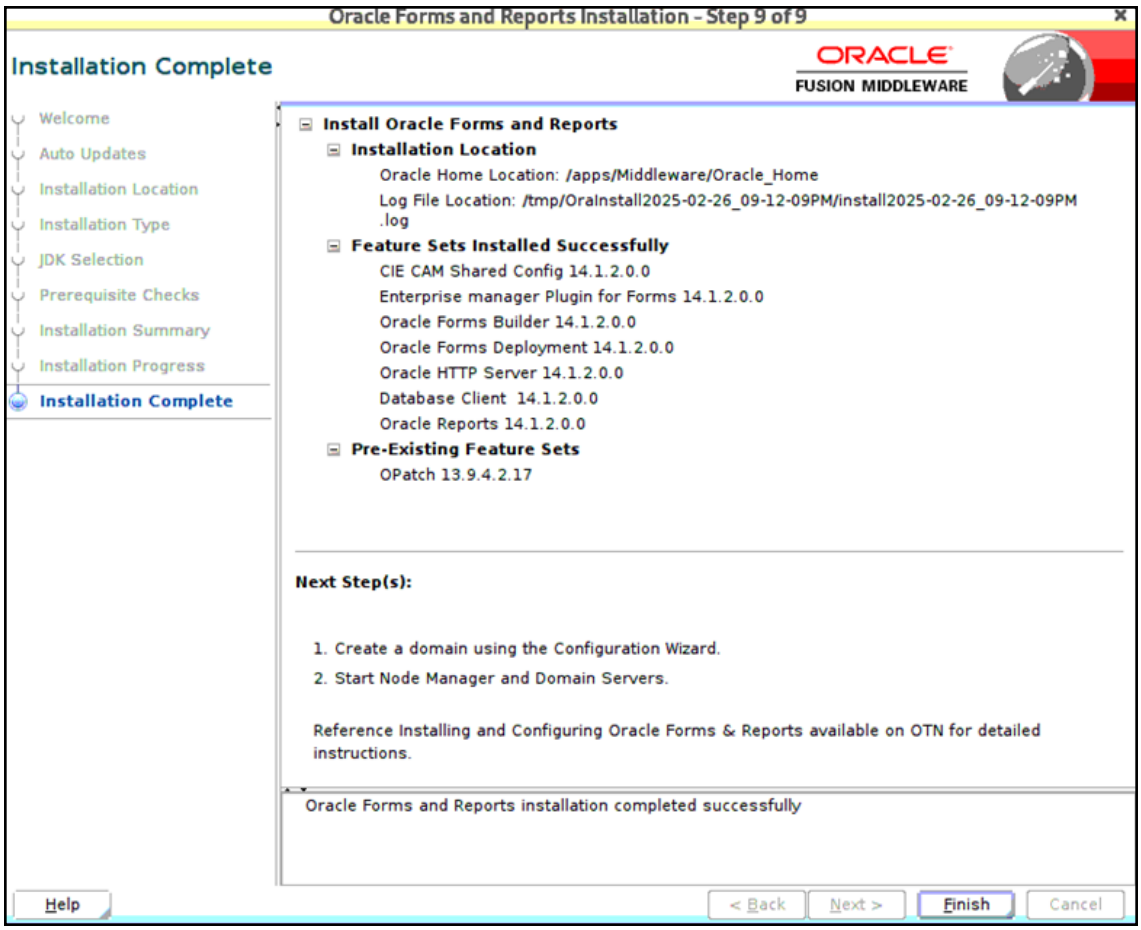


12. Click **Install** to start the Oracle forms and reports installation. The Installation Progress screen displays, as shown in the following figure.

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13. When the installation finishes successfully, click **Next**. The Installation Complete screen displays, as shown in the following figure.



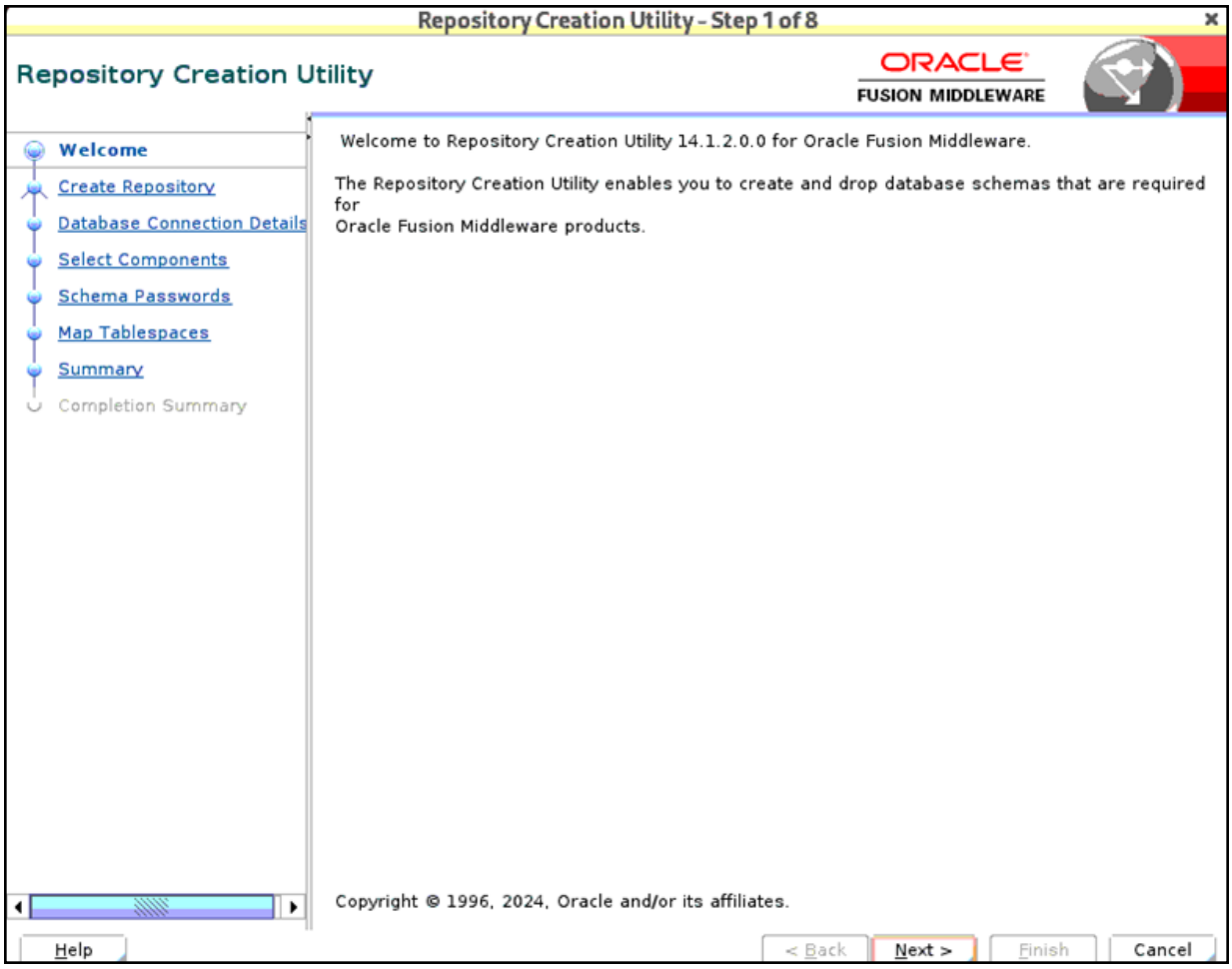
14. Click **Finish**.

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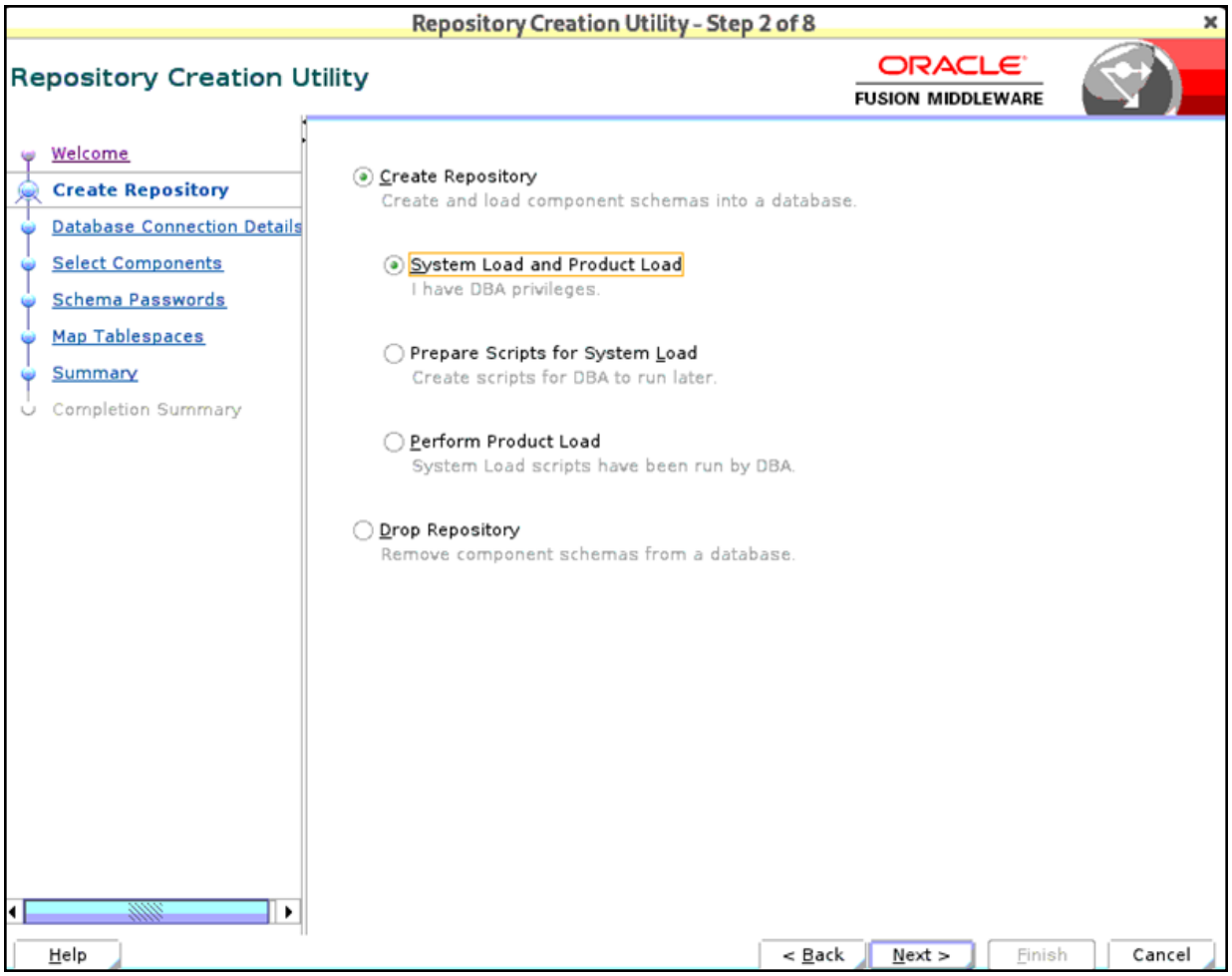
6.4 Repository Creation Utility

1. Run the following script to create the FMW Repository:
   $ `/apps/Middleware/Oracle_Home/oracle_common/bin/rcu`

   The Repository Creation Utility (RCU) launches, as shown in the following figure.

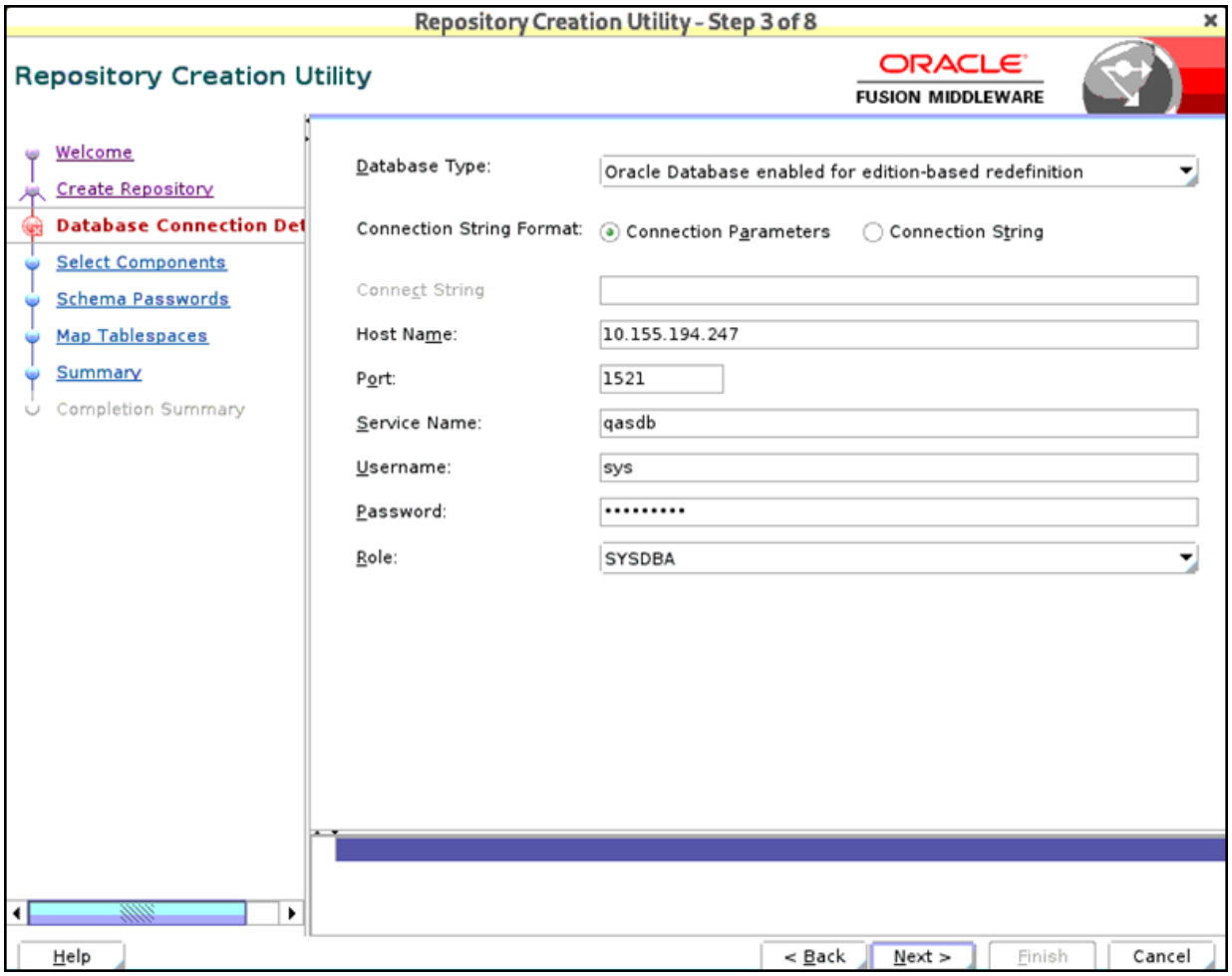


2. Click **Next**. The Create Repository screen displays, as shown in the following figure.



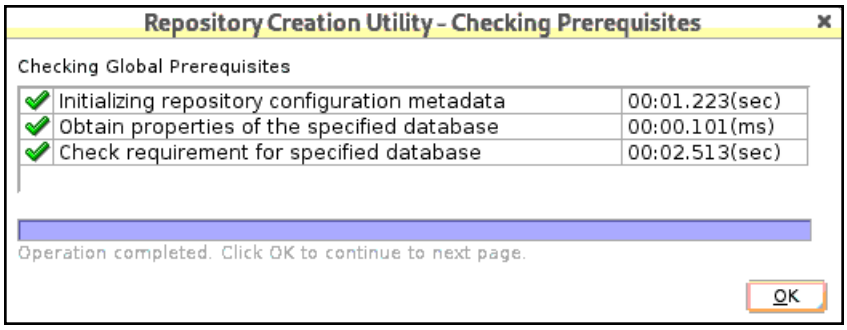
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3. Click **Next**. The Database Connection Details screen displays, as shown in the following figure.

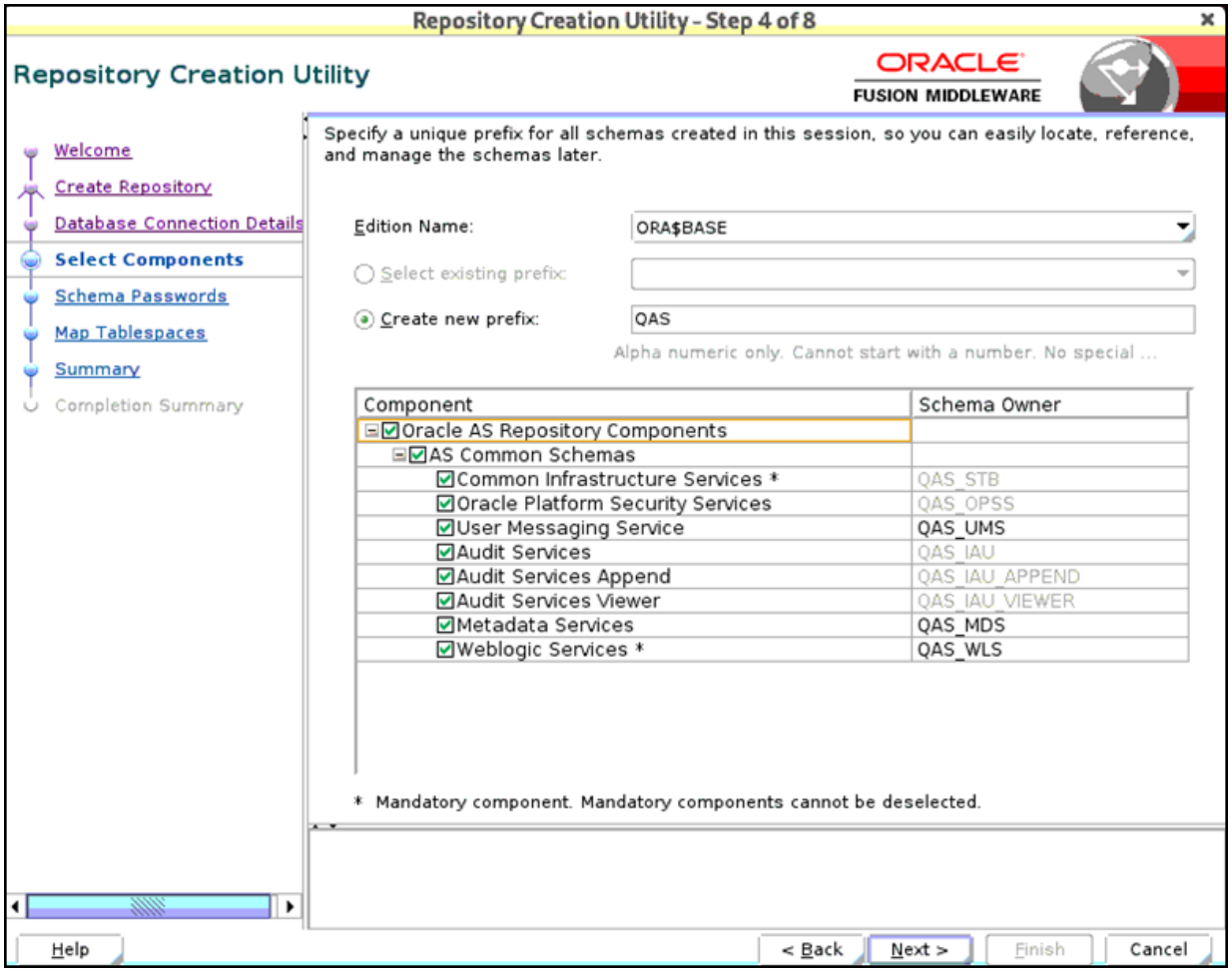


4. Verify the values that are automatically populated, and enter the following database connection values:
   - **Database Type:** Oracle Database
   - **Host Name:** Database Server's IP Address or Host name
   - **Port:** Database Port Number; default is 1521
   - **Service Name:** Blood Centers Database Name (bcprd in the example above)
   - **Username:** sys
   - **Password:** Password for sys user (for example, Password1)
   - **Role:** SYSDBA

5. Click **Next**. The RCU system checks the global prerequisites, as shown in the following figure.

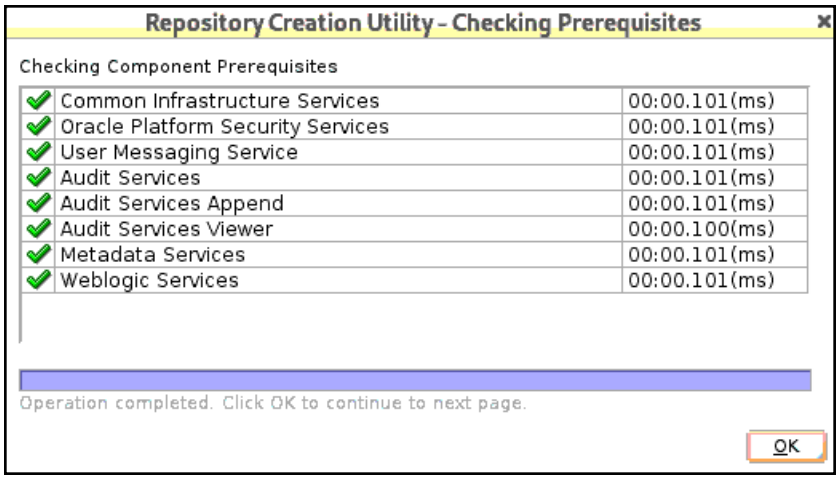


6. Click **OK** when the checks are finished. The Select Components screen displays, as shown in the following figure.

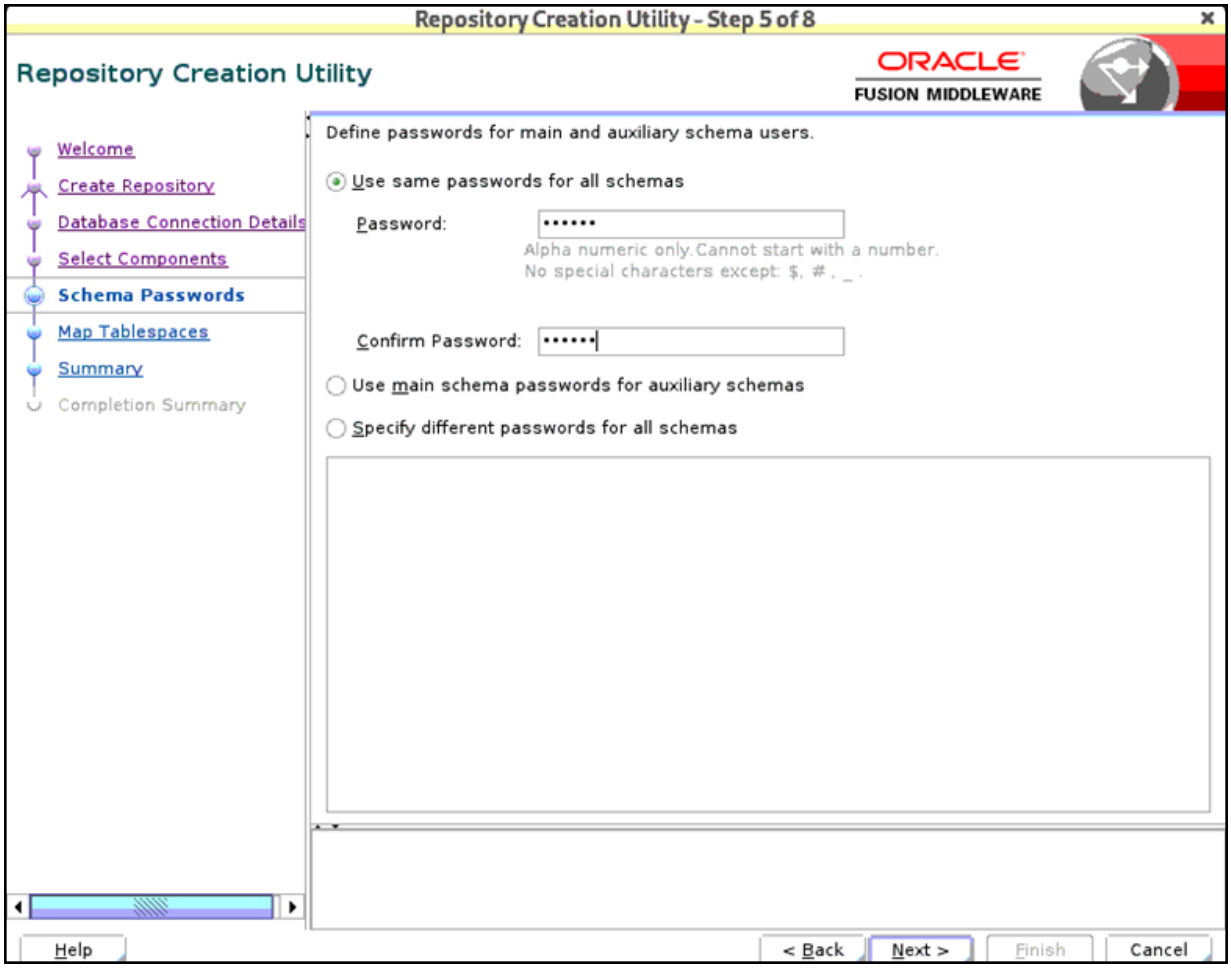


7. Select the **Create new prefix** option, and enter `BC` in the **Create new prefix** box.
8. In the Component table, select the **Oracle Platform Security Services** component.

9. Click **Next**. The RCU system checks the component prerequisites, as shown in the following figure.

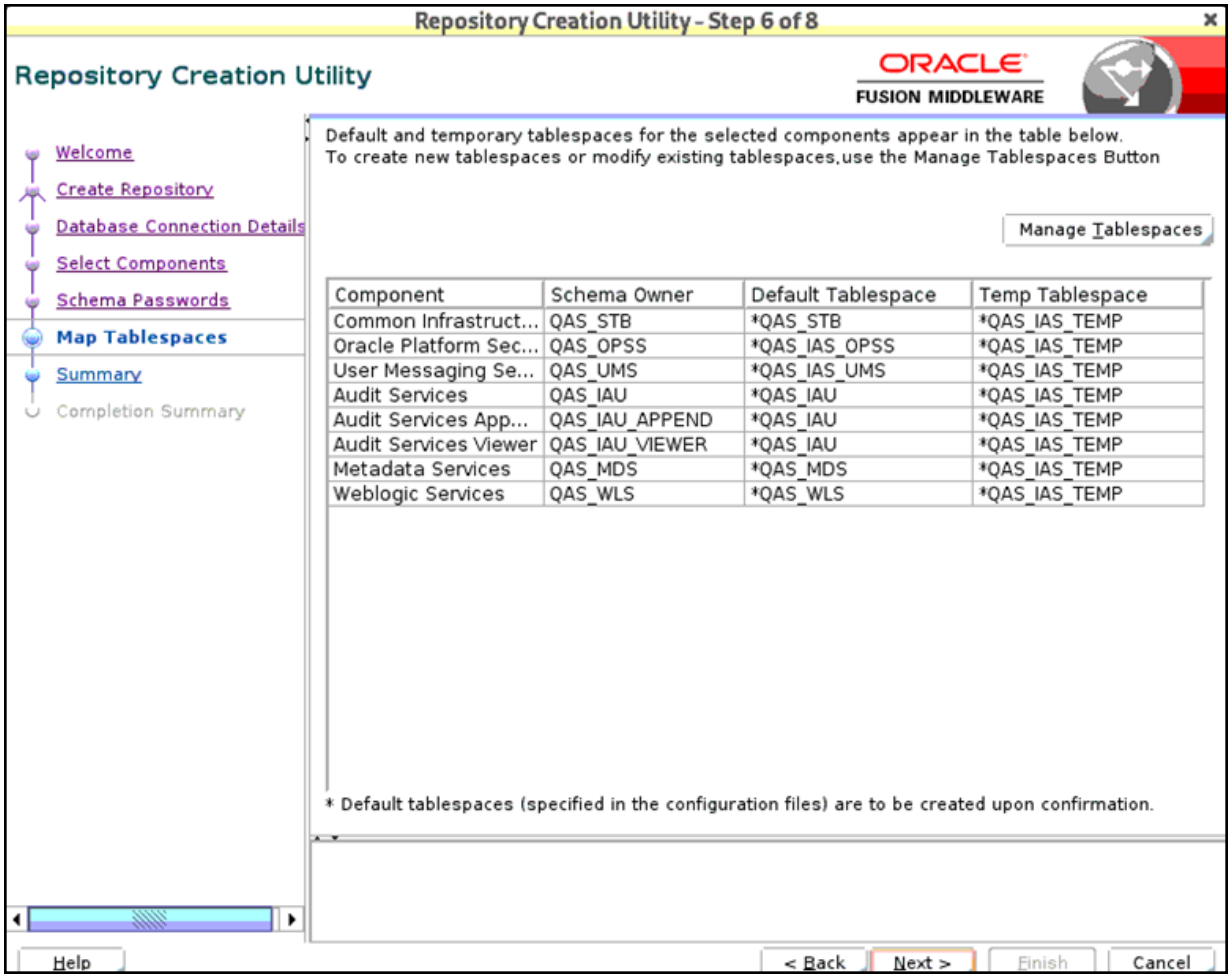


10. Click **OK** when the check is finished. The Schema Passwords screen displays, as shown in the following figure.

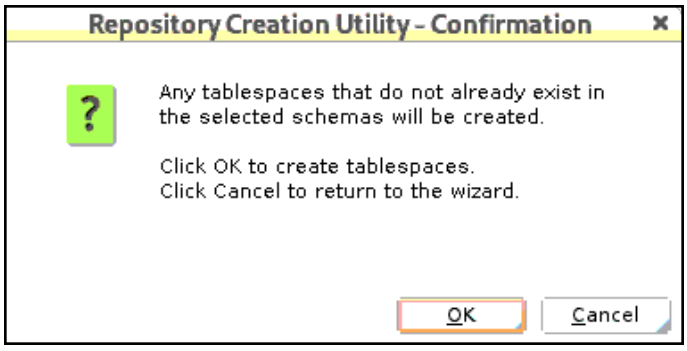


11. Select the **Use same passwords for all schemas** option.

12. Enter the password **`fmw14c`** in both the Password and Confirm Password boxes.
13. Click **Next**. The Map Tablespaces screen displays, as shown in the following figure.

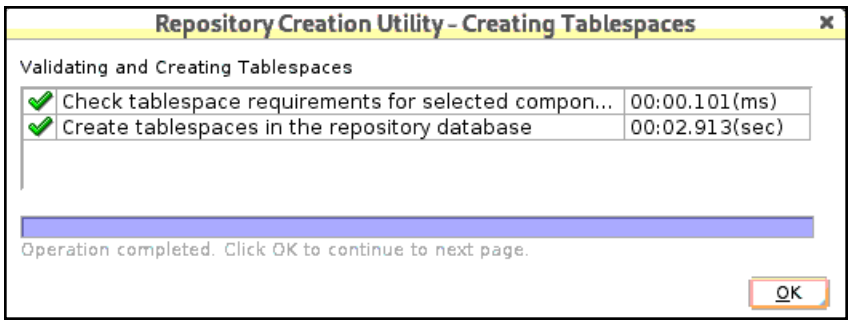


14. Click **Next**. A confirmation message appears, stating that any tablespaces that do not already exist will be created, as shown in the following figure.

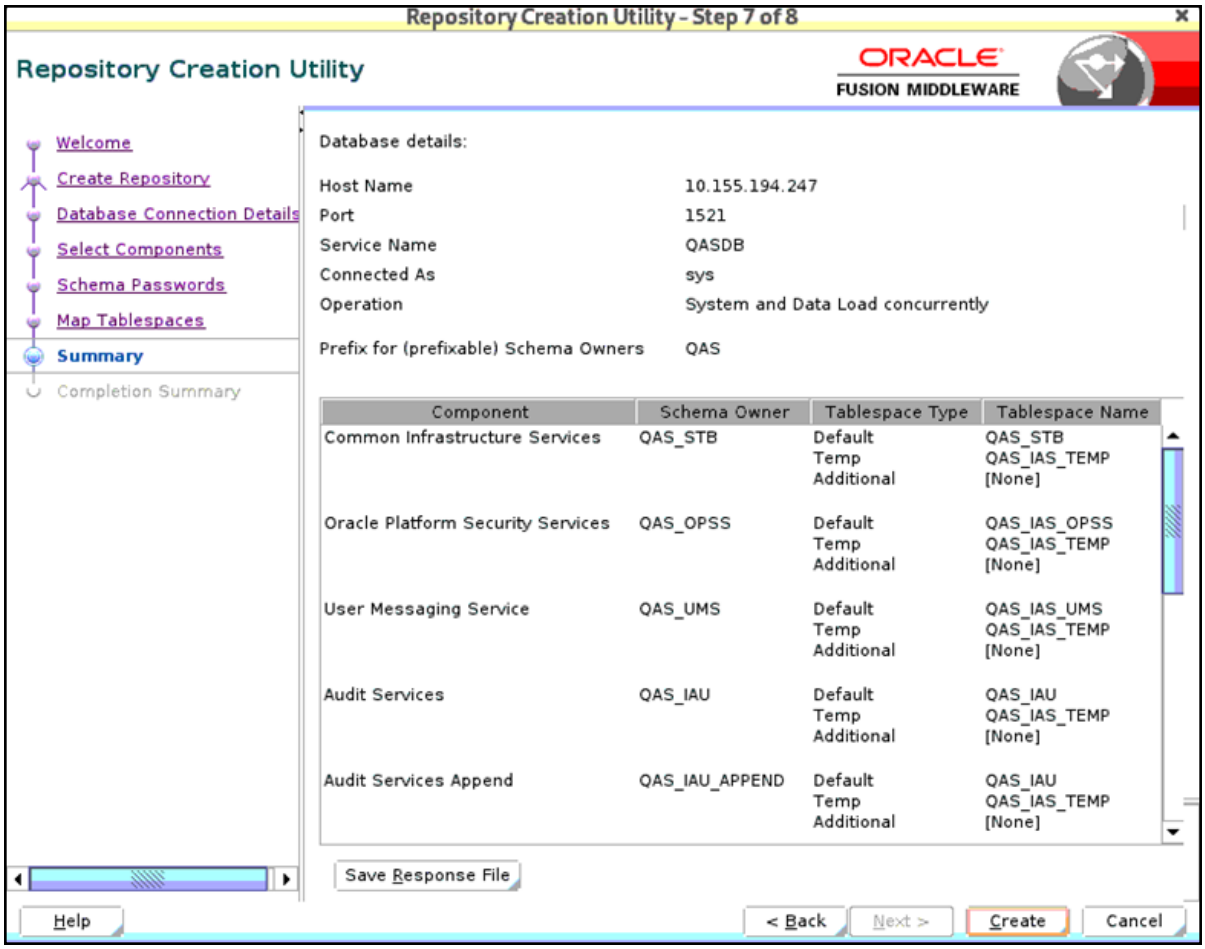


15. Click **OK**. The RCU system validates and creates the tablespaces, as shown in the following figure.

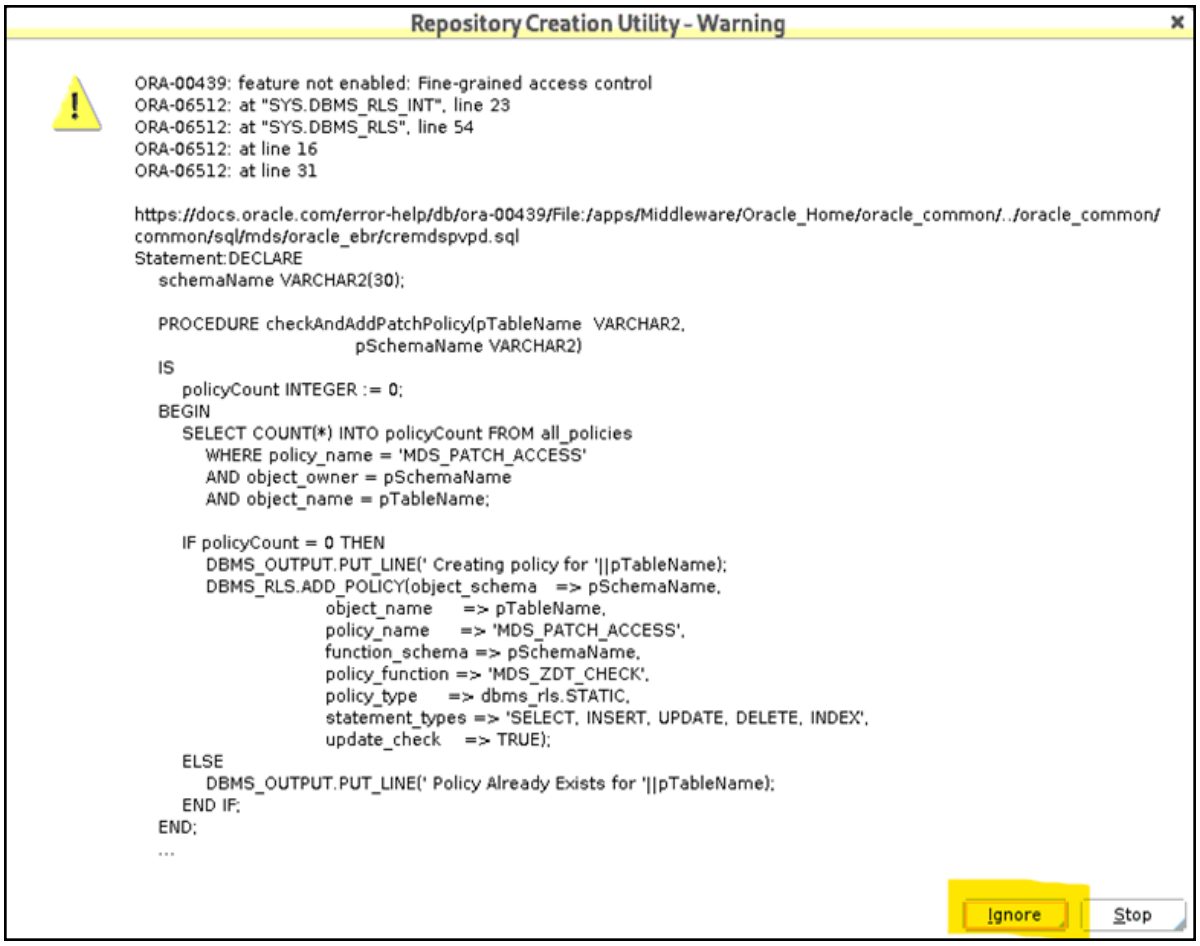
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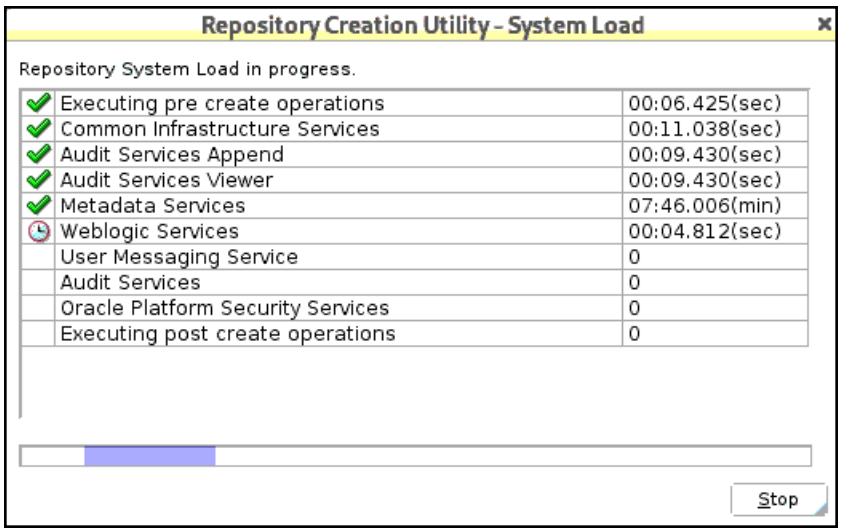
16. Click **OK** when the tablespaces validation and creation are completed. The Summary screen displays, as shown in the following figure.



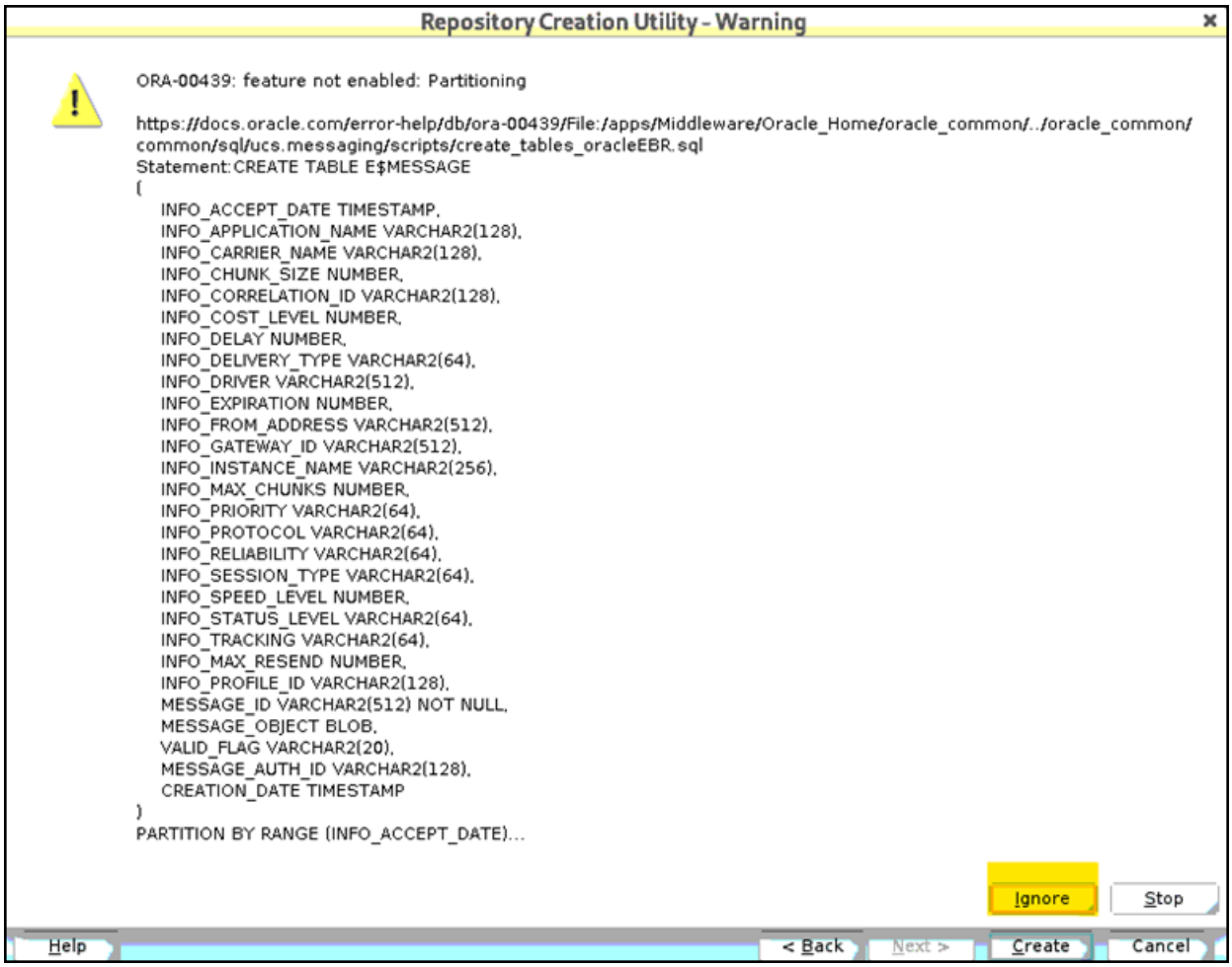
17. Click **Create** to create the Oracle repository. A Repository Creation Utility - Warning screen displays, as shown in the following figure.

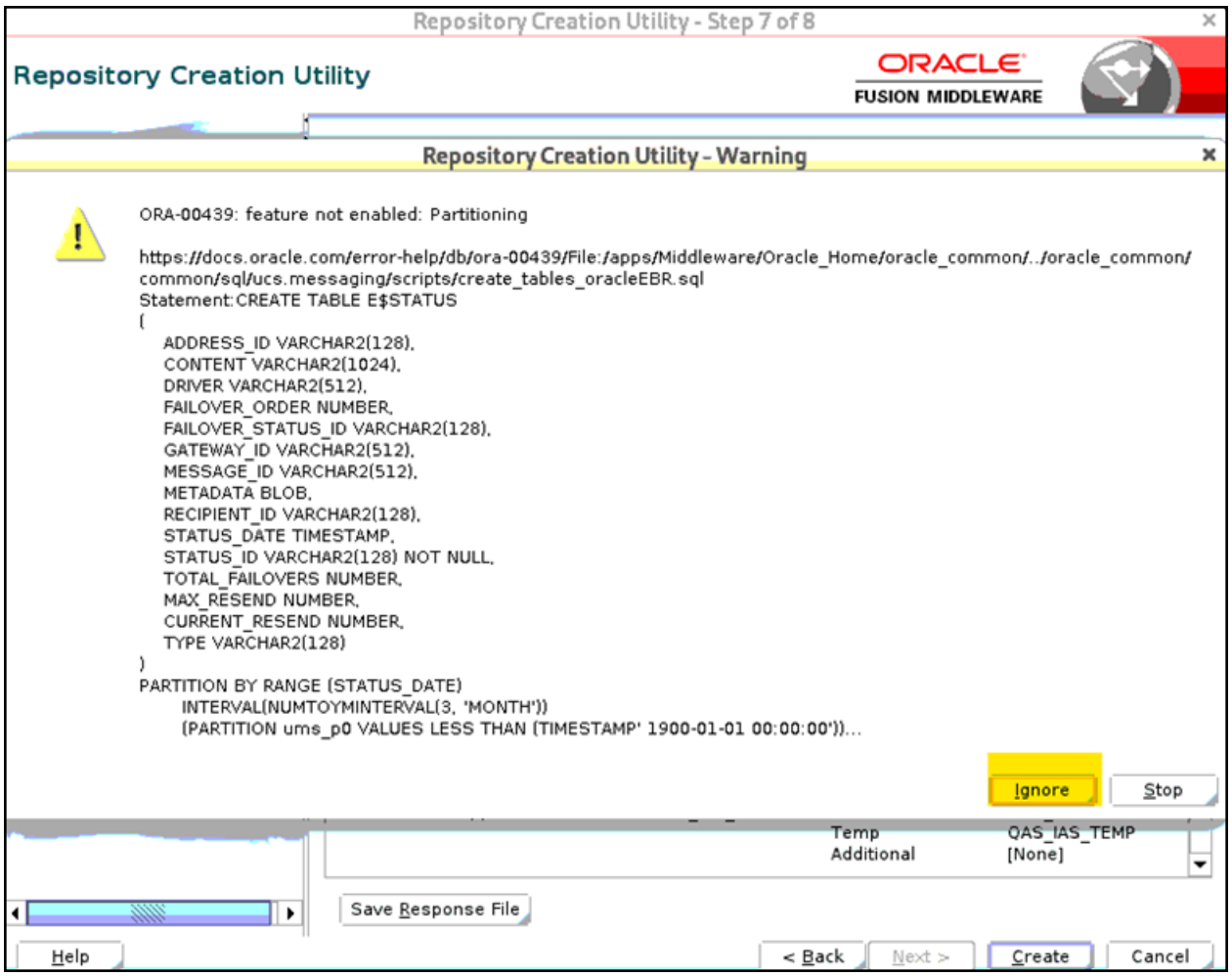


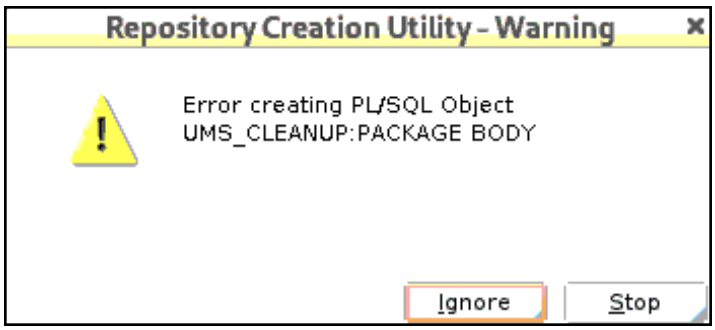
18. Click **Ignore**. Repository Creation Utility - System Load In Progress screen displays, as shown in the following figure.



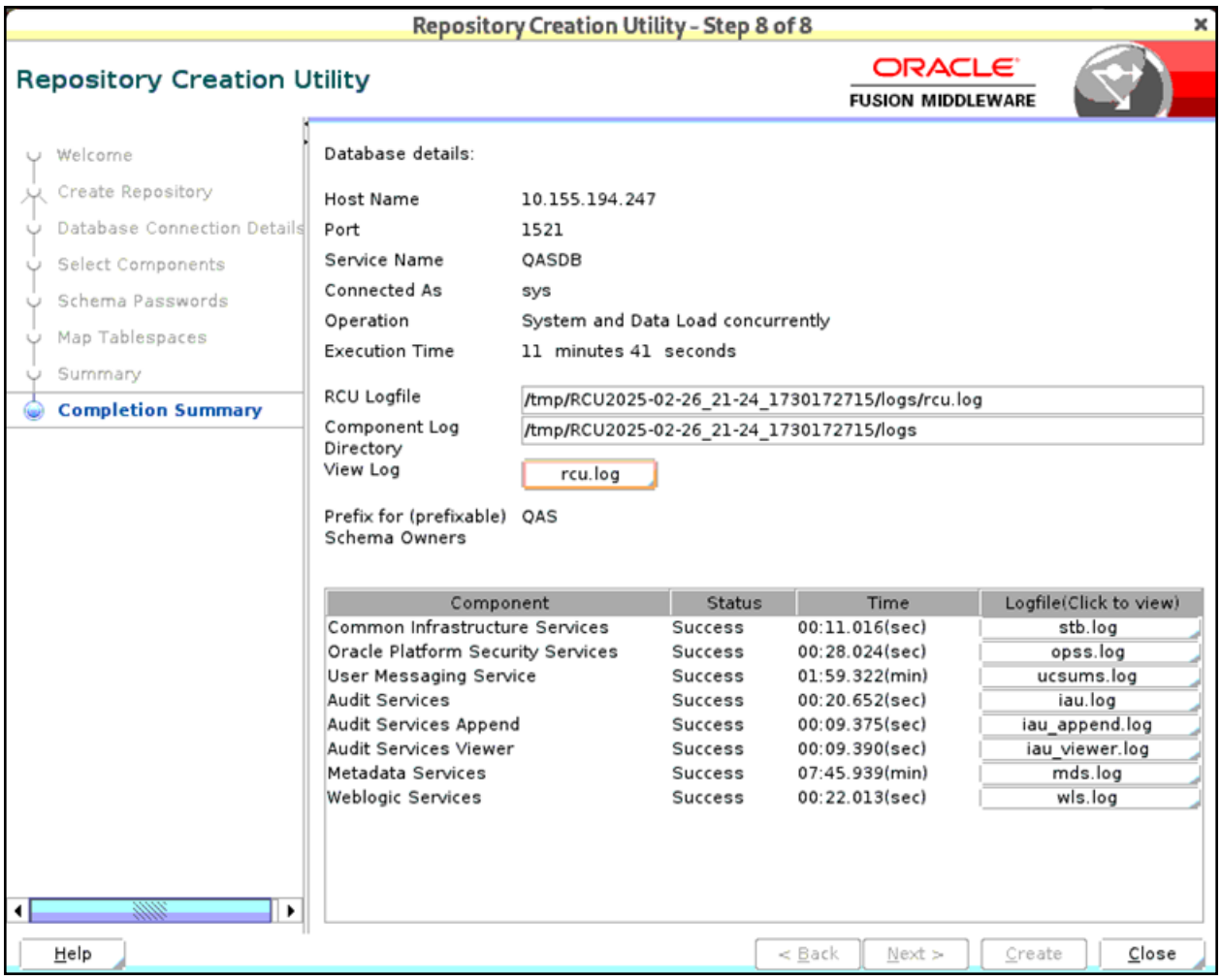
19. The Installer displays a few warnings like following figures. Click **Ignore**.







20. When the repository created, the Completion Summary screen displays, as shown in the following figure.

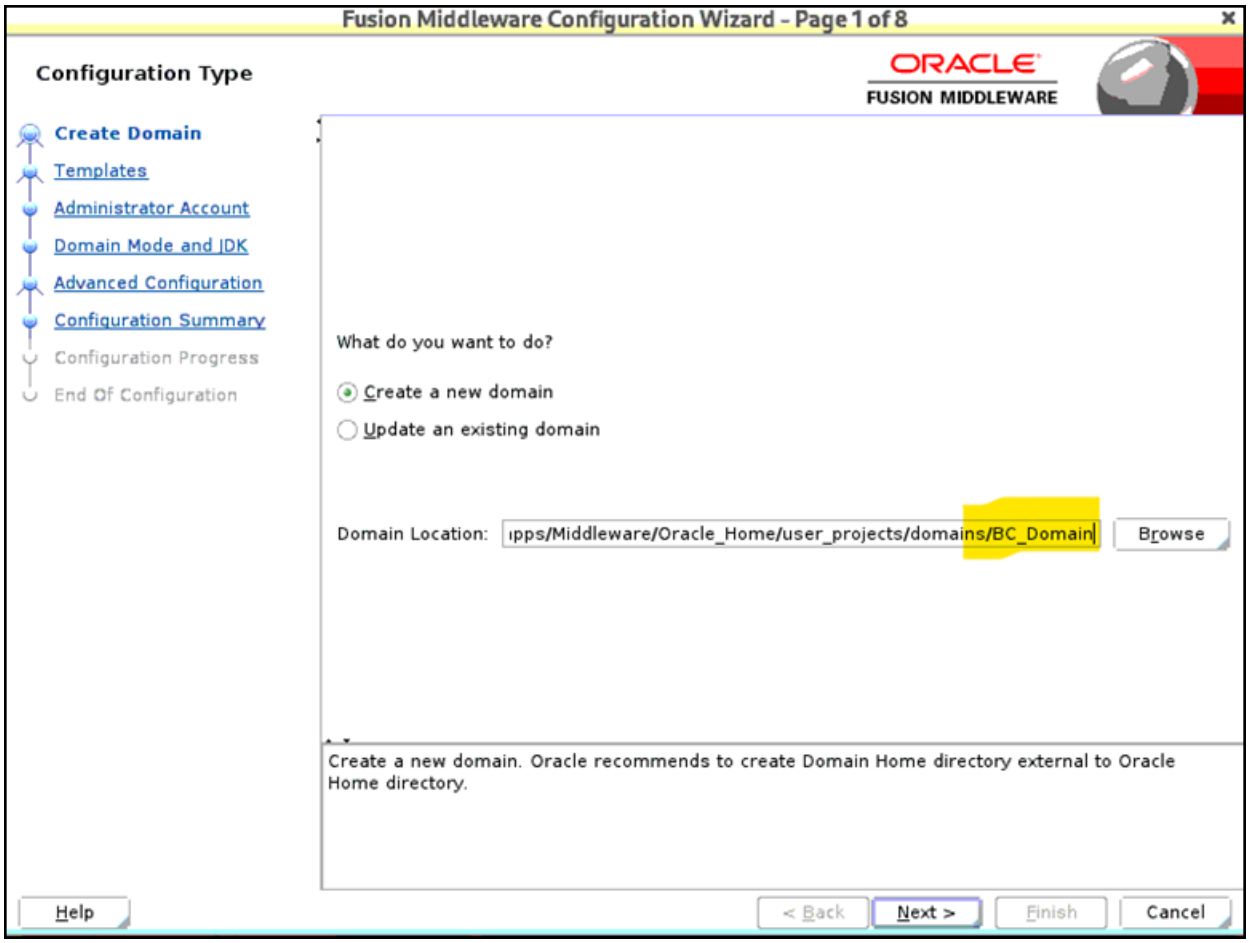


21. Click **Close** to exit the RCU.

6.5 FMW Configuration Wizard

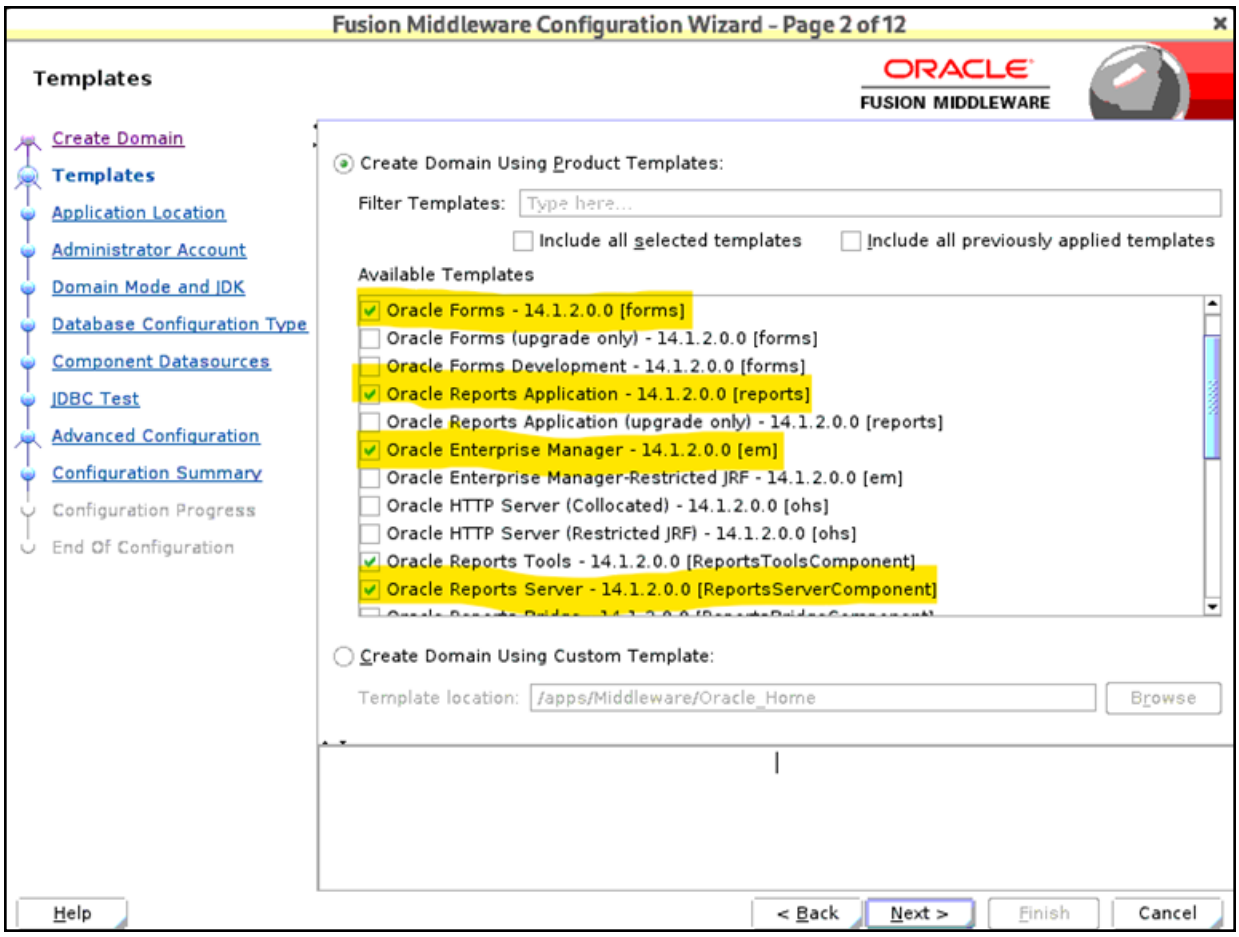
1. Run the following command to launch the FMW configuration:
   `$ /apps/Middleware/Oracle_Home/oracle_common/common/bin/config.sh`

   The Fusion Middleware Configuration Wizard opens, as shown in the following figure.



2. On the Create Domain screen, select the **Create a new domain** option.
3. In the **Domain Location** box, change `base_domain` to **`BC_Domain`**, as shown in the example above.
4. Click **Next**. The Templates screen displays, as shown in the following figure.

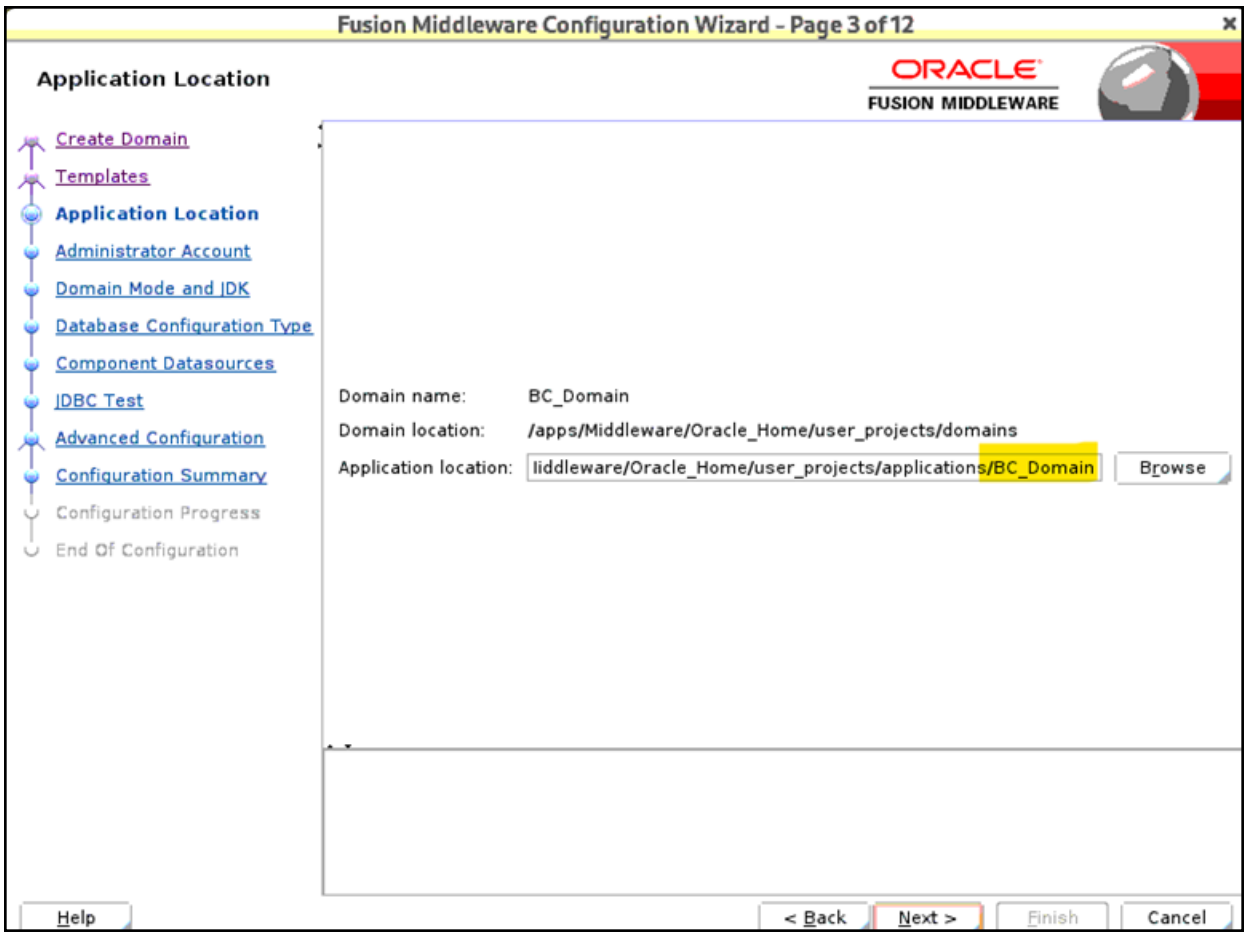
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5. Make sure the **Create Domain Using Product Templates** option is selected, and that the **All Templates** option is selected in the *Template Categories* dropdown.
6. Select the following in the *Available Templates* section:
   - **Oracle Forms -14.1.2.0.0 [forms]**
   - **Oracle Reports Application – 14.1.2.0.0[reports]**
   - **Oracle Reports Server – 14.1.2.0.0 [ReportsServerComponent]**

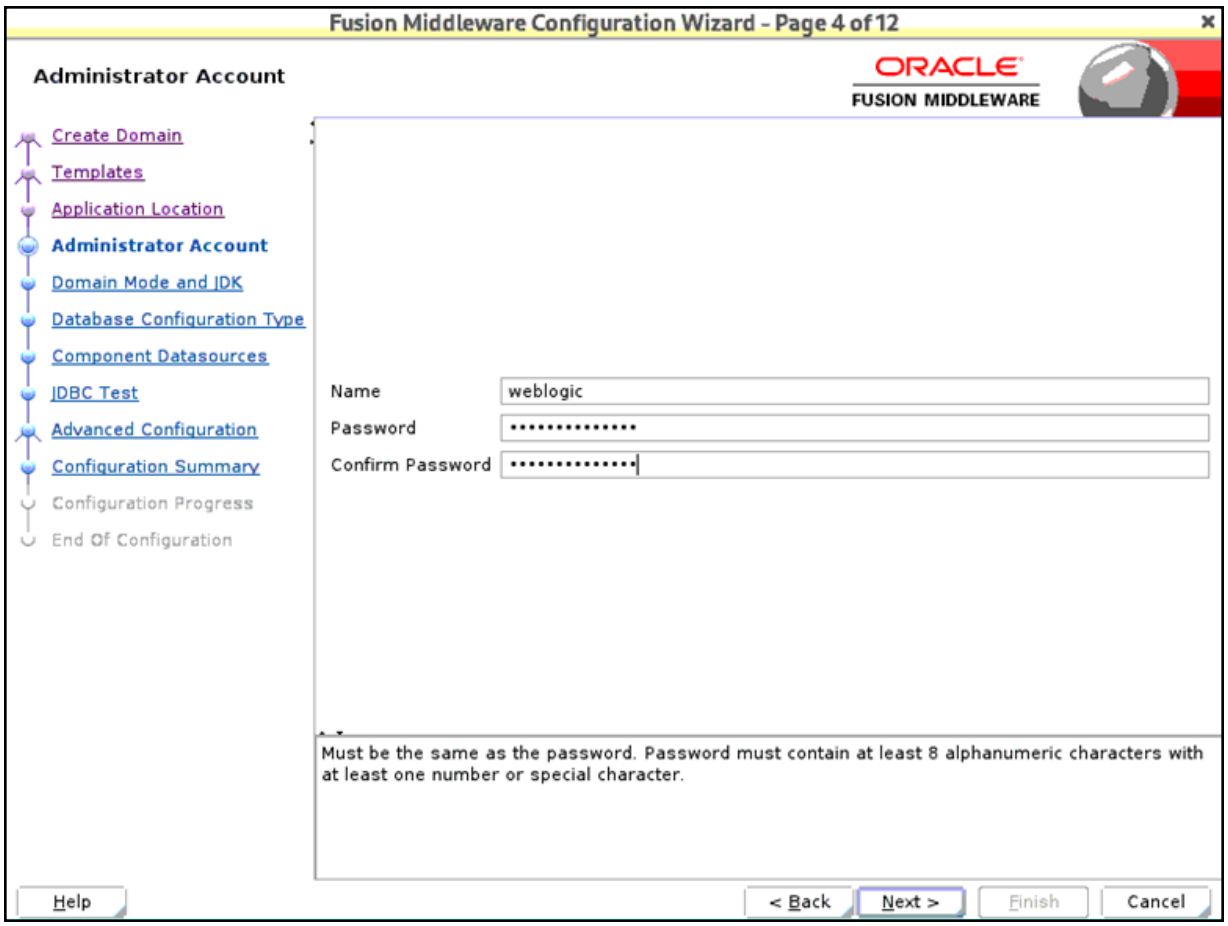
> **NOTE –** Other dependent options will be automatically selected.

7. Click **Next**. The Application Location screen displays, as shown in the following figure.

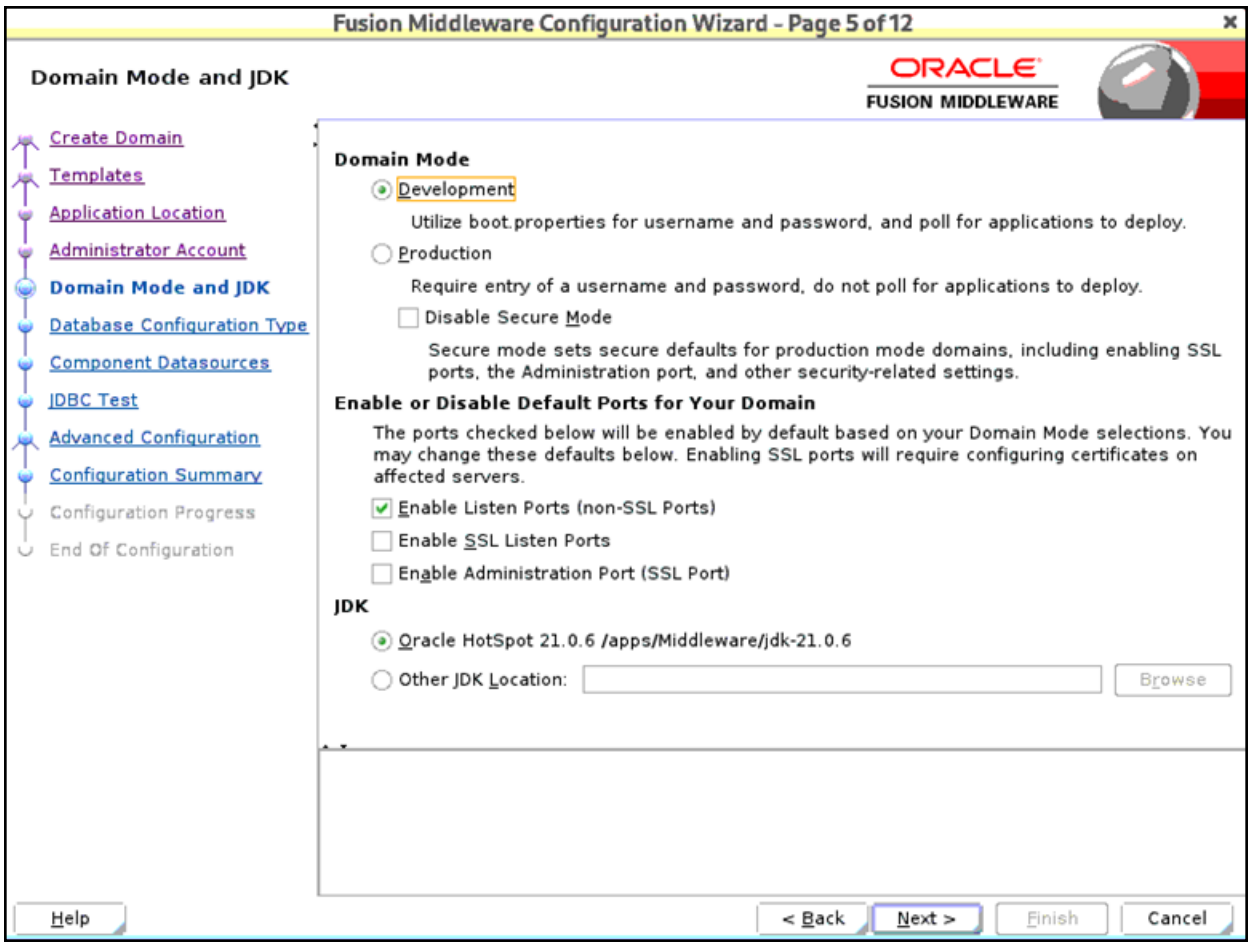


8. Verify the Domain name, Domain location, and the Application location are correct.
9. Click **Next**. The Administrator Account screen displays, as shown in the following figure.

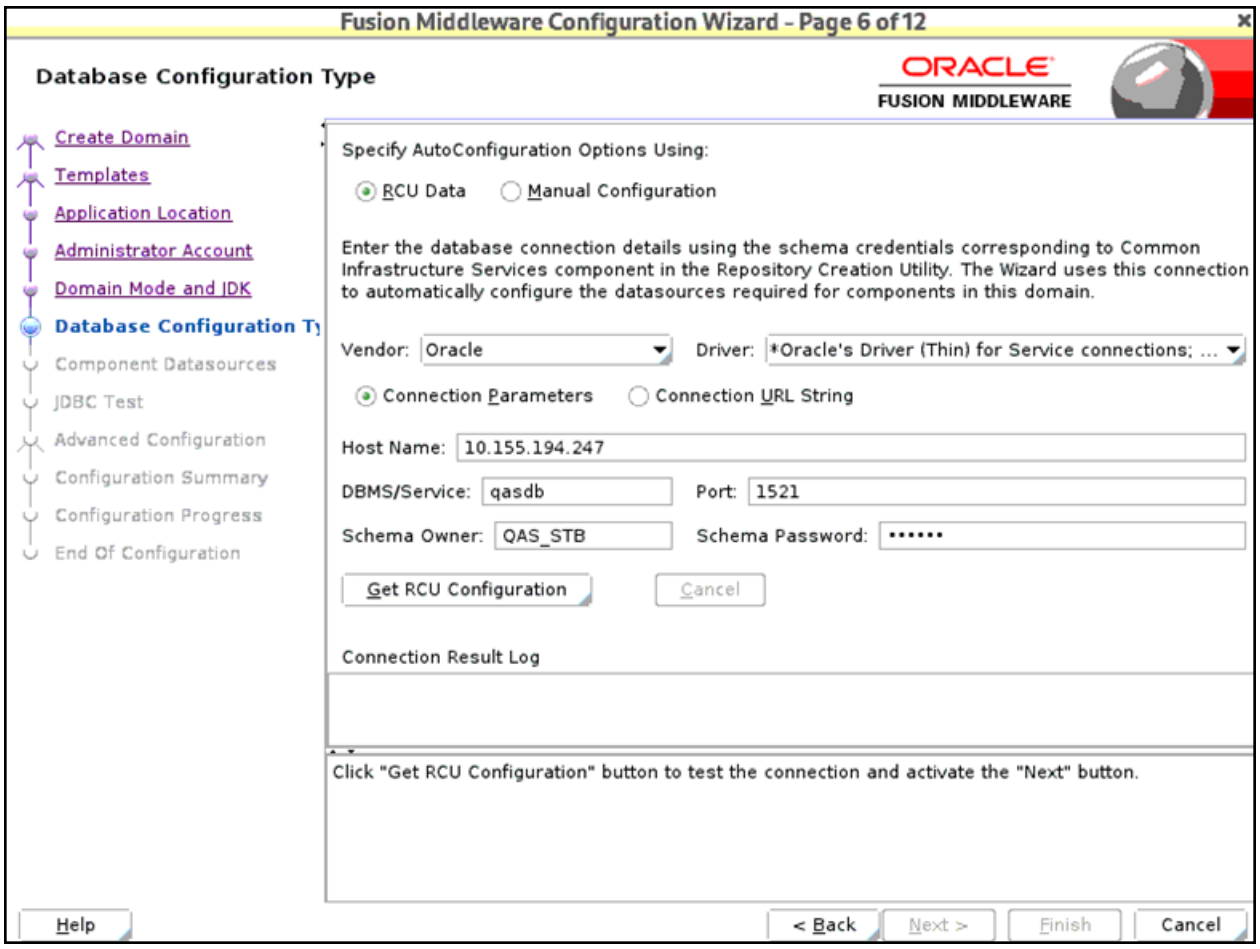
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10. Enter `weblogic` in the **Name** box.
11. Enter `WebLogicAdmin1` in the **Password** and **Confirm Password** boxes.
12. Click **Next**. The Domain Mode and JDK screen displays, as shown in the following figure.

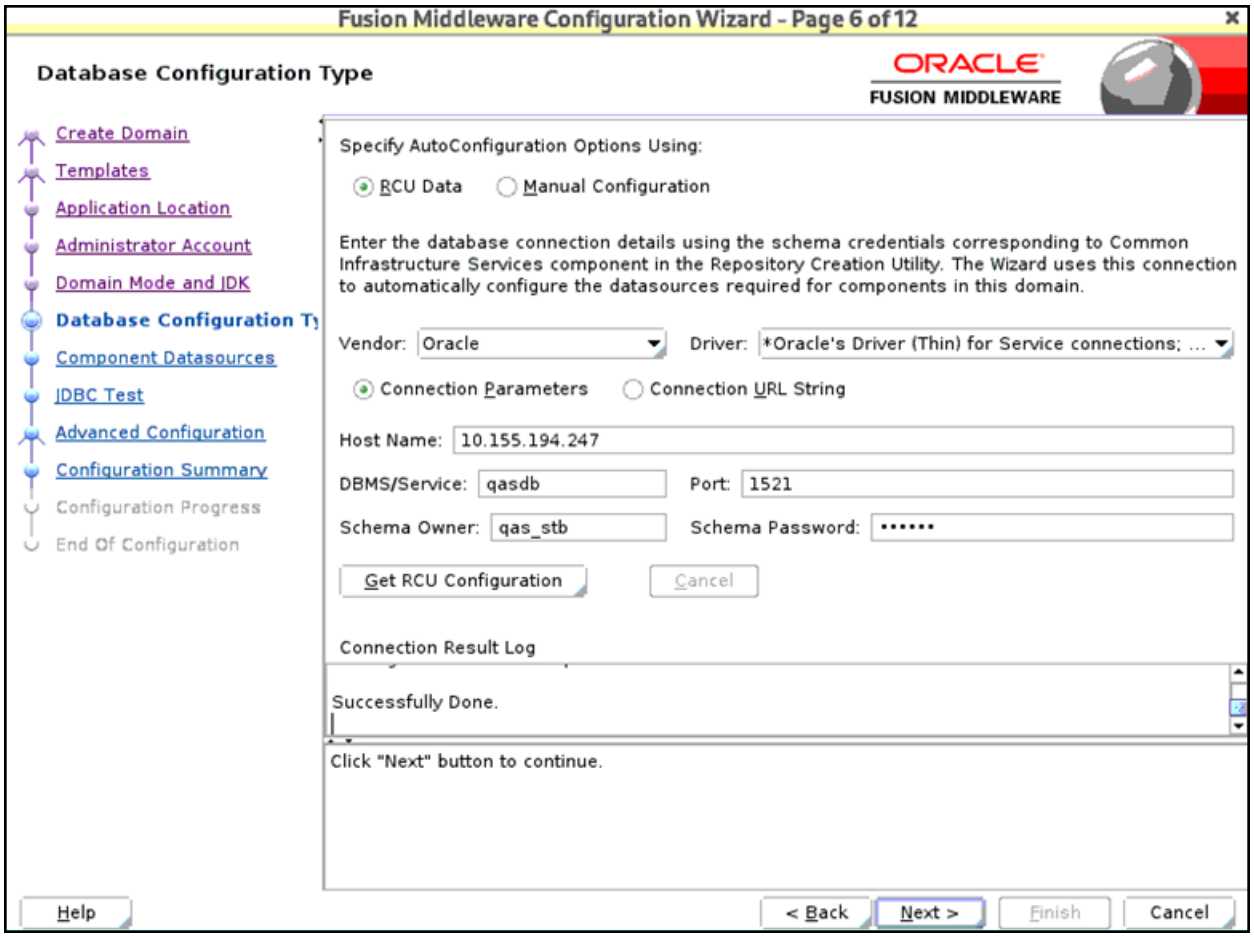


13. Verify the **Development** option is selected under *Domain Mode*.
14. Verify the **Oracle HotSpot 21.0.6 /apps/Middleware/jdk-21.0.6** option is selected under *JDK*.
15. Click **Next**. The Database Configuration Type screen displays, as shown in the following figure.



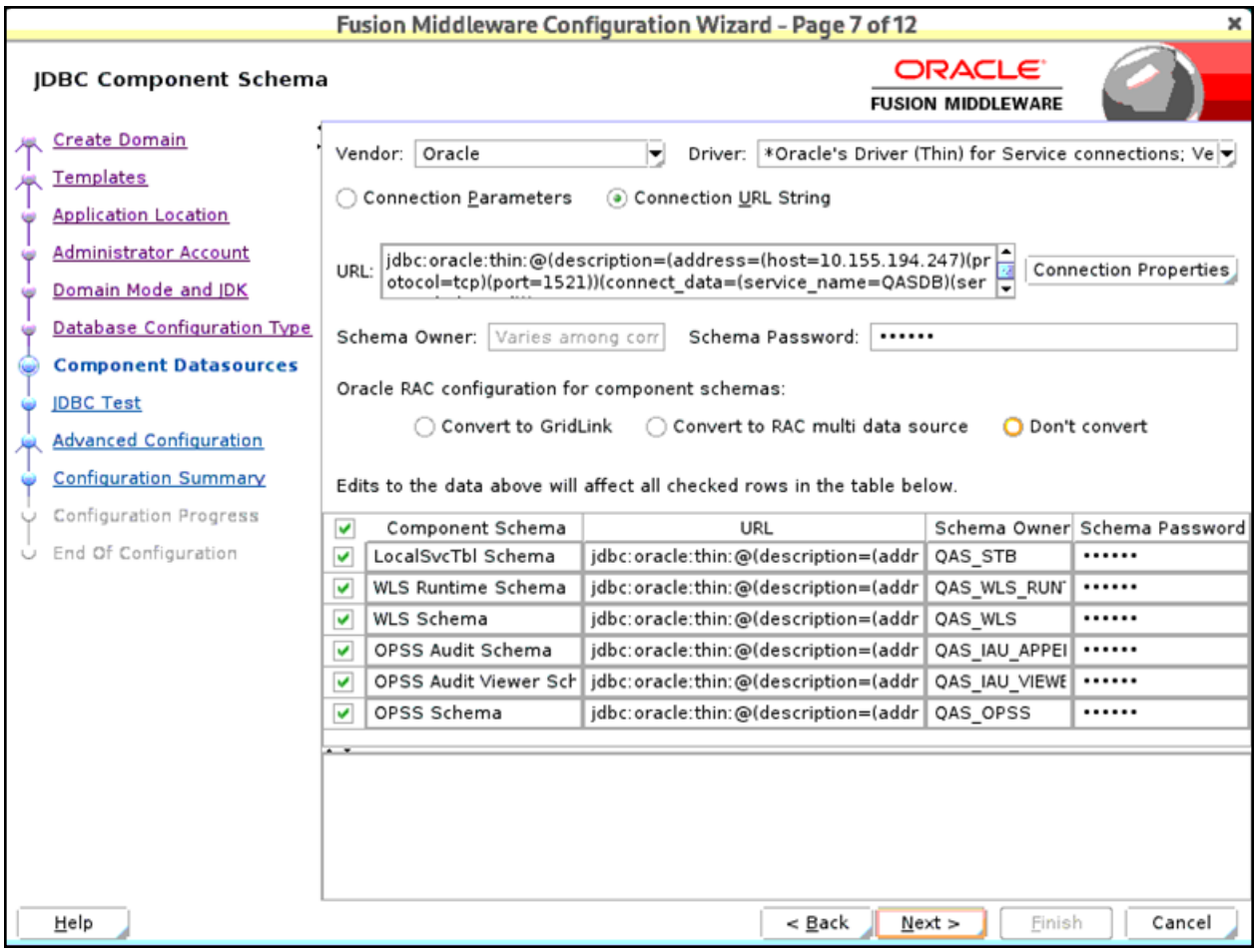
16. Under *Specify AutoConfiguration Options Using*, select the **RCU Data** option.
17. Update the following RCU Data values:
    - **Vendor:** Oracle
    - **Driver:** Oracle's Driver (Thin) for Service connections;
    - **DBMS/Service:** Blood Centers Database Name (bcprd in the above example)
    - **Host Name:** Blood Centers Database IP or Host Name (bc2026r1 in the above example)
    - **Port:** Blood Centers Database Port Number (1521 in the above example)
    - **Schema Owner:** User name of the user who owns the schema (BC_STB in the above example)
    - **Schema Password:** fmw14c
18. Click **Get RCU Configuration**.

19. Verify the connection result is successful, as shown in the following figure.

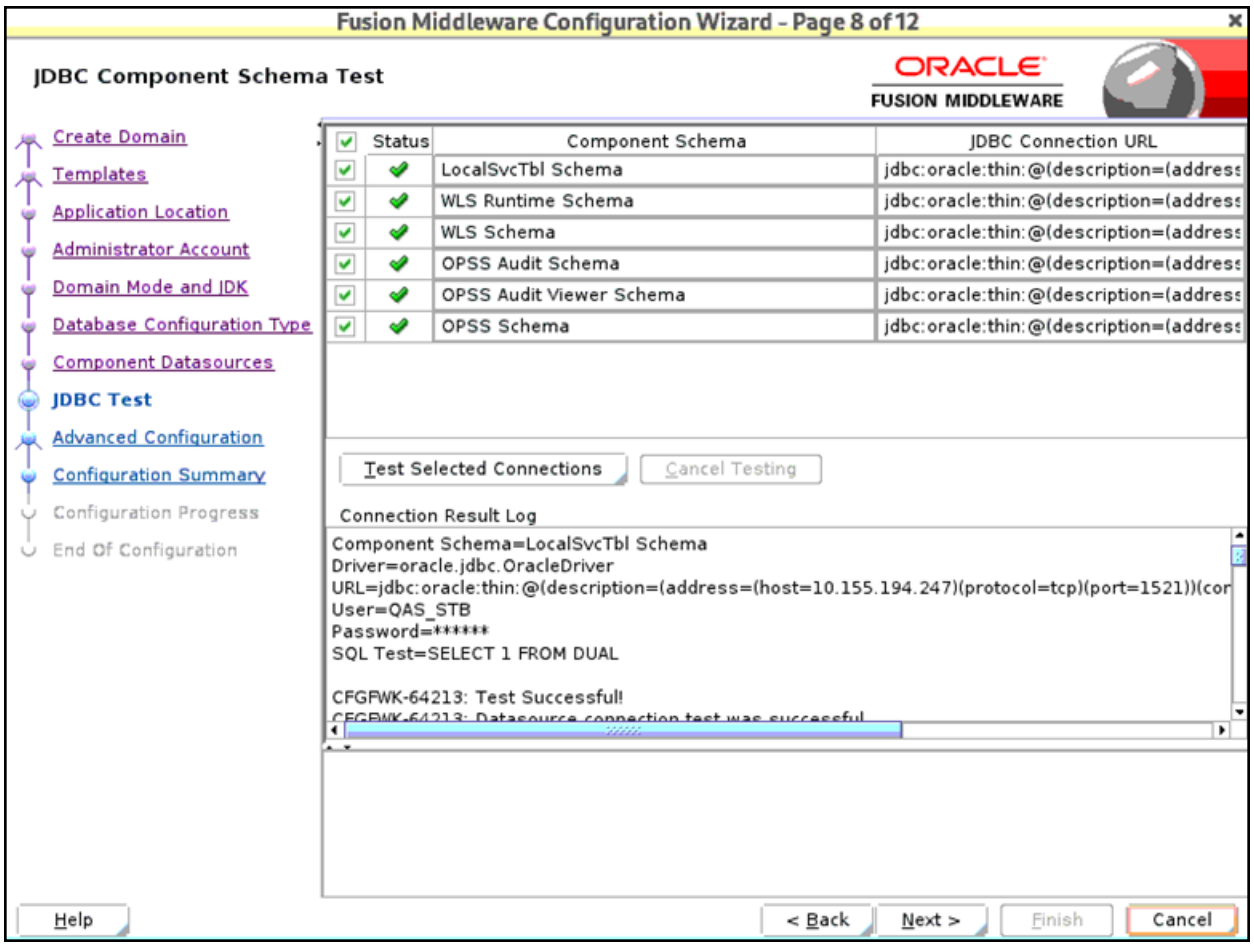


20. Click **Next**. The JDBC Component Datasources screen displays, as shown in the following figure.

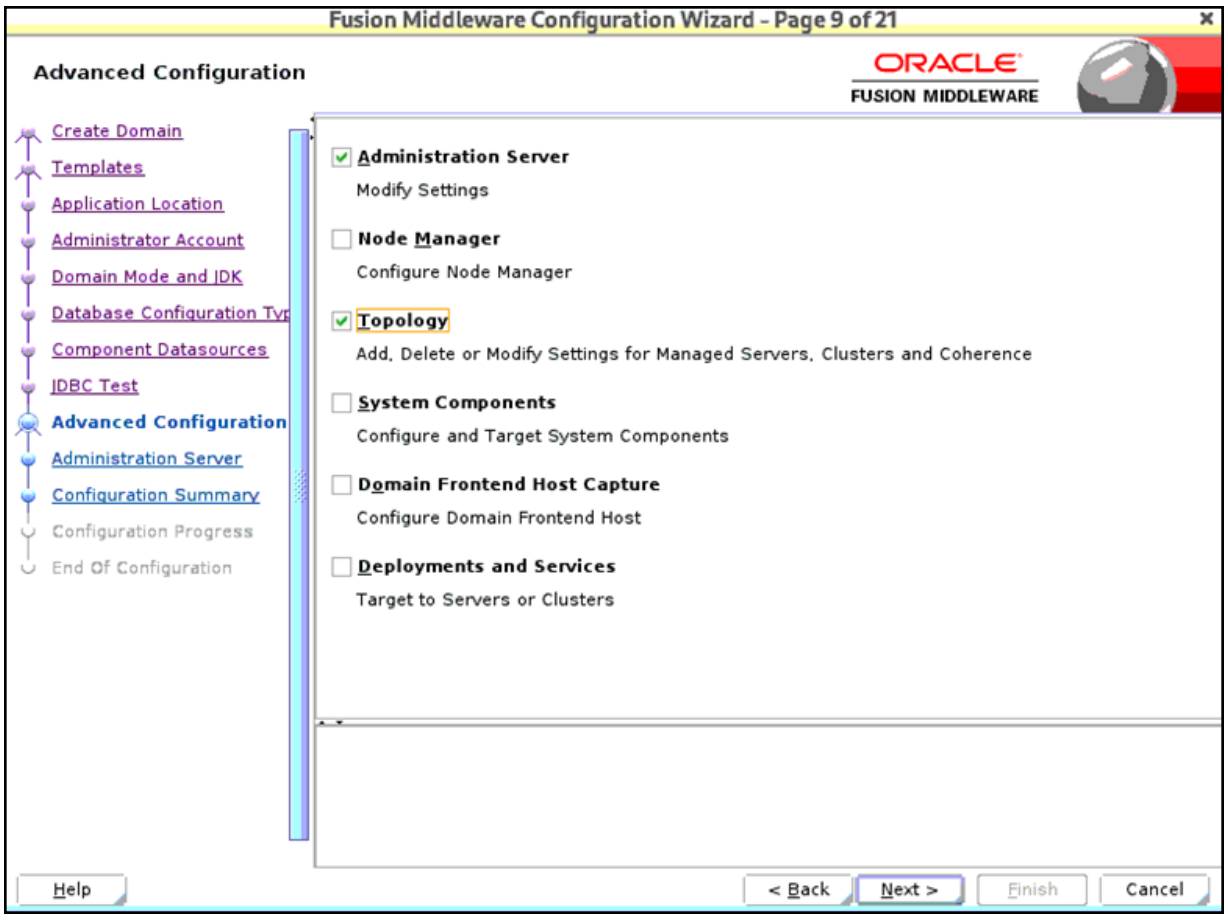
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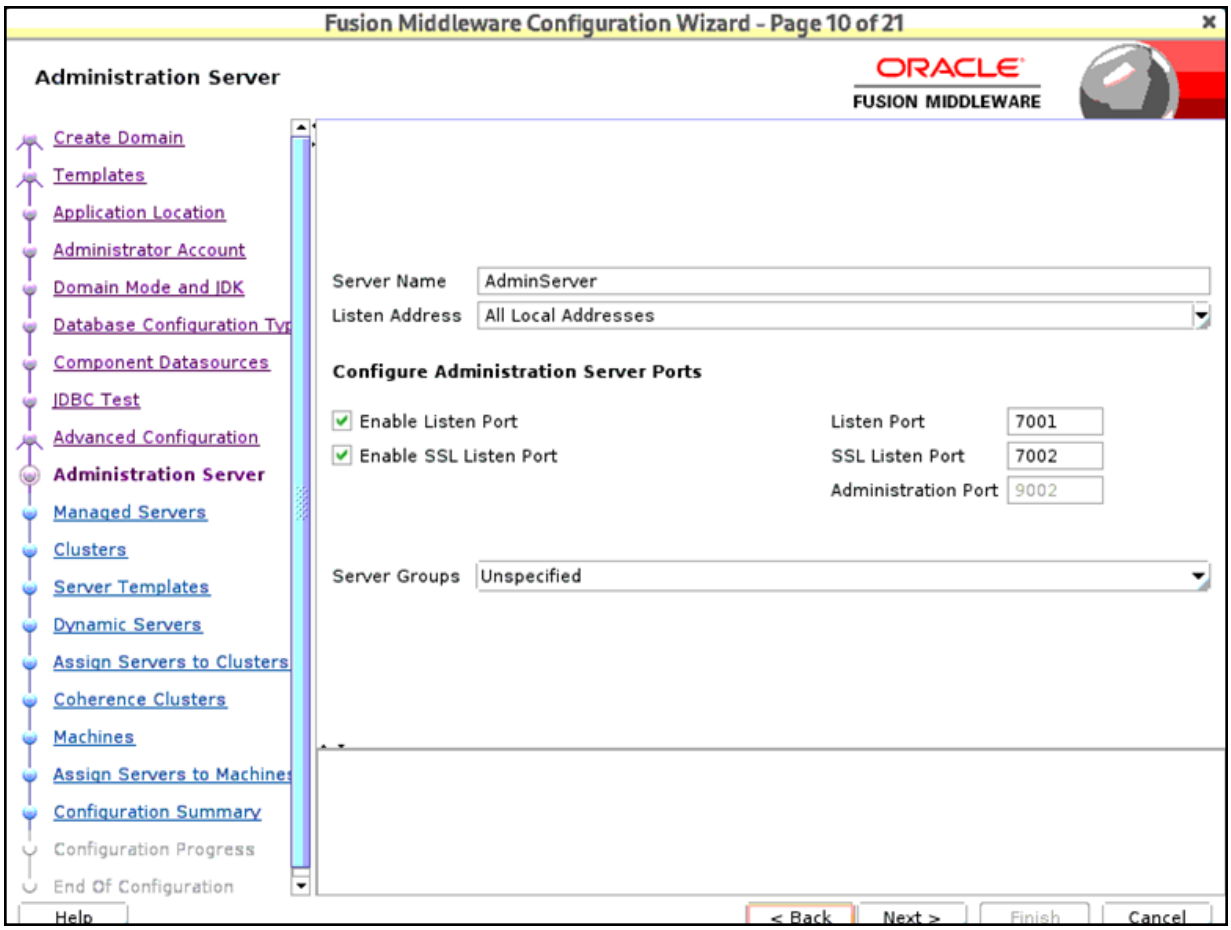
21. Select the check box next to the **Component Schema** table heading to select all of the component schema listed, as shown in the figure above.
22. Click **Next**. The JDBC Component Schema Test screen displays, as shown in the following figure.



23. Click Test Selected Connections.
24. Verify the test is successful.
25. Click Next. The Advanced Configuration screen displays, as shown in the following figure.

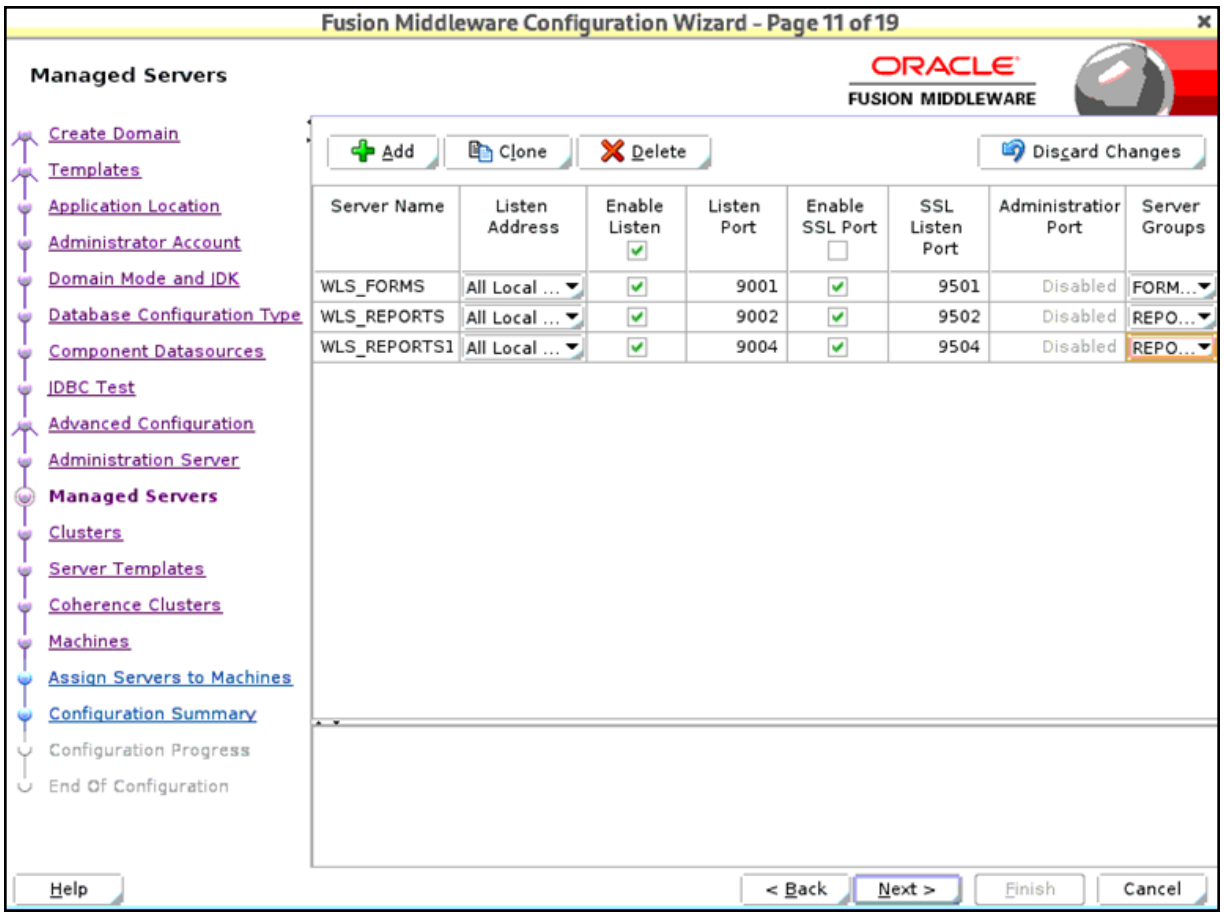


26. Select the **Administration Server**and **Topology**options.
27. Click **Next**. The Administration Server screen displays, as shown in the following figure.

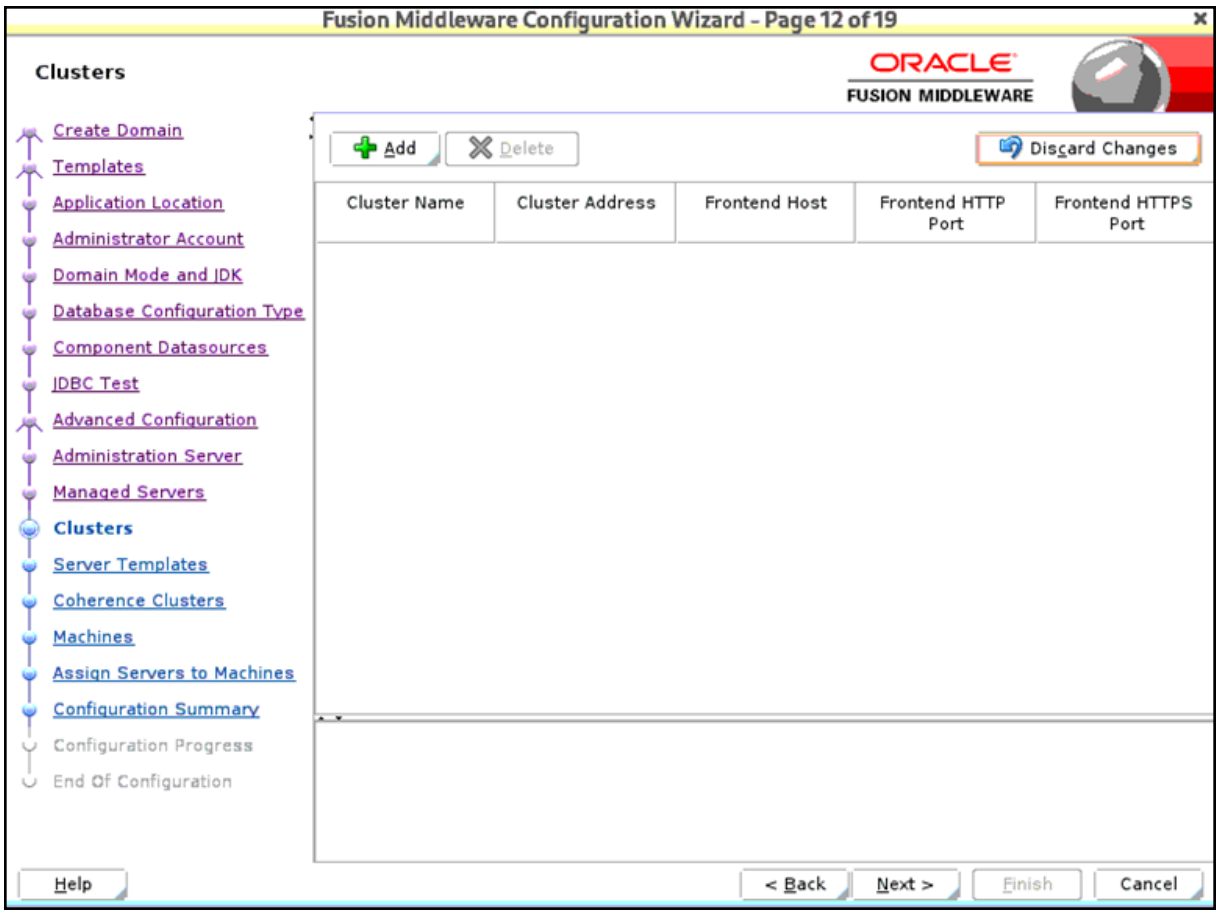


28. Verify the following:
    - The Server Name is **AdminServer**.
    - The Listen Address is **All Local Addresses**.
    - The Listen Port is **7001**.
    - SSL Listen Port is optional.
29. Click **Next**. The Managed Servers screen displays, as shown in the following figure.

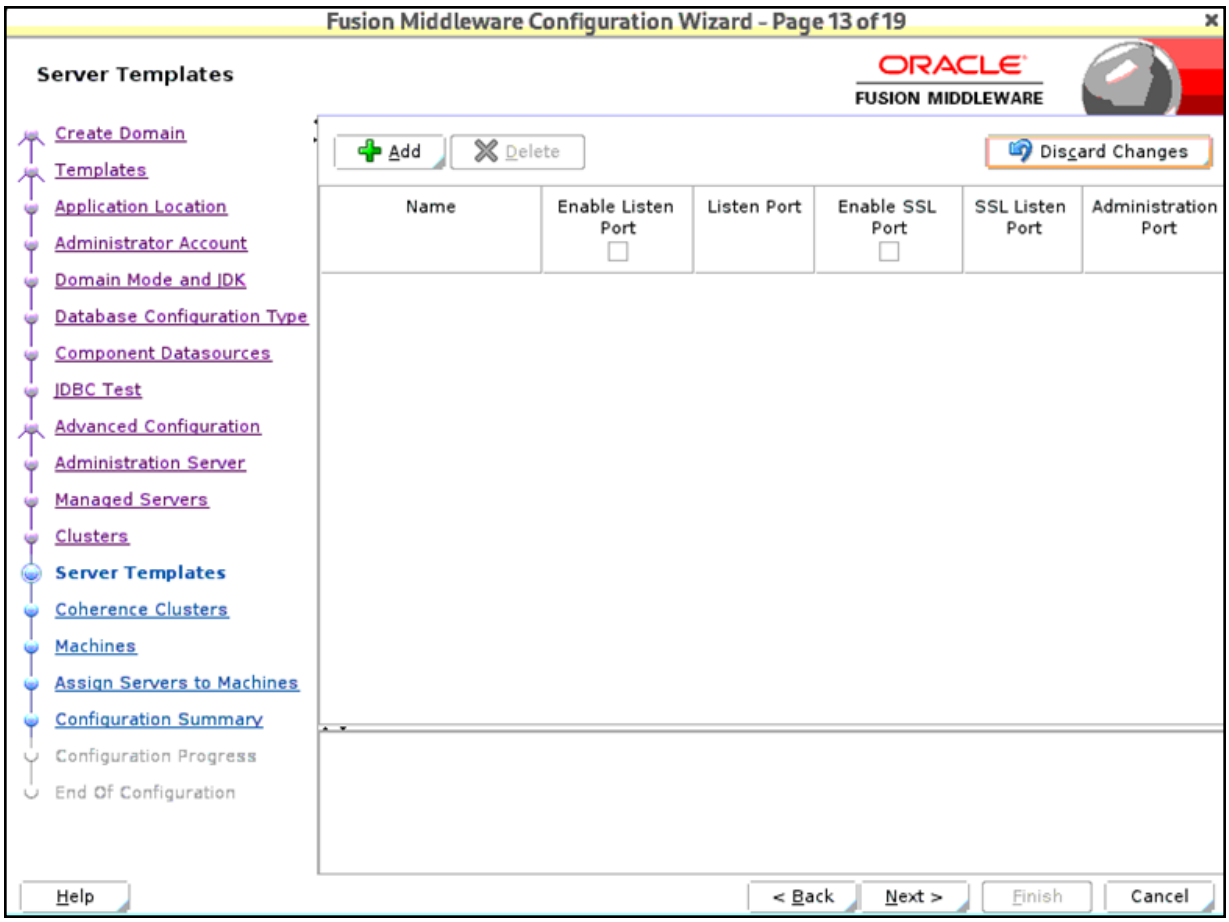
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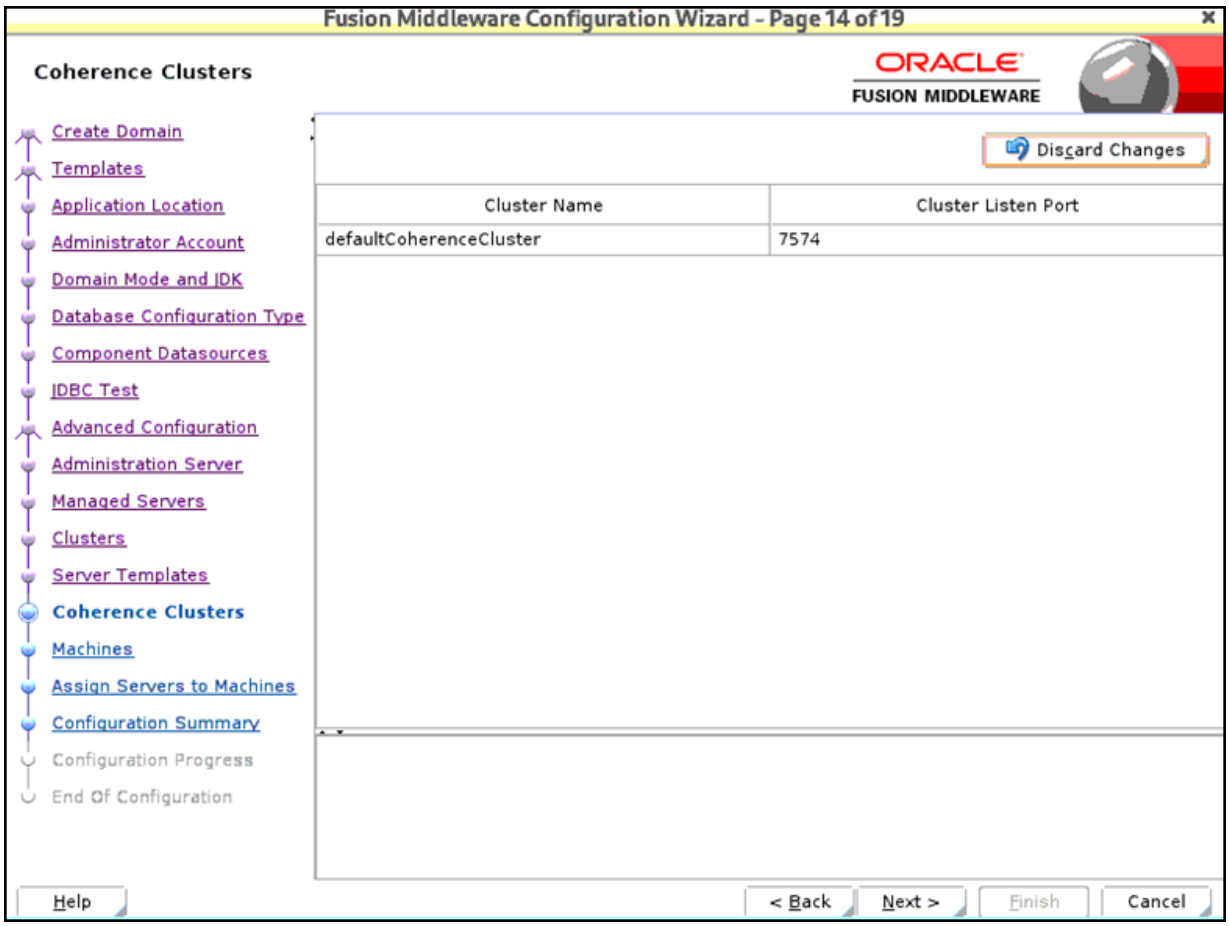
30. Click **Add**. Enter the following values for the secondary report server, as shown in the example below.
    - **Server Name:** `WLS_REPORTS1`
    - **Listen Port:** `9004`
    - **Server Groups:** `REPORTS_APP_SERVERS`
        - Update WLS_REPORTS Listen Port to `9002`.
31. Click **Next**. The Clusters screen displays, as shown in the following figure.



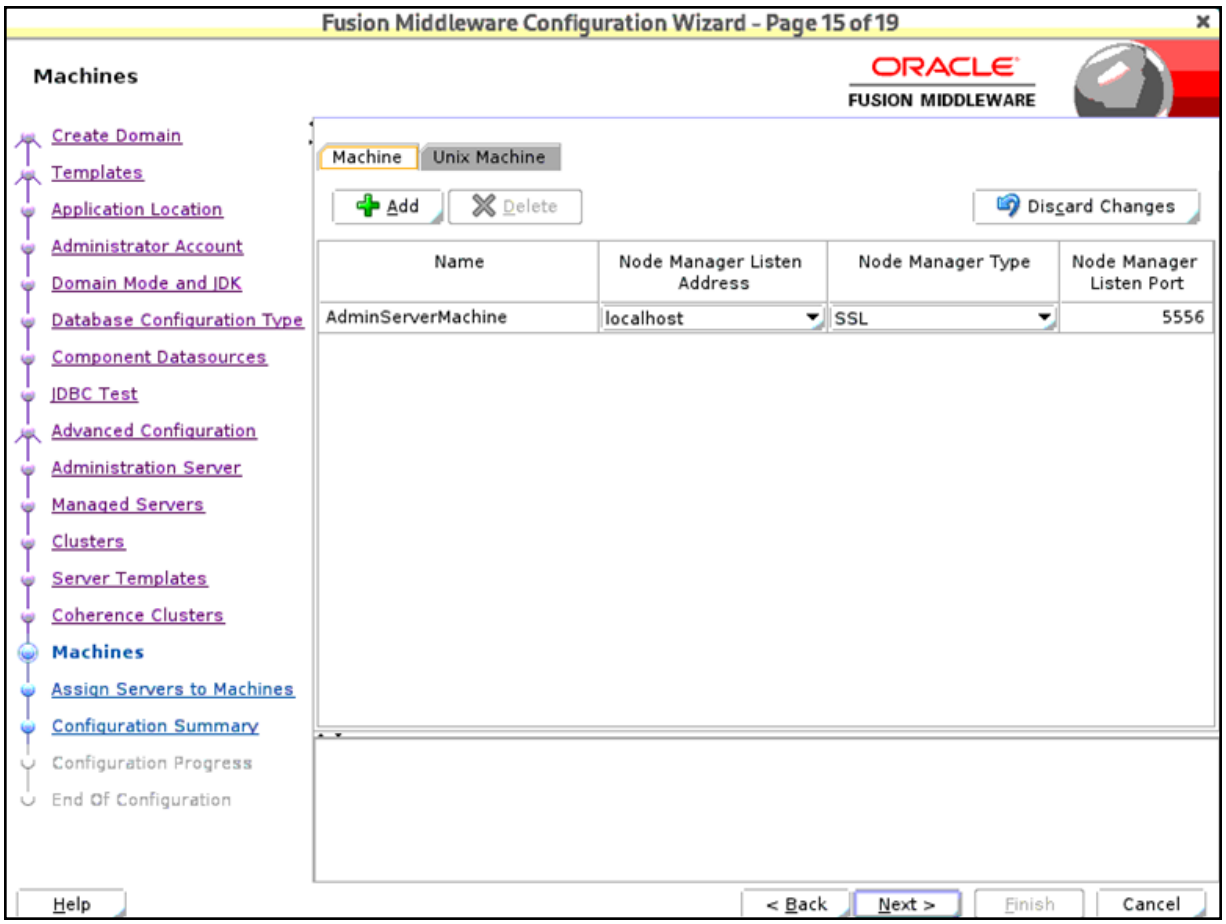
32. Delete any clusters it displays and click **Next**.
33. Click **Next**. The Server Templates screen displays, as shown in the following figure.



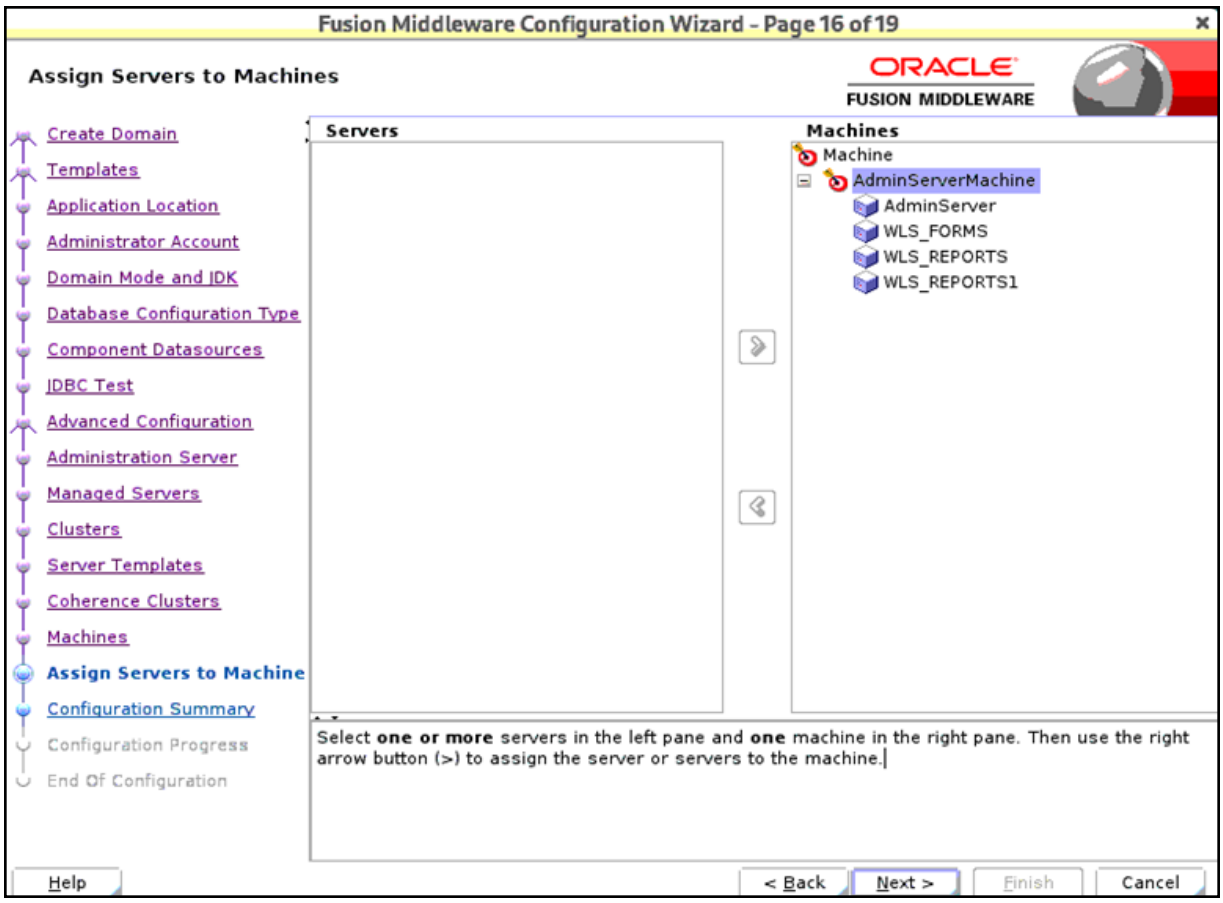
34. Click **Next**. The Coherence Clusters screen displays, as shown in the following figure.



35. Click **Next**. The Machines screen displays, as shown in the following figure.

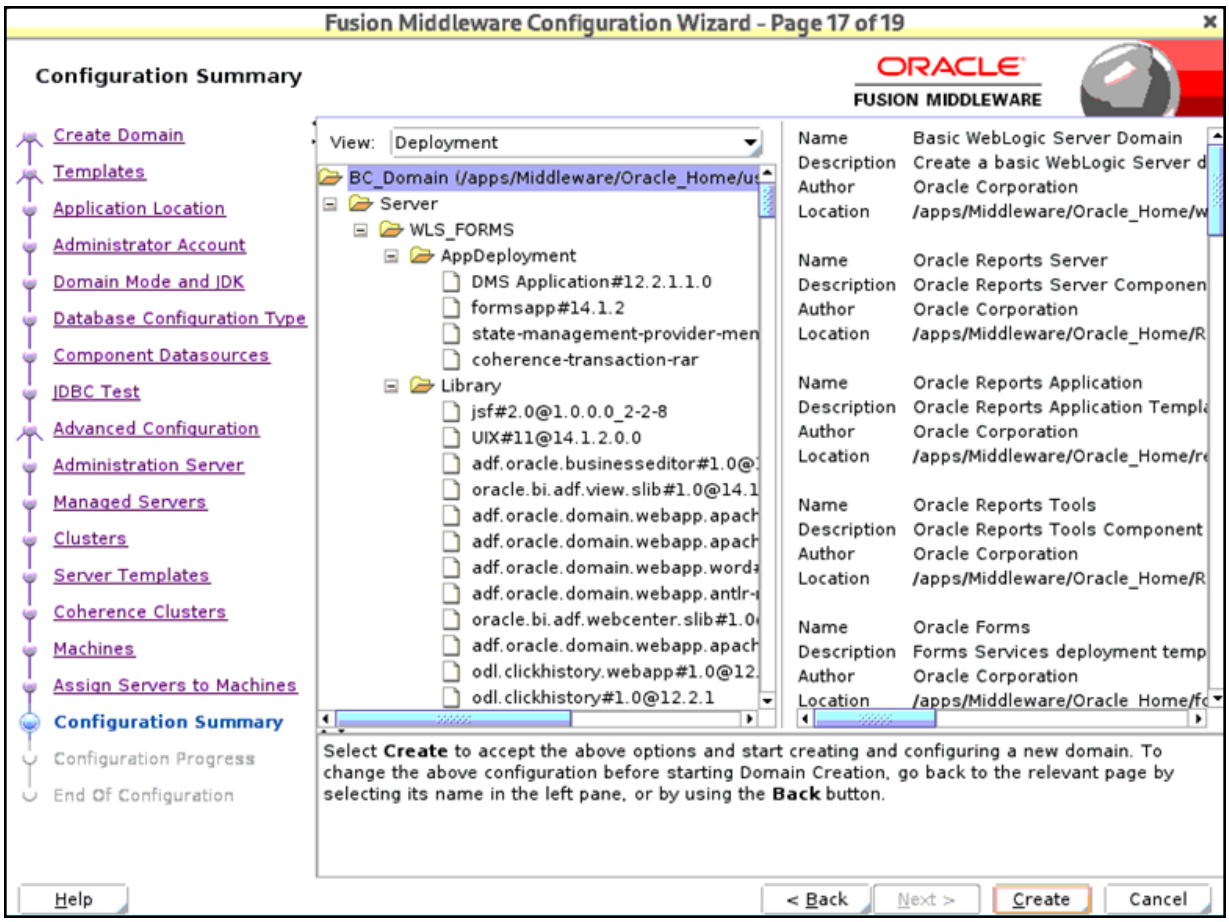


36. Click **Next**. The Assign Servers to Machines screen displays, as shown in the following figure.

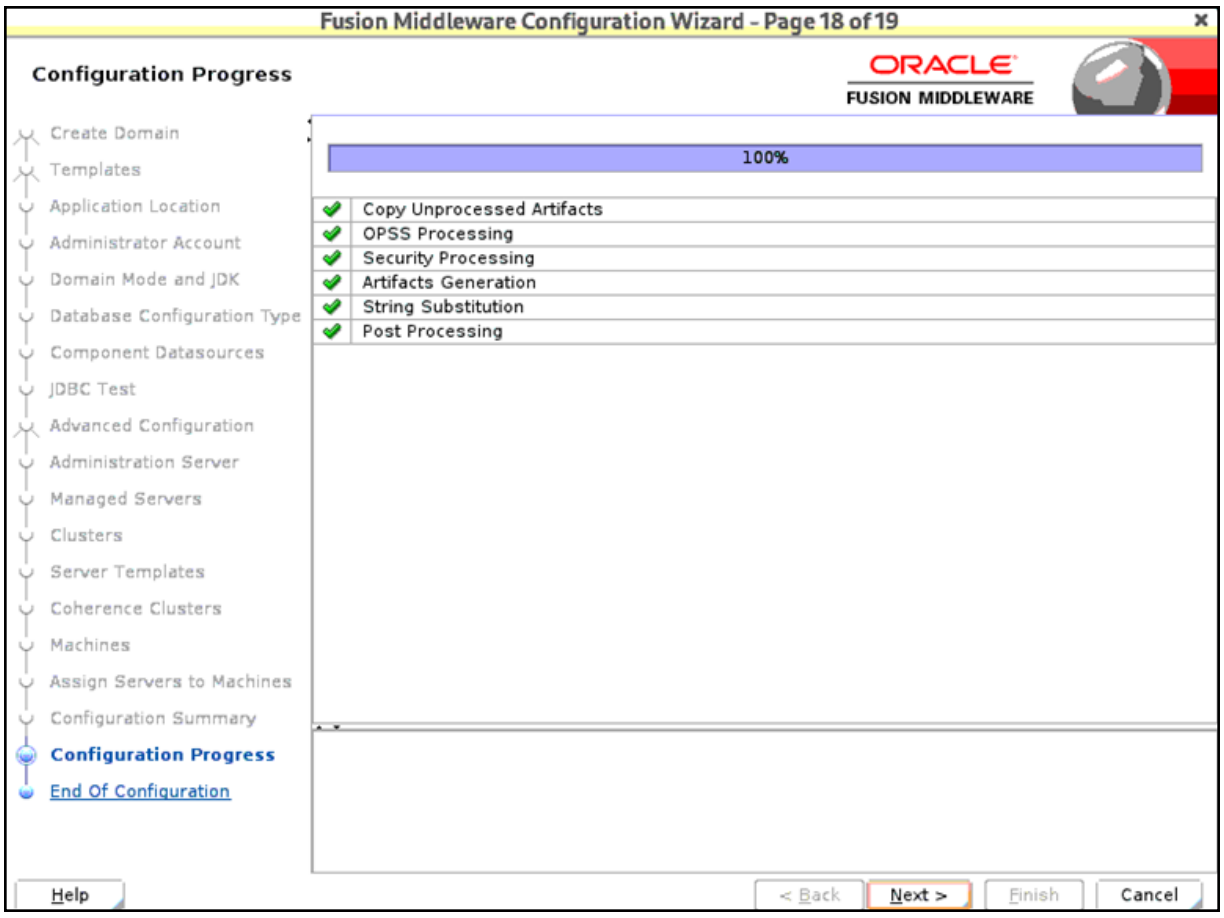


37. Verify that all of the servers are listed under AdminServerMachine in the column on the right.
38. Click Next. The Configuration Summary screen displays, as shown in the following figure.

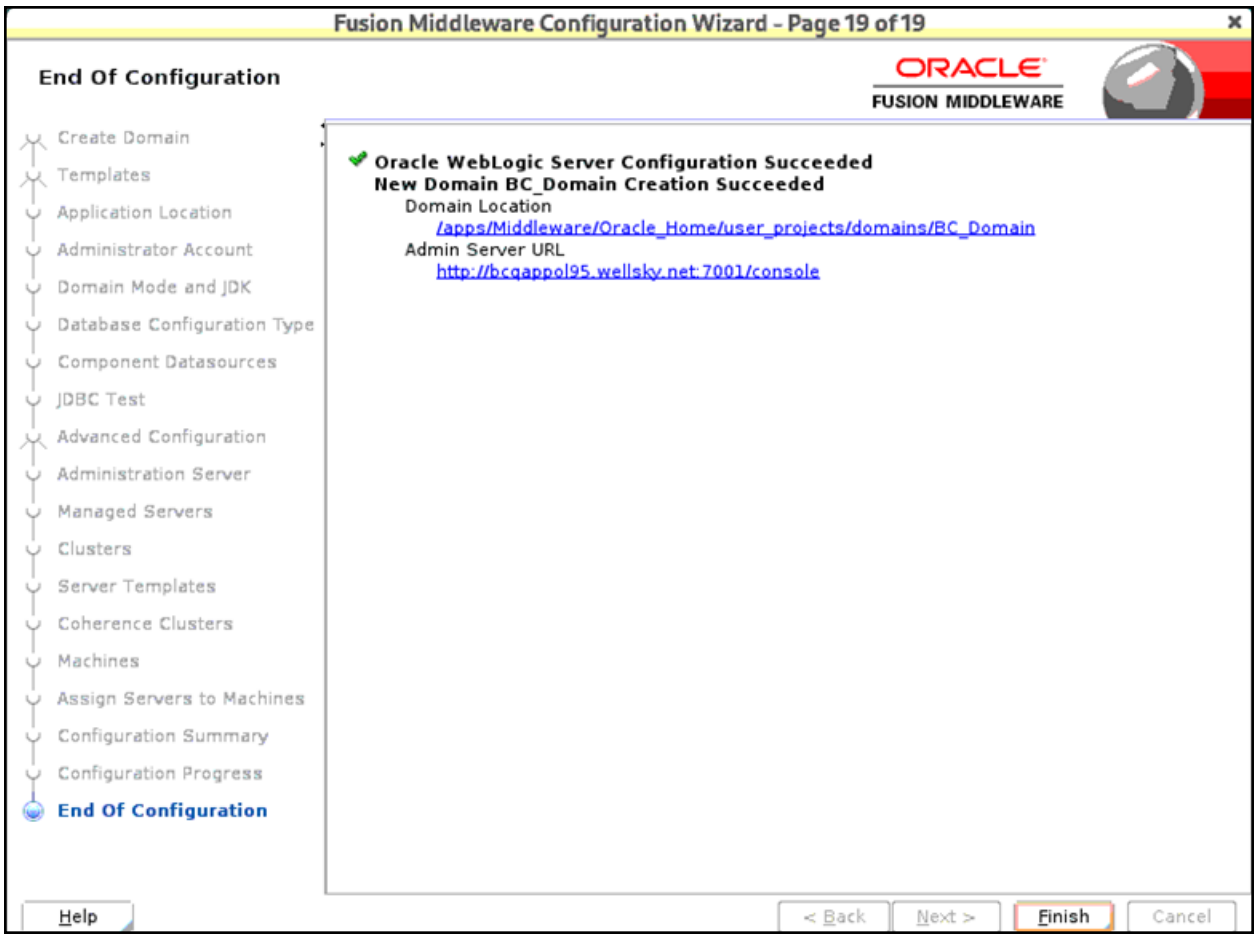
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39. Click **Create**. The Configuration Progress screen displays, as shown in the following figure.



40. When the configuration is successfully completed, click **Next**. The End Of Configuration screen displays, as shown in the following figure.



41. Click **Finish** to exit the Configuration Wizard.

6.6 Application Server Configuration

1. Run the Config_BCFMW_2026.sh script by entering the following commands:

```
$ cd /apps/Software/FMW/Config
$ sh Config_BCFMW_2026.sh
```

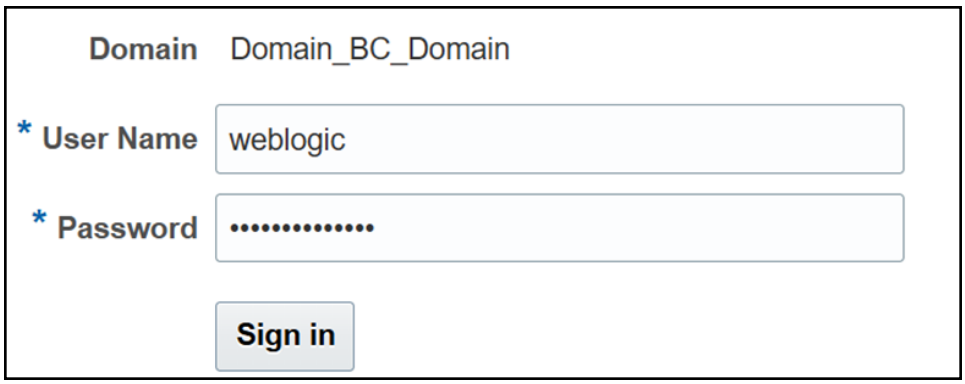
> **NOTE –** This script will run for over 10 minutes. Please do not close the terminal.

2. Open the Enterprise Manager Console by entering the following URL: `http://10.10.10.10:7001/em`.

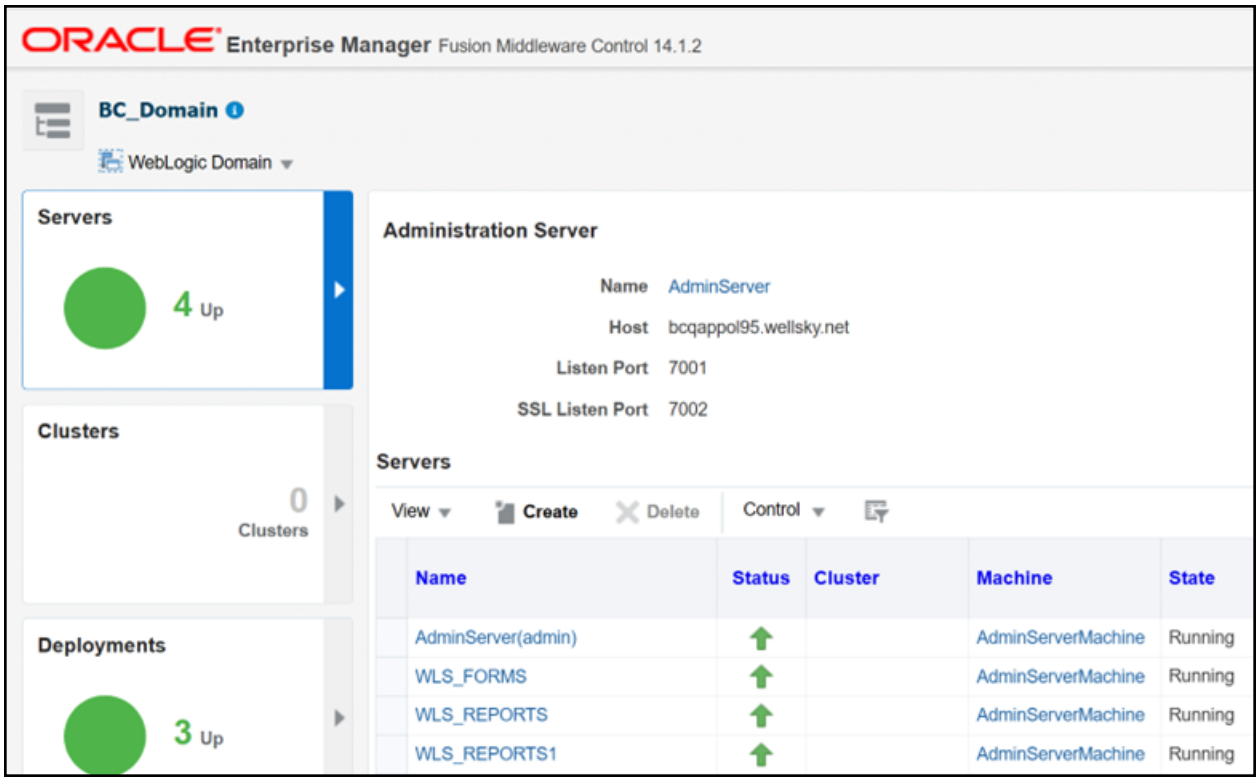
> **NOTE –** Edit the URL to use the IP address for the Application Server.



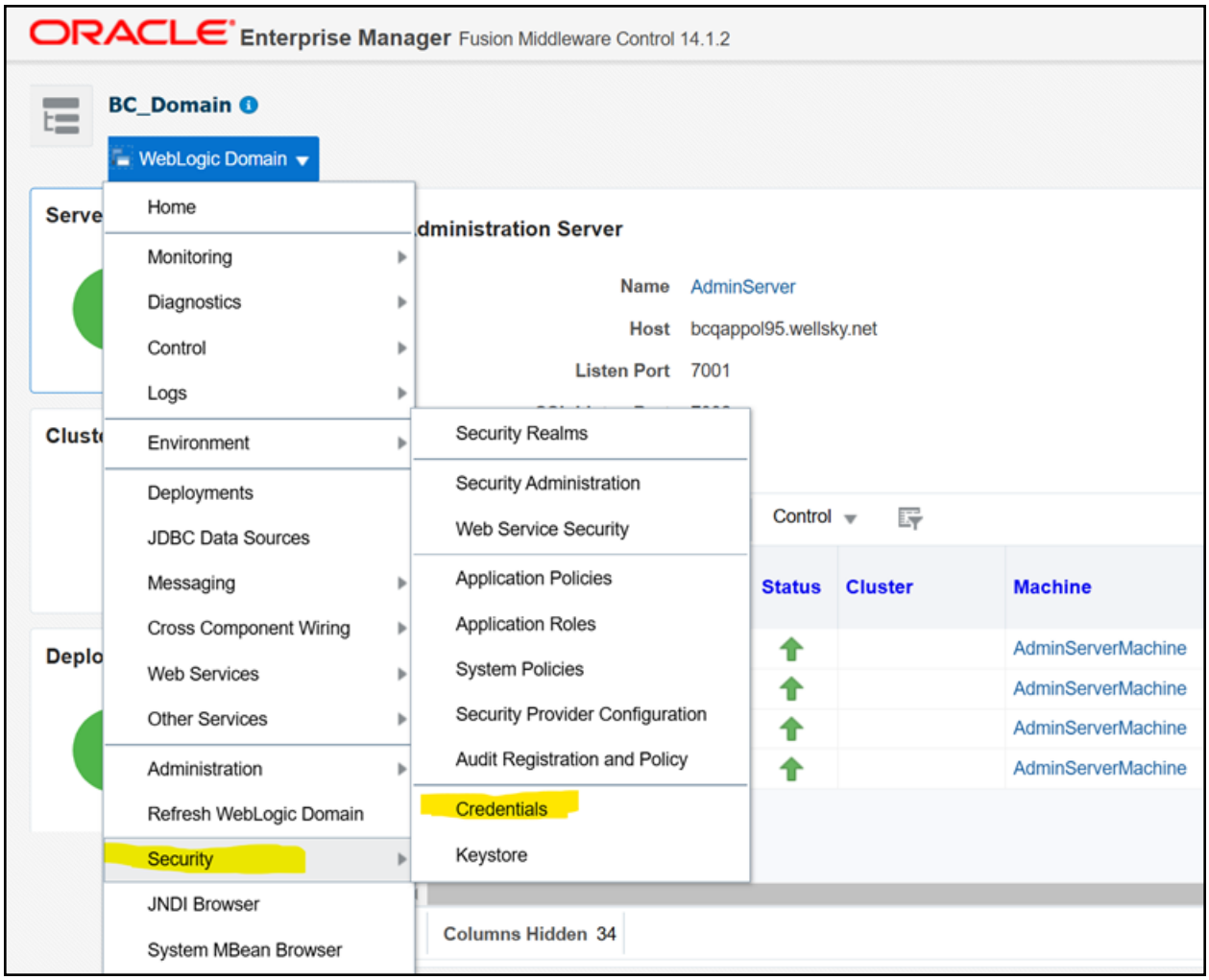
3. Log in to the Enterprise Manager Console by entering the following:
   - User Name: `weblogic`
   - Password: `WebLogicAdmin1`



4. Click Sign in. The Oracle Enterprise Manager Console displays, as shown in the following figure.



5. Click **WebLogic Domain > Security > Credentials**, as shown in the following figure.

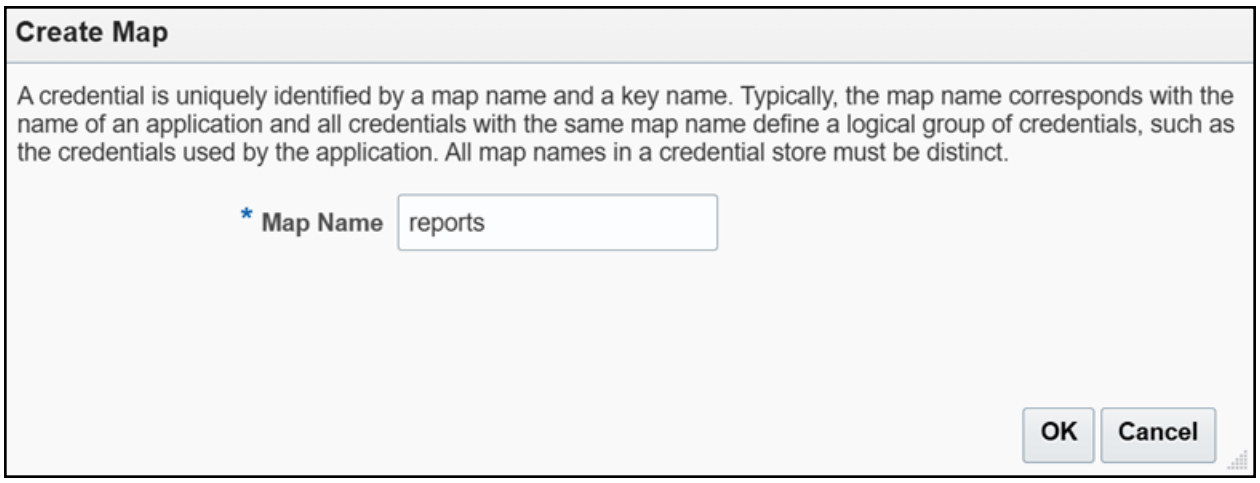


The Credentials screen displays, as shown in the following figure.

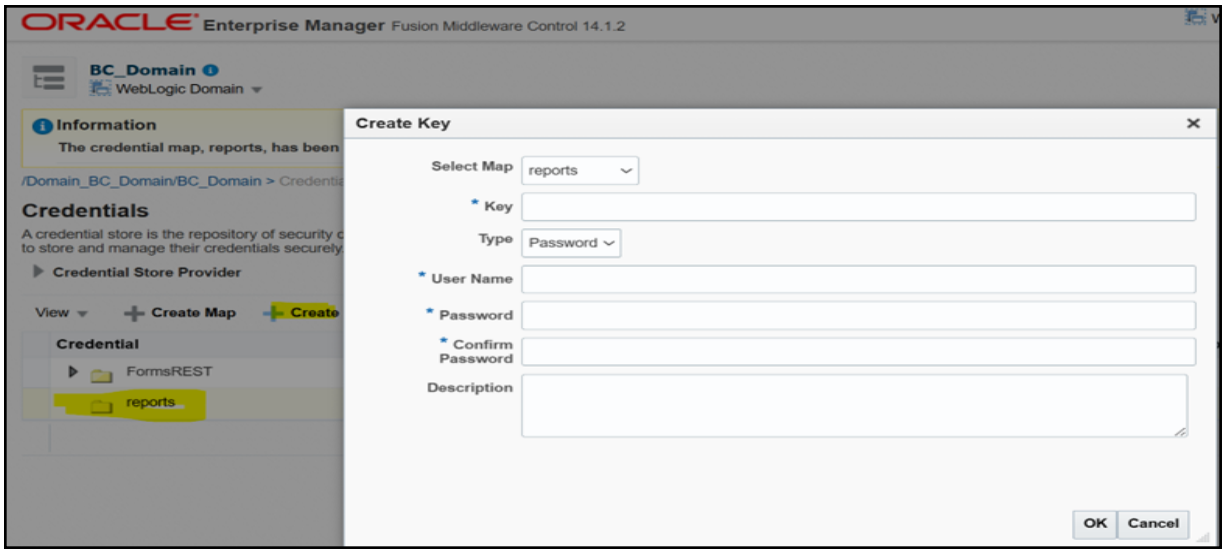


6. Click **Create Map**. The Create Map screen displays, as shown in the following figure.

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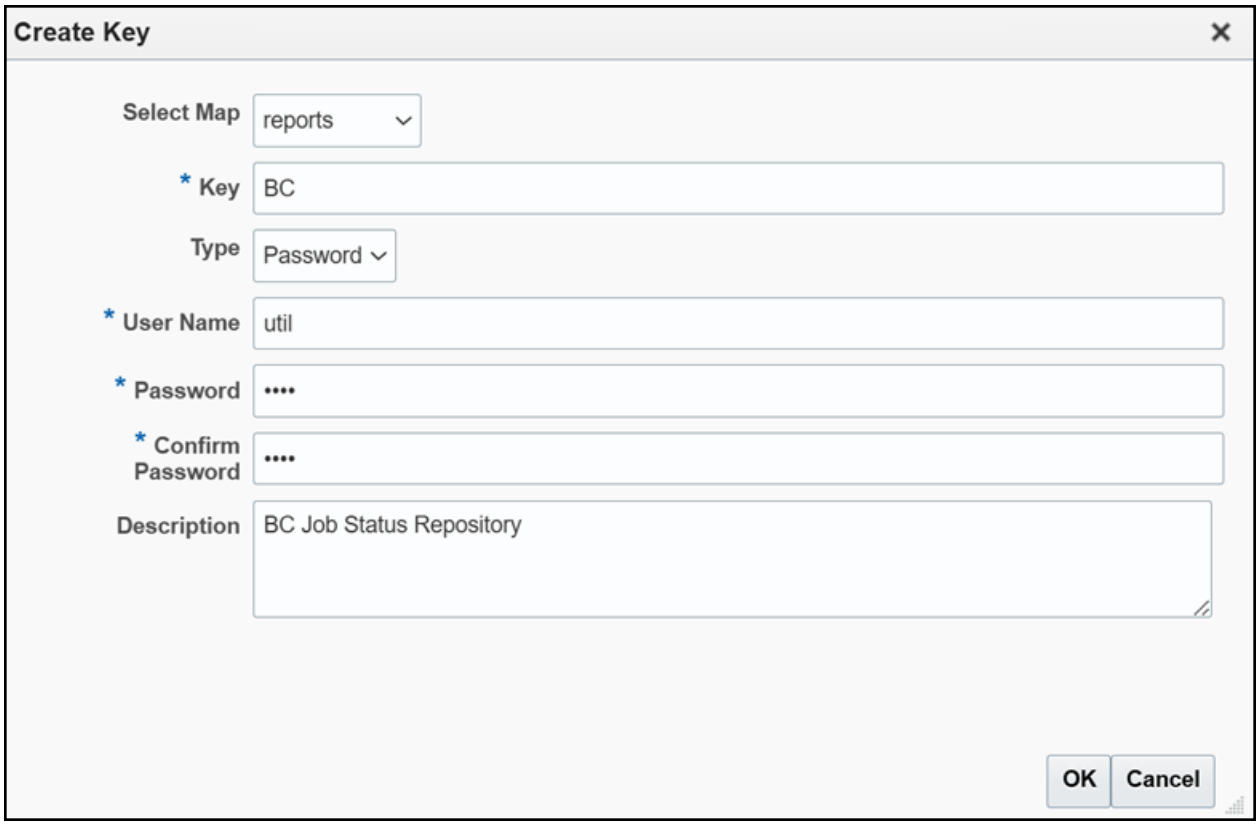


7. Enter `reports` in the **Map Name** box.
8. Click **OK**.
9. On the Credentials screen, select the **reports** credential, and click **Create Key**. The Create Key form displays, as shown in the following figure.



10. Enter the following values on the Create Key form:
    - **Key:** `BC`
    - **Type:** Password
    - **User Name:** `util`
    - **Password:** `util`
    - **Confirm Password:** `util`
    - **Description:** `BC Job Status Repository`

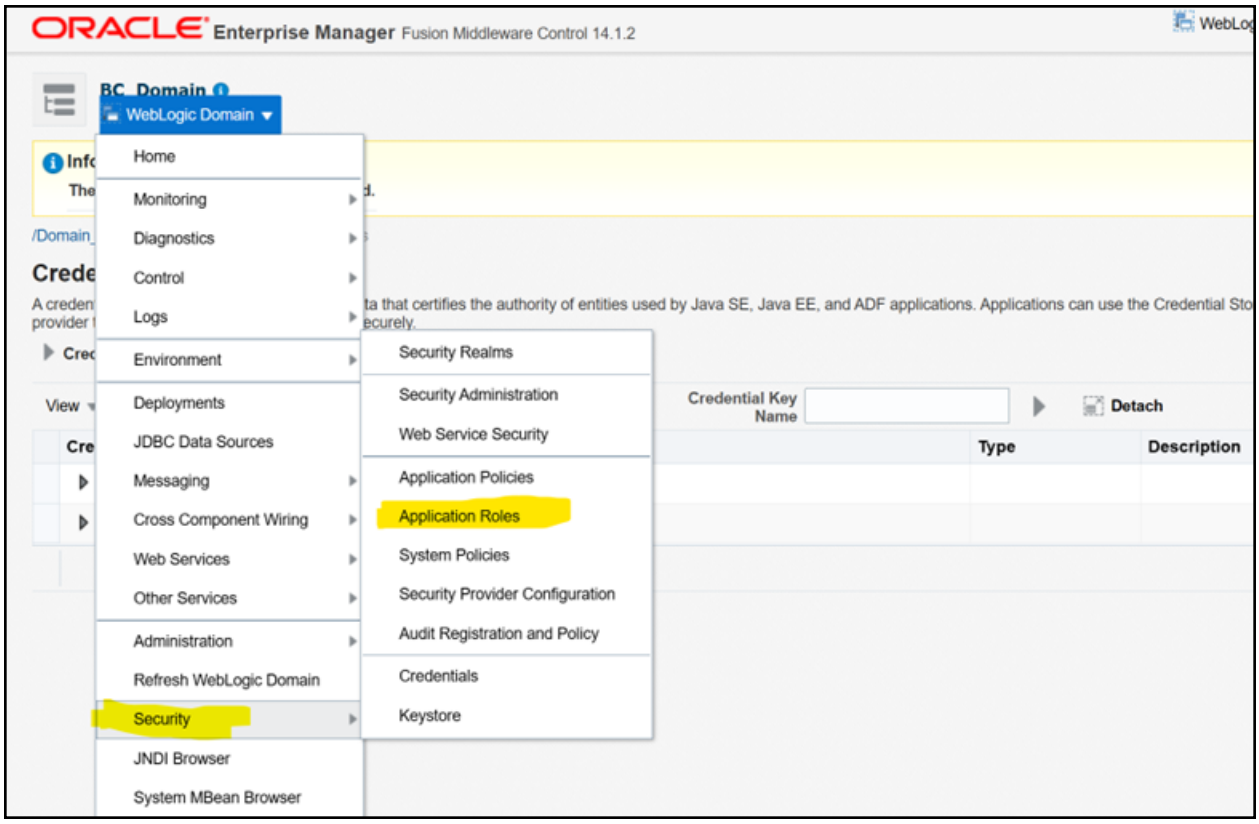
    The Create Key form should look like this:



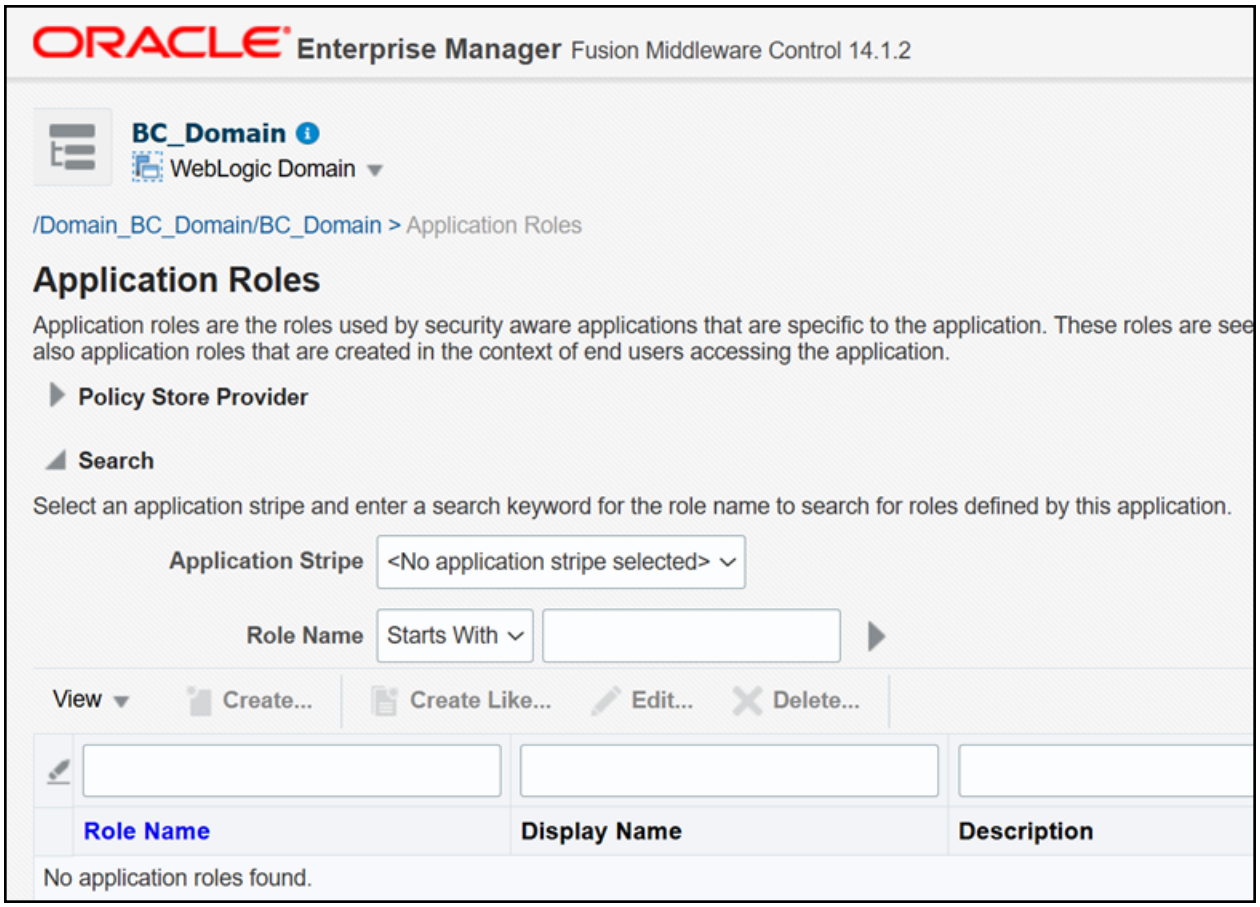
11. Click **OK**. A message displays on the Credentials screen, indicating that the key has been created.

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12. Click **WebLogic Domain > Security > Application Roles**, as shown in the following figure.

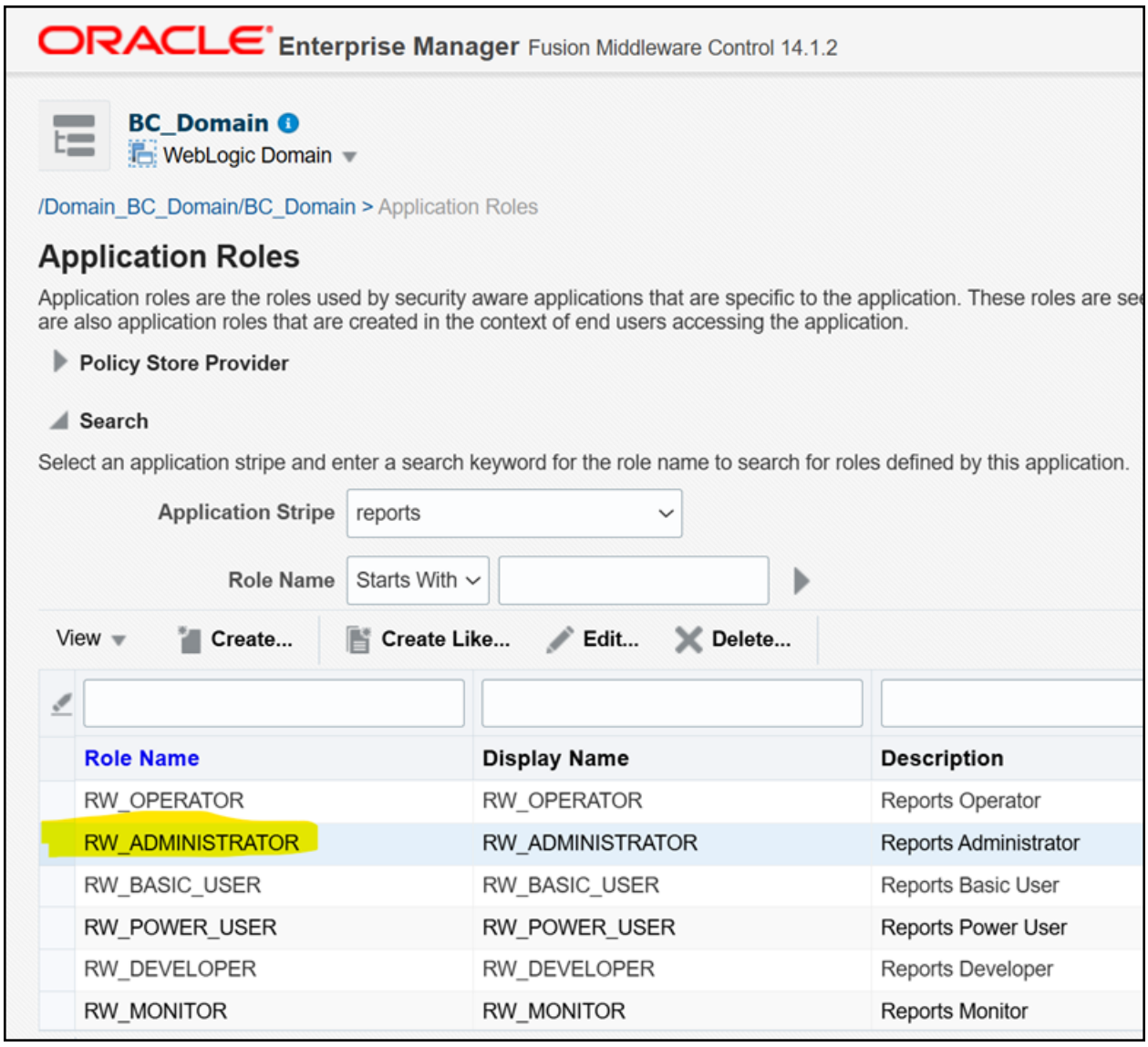


The Application Roles screen displays, as shown in the following figure.



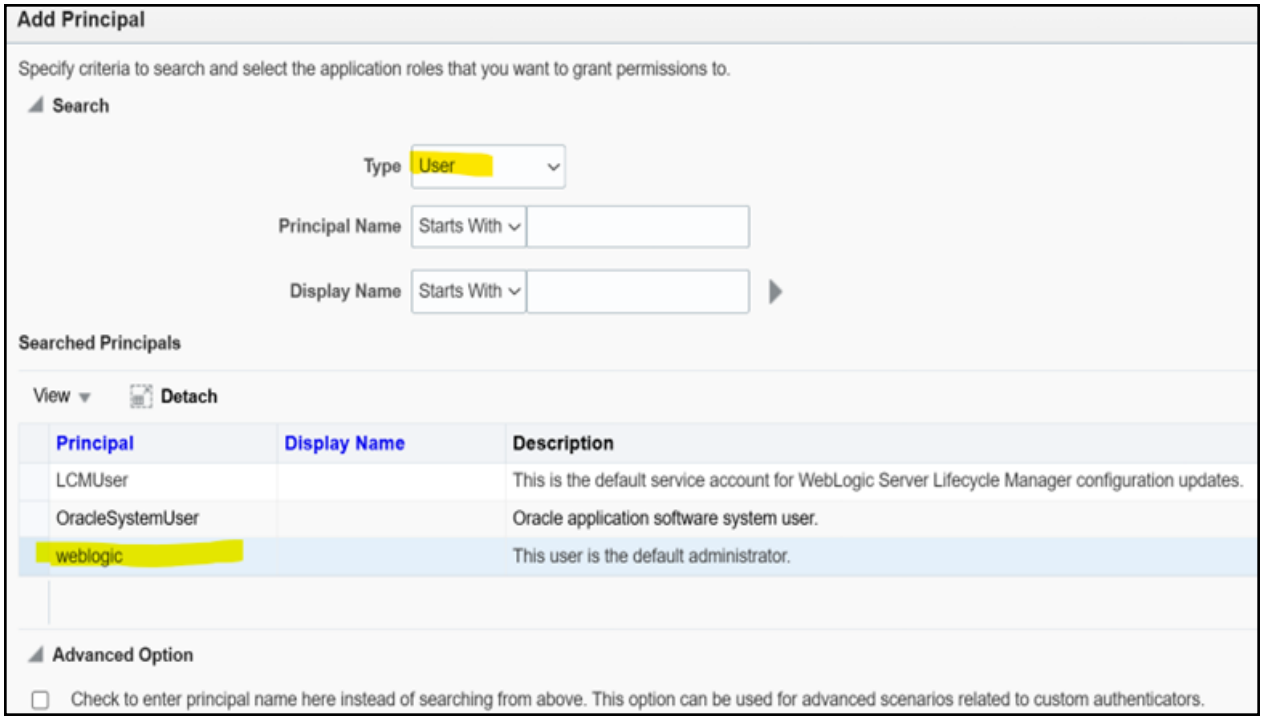
13. Select **reports** from the *Application Stripe* dropdown, and click **Search**.
    The list of reports-related roles displays, as shown in the following figure.

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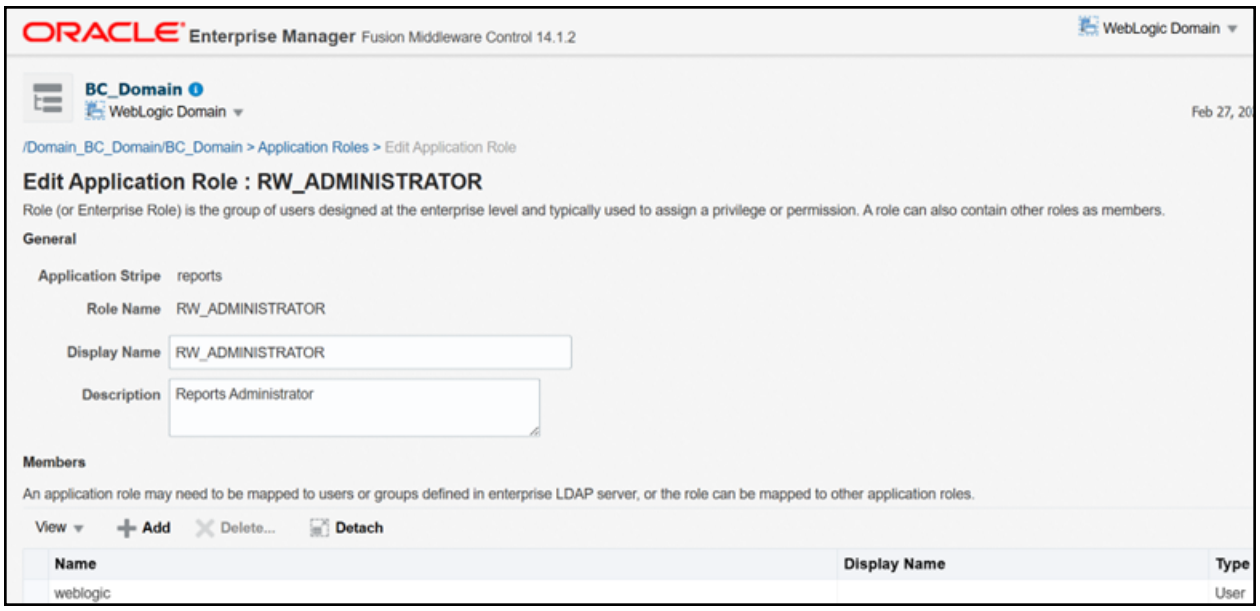


14. Select the RW_Administrator role, and click Edit.
15. On the Edit Application Role screen, under *Members*, click Add. The Add Principal screen displays.

16. On the Add Principal screen, select **User** from the *Type* drop-down, and click **Search**.
    The list of relevant principals displays, as shown in the following figure.

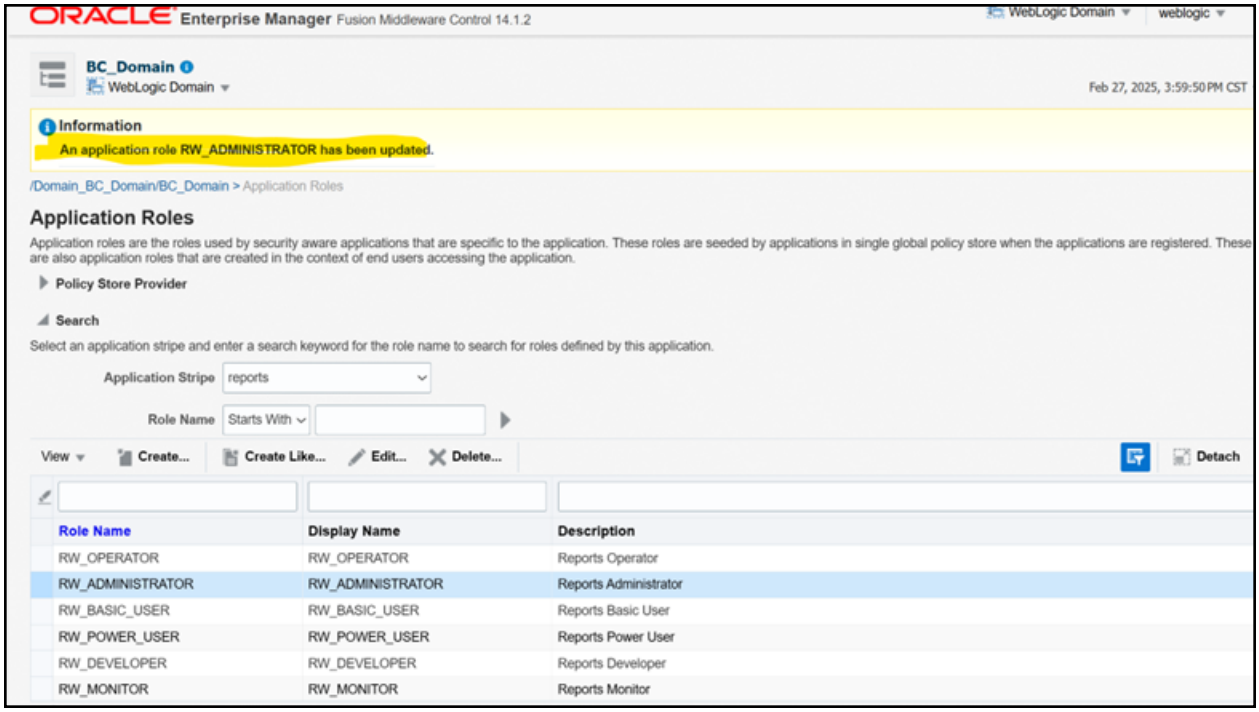


17. Select the **weblogic** principal, and click **OK**.
    On the Edit Application Role screen, the weblogic principal is added as a member of the RW_Administrator role, as shown in the following figure.



18. Click OK.

    A message displays on the Application Roles screen, indicating that the application role has been updated, as shown in the following figure.



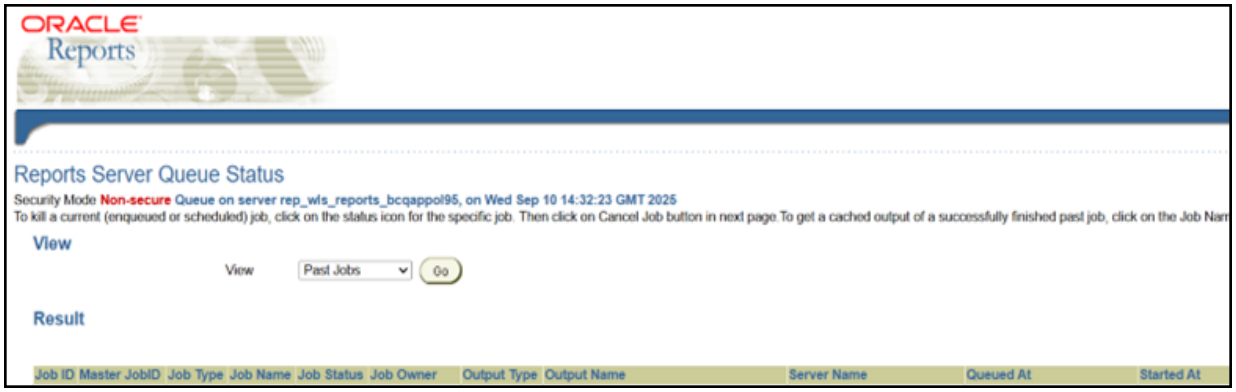
19. Using VI Editor, edit the following files one by one:

    /apps/Middleware/Oracle_Home/user_projects/domains/BC_Domain/config/fmwconfig/servers/WLS_REPORTS/applications/reports_14.1.2/configuration/rwserver.conf

    /apps/Middleware/Oracle_Home/user_projects/domains/BC_Domain/config/fmwconfig/servers/WLS_REPORTS1/applications/reports_14.1.2/configuration/rwserver.conf

20. Update the line with the following database server details:
    - Replace 10.10.10.10 with the Database Server's IP address.
    - Replace prddb with the Blood Centers Database Name, if required.
    - Change 1521 if a different port is being used.
21. If required, update the following file with Database Server Details:
    /apps/Middleware/Oracle_Home/user_projects/domains/BC_Domain/config/fmwconfig/tnsnames.ora.
22. Restart the Application Services by running the following commands:
    $ `bcorappstop`
    $ `bcorappstart`
23. Open the Reports Server by entering the following URLs one by one:
    `http://10.10.10.10:9002/reports/rwservlet/showjobs?`
    `http://10.10.10.10:9004/reports/rwservlet/showjobs?`

NOTE – Update the IP address with the Application Server's IP address.



6.7 Set Up Cron Jobs

The installation and configuration of cron jobs is unique to each client. Both required and optional cron jobs are described. Contact your WellSky Implementation Team for assistance with this task.

6.7.1 Required Cron Jobs

The following table identifies the cron jobs that must be configured for your installation.

This table includes recommended times. Timing of the cron jobs can be changed for your installation needs except for jobs that must run at midnight.

Do not set cron jobs to run too frequently since that could create multiple instances of the same cron job leading to system degradation, unpredictable results, and possible system failure.

Job Name	User	Description	Recommended
cmv_abo.sh (prior_abo_interface.sh cmv_interface.sh)	sbcsys	Interface prior ABO and deferral from LT/Donor. (associated package dnlbu.donor_to_lab_unit_data) Interface prior CMV and antigens from LT/Donor. (associated package cmvint.donor_to_lab_unit_data)	Twice daily.
dayinit.sh	donsys	Clean donor temporary tables and reports.	Daily at midnight
dayinit.sh	sbcsys	Reset chapter parameters and run analyze.	Daily at midnight.
dnlbu.sh	sbcsys	Donor To Lab Unit Interface	Twice daily.
doniddr.sh	sbcsys	Donor deferral to Distribution for Distribution to check deferrals when Donor is offline.	Once or twice a day.
donlist.sh	donsys	List generation.	Every 15 minutes
drive_int.sh	sbcsys	Interface drives from LT/Donor. (associated package don_lab_drv.interface_drives)	Twice daily.
hla.sh	donsys	Run HLA/RBC requests. (associated package don_hla_rbc.hlarbc_proc_caller)	Every 15 minutes
labclean.sh	sbcsys	Clean lab temporary tables and reports directory.	Once a day.
lbdna.sh	donsys	Interface antigen test results from LT/Lab. (associated package LBDNA.read_antigen_results)	Twice daily
lbdns.sh	donsys	Interface confirmatory and supplemental results from LT/Lab. (associated package LBDNS.read_lab_test_results)	Twice daily

Job Name	User	Description	Recommended
lbdnc.sh	donsys	Interface profile test results from LT/Lab. (associated package LBDNC.read_lab_test_results)	Twice daily.
reg_audit.sh	sbcsys	Interface nursing audit information from LT/Donor. (associated package don_lab_nai.interface_nursing_audit)	Twice daily
Rep_Server_Status.sh	sbcsys	Updates the report status for multi-report servers.	Every 30 minutes
reset_numbers.sh	sbcsys	Reset number generation tables.	Daily at midnight.
rturload.sh	sbcsys	Test result upload from samples directory.	As needed per schedule for receiving results.
schedule_report.sh	donsys	Allows the ability to print reports in the application at a specified time.	Every minute.
schedule_report.sh	sbcsys	Allows the ability to print reports in the application at a specified time.	Every 60 seconds.
update_dst.sh	donsys	Updates the daylight savings time	Every 30 minutes on the hour and half hour.
update_dst_lab.sh	sbcsys	Updates the daylight savings time.	Every 30 minutes on the hour and half hour.
UPD_REP_STATUS.sh	donsys	Updates report status.	Every minute
UPD_REP_STATUS.sh	sbcsys	Updates report status.	Every minute

6.7.2 Optional Cron Jobs

The following table identifies the various cron jobs that may or may not need to be configured for your particular installation.

Job Name	Description	Optional/Required
atirdnat.sh	Chiron NAT Tracker Interface.	Optional Purchase Module
atirdoly.sh	Olympus PK7200 Interface.	Optional Purchase Module
attrbact.sh	BioMerieaux BacT Alert Interface.	Optional Purchase Module
atupaero.sh	Abbot Aeroset Interface.	Optional Purchase Module
atupastm.sh	Ortho Summit OAS Interface.	Optional Purchase Module
atupcmdr.sh	Abbott Commander Interface.	Optional Purchase Module
atupgali.sh	Galileo Interface.	Optional Purchase Module
atupk7300.sh	Olympus PK7300 Interface.	Optional Purchase Module

Job Name	Description	Optional/Required
atuprism.sh	Abbott Prism Interface.	Optional Purchase Module
atuprtur.sh	For table handling of external sample files.	Optional Purchase Interface
atupsurd.sh	Convert Surround results file to Blood Centers upload format and upload the results.	Optional Purchase Interface
atuptang.sh	Stratec Biomedical Systems Tango Interface.	Optional Purchase Module
AUTO_PRINT.sh	Automatically print reports to a designated network printer.	
BC_DONLBL_ PRINT.sh	Initiates the special donation labeling.	Every 10 minutes
CSL_FILES_MOVE.sh	Moving the plasma files to Custom folder.	Optional Purchase Module
EMFUTIL.sh	Initiates the EMF process.	Every 1 minute
EMFUTIL_RECHECK_ DONOR.sh	Updates the donor Chagas test scheduling status as part of the EMF.	Every 10 minutes
export18c.sh	To create the export files for the Mobile PC database if it is the 18c express edition.	Required only if the Mobile PC download method is used.
export19c.sh	To create the export files for the Mobile PC database if it is the 19c standard edition.	Required only if the Mobile PC download method is used.
GENBDR_Convert.sh	Converts from BDR text file to PDF, and moves the pdf to the specified location.	Every 10 minutes
LT_DONOR_ SCHEDULE	To automate the donor schedule API.	Optional Purchase Interface
LT_GENERAL_API	To Support all the APIs.	Optional Purchase Interface
lt_hcll_interface.sh	Blood Centers to Transfusion product transfer job, which transfers finished blood products to Transfusion.	Optional Purchase Interface
LT_RECRUIT_API	To Automate the recruitment API.	Optional Purchase Interface
MAS90CRON.sh	Initiates Finance interface.	The script must be modified depending on the how often the data needs to be interfaced to Financial System
MPC_API_UPLOAD.sh	Automate the mobile upload files insert into the API tables.	Required only if the Mobile PC upload method is used with the files' manual copy method in the server.
MPC_MASTER_ UPLOAD.sh	Automate the mobile upload API tables data into the Main tables.	Required only if the Mobile PC upload method is used for both the Mobile Client upload and the manual copy method.

Job Name	Description	Optional/Required
outbound_ recruitment_api.sh	Outbound Recruitment API.	Optional Purchase Module
RAPND.sh	Registration Audit Pending Hold, requires set up of associated SYSPARM in donor.	Optional
RGPREXTR.sh	Test result extract - CTS Only.	Optional Purchase Module
RTURLOAD.sh	RTURLOAD - for file handling of external sample files.	Optional Purchase Interface
schchagas.sh	Chagas Utility.	Optional Purchase Module
SCHEXTRACT.sh	Test result extract- CTS Only.	Optional Purchase Module
TRPH.sh	Test Results Pending Hold, requires set up of associated SYSPARM in donor.	Optional
tube_checkin_api.sh	Tube Checkin Interface.	Optional Purchase Module
vista_donation.sh	Vista donation interface file upload.	Optional Purchase Interface
vista_donor.sh	Generate donor demographics and eligibility for Vista.	Optional Purchase Interface

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6.8 Automatically Printing Reports

6.8.1 Overview

You can print any Blood Centers report automatically without the need to print it from the PRINT or SPRINT screen. To enable this capability, complete the following steps:

1. Verify that the network printer to which the report will be sent is configured on the Blood Centers server. Refer to the Linux Operations Guide for information.
2. Associate the network printer to the user who will print the reports automatically. Refer to the Add User (ADDUSER) screen description in the WellSky Blood Centers Administration Guide for information.
3. Configure and enable the AUTO_PRINT.sh cron job. Refer to the Cron Job section in this document.
4. Add report names to the /home/sbcsys/labexec/UPD_REP_STATUS.sql file. Refer to the following procedures for information.

6.8.2 Adding Report Names to the UPD_REP_STATUS.sql File

To configure a specific report for automatic printing, you must add the operating system (OS) file name associated with the report to the UPD_REP_STATUS.sql file, located in the /home/sbcsys/labexec directory. Use the following procedures to first find the OS file name associated with the report name and then edit the UPD_REP_STATUS.sql to add or remove these OS file names.

Finding the OS File Name

1. Using PuTTY or Reflection, log into to the Blood Centers Server. Use the operating system user name sbcsys, as shown:

```
login as: sbcsys
sbcsys@172.16.12.170's password:
Last login: Wed Dec 18 21:01:06 2013 from 10.2.64.7
[sbcsys@chdevsrv10 ~]$
[sbcsys@chdevsrv10 ~]$
```

2. Set the oracle environment variable for the Blood Centers database (for example, qtrak), as shown:

```
[sbcsys@chdevsrv10 ~]$
[sbcsys@chdevsrv10 ~]$ . oraenv
ORACLE_SID = [sbcsys] ? qtrak
The Oracle base has been set to /apps/product/11.2.0
[sbcsys@chdevsrv10 ~]$
```

3. Enter the following command:
   sql> `select lower (p1_value) from menu_item where memonic=<report keyword>';`
   Example:
   To find the OS name for the report PACKLIST, enter the following command:

```
SQL>
SQL> select lower(p1_value) from menu_item where mnemonic='PACKLST';

LOWER(P1_VALUE)
--------------------
shrpklst

SQL>
```

   The OS name, shrpklst, appears below the dotted line, as shown.

Editing the UPD_REP_STATUS.sql File

1. Open the UPD_REP_STATUS.sql file for editing by entering the following command:
   $ `gedit /home/sbcsys/labexec/UPD_REP_STATUS.sql`
2. Locate the v_job_name variable in the file, as shown:

```
            -- the following insert is to auto print
            If v_job_name in
('dprmerge','drrdfbat','dprnuhld','drrhlddt') then

              Begin
```

3. Add or remove OS file names that correspond to the reports you want to automatically print. These are the OS file names determined by the Finding the OS File Name procedure. The entire list must be enclosed in parentheses , each OS file name must be enclosed in single quotes, and each OS file name must be separated by a comma.
   Example:

```
If v_job_name in ('oerorslp','scrpicks','scrunpsl','scrtrafr','shrpklst',
                  'imrrtlst','scroutdt','imrckosl','csrdclst',
                  'cprckosl','cprpool','mcrchblt','mcrcold4') then
```

   In this example, the OS file names correspond to the following report keywords:

OS File Name	Report Keyword	Module
cprpool	CPRPOOL	Lab
cprckosl	CPRCKOSL	Lab
mcrcold4	MCRCOLD	Lab
csrdclst	CDSLIP	Distribution
imrrtlst	IMRRTLST	Distribution
scrunpsl	SCRUNPSL	Distribution
scrtrafr	SCRTRAFR	Distribution

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OS File Name	Report Keyword	Module
scroutdt	SCROUTDT	Distribution
scrpicks	PICK	Distribution
oerorslp	OERORSLP	Distribution
imrckosl	IMRCKOSL	Distribution
mcrchblt	MCRCHBLT	Lab

NOTE – You can mix Lab and Distribution reports in the same OS file name list.

6.9 Linux NFS Setup Between the Oracle Application and the Database Server

This is an optional procedure. The Database Administrator needs to share the Oracle application server's /home/bcorapp directory to the Oracle database server. This directory can share to all of the Oracle application servers, including if the Blood Center server is set up with a multi-node application server.

Perform the following procedure.

1. Create a NFS shared directory in the Linux 8 application server.
    a. Install the NFS services using the following command as a root user:
    ```
    # dnf install -y rpcbind nfs-utils
    ```
    b. Configure the following services so they automatically start at boot time:
    ```
    # systemctl enable rpcbind
    # systemctl enable nfs-server
    ```
    c. Once the services have been enabled, start them as follows:
    ```
    # systemctl start rpcbind
    # systemctl start nfs-server
    ```
    d. Specify the **/home/bcorapp** directory in the **/etc/exports** directory, as shown below:
    ```
    /home/bcorapp 10.10.11.11(ro)
    ```

    NOTE – In the above example, 10.10.11.11 is the database server's IP address. You can update with multiple IP addresses or a wild card by using *.

    e. Update the NFS server for the exported file system using the following command:
    ```
    # exportfs -a
    ```
    f. Run the following command to view the shared settings:
    ```
    # exportfs
    ```

    It will generate the following output:
    ```
    /home/bcorapp
    ```

2. Access the Linux 8 application server's NFS shared directory **/home/bcorapp** in the database server.
    a. Install the NFS util package using the following command:
    ```
    # dnf install -y nfs-utils
    ```
    b. Make sure the **/home/bcorapp** directory exists on the database server, otherwise create it using the following command:
    ```
    # mkdir /home/bcorapp
    ```
    c. Enter the following command to mount the remote folder:
    ```
    # mount -t nfs 10.10.10.10 /home/bcorapp /home/bcorapp
    ```


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NOTE – Here 10.10.10.10 is the remote application server's IP address.

d. Configure the NFS filesystem on system startup by editing **/etc/fstab** and appending the line like the following example:

```
10.10.10.10:/home/bcorapp /home/bcorapp nfs ro 0 0
```

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7

Chapter 7 - Post Donor Application and Database Checklist

7.1 Update the Donor Database Tables with Donor Application Server Configuration 148
7.1.1 Configuration 148
7.2 Blood Centers Client Setup 150
7.2.1 Client Setup 150
7.2.2 Installing the WebCam 151
7.2.3 Setting Up the Internal Camera for On-Site 155
7.3 HemaTrax-UNITY Printing 158
7.3.1 Install HemaTrax-UNITY 158
7.3.2 Configuring the HemaTrax Client Utility 164
7.3.3 Configuring Label Printers in Blood Centers 168
7.3.4 HemaTrax Print Server 169
7.4 Installation Cleanup 171

7.1 Update the Donor Database Tables with Donor Application Server Configuration

NOTE – If you are installing Blood Centers on two servers (one for the database and one for the application), this procedure is completed on the database server.

7.1.1 Configuration

1. Log on as **oracle**.
2. Right-click on the desktop and select **Open Terminal**.
3. Enter the following command to set the environment variables, connect to the database, and run the update_bcappdb.sql script for updating the database:
   ```
   $ . oraenv
   ```

   When prompted, enter the name of the database (ex: bcprd)

   Example:
   ```
   $ cd / oradata/bc2024/BC_SERVER_INSTALL/bloodc
   $ sqlplus "/as sysdba"
   SQL> @update_bcappdb.sql
   ```

   ```
   [oracle@bc2024rl bloodc]$
   [oracle@bc2024rl bloodc]$ . oraenv
   ORACLE_SID = [oracle] ? bcprd
   The Oracle base has been set to /apps/product/19.0.0
   [oracle@bc2024rl bloodc]$ sqlplus "/as sysdba"

   SQL*Plus: Release 19.0.0.0.0 - Production on Wed Jan 19 18:20:21 2022
   Version 19.3.0.0.0

   Copyright (c) 1982, 2019, Oracle.  All rights reserved.


   Connected to:
   Oracle Database 19c Standard Edition 2 Release 19.0.0.0.0 - Production
   Version 19.3.0.0.0

   SQL> @update_bcappdb.sql
   ```

4. When prompted, enter the following values one by one. Press **Enter** after each entry.
   - Blood Centers hostname: Hostname Blood Centers Apps Server
   - Blood Centers ipaddress: IP Address of Blood Centers Apps Server

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- Ktrakservername: KTRAK Server Name
- Blood Centers databasename: Blood Centers Database Name

```
SQL> @update_bcappdb.sql
Enter value for bc_apps_hostname: bc2024r1
Enter value for bc_apps_ipaddress: 3.145.99.5
Enter value for bc_ktrakservername: KCMISKTSRV02.mis.mediware.com
Enter value for bc_database: bcprd
Connected.
```

5. Change the directory to /home/bcorapp/donor/frmx using the following command:
   $ `cd /home/bcorapp/donor/frmx`
6. Edit the following files with VI editor to replace bc2024r1.wellsky.com with the application server's hostname.
   - cac_login.css
   - forms_laf.css
   - lab.css

```
cac_login.css:      image1: http://bc2024r1.wellsky.com:9001/forms/java/images/WS_Blood_Centers_Al
forms_laf.css:        image1:http://bc2024r1.wellsky.com:9001/forms/java/images/WS_Blood_Centers_S
forms_laf.css:    image1:http://bc2024r1.wellsky.com:9001/forms/java/images/WS_Blood_Centers_Stack
lab.css:      image1: http://bc2024r1.wellsky.com:9001/forms/java/images/WS_Blood_Centers_Stacked.
```

7. Launch the following Oracle Consoles:
   - WebLogic Server Enterprise Manager Link: http://10.10.10.10:7001/em
   - WebLogic Server Administration Console Link: http://10.10.10.10:7001/console
8. Both links will prompt you for a User name and Password:
   User Name: weblogic
   Password: WebLogicAdmin1
9. Contact WellSky for assistance running the scripts under the BC_Suite/Database/Scripts/WellSky_Restricted directory.
10. If you purchased any add-on products that utilize an API, contact WellSky for assistance completing the installation and configuration, as the process is unique for each client.

7.2 Blood Centers Client Setup

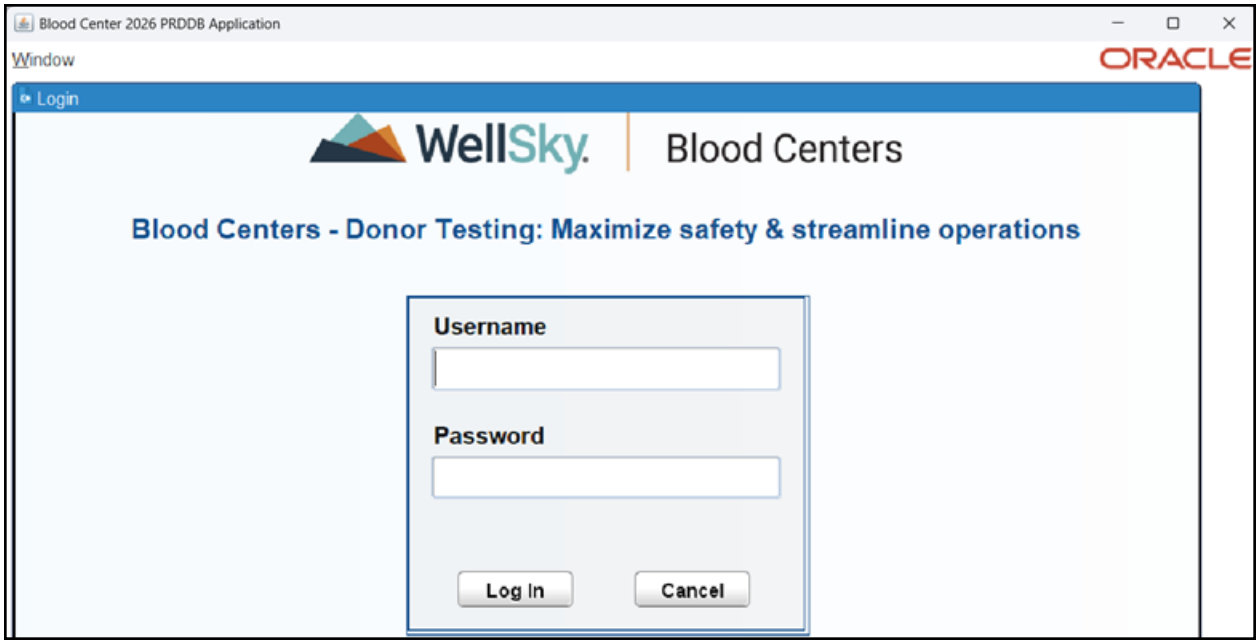
The following information provides instructions on running Blood Centers on a client PC with a Windows OS.

7.2.1 Client Setup

1. Right click on notepad & run as administrator.
2. Navigate to C:\windows\system32\drivers\etc\hosts file.
3. Add an entry for the Blood Centers Application Server as follows: IP address of the server followed by the hostname.domain name of the sever.
   10.10.10.10 bc2026r1.wellsky.com

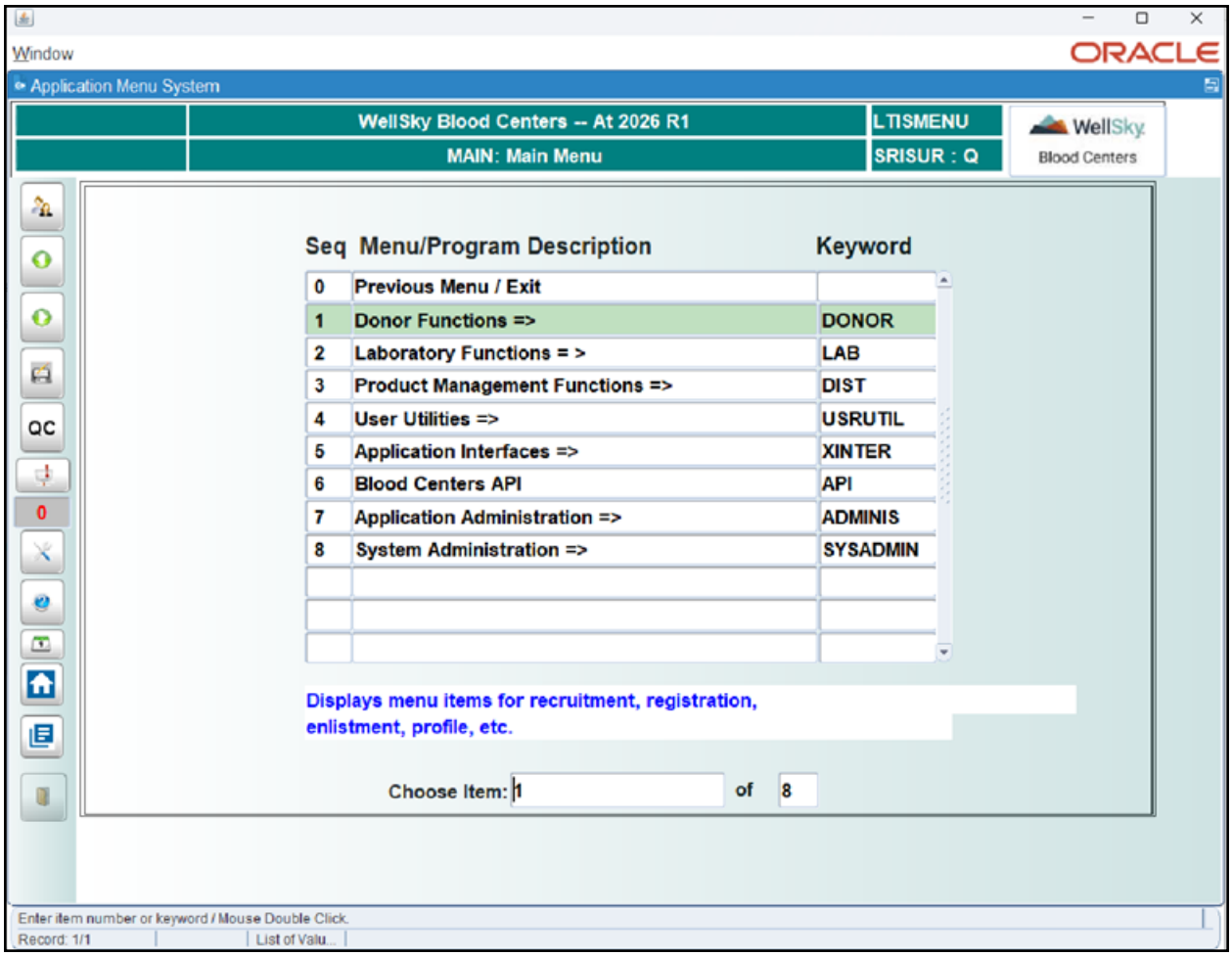
> **NOTE –** In the example above, you would replace bc2026r1.wellsky.com with the hostname.domainname of your Blood Centers application server and IP address with BC application server's IP address.

4. Save and close the hosts file.
5. Copy the BC2026.zip file from Software CD -> BC2026R1/BC_SERVER_INSTALL/client directory to the application users work station and unzip.
6. Copy the extracted BC2026 directory in C:\ drive as C:\BC2026.
7. Edit the C:\BC2026\BC2026_PRDDB_FORMS_APP.bat file with notepad and update the Application server's IP or hostname with 10.10.10.10 IP and save the file.
8. Create the shortcut for C:\BC2026\BC2026_PRDDB_FORMS_APP.vbs file on the desktop and double-click to launch the BC2026 application.



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9. Enter the application Username and password and click **Log in**.

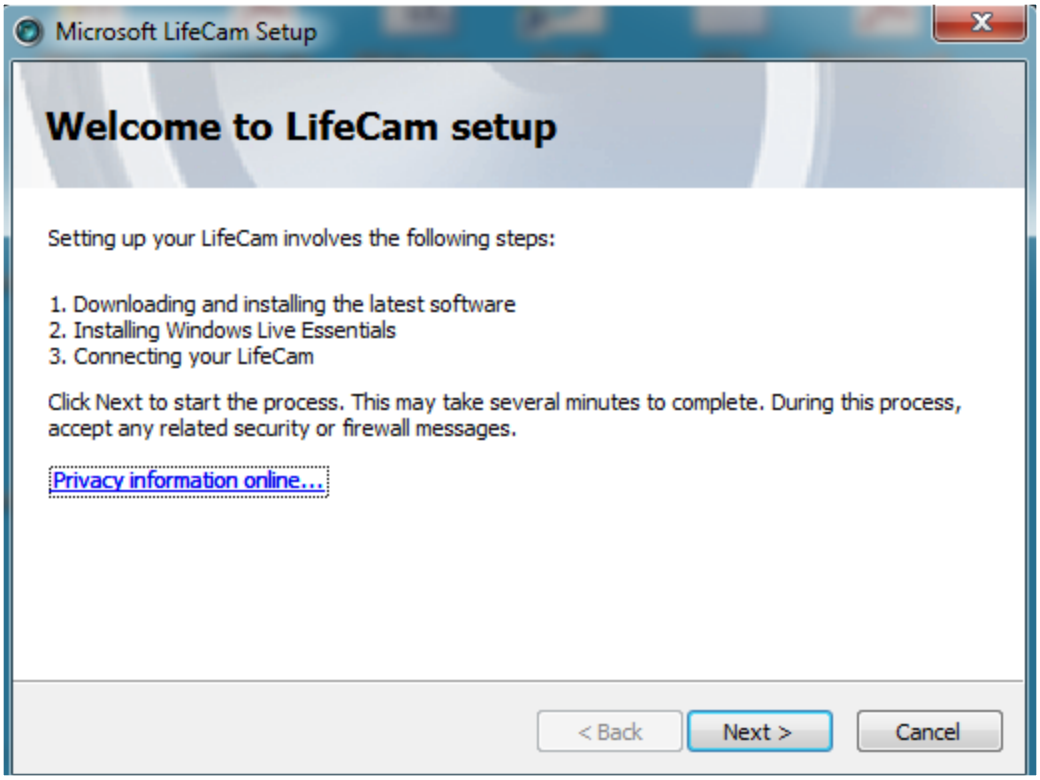


7.2.2 Installing the WebCam

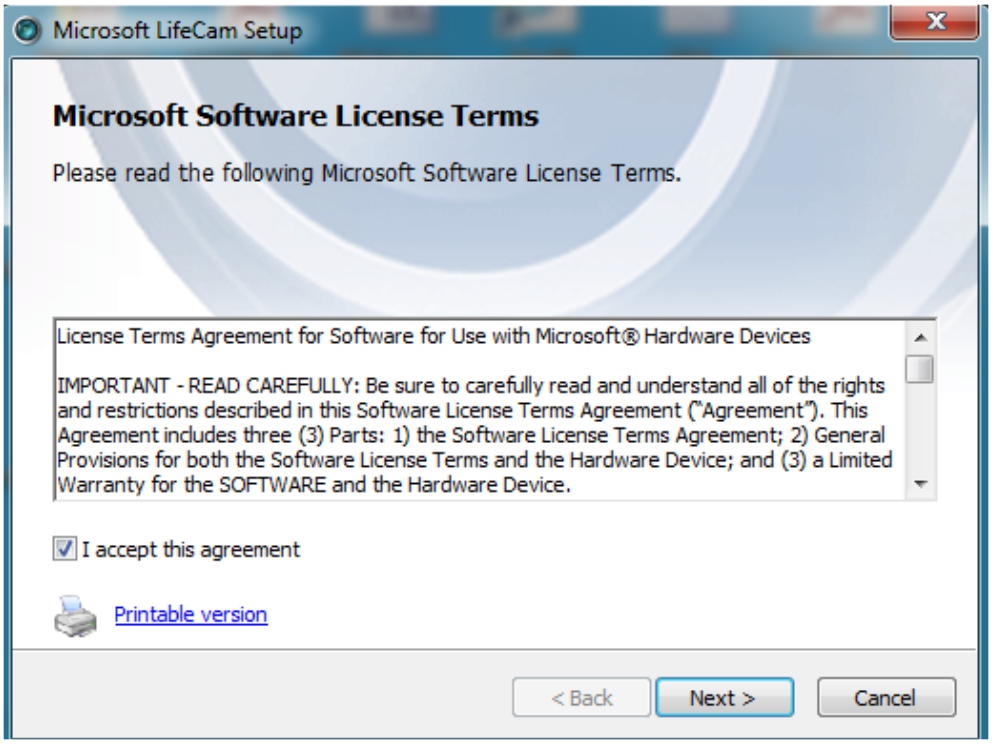
Important! This procedure is to install the Microsoft LifeCam software. This is the webcam recommended by WellSky for the Donor application. Due to a limitation in Oracle forms, if you use a picture taken from a digital camera, you may need to reformat the image file before it can be uploaded. Refer to the "READ_IMAGE_FILE Fails With FRM-47100 For JPG's Stored In Exif Type Format" article (Doc ID402773.1) on the Oracle website for more information.

1. Navigate to the BC_SERVER_INSTALL/sigplus directory.
2. Copy the LifeCam3.60.exe file to your desktop.

3. Double-click **LifeCam3.60.exe**. The following window opens:

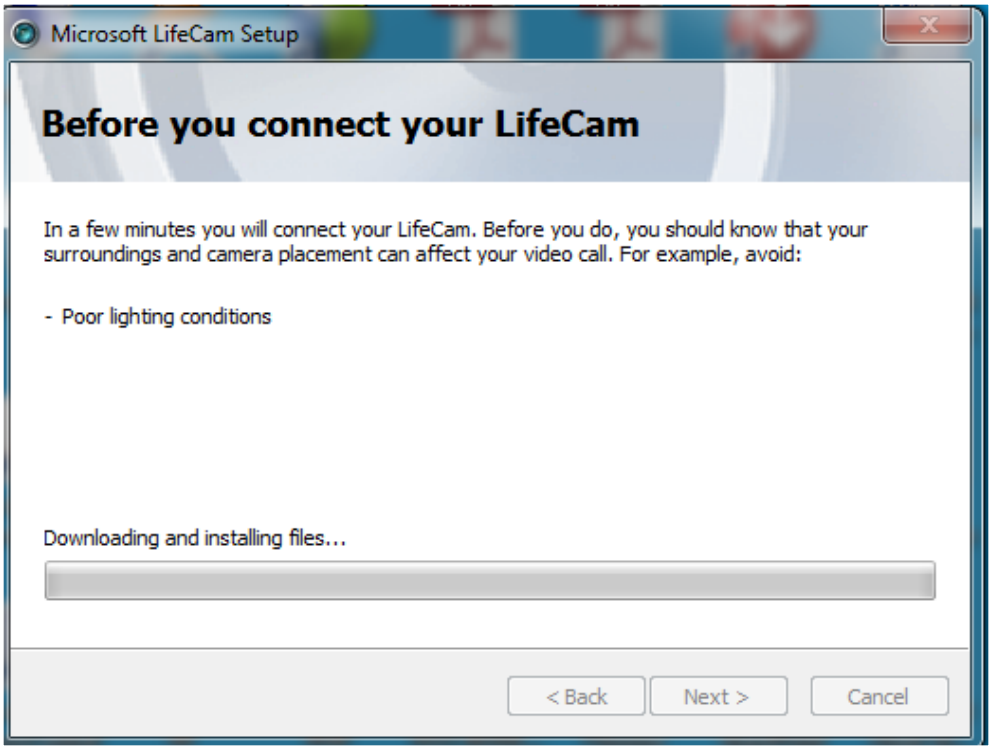


4. Click **Next**. The following window opens:

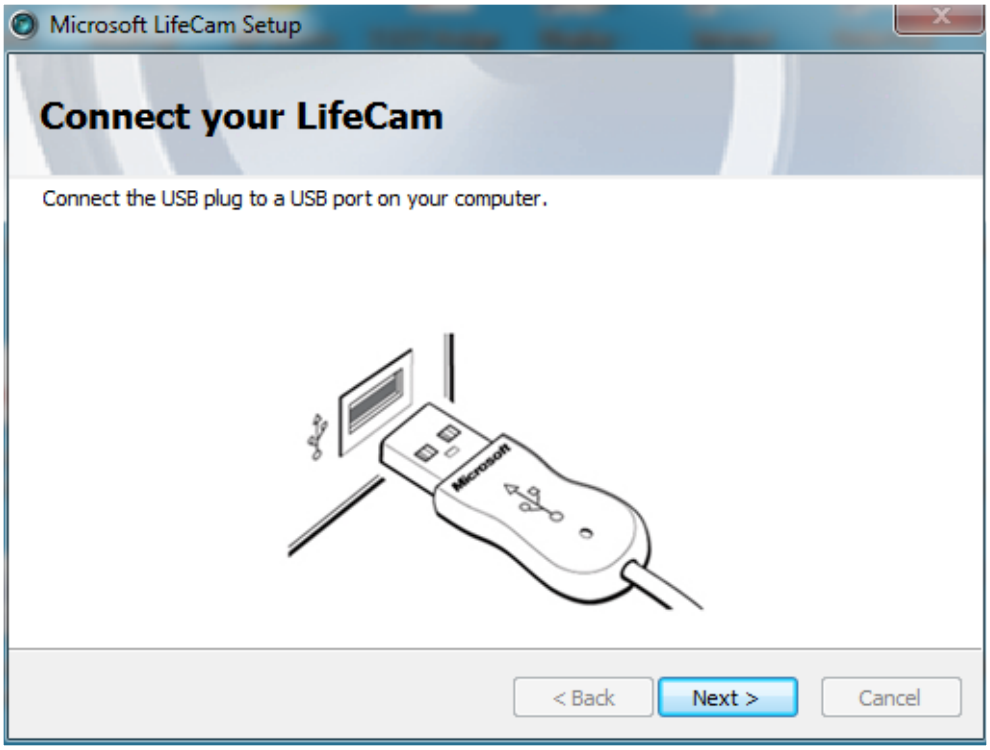


   - Select the **I accept this agreement** check box.

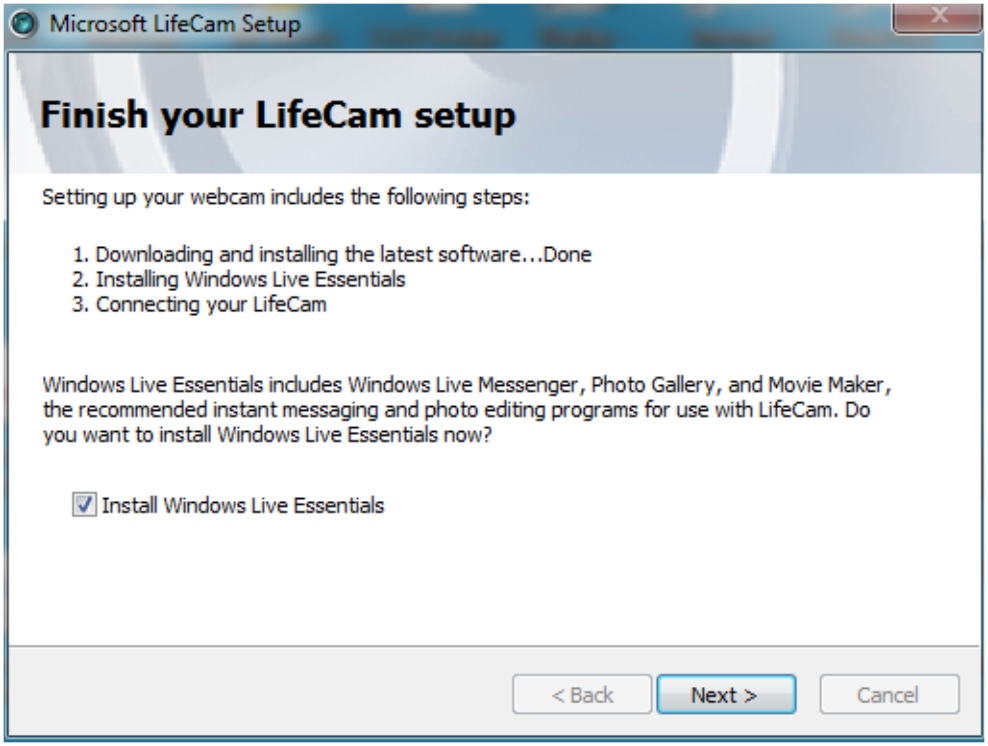
5. Click **Next**. The following window opens:



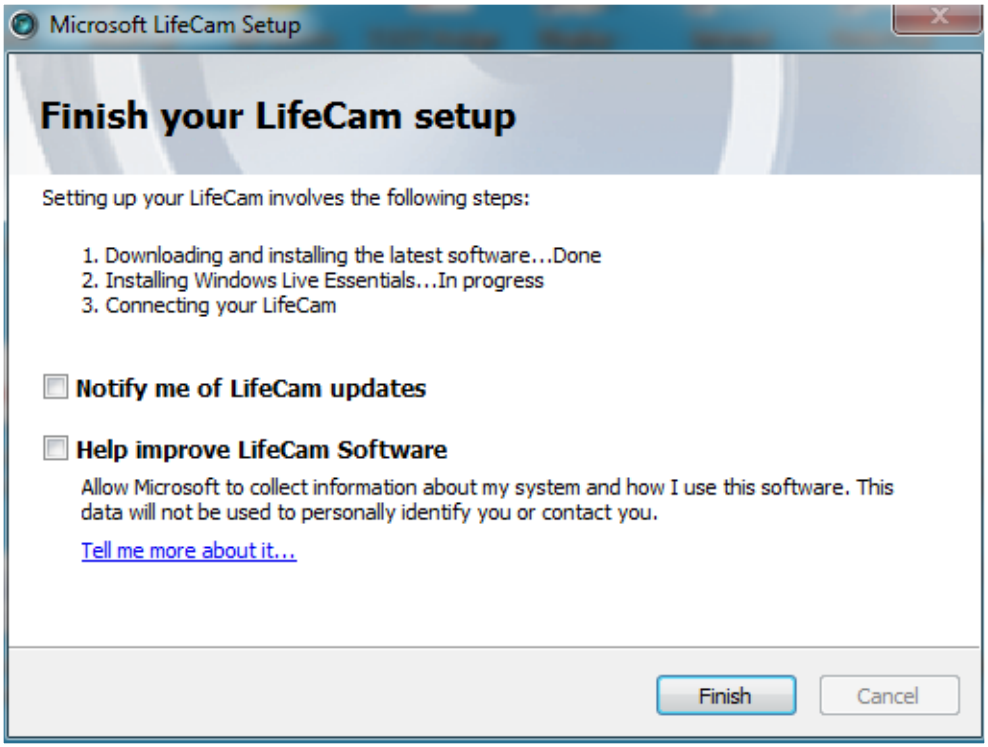
6. After the installation completes, the following window opens:



7. Connect the Web cam to one of your computer's USB port and click **Next**. The following window opens:



8. Click **Next**. The following window opens:



9. Click **Finish** to complete the installation. The Welcome screen opens, as shown:



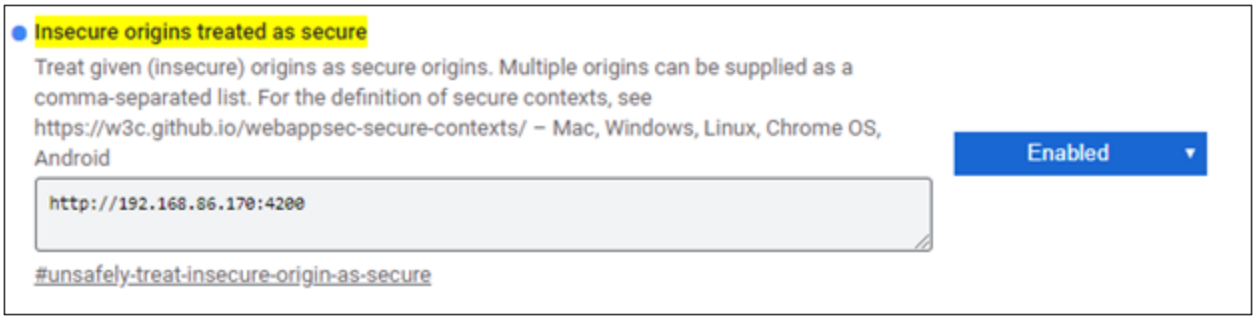
Your WebCam is now ready for use.

7.2.3 Setting Up the Internal Camera for On-Site

Use MediaDevices API to access the camera with permission. It works only with https (secure origin), and not over http (unsecure origin).

The following are the options for non-secure test server that work for the camera component:

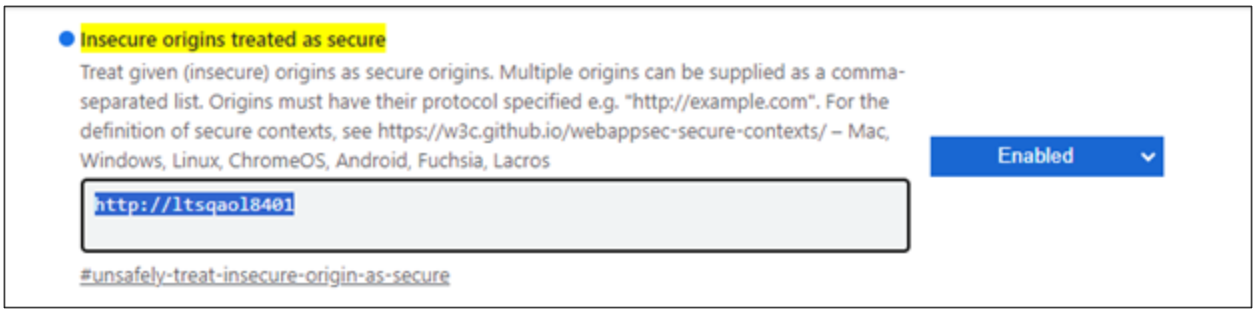
- Localhost works, as it is treated as secure over http.
- For server testing, run Chrome to set a flag.
  - Enter the following URL: chrome://flags/#unsafely-treat-insecure-origin-as-secure.
  - Select the **Insecure Origins treated as secure** option.
  - Add the server IP address, which needs to be treated as secure.



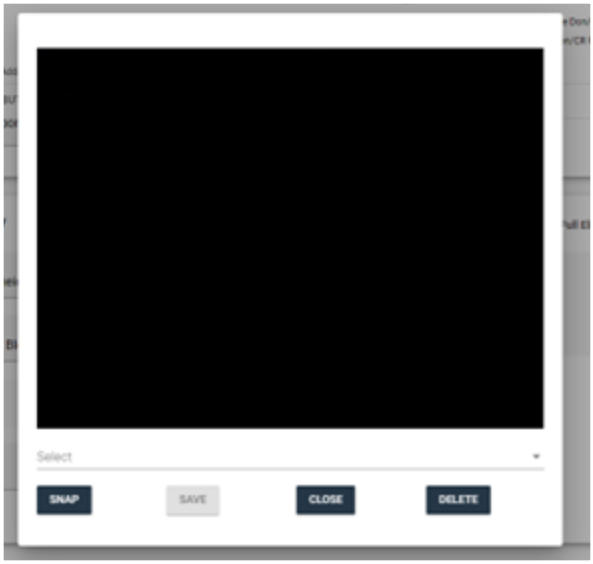
Reference: https://www.chromium.org/Home/chromium-security/deprecating-powerful-features-on-insecure-origins/.

Steps to get Camera Working in SQA or Production

1. Open the Chrome browser, and enter the following URL: chrome://flags/#unsafely-treat-insecure-origin-as-secure.
2. Select the **Insecure origins treated as secure option**.
3. Enter the SQA or Production URL (e.g., http://ltsqaol8401) in the URL box.
4. Change the setting to **Enabled**.



5. Open the Camera window from the Registration screen.



6. Select a camera from the **Select** drop-down menu.

> **NOTE –** All of the camera devices on the client machine will display in the list (includes both internal and external cameras).



7. Click **SNAP** to take a picture.

> **NOTE –** To re-take the picture, click **DISCARD**, then repeat step 7.

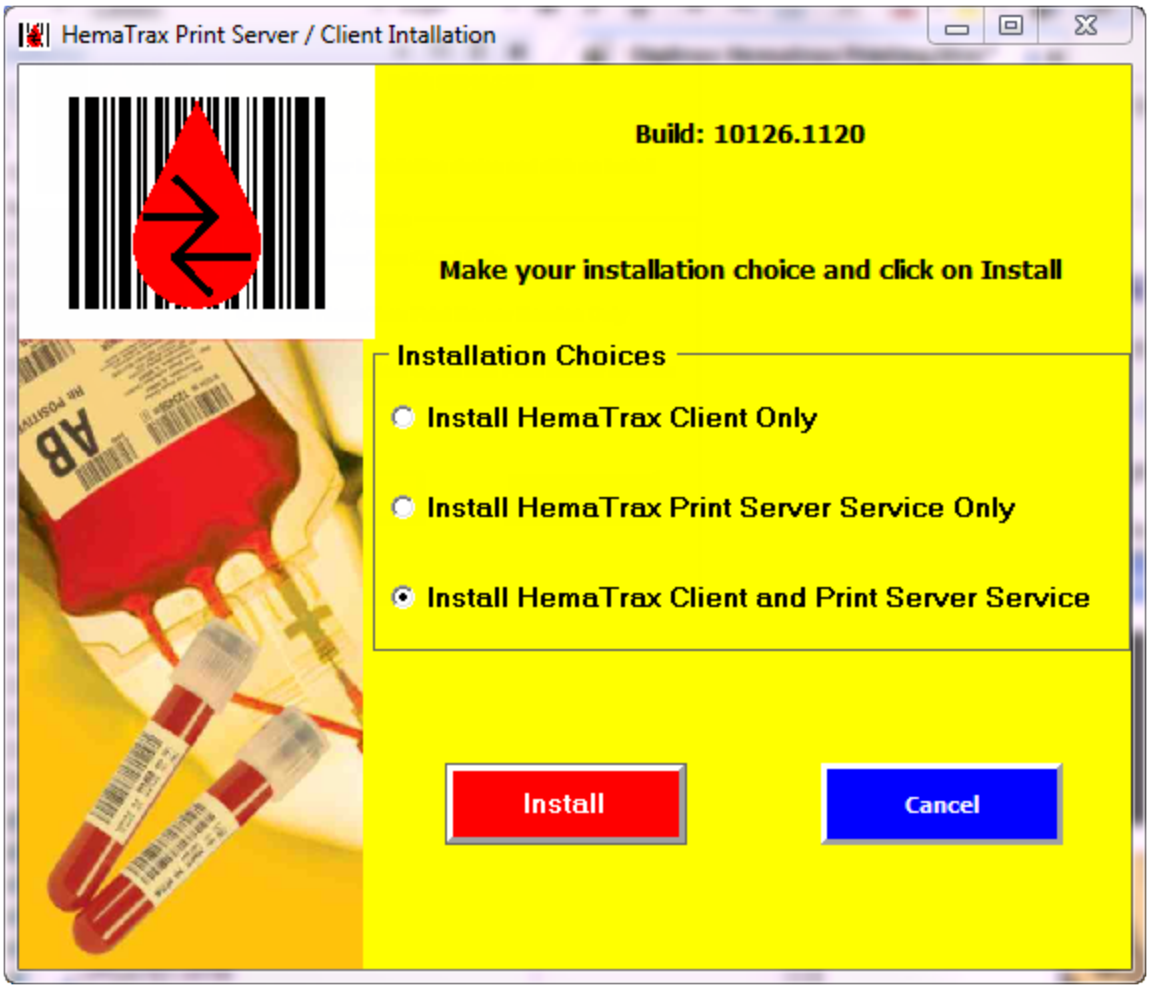
8. Click **SAVE** to upload the picture to the database. The image will be displayed in the Donor Header banner.

7.3 HemaTrax-UNITY Printing

Follow these procedures to install and configure the HemaTrax-UNITY Print Server Software on a Windows based OS system.

7.3.1 Install HemaTrax-UNITY

1. Load the HemaTrax-UNITY Installation CD into your CD-ROM drive.
2. Open **Windows Explorer**.
3. Navigate to your CD-ROM drive.
4. Double-click **setup.exe**. The primary installation window appears.

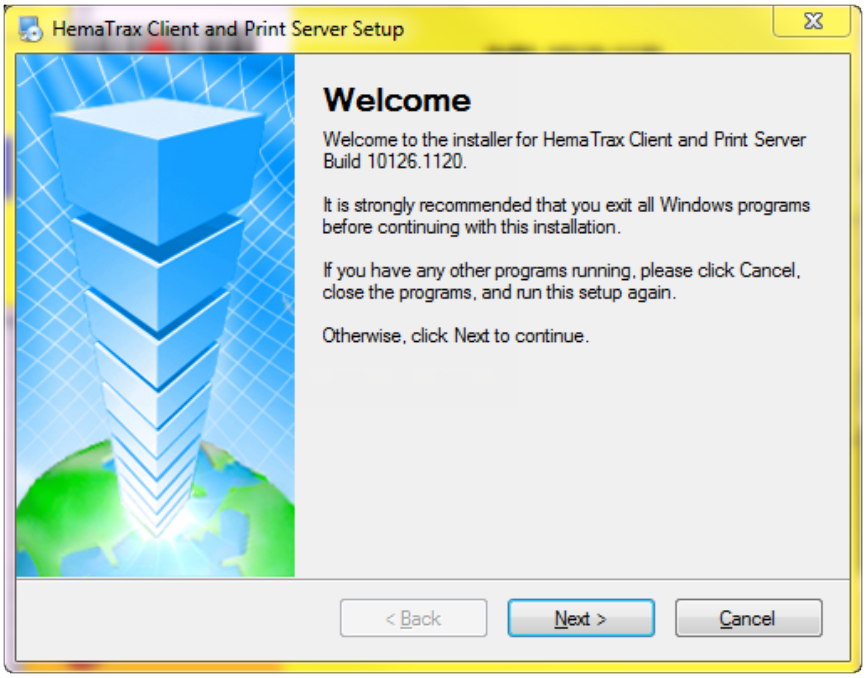


Verify **Install HemaTrax Client and Print Server Service** is selected. If not, select it.

5. Click **Install** to begin the install sequence.

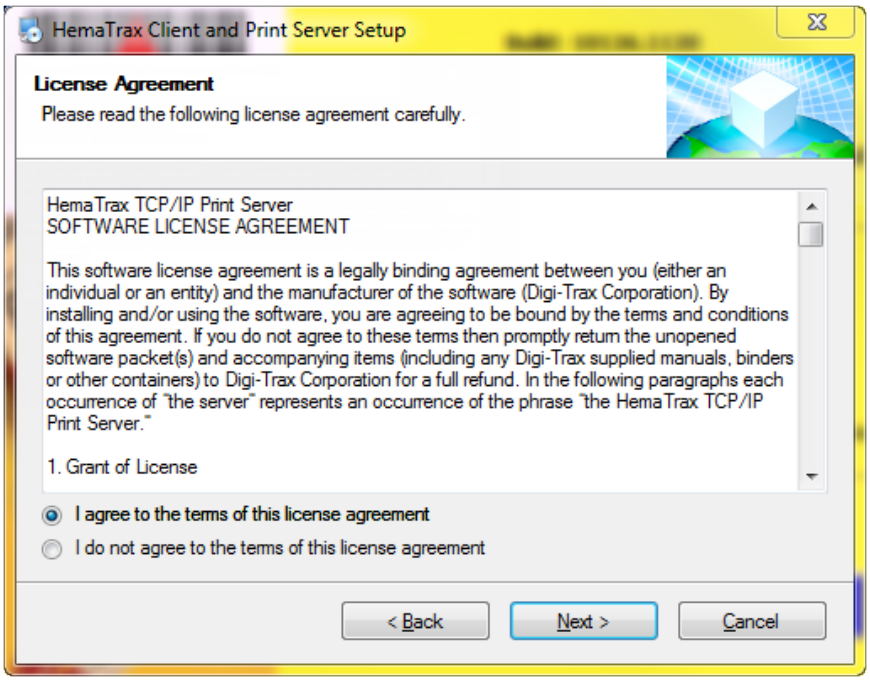
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6. The Welcome window appears.



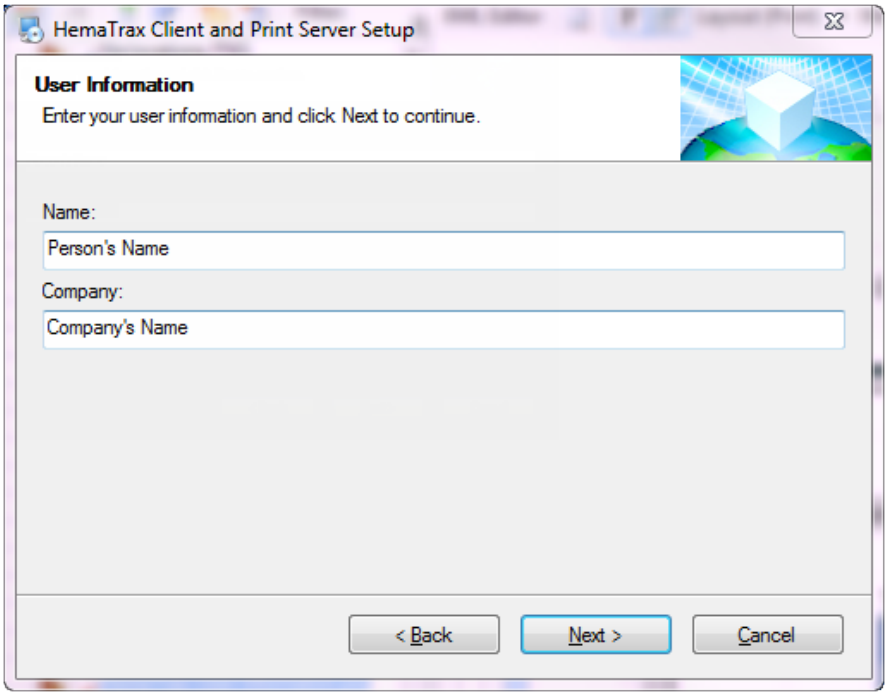
- Click **Next**.

7. The License Agreement window appears.



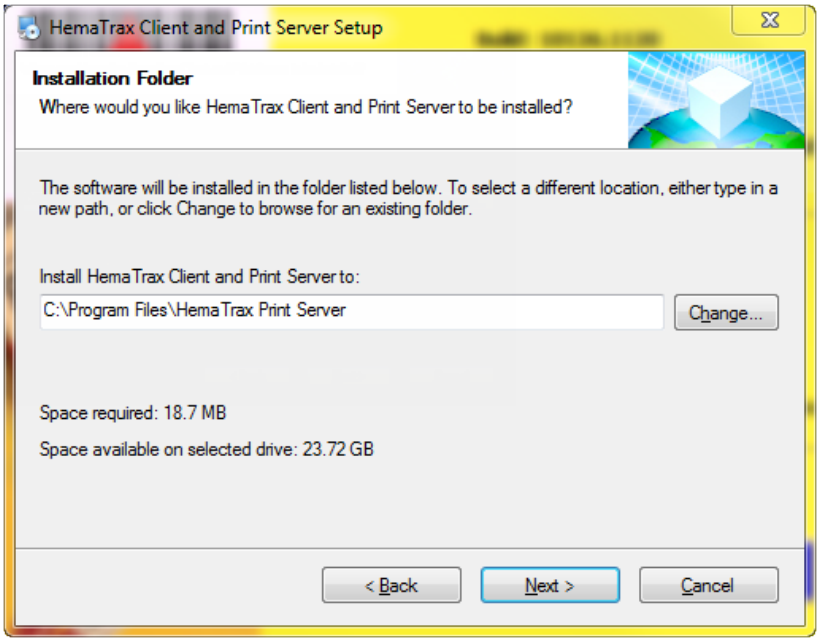
- Select **I agree to the terms of this license agreement**.
- Click **Next**.

8. The User Information window appears.



- Enter your name in the Name text box.
- Enter the name of your company in the Company text box.
- Click **Next**.

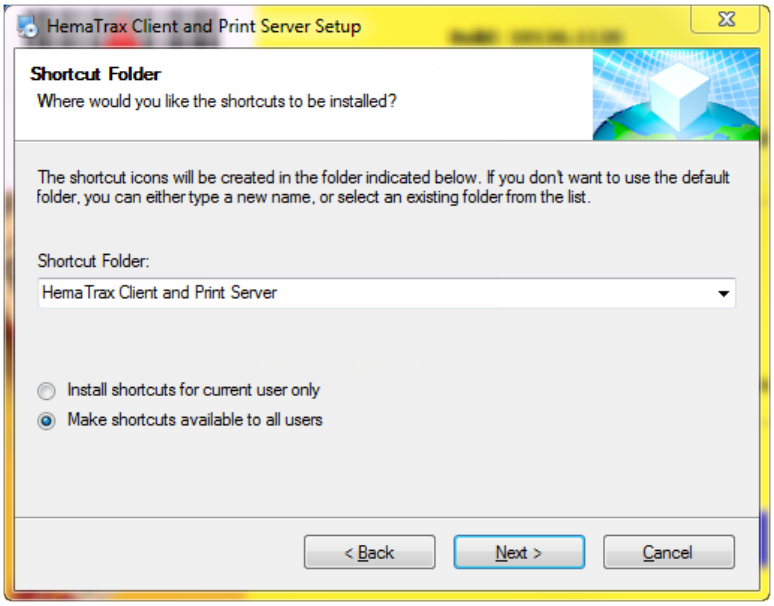
9. The Installation Folder window appears.



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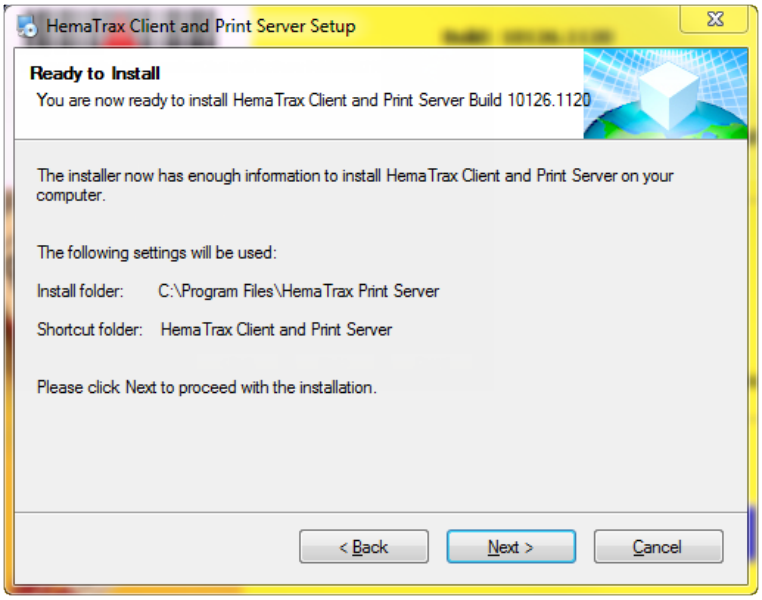
- If the default install directory listed is not acceptable, click **Change** to select an alternate directory.
- Click **Next**.

10. The Shortcut Folder window opens.



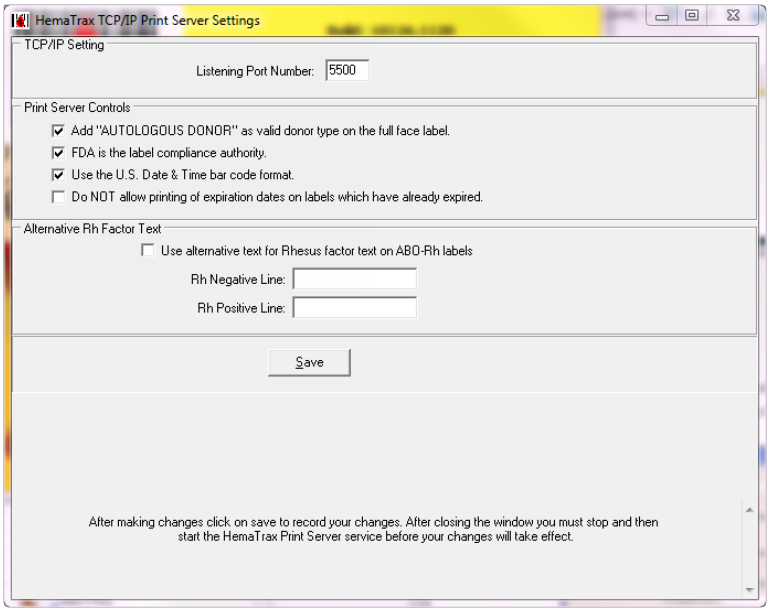
- Select **Make shortcuts available to all users**.
- Click **Next**.

11. The Ready to Install window appears.



- Click **Next**.

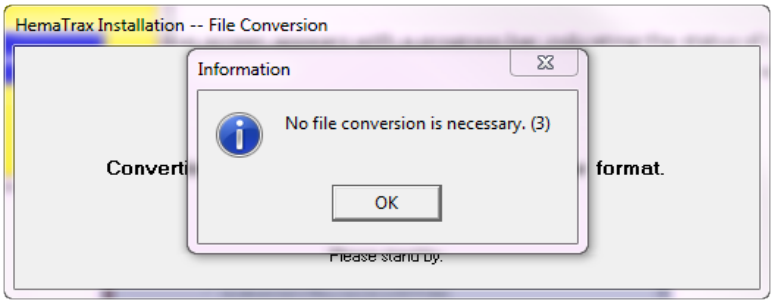
12. The Setup screen appears with a progress bar indicating the status of the software load on your system. When the process is finished, the HemaTrax TCP/IP Print Server Settings window appears.



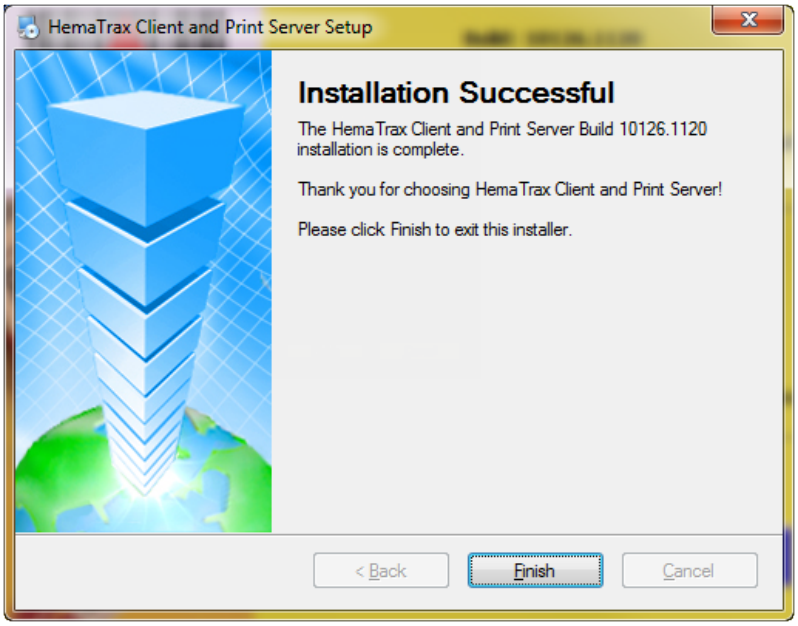
NOTE – 5500 is the default setting for the Listening Port Number field.

- Select the **Add 'AUTOLOGOUS DONOR' as valid donor type on full face label** check box.
- Click **Save**.
- Close the window.

13. The install program checks for any conversions that need to be made and performs them at this time. A window appears when the conversion check is complete. Click **OK** to continue.



14. The Installation Successful window appears.



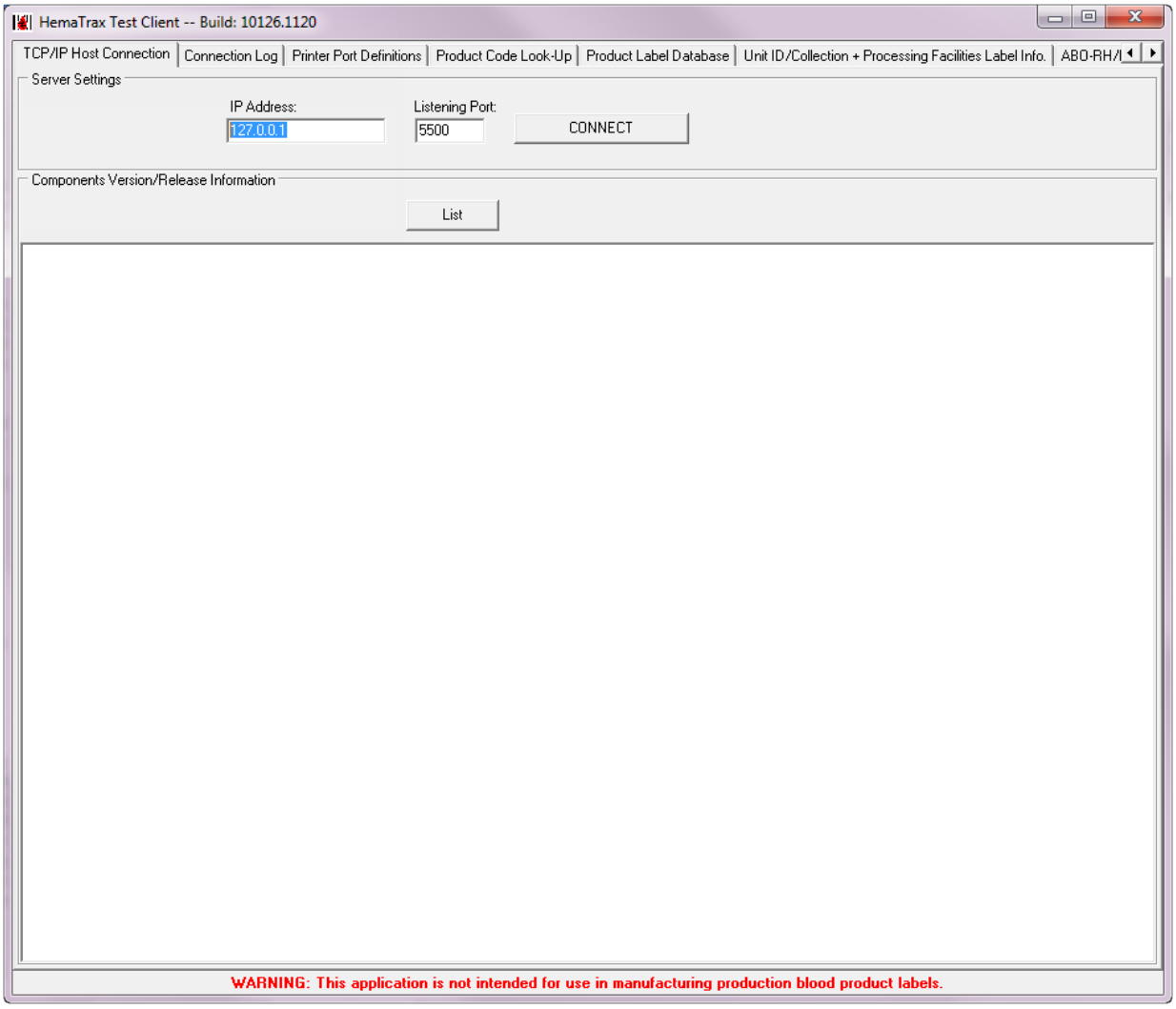
- Click **Finish**.

7.3.2 Configuring the HemaTrax Client Utility

NOTE – Before proceeding with the following steps, verify that all HemaTrax-UNITY printers intended for use with this setup are installed on your network and respond to ping.

To configure the HemaTrax client utility, follow these steps.

1. On your Windows desktop, navigate to Start > Programs > HemaTrax Client and Print Server > HemaTrax Client.
2. The HemaTrax Client screen appears.

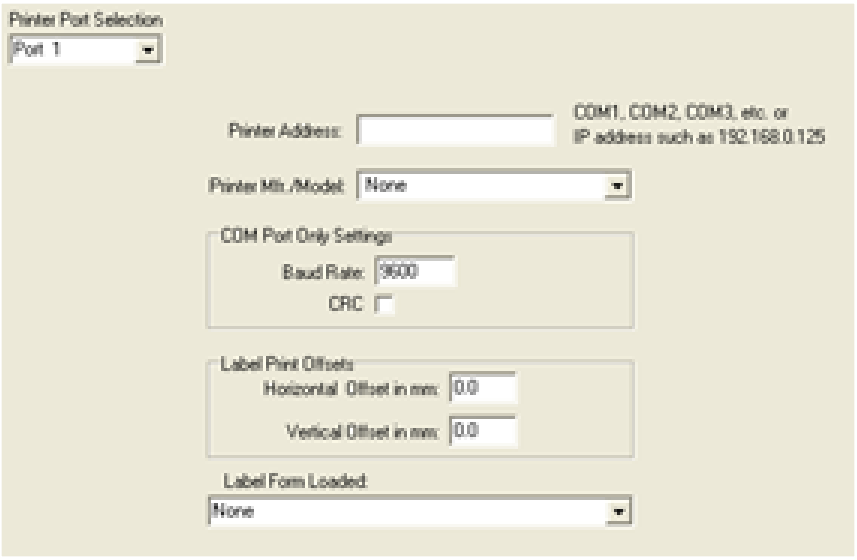


- On the TCP/IP Host Connection tab, record the IP Address listed for future use.
- Click CONNECT. The button changes to DISCONNECT.

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3. Select the **Printer Port Definitions** tab.

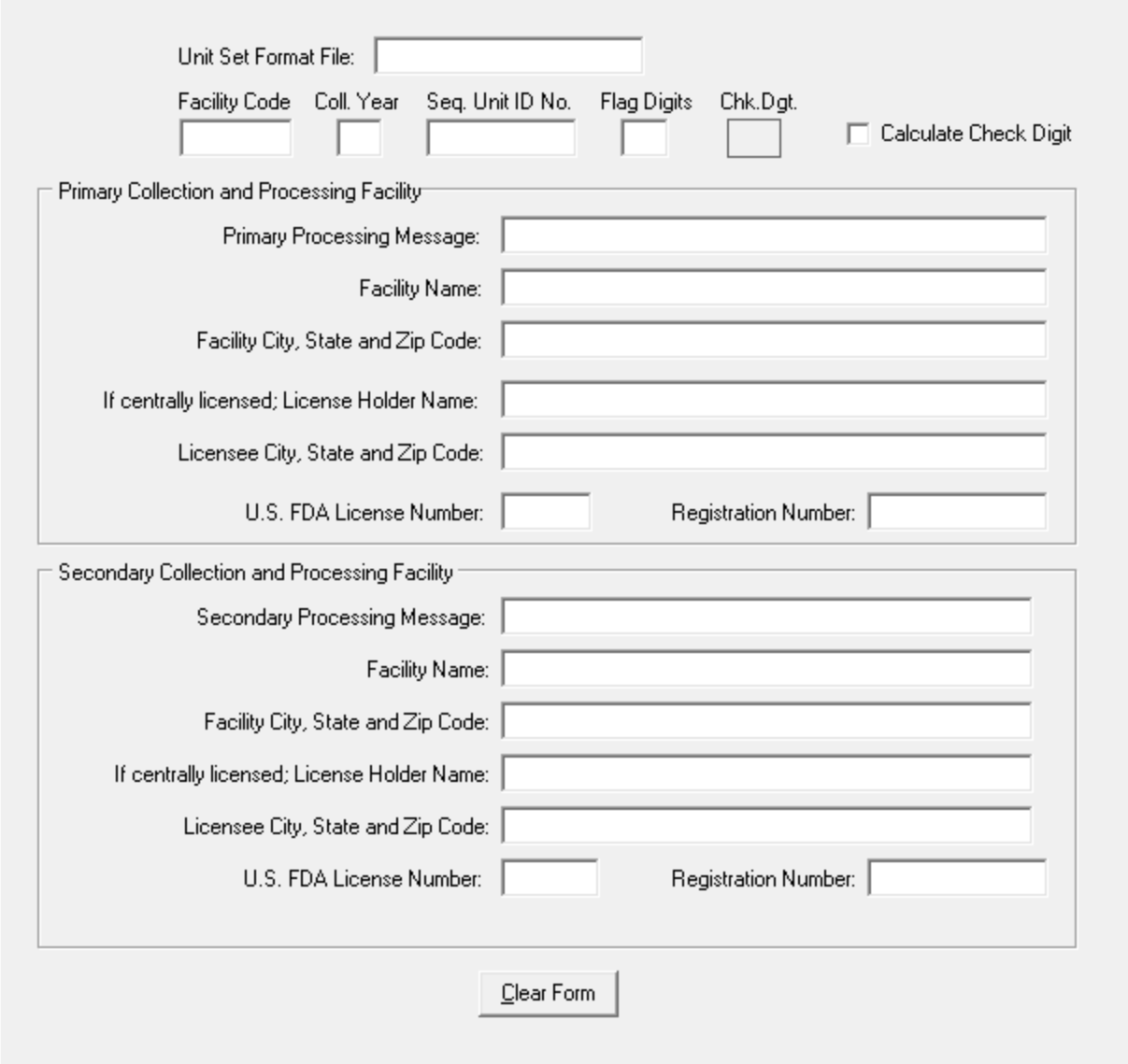


- Open the Printer Port Selection drop-down and select a port.

> **NOTE –** Each printer you configure will need a unique port.

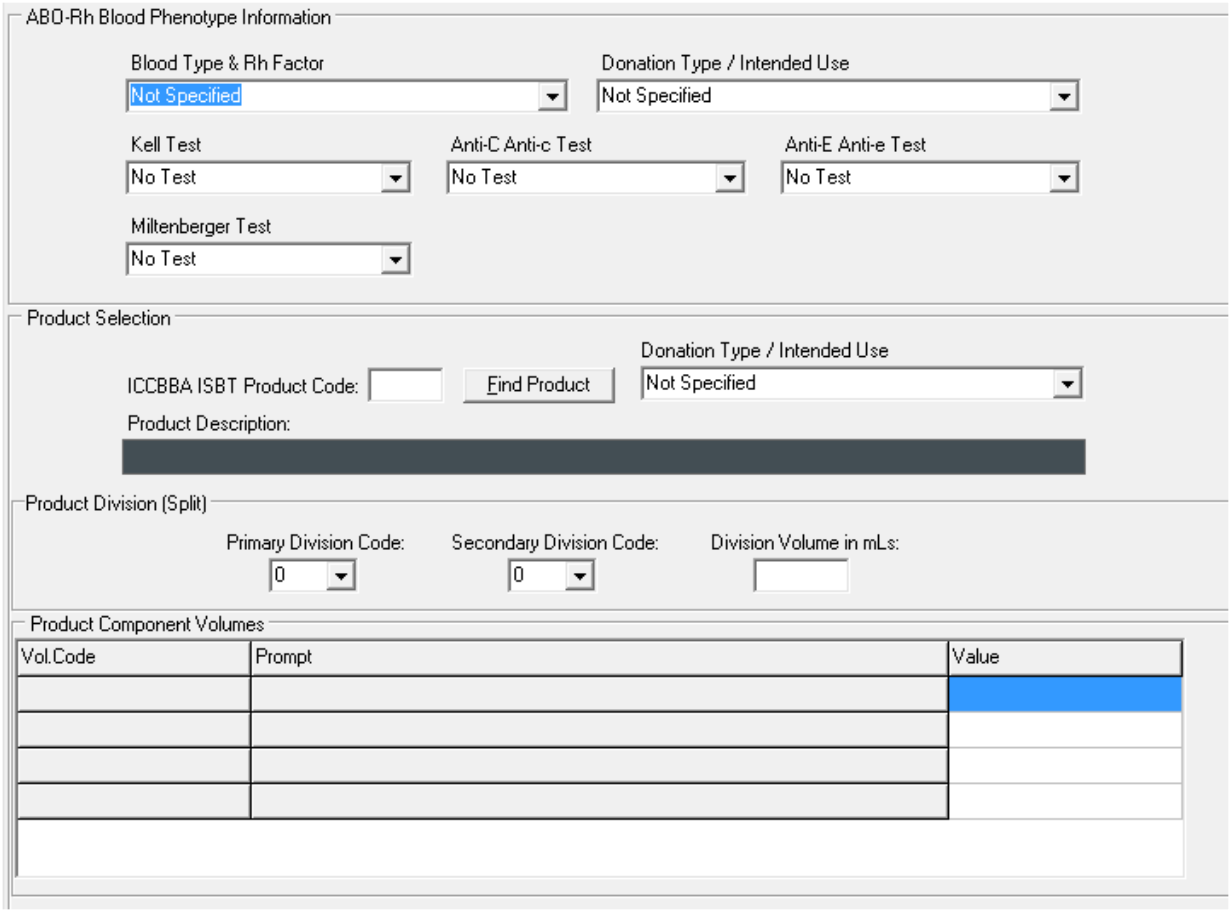
- In the Printer Address text box, enter the IP address of the barcode printer.
- Open the Printer Mfr/Model drop-down and select an appropriate printer.
- Open the Label Form Loaded drop-down and select an appropriate label form and size.

4. Select the **Unit ID/Collection & Processing Facilities Label Info** tab.



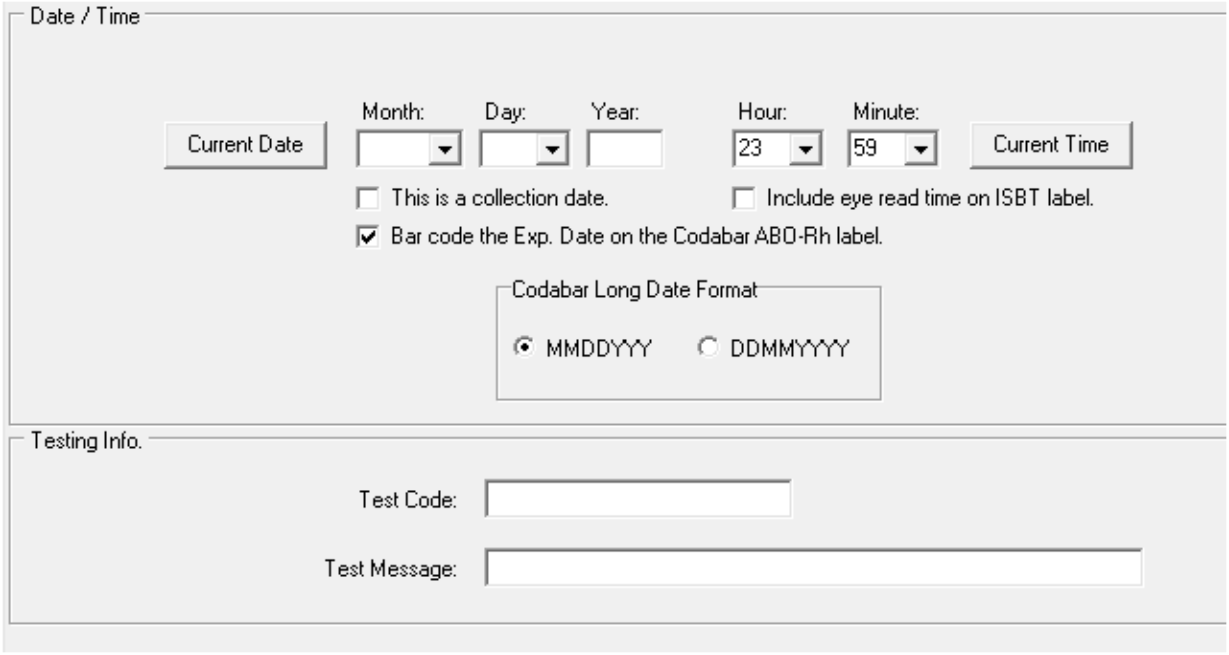
NOTE – Not all tabs are visible. Use the arrow buttons located in the upper right-hand corner of the window to scroll the tabs.

5. Edit the information to reflect your facility's specific needs.
6. Select the **ABO-RH Product Label Info** tab.



7. Edit the information to reflect your facility's specific needs.

8. Select the **Date/Time & Testing Label Info** tab.



9. Edit the information to reflect your facility's specific needs.
10. Select the **Printer Port Definitions** tab.
11. Click **Set**.
12. Select the **TCP/IP Host Connection** tab
13. Click **DISCONNECT**. The DISCONNECT button changes to CONNECT.
14. Close the window.
15. Repeat steps 1-14 for each HemaTrax-UNITY printer you will be using.

7.3.3 Configuring Label Printers in Blood Centers

1. Log into Blood Centers.
2. Press DOWN  to select Blood Centers and press **F12** (Commit).
3. If your system is set up with multiple warehouses, select your Warehouse and press **F12** (Commit).
4. On the main menu, enter `RMASTER` in the Choose Item text box and press **ENTER**.
5. In the Type text box, enter `LBLPRT` and press **TAB**.
6. In the Code text box, enter a unique code for the printer and press **TAB**.
7. In the Description text box, enter a description for the printer and press **TAB**.
8. In the Command/IP Address text box, enter the IP address of the Windows server where HemaTrax is installed and press **TAB**.

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9. In the Port # text box, enter the listening port that was configured when HemaTrax was installed. The default port is 5500.
10. In the Seq text box, enter the Printer Port (ex: 1, 2, etc.) you selected when setting up the printer in HemaTrax. (Refer to Configuring the HemaTrax Client Utility, step 3).
11. In the L text box, enter the Warehouse code.
12. Press F12 (Commit).

7.3.4 HemaTrax Print Server

To check if the HemaTrax Print Server is loaded, started, or stopped: from the Windows desktop, navigate to Start > Control Panel > Administrative Tools > Services to view all loaded services and their status.

The Services window allows you to view the current status (started or stopped) of the displayed services, including the HemaTrax Print Server. However, the Services window does not permit you to add or edit printer definitions, nor query the product database maintained by the print server. This level of access can be obtained via the "HTClient.exe" program, which is provided as part of the installation.

Print Server Start

When the HemaTrax Print Server service is started, the program examines the Windows registry to determine where its printer definitions and database tables are located. Specifically, the program looks for the following registry path:

- HKEY_LOCAL_MACHINE\SOFTWARE\Digi-Trax Corporation\HemaTrax Print Server

In the HemaTrax Print Server key, a string value named "Path" is defined. This value represents the target drive and the fully defined directory/subdirectory path where the HemaTrax Print Server program and its primary files and subdirectories are located. This registry path and string value are created as part of the installation process. Without this entry, a default Windows system directory path would be provided to the starting service and the program start up would fail.

An example of the "Path" value as it could be represented in the registry key defined above is:

- Path REG_SZ C:\Program Files\HemaTrax Print Server

Using the directory path retrieved from the registry, the print server program opens an initialization file called "HTPrintServer.ini". This file contains:

- the listening port number (which defaults to 5500)
- an alternate data file server path
- user name and password to access the alternate server path
- four flags which determine how the print server treats the donor type for autologous donations
- if the compliance authority is FDA
- which date/time barcode structure is used
- if the server tests for invalid expiration dates

The print server program next calls the HTOpen function of the “HemaTraxLIB.dll”. If the call to the HTOpen function is not successful, a warning message is posted to the event log, and the service will not start. However, if the call to the HTOpen function is successful, the release date of the HemaTrax product label table is read and saved. The server will then open a TCP/IP socket with the specified port and begin listening for messages.

Print Server Settings

During installation, a window displays allowing print server settings to be made, including the remote server path, user name, user password, etc. At times, it may be desirable to change these settings; a program ("HTPrtSrvrSettings.exe") is installed with the server for this purpose. Using this application, a user may access the “HTPrintServer.ini” file, and change any settings stored there. Once these settings are saved, the HemaTrax print server service must be stopped and restarted before the changes will take effect.

Listening Port Number

This is the port number the client side must use in conjunction with an IP address to establish a TCP/IP socket connection to the server. Select a port number that will not interfere with other “well-known” port numbers falling within the range: 1 – 1023. These numbers are assigned by the Internet Assigned Numbers Authority (IANA).

When assigning a port number for the HemaTrax Print Server, it is best to select a number in the range: 5001 – 10,000. However, the HemaTrax Print Server automatically defaults to a listening port address of “5500” if one is not manually assigned.

Autologous Donor

By selecting the **Add ‘AUTOLOGOUS DONOR’ as a valid donor type on the full face label** check box, the print server is directed to print AUTOLOGOUS DONOR on the full face label versus the generally accepted "VOLUNTEER DONOR" text. This feature is provided for previous customers that have applied for this labeling variance with the FDA.

Label Compliance Authority

By selecting the **FDA is the label compliance authority** check box, the print server is directed to include certain FDA/U.S. Consensus Standard label text adopted for labeling blood in the U.S.

U.S. Date & Time Format

By selecting the **Use the U.S. Date & Time bar code format** check box, the print server is directed to use the U.S. Consensus Standard specification for formatting the barcode and text of collection/expiration dates on the date/time label quadrant in place of ICCBBA ISBT-128 standard specifications.

Expiration dates that have expired

By selecting the **Do NOT allow printing of expiration dates on labels that have already expired** check box, the print server prohibits the printing of ISBT-128 date/time quadrant labels with expired dates (based on current server system dates).

Save

For settings to take effect, you must click **Save**. The HemaTrax Print Server must then be stopped and rebooted by going to **Start > Control Panel > Administrative Tools > Services**, selecting the HemaTrax Print Server service and selecting **Action > Restart** from the menu bar.

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7.4 Installation Cleanup

After completing all of the installation steps, verify the operation of the system.

If desired, once the install has been successfully verified you may remove the /ltdonlab2026 R1 directory (under /oradata).

If desired, once the install has been successfully verified you may remove the /oradata/bc2024 directory.

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Chapter 8 - Upgrading Blood Centers

8.1 Overview 174
8.2 Upgrading Blood Centers 175
8.2.1 Upgrade Procedure in BC2026R1 Database Server 175

8.1 Overview

Due to the fundamental changes in the underlying architecture for Oracle 19c, there is not an upgrade path. Clients currently running an older version of Blood Centers will need to perform a new installation of the following software:

- Oracle Linux 9.6
- Oracle Database 19c
- The new application server components (Fusion Middleware and Web logic server)

You will then need to migrate your existing data.



NOTE – If needed, contact your WellSky support team to inquire about assistance with this process.

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8.2 Upgrading Blood Centers

NOTE – To upgrade to Blood Centers 2026 R1, Blood Centers databases should be on the Blood Centers 2022 R1 version.

NOTE – If there are any previous releases of Blood Centers software, contact the WellSky Technical Support team.

NOTE – Take the complete COLD and export backup of the existing BC2022R1 databases with the latest service packs, and place them in the backup location.

8.2.1 Upgrade Procedure in BC2026R1 Database Server

Perform the following procedure.

1. Export the BC2022R1 database (bcprd) using the command below. You may need to follow the bc2026/PRE_UPGRADE_BC2022 procedure based on existing Blood Center version. If so, contact the WellSky Technical Support team.
   ```
   $ exp util/util file=exp_bc2022r1.dmp log=exp_bc2022r1.log
   owner=agmgr,donsys,sbcsys,srvsys,util,ltapi,ltweb,sfapi,vista statistics=none
   ```

   NOTE – You may need to add or delete a few optional database schemas in the above export command.

2. Copy the exported bc2022r1 dump file into the BC2026R1 database server's /oradata/bc2026 directory.
3. Complete steps 1-11 in Section 5.1 "Database Configuration Procedure" on page 72.
4. Import the bc2022r1 dump into the bc2026r1 database using below command.
   ```
   $ imp util/util file=exp_bc2022r1.dmp log=imp_bc2022r1._into_bc2026r1.log
   fromuser= agmgr,donsys,sbcsys,srvsys,util,ltapi,ltweb,sfapi,vista ignore=y
   grants=n buffer=1048576
   ```
5. Change the directory to /oradata/bc2026r1/BC_Suite/Database/Scripts.
6. Run the following command and set the environment variable:
   ```
   $ . oraenv
   ```
7. Enter the database name. For example, **prddb**.
8. Connect to the database as sys user, using the following command:
   ```
   $ sqlplus "/as sysdba"
   ```
9. Run the following upgrade scripts to upgrade from the 2022 R1 databases to the 2026 R1 version:

```
SQL>@UPG_UPD_BC2022R2.sql
SQL>@UPG_UPD_BC2022R3.sql
SQL>@UPG_UPD_BC2022R4.sql
SQL>@UPG_UPD_BC2024R1.sql
SQL>@UPG_UPD_BC2026R1.sql
```

NOTE – UPG_UPD_BC2022R2.sql contains 2022R2.1 and 2022R2.2 service packs as well. UPG_UPD_BC2022R4.sql contains 2022R4.1 and 2022R4.2 service packs as well. UPG_UPD_BC2026R1.sql contains 2024R1.1 and 2024R2 database updates as well.

10. Enter the following commands to run the Blood Centers 2026 Packages and Triggers scripts:
    ```
    $ cd /oradata/bc2026/BC_Suite/Database/Packages_Triggers
    $ . oraenv
    ```
11. When prompted, enter the Blood Centers Database Name. For example, **prddb**.
12. Connect to the database as sys user, using the following command:
    ```
    $ sqlplus "/as sysdba"
    ```
13. At the SQL prompt, enter the following command:
    ```
    SQL>@BC_Packages_Triggers.sql
    ```
14. Enter the following commands to run the bc2026_Grants.sql script:
    ```
    $ cd /oradata/bc2026/BC_SERVER_INSTALL/bloodc
    $ . oraenv
    ```
15. When prompted, enter the Blood Centers Database Name. For example, **prddb**.
16. Connect to the database as sys user, using the following command:
    ```
    $ sqlplus "/as sysdba"
    ```
17. At the SQL prompt, enter the following command:
    ```
    SQL>@BC2026R1_Grants.sql
    ```
18. Complete the procedures in chapters 6 and 7.

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Chapter 9 - ASP.NET Installation and Configuration

9.1 ASP.NET Installation and Configuration Procedure 178
9.2 Deploy Blood Centers On-Site Application 180
9.2.1 Gallon Donor Image File Location 181
9.2.2 Multi-Language Files 182
9.3 Deploy Blood Centers FastTrak Application 184
9.3.1 Multi-Language Files 186
9.4 Deploy Blood Centers Donor Card Application 188
9.5 HTTPS Setup for Blood Centers Applications 189
9.6 External Proxy Setup for FastTrak Application 191
9.7 Deploy Blood Centers Donor Survey Application 192
9.8 Postfix Setup Instructions 193

9.1 ASP.NET Installation and Configuration Procedure

Perform the following procedure to install and configure ASP.NET.

1. Copy the **BC_DOT_NET** Directory from the software CD to the Blood Centers On-Site DOTNET Application (Linux) Server as a root user. The BC_DOT_NET software directory contains the On-Site software and configuration directories listed below:
    - BC_SCRIPTS
    - DONOR_CARDS
    - FASTTRAK
    - ONSITE
2. Change the **BC_SCRIPTS** directory under the BC_DOT_NET directory. This sub-directory contains all of the required installation and configuration script files.
3. Run the **INSTALL_APACHE_DOTNET_Software.sh** script as a root user. The script will install Linux Apache and Linux DOTNET, as well as configure Blood Center's On-Site, FastTrak, and Donor Cards services.

```
# cd BC_DOT_NET/BC_SCRIPTS
# sh INSTALL_APACHE_DOTNET_Software.sh
```


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```
[root@cac2026 BC_SCRIPTS]# sh INSTALL_APACHE_DOTNET_Software.sh
Last metadata expiration check: 1:40:24 ago on Wed 17 Sep 2025 08:38:54 AM EDT.
Dependencies resolved.
================================================================================
 Package                 Architecture  Version                    Repository                 Size
================================================================================
Installing:
 httpd                   x86_64        2.4.62-4.0.1.el9_6.4       ol9_appstream              64 k
 mod_ssl                 x86_64        1:2.4.62-4.0.1.el9_6.4     ol9_appstream             116 k
Installing dependencies:
 apr                     x86_64        1.7.0-12.el9_3             ol9_appstream             131 k
 apr-util                x86_64        1.6.1-23.el9               ol9_appstream              99 k
 apr-util-bdb            x86_64        1.6.1-23.el9               ol9_appstream              12 k
 httpd-core              x86_64        2.4.62-4.0.1.el9_6.4       ol9_appstream             1.8 M
 httpd-filesystem        noarch        2.4.62-4.0.1.el9_6.4       ol9_appstream              11 k
 httpd-tools             x86_64        2.4.62-4.0.1.el9_6.4       ol9_appstream              93 k
 oracle-logos-httpd      noarch        90.4-1.0.1.el9             ol9_baseos_latest          37 k
Installing weak dependencies:
 apr-util-openssl        x86_64        1.6.1-23.el9               ol9_appstream              14 k
 mod_http2               x86_64        2.0.26-4.el9_6.1           ol9_appstream             171 k
 mod_lua                 x86_64        2.4.62-4.0.1.el9_6.4       ol9_appstream              50 k

Transaction Summary
================================================================================
Install  12 Packages

Total download size: 2.6 M
Installed size: 6.3 M
Downloading Packages:
(1/12): apr-1.7.0-12.el9_3.x86_64.rpm                          1.0 MB/s | 131 kB     00:00
(2/12): apr-util-1.6.1-23.el9.x86_64.rpm                       737 kB/s |  99 kB     00:00
(3/12): oracle-logos-httpd-90.4-1.0.1.el9.noarch.rpm           261 kB/s |  37 kB     00:00
(4/12): apr-util-openssl-1.6.1-23.el9.x86_64.rpm               1.3 MB/s |  14 kB     00:00
(5/12): httpd-2.4.62-4.0.1.el9_6.4.x86_64.rpm                  7.3 MB/s |  64 kB     00:00
(6/12): apr-util-bdb-1.6.1-23.el9.x86_64.rpm                   465 kB/s |  12 kB     00:00
(7/12): httpd-filesystem-2.4.62-4.0.1.el9_6.4.noarch.rpm       1.3 MB/s |  11 kB     00:00
(8/12): httpd-tools-2.4.62-4.0.1.el9_6.4.x86_64.rpm            7.0 MB/s |  93 kB     00:00
(9/12): mod_http2-2.0.26-4.el9_6.1.x86_64.rpm                   12 MB/s | 171 kB     00:00
(10/12): mod_lua-2.4.62-4.0.1.el9_6.4.x86_64.rpm               4.5 MB/s |  50 kB     00:00
(11/12): mod_ssl-2.4.62-4.0.1.el9_6.4.x86_64.rpm               9.7 MB/s | 116 kB     00:00
(12/12): httpd-core-2.4.62-4.0.1.el9_6.4.x86_64.rpm             30 MB/s | 1.0 MB     00:00
--------------------------------------------------------------------------------
Total                                                           12 MB/s | 2.6 MB     00:00
Running transaction check
Transaction check succeeded.
```

```
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Installing       : dotnet-host-9.0.9-1.0.1.el9_6.x86_64                   1/4
  Installing       : dotnet-hostfxr-8.0-8.0.20-1.0.1.el9_6.x86_64           2/4
  Installing       : dotnet-runtime-8.0-8.0.20-1.0.1.el9_6.x86_64           3/4
  Installing       : aspnetcore-runtime-8.0-8.0.20-1.0.1.el9_6.x86_64       4/4
  Running scriptlet: aspnetcore-runtime-8.0-8.0.20-1.0.1.el9_6.x86_64       4/4
  Verifying        : aspnetcore-runtime-8.0-8.0.20-1.0.1.el9_6.x86_64       1/4
  Verifying        : dotnet-host-9.0.9-1.0.1.el9_6.x86_64                   2/4
  Verifying        : dotnet-hostfxr-8.0-8.0.20-1.0.1.el9_6.x86_64           3/4
  Verifying        : dotnet-runtime-8.0-8.0.20-1.0.1.el9_6.x86_64           4/4

Installed:
  aspnetcore-runtime-8.0-8.0.20-1.0.1.el9_6.x86_64    dotnet-host-9.0.9-1.0.1.el9_6.x86_64    dotnet-hostfxr-8.0-8.0.20-1.0.1.el9_6.x86_64    dotnet-runtime-8.0-8.0.20-1.0.1.el9_6.x86_64

Complete!
Last metadata expiration check: 0:00:15 ago on Wed 17 Sep 2025 10:19:28 AM EDT.
Dependencies resolved.
================================================================================
 Package                          Architecture  Version                  Repository                 Size
================================================================================
Installing:
 dotnet-sdk-8.0                   x86_64        8.0.414-1                packages-microsoft-com-prod 149 M
Installing dependencies:
 aspnetcore-targeting-pack-8.0    x86_64        8.0.20-1.0.1.el9_6       ol9_appstream              2.2 M
 dotnet-apphost-pack-8.0          x86_64        8.0.20-1.0.1.el9_6       ol9_appstream              4.1 M
 dotnet-targeting-pack-8.0        x86_64        8.0.20-1.0.1.el9_6       ol9_appstream              3.2 M
 netstandard-targeting-pack-2.1   x86_64        9.0.110-1.0.1.el9_6      ol9_appstream              1.4 M

Transaction Summary
================================================================================
Install  5 Packages
```

```
Running transaction
  Preparing        :
  Installing       : netstandard-targeting-pack-2.1-9.0.110-1.0.1.el9_6.x86_64
  Installing       : dotnet-targeting-pack-8.0-8.0.20-1.0.1.el9_6.x86_64
  Installing       : dotnet-apphost-pack-8.0-8.0.20-1.0.1.el9_6.x86_64
  Installing       : aspnetcore-targeting-pack-8.0-8.0.20-1.0.1.el9_6.x86_64
  Installing       : dotnet-sdk-8.0-8.0.414-1.x86_64
  Running scriptlet: dotnet-sdk-8.0-8.0.414-1.x86_64
  Verifying        : dotnet-sdk-8.0-8.0.414-1.x86_64
  Verifying        : aspnetcore-targeting-pack-8.0-8.0.20-1.0.1.el9_6.x86_64
  Verifying        : dotnet-apphost-pack-8.0-8.0.20-1.0.1.el9_6.x86_64
  Verifying        : dotnet-targeting-pack-8.0-8.0.20-1.0.1.el9_6.x86_64
  Verifying        : netstandard-targeting-pack-2.1-9.0.110-1.0.1.el9_6.x86_64

Installed:
  aspnetcore-targeting-pack-8.0-8.0.20-1.0.1.el9_6.x86_64    dotnet-apphost-pack-8.0-8.0.20-1.0.1.el9_6.x86_64    dotnet-sdk-8.0-8.0.414-1.x86_64
  dotnet-targeting-pack-8.0-8.0.20-1.0.1.el9_6.x86_64        netstandard-targeting-pack-2.1-9.0.110-1.0.1.el9_6.x86_64

Complete!
```

9.2 Deploy Blood Centers On-Site Application

Perform the following procedure to deploy the Blood Centers On-Site application.

1. Change the directory to **BC_DOT_NET/ONSITE**, and copy the following directories to the **/var/www/html** directory.
   BC_Onsite.API
   BC_OnsiteAuth.API
   BCOnsiteUI

   ```
   # cd BC_DOT_NET/ONSITE
   # cp -rf * /var/www/html
   ```

2. Using a text (vi) editor, update the following files for the error-logging file path and DB connection:
   /var/www/html/BC_Onsite.API/appsettings.json
   /var/www/html/BC_OnsiteAuth.API/appsettings.json

   a. In Line 15, the default settings for the path section is listed as: "path": Logs/OnsiteApp_.json"

      The Logs directory will be in the parent directory for both API folders, like the following:
      /var/www/html/BC_Onsite.API/Logs
      /var/www/html/BC_OnsiteAuth.API/Logs

      The Logging file will generate in the Logs directory with json extension format (for example, OnsiteApp_**20210601**.json). In this example, the appended number is year, month, and date. Every day it will generate a new file with the appended date format.

   b. In Line 18, "retainedFileCountLimit": 7
      As per the retainedFileCountLimit setting in the appsettings.json configuration, only the latest 7 log files are retained on the server at any time.

   c. In Line 29, update the DB parameters in Default connection.

3. Run the **bcapstart** command for starting the DOTNET and Apache Services.
4. Run the **bcapstatus** command for checking the status of DOTNET and Apache Services.
5. Run the **bcapstop** command for stopping the DOTNET and Apache Services.



NOTE – These scripts are located under the **/usr/local/bin** directory. These scripts include On-Site, FastTrak, and Donor Card Services as well. You can comment them out if you do not need any of these specific application services.

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6. Open the BC Onsite application in Google Chrome with the following URL:
   http://bconsite.wellsky.com/BCOnsiteUI/index.html

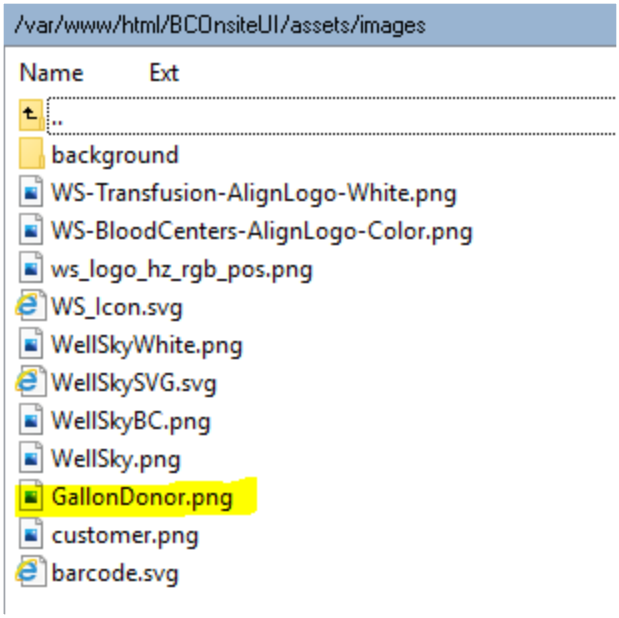
> **NOTE –** Replace bconsite.wellsky.com with the name of Blood Center's Onsite Application Server.

> **NOTE –** If you have any issues, please contact the WellSky Technical Team.

9.2.1 Gallon Donor Image File Location

The gallon donor image for the .NET application does not use the system parameter GALLON_IMG. The .NET application has the following requirements:

- The file must be named GallonDonor.
- The image type can be a .png, .jpg, or .svg image.
- The file must be placed in the following location: /var/www/html/BCOnsiteUI/assets/images.



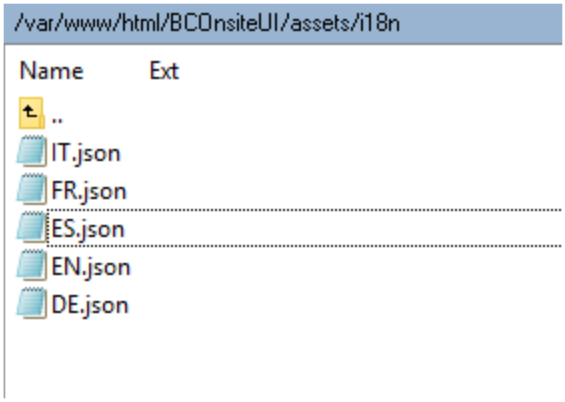
> **NOTE –** The size of the file has not been an issue, but after loading the file you should view it and make sure it fits within the location in the Donor Header.

9.2.2 Multi-Language Files

The multi-language file is used for the health history questions, donor survey question and answers, Donor Consents, FastTrak Login configuration, and the Donor Response initial displays information.

The files require the following:

- File has to be JSON.
- JSON file has to be labeled based on the "List of ISO 639-1 codes" (https://en.wikipedia.org/wiki/List_of_ISO_639-1_codes) for languages.
- The files must follow a specific format. Example files can be supplied by your Clinical Consultant, or by contacting Technical Support. Below is an example of a JSON file for Spanish.
- A file must be created for each language that you wish to support.
- The files must be placed in the following location: /var/www/html/BCOnsiteUI/assets/i18n/.



Example of JSON File for Spanish:

ES - Notepad

File Edit Format View Help


```
{
  "46-59": "46 a 59",
  "BackToQuestions": "Volver a las preguntas",
  "CANCEL": "CANCELAR",
  "CLEAR": "LIMPIAR",
  "CONTINUE": "CONTINUAR",
  "Chocolate Chip": "Chispas de chocolate",
  "DECLINE": "DECLINAR",
  "DECLINED": "DECLINADO",
  "DS0001": {
    "FIRST": "Donante por primera vez",
    "YES": "Sí",
    "text": "¿Eres un donante recurrente?"
  },
  "DS0002": {
    "NO": "No",
    "WARM": "Prefiero manta calentada",
    "YES": "Sí",
    "text": "¿Usas una manta cuando donas?"
  },
  "DS0003": {
    "1": "1-5 veces",
    "2": "6-10 veces",
    "FIRST": "Donante por primera vez",
    "text": "¿Cuántas veces dona en un año?"
  },
  "DS0004": {
    "NO": "No",
    "YES": "Sí",
    "text": "¿Eres un donante de féresis?"
  },
  "DeclineConfirmMessage": "¿Está seguro de que quiere rechazar este consentimiento?",
  "DonorSignature": "Firma del Donante",
  "FINISH": "FINALIZAR",
  "Finish": "Terminar",
  "FinishedWithDRMessage": "Gracias por sus respuestas. Haga clic en FINALIZAR y devuelva el dispositivo al Personal.",
  "GVT0001": {
    "60": "60+",
    "1721": "17-21",
    "2235": "22-35",
    "3645": "36-45",
    "4659": "46-59",
    "text": "¿Qué grupo de edad tienes?"
  },
  "HQ0010": "¿Es usted alérgico al Iodo y/o Látex?",
  "HQ0010-A": "A Desde su última donación, ha adquirido una alergia al yodo y / o ¿látex? - ABREVIADO",
  "HQ0010-D": "¿Alguna vez ha tenido una reacción alérgica o irritación de la piel al látex o productos con látex como los vendajes? Si es así, responda SÍ y el investigador lo discutirá con usted",
  "HQ0020": "¿Te sientes bien y saludable hoy?",
  "HQ0020-D": "No tenga ningún problema, como dolores de cabeza, dolor de espalda o estrés por donar.",
  "HQ0030": "En las últimas 8 semanas, ¿ha dado sangre, plasma o plaquetas?",
  "HQ0040": "¿Alguna vez ha sido rechazado como donante de sangre o le han dicho que no done sangre?",
  "HQ0071": "En las últimas 4 semanas, ¿ha tomado alguna píldora o medicamento?",
  "DCONPP_v2": {
    "content": "Entiendo que la donación de sangre para una féresis plaquetaria es un acto totalmente voluntario y no se ha ofrecido ningún incentivo o remuneración. La donación de sangre / componentes es un procedimiento médico y que mediante la donación voluntaria. Acepto los riesgos asociados con este procedimiento. Se analizará mi sangre para detectar hepatitis B / C, parásito de la malaria, VIH / SIDA y enfermedades venéreas, además de cualquier otra prueba de detección requerida para garantizar la seguridad de la sangre. Prohíbo que cualquier información proporcionada por mí o sobre mi donación se divulgue a cualquier persona o agencia gubernamental sin ningún permiso previo.",
    "title": "Consentimiento de féresis plaquetaria"
  },
  "DCONAUTO_v1": {
    "content": "Entiendo que la donación de sangre para mí (autólogo) es un acto totalmente voluntario y no se ha ofrecido ningún incentivo o remuneración. La donación de sangre / componentes es un procedimiento médico y que mediante la donación voluntaria. Acepto los riesgos asociados con este procedimiento. Se analizará mi sangre para detectar hepatitis B / C, parásito de la malaria, VIH / SIDA y enfermedades venéreas, además de cualquier otra prueba de detección requerida para garantizar la seguridad de la sangre. Prohíbo que cualquier información proporcionada por mí o sobre mi donación se divulgue a cualquier persona o agencia gubernamental sin ningún permiso previo..",
    "title": "Consentimiento del donante Donación autóloga"
  },
  "DCONTST_v3": {
    "content": "Doy mi permiso para tipear mis células sanguíneas y para pruebas de laboratorio detalladas de mi sangre, incluidas las pruebas de VIH (el virus del SIDA), hepatitis, pruebas de anticuerpos de glóbulos blancos y pruebas de investigación relacionadas con la seguridad de la sangre. Si un componente de mi sangre se almacena en estado congelado, doy mi permiso para realizar cualquier prueba de laboratorio adicional que se requiera en el futuro para aclarar la elegibilidad para donaciones pasadas o futuras. Prueba de comillas dobles \\'derecha e izquierda\", comillas simples `derecha e izquierda ', y & y barra diagonal /<br>Colon :<br>Semi colon;<br>Corchetes [derecha e izquierda]<br>parèntesis (derecha e izquierda)<br>Punto de exclamación !<br>A la señal @<br>Hash #<br>Dólar $<br>Porcentaje%<br>Carot ^<br>Asteriscos *<br>guión menos<br>Más +<br>Igual SALIDA=<br>Guion bajo _<br>Mayor que y menor que <><br>Barra diagonal \\.<br>Tilde ~<br>Marca de acento grave<br>llaves (derecha e izquierda)<br>barra vertical |",
    "title": "Consentimiento del donante"
  },
  "DCONGEN_v1": {
    "content": "Tengo entendido que la donación de sangre es un acto totalmente voluntario y no se ha ofrecido ningún incentivo o remuneración. La donación de sangre / componentes es un procedimiento médico y que mediante la donación voluntaria. Acepto los riesgos asociados con este procedimiento. Se analizará mi sangre para detectar hepatitis B / C, parásito de la malaria, VIH / SIDA y enfermedades venéreas, además de cualquier otra prueba de detección requerida para garantizar la seguridad de la sangre. Prohíbo que cualquier información proporcionada por mí o sobre mi donación se divulgue a cualquier persona o agencia gubernamental sin ningún permiso previo.",
    "title": "Consentimiento general para donación"
  }
},
"donorHint": "Pida asistencia al encuestador si no comprende el proceso.",
"donorSurvey": {
  "back": "ATRÁS",
  "complete": "Completar",
  "describe": "Por favor describa",
  "deviceMsg": "Por favor devuelve el dispositivo a un empleado",
  "greetings": "Hola {{name}}, ¡Gracias por tu donación!",
  "next": "SIGUIENTE",
  "okay": "ACEPTAR",
  "thankYouMsg": "Gracias por tu retroalimentación!"
},
"greeting": "Estimado donante",
"letterContent": "Nuestro centro de donación de sangre ha introducido la funcionalidad de preguntas contestadas por el propio donante, para mantener la privacidad del donante.",
"of": "de",
"preferedLanguageTitle": "Idioma preferido",
"verifyCommunicationQuestion": "Es usted capaz de leer en Español!",
"incompleteQuestionnaireMessage": "No has revisado todas las preguntas. ¿Quieres finalizar el proceso?",
"completeQuestionnaireMessage": "Has revisado todas las preguntas. ¿Desea finalizar el proceso e imprimir el código de barras?",
"logoutWarningMessage": "Sus respuestas no se guardarán si no imprime o envía por correo electrónico el código QR que se muestra al final del cuestionario. ¿Desea continuar?",
"WEB_THANQU": "Gracias por sus respuestas. Imprima el código de barras QR y llévelo a su Centro de sangre dentro de las 24 horas.",
"SENDEMAIL": "Enviar correo electrónico",
"EXIT": "SALIDA"
}
```

9.3 Deploy Blood Centers FastTrak Application

NOTE – Please contact the WellSky Technical Team before working on this section. This section is both optional and customized.

Perform the following procedure.

1. Change the directory to BC_DOT_NET/FASTTRAK, and copy the following directories to the /var/www/html directory:
   fasttrak
   fasttrakapi

   ```
   # cd BC_DOT_NET/FASTTRAK
   # cp -rf * /var/www/html
   ```

2. Copy the Kestrel_fasttrakapi.service from the BC_DOT_NET/BC_SCRIPTS directory to the /etc/systemd/system directory.

3. Using vi editor,

   - Update the /var/www/html/fasttrak/config.json file for the FastTrak server's hostname in the bcFastTrakAPIUrl section.
   - Update the /var/www/html/fasttrakapi/appsettings.json file for the Host section with the FastTrak server's hostname.
   - Update the /var/www/html/fasttrakapi/appsettings.json file for the AllowedOrigins section with the FastTrak server's hostname, and any external proxy server's hostname(s).
   - Update the /var/www/html/fasttrakapi/settings/appsettings.DatabaseConnection.json file for DB Parameters in the ConnectionString section.
   - Update the /var/www/html/fasttrakapi/settings/appsettings.EmailConfiguration.json file for smtp server-related variables for the following parameters:
     - SmtpServer
     - SmtpPort
     - SmtpUsername
     - SmtpPassword
     - Address
   - If you want to bypass the email server authentication, please update the AnonymousConnection section with Y in the /var/www/html/fasttrakapi/settings/appsettings.EmailConfiguration.json file.

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4. Update the following ProxyPass parameters for the FastTrak API internal proxy configuration in the **/etc/httpd/conf/httpd.conf** file (http) in the **/etc/httpd/conf.d./ssl.conf** (https) mode.

   ##### Below bcfast proxy is for FasTrak application; this setup uses port 5004 internally

   ```
   ProxyPass /bcfast/ http://bcfastrak.wellsky.com:5004
   ProxyPassReverse /bcfast/ http://bcfastrak.wellsky.com:5004
   ```

5. Update the FastTrak API service start/stop/status commands in the bcapstart, bcapstop, and bcapstatus scripts in the **/usr/local/bin** directory.

Start:	`service Kestrel_fasttrakapi.service start`
Stop:	`service Kestrel_fasttrakapi.service stop`
Status:	`service Kestrel_fasttrakapi.service status`

6. Start the application using the above start command.
7. Open the BC FastTrak application in Google Chrome with one of the following URLs:
   http://bcfastrak.wellsky.com/fasttrak
   https://bcfastrak.wellsky.com/fasttrak

   NOTE – Replace **bcfastrak.wellsky.com** with the name of Blood Center's FastTrak Application Server.

   NOTE – If you have any issues, please contact the WellSky Technical Team.

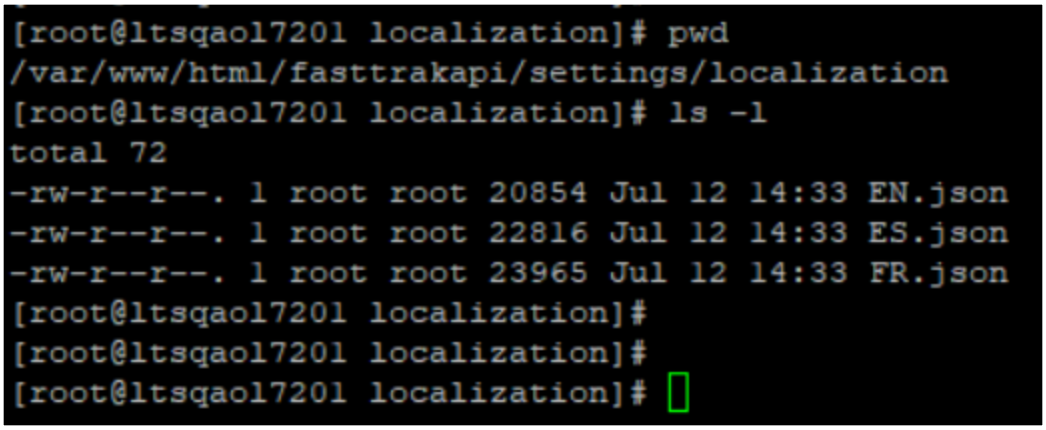
8. Update the customer logo file in the **/var/www/html/fasttrak/assets/images/logos** directory.

9.3.1 Multi-Language Files

The multi-language file is used for the health history questions, donor survey questions and answers, Donor Consents, FastTrak Login configuration, and the Donor Response initial displays information.

The files require the following:

- File has to be JSON.
- JSON file has to be labeled based on the "List of ISO 639-1 codes" (https://en.wikipedia.org/wiki/List_of_ISO_639-1_codes) for languages.
  - ES = Spanish
  - FR = French
  - IT = Italian
  - DE = German
- The JSON files MUST BE defined in upper case.
- The files must follow a specific format. Example files can be supplied by your Clinical Consultant, or by contacting Technical Support. Below is an example of a JSON file for Spanish.
- A file must be created for each language that you wish to support.
  - EN.json is an empty JSON file. Define the file, but the information does not need to be present in the file.
- Place the file for the FastTrak application in the following location: /var/www/html/fasttrakapi/settings/localization.



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Example of JSON File for Spanish:

ES - Notepad

File Edit Format View Help


```
{
  "46-59": "46 a 59",
  "BackToQuestions": "Volver a las preguntas",
  "CANCEL": "CANCELAR",
  "CLEAR": "LIMPIAR",
  "CONTINUE": "CONTINUAR",
  "Chocolate Chip": "Chispas de chocolate",
  "DECLINE": "DECLINAR",
  "DECLINED": "DECLINADO",
  "DS0001": {
    "FIRST": "Donante por primera vez",
    "YES": "Sí",
    "text": "¿Eres un donante recurrente?"
  },
  "DS0002": {
    "NO": "No",
    "WARM": "Prefiero manta calentada",
    "YES": "Sí",
    "text": "¿Usas una manta cuando donas?"
  },
  "DS0003": {
    "1": "1-5 veces",
    "2": "6-10 veces",
    "FIRST": "Donante por primera vez",
    "text": "¿Cuántas veces dona en un año?"
  },
  "DS0004": {
    "NO": "No",
    "YES": "Sí",
    "text": "¿Eres un donante de féresis?"
  },
  "DeclineConfirmMessage": "¿Está seguro de que quiere rechazar este consentimiento?",
  "DonorSignature": "Firma del Donante",
  "FINISH": "FINALIZAR",
  "Finish": "Terminar",
  "FinishedWithDRMessage": "Gracias por sus respuestas. Haga clic en FINALIZAR y devuelva el dispositivo al Personal.",
  "GVT0001": {
    "60": "60+",
    "1721": "17-21",
    "2235": "22-35",
    "3645": "36-45",
    "4659": "46-59",
    "text": "¿Qué grupo de edad tienes?"
  },
  "HQ0010": "¿Es usted alérgico al Iodo y/o Látex?",
  "HQ0010-A": "A Desde su última donación, ha adquirido una alergia al yodo y / o ¿látex? - ABREVIADO",
  "HQ0010-D": "¿Alguna vez ha tenido una reacción alérgica o irritación de la piel al látex o productos con látex como los vendajes? Si es así, responda SÍ y el investigador lo discutirá con usted",
  "HQ0020": "¿Te sientes bien y saludable hoy?",
  "HQ0020-D": "No tenga ningún problema, como dolores de cabeza, dolor de espalda o estrés por donar.",
  "HQ0030": "En las últimas 8 semanas, ¿ha dado sangre, plasma o plaquetas?",
  "HQ0040": "¿Alguna vez ha sido rechazado como donante de sangre o le han dicho que no done sangre?",
  "HQ0071": "En las últimas 4 semanas, ¿ha tomado alguna píldora o medicamento?",
  "DCONPP_v2": {
    "content": "Entiendo que la donación de sangre para una féresis plaquetaria es un acto totalmente voluntario y no se ha ofrecido ningún incentivo o remuneración. La donación de sangre / componentes es un procedimiento médico y que mediante la donación voluntaria. Acepto los riesgos asociados con este procedimiento. Se analizará mi sangre para detectar hepatitis B / C, parásito de la malaria, VIH / SIDA y enfermedades venéreas, además de cualquier otra prueba de detección requerida para garantizar la seguridad de la sangre. Prohíbo que cualquier información proporcionada por mí o sobre mi donación se divulgue a cualquier persona o agencia gubernamental sin ningún permiso previo.",
    "title": "Consentimiento de féresis plaquetaria"
  },
  "DCONAUTO_v1": {
    "content": "Entiendo que la donación de sangre para mí (autólogo) es un acto totalmente voluntario y no se ha ofrecido ningún incentivo o remuneración. La donación de sangre / componentes es un procedimiento médico y que mediante la donación voluntaria. Acepto los riesgos asociados con este procedimiento. Se analizará mi sangre para detectar hepatitis B / C, parásito de la malaria, VIH / SIDA y enfermedades venéreas, además de cualquier otra prueba de detección requerida para garantizar la seguridad de la sangre. Prohíbo que cualquier información proporcionada por mí o sobre mi donación se divulgue a cualquier persona o agencia gubernamental sin ningún permiso previo..",
    "title": "Consentimiento del donante Donación autóloga"
  },
  "DCONTST_v3": {
    "content": "Doy mi permiso para tipear mis células sanguíneas y para pruebas de laboratorio detalladas de mi sangre, incluidas las pruebas de VIH (el virus del SIDA), hepatitis, pruebas de anticuerpos de glóbulos blancos y pruebas de investigación relacionadas con la seguridad de la sangre. Si un componente de mi sangre se almacena en estado congelado, doy mi permiso para realizar cualquier prueba de laboratorio adicional que se requiera en el futuro para aclarar la elegibilidad para donaciones pasadas o futuras. Prueba de comillas dobles \\'derecha e izquierda\", comillas simples `derecha e izquierda`, y & y barra diagonal /<br>Colon :<br>Semi colon;<br>Corchetes [derecha e izquierda]<br>parèntesis (derecha e izquierda)<br>Punto de exclamación !<br>A la señal @<br>Hash #<br>Dólar $<br>Porcentaje%<br>Carot ^<br>Asteriscos *<br>guión menos<br>Más +<br>Igual SALIDA=<br>Guion bajo _<br>Mayor que y menor que <><br>Barra diagonal \\.<br>Tilde ~<br>Marca de acento grave<br>llaves (derecha e izquierda)<br>barra vertical |",
    "title": "Consentimiento del donante"
  },
  "DCONGEN_v1": {
    "content": "Tengo entendido que la donación de sangre es un acto totalmente voluntario y no se ha ofrecido ningún incentivo o remuneración. La donación de sangre / componentes es un procedimiento médico y que mediante la donación voluntaria. Acepto los riesgos asociados con este procedimiento. Se analizará mi sangre para detectar hepatitis B / C, parásito de la malaria, VIH / SIDA y enfermedades venéreas, además de cualquier otra prueba de detección requerida para garantizar la seguridad de la sangre. Prohíbo que cualquier información proporcionada por mí o sobre mi donación se divulgue a cualquier persona o agencia gubernamental sin ningún permiso previo.",
    "title": "Consentimiento general para donación"
  }
},
"donorHint": "Pida asistencia al encuestador si no comprende el proceso.",
"donorSurvey": {
  "back": "ATRÁS",
  "complete": "Completar",
  "describe": "Por favor describa",
  "deviceMsg": "Por favor devuelve el dispositivo a un empleado",
  "greetings": "Hola {{name}}, ¡Gracias por tu donación!",
  "next": "SIGUIENTE",
  "okay": "ACEPTAR",
  "thankYouMsg": "Gracias por tu retroalimentación!"
},
"greeting": "Estimado donante",
"letterContent": "Nuestro centro de donación de sangre ha introducido la funcionalidad de preguntas contestadas por el propio donante, para mantener la privacidad del donante.",
"of": "de",
"preferedLanguageTitle": "Idioma preferido",
"verifyCommunicationQuestion": "Es usted capaz de leer en Español!",
"incompleteQuestionnaireMessage": "No has revisado todas las preguntas. ¿Quieres finalizar el proceso?",
"completeQuestionnaireMessage": "Has revisado todas las preguntas. ¿Desea finalizar el proceso e imprimir el código de barras?",
"logoutWarningMessage": "Sus respuestas no se guardarán si no imprime o envía por correo electrónico el código QR que se muestra al final del cuestionario. ¿Desea continuar?",
"WEB_THANQU": "Gracias por sus respuestas. Imprima el código de barras QR y llévelo a su Centro de sangre dentro de las 24 horas.",
"SENDEMAIL": "Enviar correo electrónico",
"EXIT": "SALIDA"
}
```

9.4 Deploy Blood Centers Donor Card Application

NOTE – Please contact the WellSky Technical Team before working on this section. This section is both optional and customized.

Perform the following procedure.

1. Change the directory to BC_DOT_NET/DONOR_CARDS, and copy the following directories to the /var/www/html directory:
   BC_DonorCards.API
   BCDonorCardsUI

   ```
   # cd BC_DOT_NET/DONOR_CARDS
   # cp -rf * /var/www/html
   ```

2. Using vi editor, update the following file for DB Parameters in the DefaultConnection section, line 9:
   /var/www/html/BC_DonorCards.API/appsettings.json
3. The main Blood Centers On-Site application connects internally to the Blood Centers Donor Card application.

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9.5 HTTPS Setup for Blood Centers Applications

NOTE – This section is optional, but common for all of the applications deployed above.

Perform the following procedure.

1. Download and copy the server's wild card Personal Information Exchange file (**certificate.pfx**) from the security vendor with password information into the **/var/www/html/SSL_Cert** directory on the ASP.NET Linux Server.

   Example:

   ```
   [root@ltsqaol7205 SSL_Cert]# pwd
   /var/www/html/SSL_Cert
   [root@ltsqaol7205 SSL_Cert]# ll
   total 8
   -rw-r--r--. 1 root root 5897 Sep 25 14:27 wildcard.wellsky.pfx
   [root@ltsqaol7205 SSL_Cert]#
   ```

2. Extract the certificate file and Private Key from the above file using the **openssl** command:

   ```
   # cd /var/www/html/SSL_Cert
   # openssl pkcs12 -in wildcard.wellsky.pfx -out ltsqaol7205.txt -nodes
   ```

3. Enter the password at the prompt.

   Example:

   ```
   [root@ltsqaol7205 SSL_Cert]#
   [root@ltsqaol7205 SSL_Cert]# openssl pkcs12 -in /var/www/html/SSL_Cert/wildcard.wellsky.pfx -out ltsqaol7205.txt -nodes
   Enter Import Password:
   MAC verified OK
   [root@ltsqaol7205 SSL_Cert]# ll
   total 16
   -rw-r--r--. 1 root root 8000 Oct  3 12:21 ltsqaol7205.txt
   -rw-r--r--. 1 root root 5897 Sep 25 14:27 wildcard.wellsky.pfx
   [root@ltsqaol7205 SSL_Cert]#
   ```

4. A text file is generated. Open the text file, and copy the entire contents, from the ""-----BEGIN PRIVATE KEY-----" line to the "-----END PRIVATE KEY-----" line, to a new file called **<Server_Name>.key**.

5. Copy the **<Server_Name>.key** file into the **/etc/pki/tls/private** directory.

   ```
   [root@ltsqaol7205 SSL_Cert]#
   [root@ltsqaol7205 SSL_Cert]# ll /etc/pki/tls/private/ltsqaol7205.key
   -rw-r--r--. 1 root root 1704 Sep 30 14:58 /etc/pki/tls/private/ltsqaol7205.key
   [root@ltsqaol7205 SSL_Cert]#
   ```

6. Open the above text file, and copy the entire contents, from the ""-----BEGIN CERTIFICATE-----" line to the "-----END CERTIFICATE-----" line. There will be multiple chain certificate contents. Copy one by one into a new certificate file called **<Server_Name>.crt**.

7. Copy the **<Server_Name>.crt** file into the **/etc/pki/tls/certs** directory.

```
[root@ltsqaol7205 SSL_Cert]#
[root@ltsqaol7205 SSL_Cert]# ll /etc/pki/tls/certs/ltsqaol7205.crt
-rw-r--r--. 1 root root 5428 Sep 30 14:58 /etc/pki/tls/certs/ltsqaol7205.crt
[root@ltsqaol7205 SSL_Cert]#
[root@ltsqaol7205 SSL_Cert]#
```

8. Make a backup of the existing ssl.conf file in **/etc/httpd/conf.d** as **ssl.conf.ORG**.
9. Copy the **ssl.conf** file from the **/var/www/html/SSL_Cert** directory to the **/etc/httpd/conf.d** directory.
10. Edit the **/etc/httpd/conf.d/ssl.conf** file to update the following parameter values with server values:
    - ServerName Linux Server Name
    - SSLCertificateFile Server_Name.crt
    - SSLCertificateKeyFile Server_name.key
11. Restart the application services by running the following commands:
    - `bcapstop`
    - `bcapstart`
    - `bcapstatus`
12. Open the secured http applications using the following URLs, and make sure everything works without any issues.
    On-Site: https://bconsite.wellsky.com/BCOnsiteUI/index.html
    FastTrak: https://bconsite.wellsky.com/BCFastTrakUI/index.html

> **NOTE –** Replace **bconsite.wellsky.com** with the Application Server name for Blood Centers On-Site application.

> **NOTE –** If you have any issues, please contact the WellSky Technical Team.

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9.6 External Proxy Setup for FastTrak Application

NOTE – This section is optional. However, a separate Linux server is needed if you want to set up FastTrak application services as public facing.

Perform the following procedure.

1. Install the Apache modules on the Linux server using the following command:

   ```
   # yum -y install httpd mod_ssl
   ```

2. Refer to steps 1-9 in the HTTPS Setup for Blood Centers Applications section above.
3. Copy **sslproxy.conf** from the software CD's **BC_DOT_NET/BC_SCRIPTS** directory to the **/etc/httpd/conf.d** directory as **ssl.conf**.
4. Verify the following parameters, and update according to the required values.

   - ServerName — Linux Public Facing Server Name
   - SSLCertificateFile — Server_Name.crt
   - SSLCertificateKeyFile — Server_name.key

5. For the ProxyPass and ProxyPassReverse lines, replace the FastTrak Application Server name in place of **bcft2019.wellsky.com**.
6. Make sure there is network access for an External Proxy server to the FastTrak Application server for the HTTPS (443) port.
7. Launch and test the FastTrak application using the following URL:
   https://ftrakproxy.wellsky.com/fasttrak

   NOTE – Replace the **ftrakproxy.wellsky.com** with the External Apache Proxy Server name.

   NOTE – If you have any issues, please contact the WellSky Technical Team.

9.7 Deploy Blood Centers Donor Survey Application

NOTE – Please contact the WellSky Technical Team before working on this section. This section is both optional and customized.

Perform the following procedure.

1. Change the directory to BC_DOT_NET/DONOR_SURVEY, and copy the following directories to the /var/www/html directory:

   BC_DonorSurvey.API
   BCDonorSurveyUI

   ```
   # cd BC_DOT_NET/DONOR_SURVEY
   # cp -rf * /var/www/html
   ```

2. Using a text (vi) editor, update the following file for DB Parameters in the DefaultConnection section, line 9: /var/www/html/BC_DonorCards.API/appsettings.json

   The main Blood Centers On-Site application connects internally to the Blood Centers Donor Survey application.

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9.8 Postfix Setup Instructions

1. Install the Postfix packages, and start the services by running the following commands as a **root** user:
```
# dnf install -y postfix mailx cyrus-sasl cyrus-sasl-plain
# systemctl enable postfix
# systemctl start postfix
# systemctl status postfix
```

> **NOTE –** All the Postfix configuration files will be in the **/etc/postfix** directory.

2. Gather the mail server connection details, and adjust the values according to your SMTP server.
   The following is an example of configuring the GMAIL server.

MAIL_SMTP	"smtp.gmail.com"
MAIL_PORT	587
MAIL_USERNAME	"service-user@gmail.com"
MAIL_PASSWORD	"ServiceUserPassword"

> **NOTE –** The GMAIL server is not recommended; it is used here only as an example.

3. Update the **/etc/postfix/sasl_passwd** file with email setup using the following command:
```
cat >> /etc/postfix/sasl_passwd <<EOF
[smtp.gmail.com]:587 service-user@gmail.com: ServiceUserPassword
EOF
```

4. Run the following command to create a hash file for the Postfix password file:
```
postmap /etc/postfix/sasl_passwd
```

5. Open the **/etc/postfix/main.cf** file and search for the **smtp_tls_CAfile** parameter, and comment it.

6. Add the following content at the end of the **/etc/postfix/main.cf** file:
```
# Mail Relay
myorigin = gmail.com
relayhost = [smtp.gmail.com]:587
smtp_use_tls = yes
smtp_sasl_auth_enable = yes
smtp_sasl_password_maps = hash:/etc/postfix/sasl_passwd
smtp_tls_CAfile = /etc/postfix/cacert.pem
smtp_sasl_security_options = noanonymous
smtp_sasl_tls_security_options = noanonymous
# End Mail Relay
```

7. Use the following commands to restart the Postfix services:
```
systemctl restart postfix
systemctl status postfix
```

8. Log in to the 2022R4 database as SYS user, and run the following SQL statements to set up the Database UTIL_MAIL package on the database:

```
$ sqlplus "/as sysdba"
SQL> @/apps/product/19.0.0/dbhome_1/rdbms/admin/utlmail.sql
SQL> @/apps/product/19.0.0/dbhome_1/rdbms/admin/prvtmail.plb
SQL> grant execute on UTL_MAIL to public;
SQL> alter system set smtp_out_server='localhost:25' scope=both sid='*';
```

9. Provide CONNECT privilege to the UTIL and DBADMN users. Use the same script and replace it with the username for the DBADMN user.

```
DECLARE
l_principal VARCHAR2(20) := 'UTIL';
BEGIN
  DBMS_NETWORK_ACL_ADMIN.append_host_ace (
    host       => 'localhost',
    lower_port => 25,
    upper_port => 25,
    ace        => xs$ace_type(privilege_list => xs$name_list('connect'),
                              principal_name => l_principal,
                              principal_type => xs_acl.ptype_db));
EXCEPTION WHEN OTHERS THEN
  DBMS_OUTPUT.PUT_LINE('Error granting ACL :' || SQLERRM);
END;
```


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Appendix A - Setup Guides

A.1 Blood Centers 2026 R1 SSL Setup Guide 196
A.1.1 Part I: Create a Java Keystore which contains an SSL Certificate 196
A.1.2 Part II: Configure the WebLogic Server (WLS_Forms) for SSL 200
A.1.3 Part III: Configure the Admin Server Trust Keystore 212
A.2 LT_APPLICATION_CONFIG Setup Guide 214
A.3 CAC Setup and Configuration 216
A.3.1 Setup and Configuration 216
A.4 Blood Centers Application Links 219
A.4.1 .NET-Based Blood Centers Applications 219
A.5 AGMGR Setup Instructions 220
A.5.1 Set Up the Oracle Gateway Software 220
A.5.2 Install and Configure the AGMGR Schema 228
A.5.3 Setting Up the Zebra Label Printer 230
A.6 Oracle Wallet Setup and Configuration 237
A.6.1 Oracle Wallet Setup in Database Server Environment 237
A.6.2 Oracle Wallet Setup in BC Forms and Reports Application Server 242
A.6.3 Oracle Wallet Setup in BC Onsite Application Server 243
A.6.4 Oracle Wallet files modification and updates 243
A.7 OKTA Setup and Configuration 245

A.1 Blood Centers 2026 R1 SSL Setup Guide

NOTE – If you do not need an SSL certificate for your installation of Blood Centers, skip this section.

NOTE – If this procedure has already been performed on the server, it need not be performed a second time. Proceed to the next section.

NOTE – If you are installing Blood Centers on two servers (one for the database and one for the application), this procedure is completed on the application server.

A.1.1 Part I: Create a Java Keystore which contains an SSL Certificate

1. Log into the application server as Oracle.
2. Right-click on the desktop and select Open Terminal.
3. Set the environment for Java Keytool by entering the following command:
   $ `. /apps/Middleware/user_projects/domains/LT_Domain/bin/setDomainEnv.sh`

   Example:

   ```
   [oracle@lifetrak2012 ~]$
   [oracle@lifetrak2012 ~]$ . /apps/Middleware/user_projects/domains/LT_Domain/bin/setDomainEnv.sh
   [oracle@lifetrak2012 LT_Domain]$
   [oracle@lifetrak2012 LT_Domain]$
   ```

4. Enter the following command to create a directory for storing the keys:
   $ `mkdir /apps/Middleware/keystores`

   Example:

   ```
   [oracle@lifetrak2012 LT_Domain]$ mkdir /apps/Middleware/keystores
   ```

5. Change to the newly created directory by entering the following command:
   $ `cd /apps/Middleware/keystores`

   Example:

   ```
   [oracle@lifetrak2012 LT_Domain]$ cd /apps/Middleware/keystores
   ```


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6. Generate the Java key by entering the following command:
   ```
   $ keytool -genkey -alias lifetrak -keyalg RSA -keysize 2048 -dname "CN=lifetrak.mediware.com,OU=Support,O=Mediware,L=oak brook,ST=illinois,C=us" -keypass welcome -keystore keystore.jks -storepass welcome
   ```

   NOTE – Make the following changes to the command to reflect your install:
   - Replace lifetrak.mediware.com with the hostname.domainname of your application server.
   - Replace the *OU*, *O*, *L*, *ST*, and *C* variables for your location.
   - Change the *alias* to reflect your alias name (shown as lifetrak in the example command above).
   - If needed, change the *keypass* and *storepass* variables (shown as welcome in the command above).

   NOTE – Even though the command above displays on multiple lines, it should be entered as one line.

7. A file labeled keystore.jks was created in the /apps/Middleware/keystores directory. Make a backup of the file for future purposes.

8. Enter the following command to create a Certificate of Signing Request (CSR):
   ```
   $ keytool –certreq –v -alias lifetrak –file lifetrak.csr –keypass welcome –storepass welcome –keystore keystore.jks
   ```

   NOTE – Even though the command above displays on multiple lines, it should be entered as one line.

   NOTE – Make sure you use the same *alias*, *keypass*, and *storepass* values as you entered in step 6.

   NOTE – Running this command creates a file named lifetrak.csr.

Example:

```
-----BEGIN NEW CERTIFICATE REQUEST-----
MIIBtzCCASACAQAwdzELMAkGA1UEBhMCR0IxEjAQBgNVBAgTCUJlcmtzaGlyZTEQMA
4GA1UEBxMHUmVhZGluZzEPMA0GA1UEChMGT3JhY2xlMRAwDgYDVQQLEwdTdXBwb3J0
MR8wHQYDVQQDExZtYXJzaGFsbC5lay5vcmFjbGUuY29tMIGfMA0GCSqGSIb3DQEBA
QUAA4GNADCBiQKBgQCEopgMZp1lI6jWXxb1rMlkWIcll8bhiV/0UTcsdKzeaSHxbO
SLO3Ed9kxNWAZgXaR9f5FBlwkaRJ+IR163e64v3SplHenxHfVRaHYWPZx4KlJz/6p
YdlfAlF0PdQm1DNoFtKmCHVk/cRuvGRpsp3817K2mYlyQ+GxH3811S7g3owIDAQAB
oAAwDQYJKoZIhvcNAQEFBQADgYEAD/sG1+rSI76OjihHg3WezT+VIbSRJxyly9nbx
4uwXbDHh8DGgQLAXV51C9icaMrm+dM0eygVDDMESXFxvJiYipS/pphgYt1xDBgnEH
GcNiX3BnTaLNtzYlc5eAMsmbDlpk/qOxvQiH3bKN+UKYQlBXJZWPL6FusXu2LMTrk
zsY=
-----END NEW CERTIFICATE REQUEST-----
```

9. Send the **/apps/Middleware/keystores/lifetrak.csr** file to a Certificate Authority (CA) of your choice. You will receive the certificate(s) from the selected Certificate Authority. Although it depends on which CA you use, normally you will receive Intermediate CA & Root CA certificates. The Certificate Authority will give specific instructions on their use.
10. You will need to merge the two certificates into a single file named trustCA.cer by opening the files with your desired editor. The Intermediate CA should be first in the file, followed by the Root CA.

NOTE – Make sure there is not space or extra characters in each certificate, and/or between them when they are put together.

NOTE – You will need to merge them if the CA provided two or more Root certificates in the same order.

11. Save this merged file to **/apps/Middleware/keystores directory/trustCA.cer**.
12. Return to the command window and set the environment for the Java Keytool by entering the following command.
    `$ . /apps/Middleware/user_projects/domains/LT_Domain/bin/setDomainEnv.sh`

NOTE – Even though the command above displays on multiple lines, it should be entered as one line.

Example:

```
[oracle@lifetrak2012 ~]$
[oracle@lifetrak2012 ~]$ . /apps/Middleware/user_projects/domains/LT_Domain/bin/setDomainEnv.sh
[oracle@lifetrak2012 LT_Domain]$
[oracle@lifetrak2012 LT_Domain]$
```

13. Change to the Keystore directory by entering the following command:
    `$ cd /apps/Middleware/keystores`

Example:

```
[oracle@lifetrak2012 LT_Domain]$ cd /apps/Middleware/keystores
```


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14. Import the trustCA.cer file into the Java keystore file, keystore.jks, by entering the following command:
`$ keytool -import -alias lifetrak-cert-imp -file trustCA.cer -keystore keystore.jks -storepass welcome`

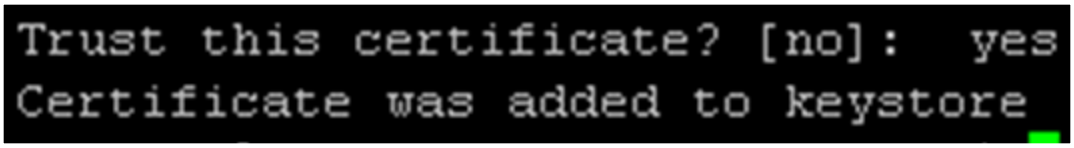
> **NOTE –** Even though the command above displays on multiple lines, it should be entered as one line.

> **NOTE –** Make sure the *alias* value is **different** than the alias used in steps 6 and 8.

> **NOTE –** Verify the *keypass* and *storepass* values are the **same** as what was entered in steps 6 and 8.

15. It will prompt for "Trust this certificate?". Type `yes` and press **Enter**.

Example:



16. Import the server certificate if any (optional), by entering following command:
`$ keytool -import -v -alias server_cert -file server.cer -keystore keystore.jks -keypass welcome -storepass welcome`

In this command, `server.cert` is the server certificate from Certificate Authority.

17. Make another backup copy of the **/apps/Middleware/keystores/keystore.jks** file, making sure to give it a **different** name from that used in step 7.

A.1.2 Part II: Configure the WebLogic Server (WLS_Forms) for SSL

1. Open a browser and enter `http://<server IP address>:7001/console` to open the Oracle WebLogic Server Administration Console (replacing <server IP address> with the IP address of your Blood Centers server).
2. The Oracle WebLogic Server Administration Console Login window opens.

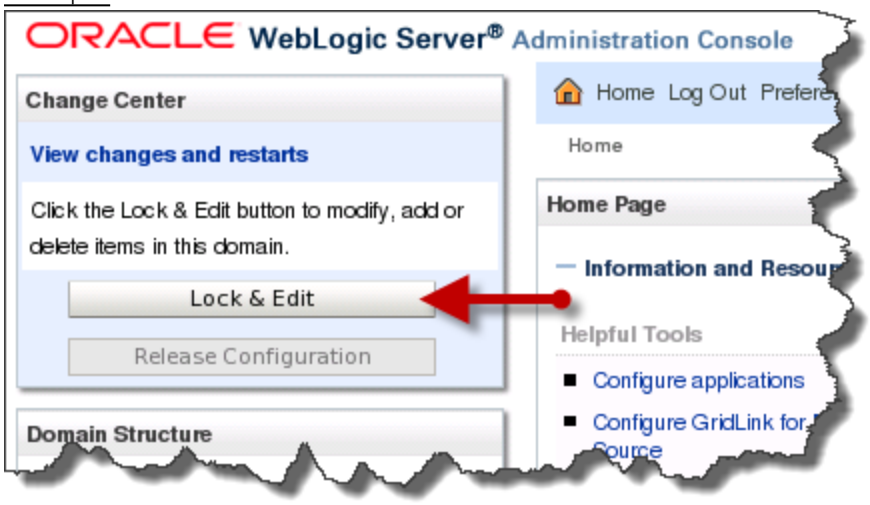
   Example:



   a. In the *Username* box, enter `weblogic`.
   b. In the *Password* box, enter `WebLogicAdmin1`.
   c. Click **Login**.

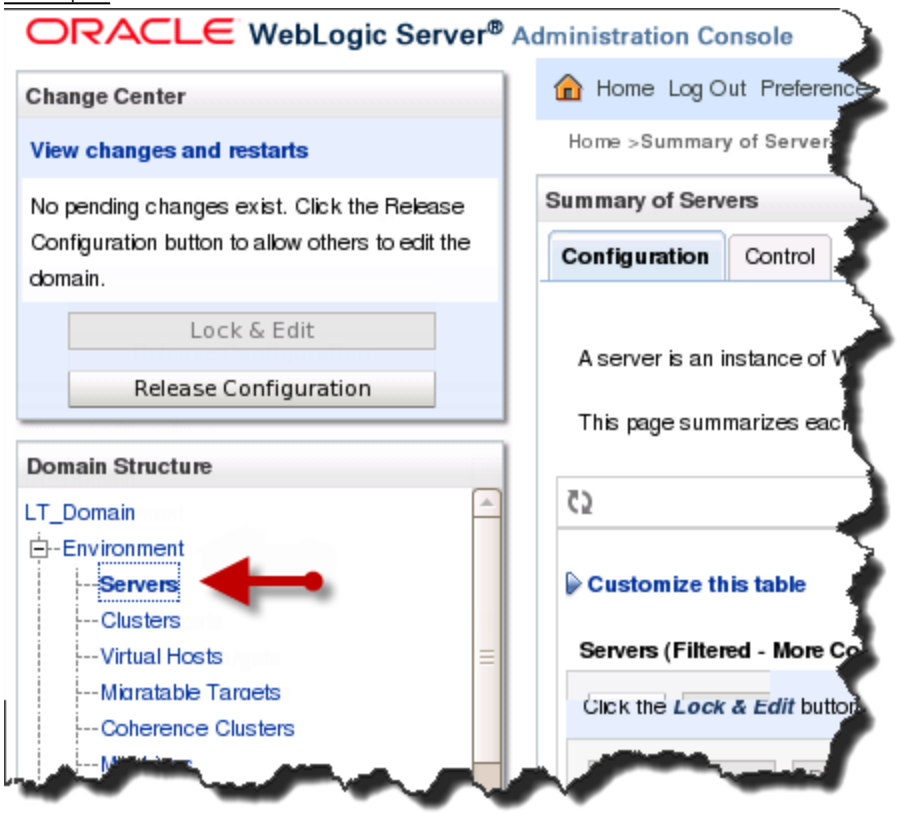
3. In the pane on the left under *Change Center*, click **Lock & Edit**.

   Example:



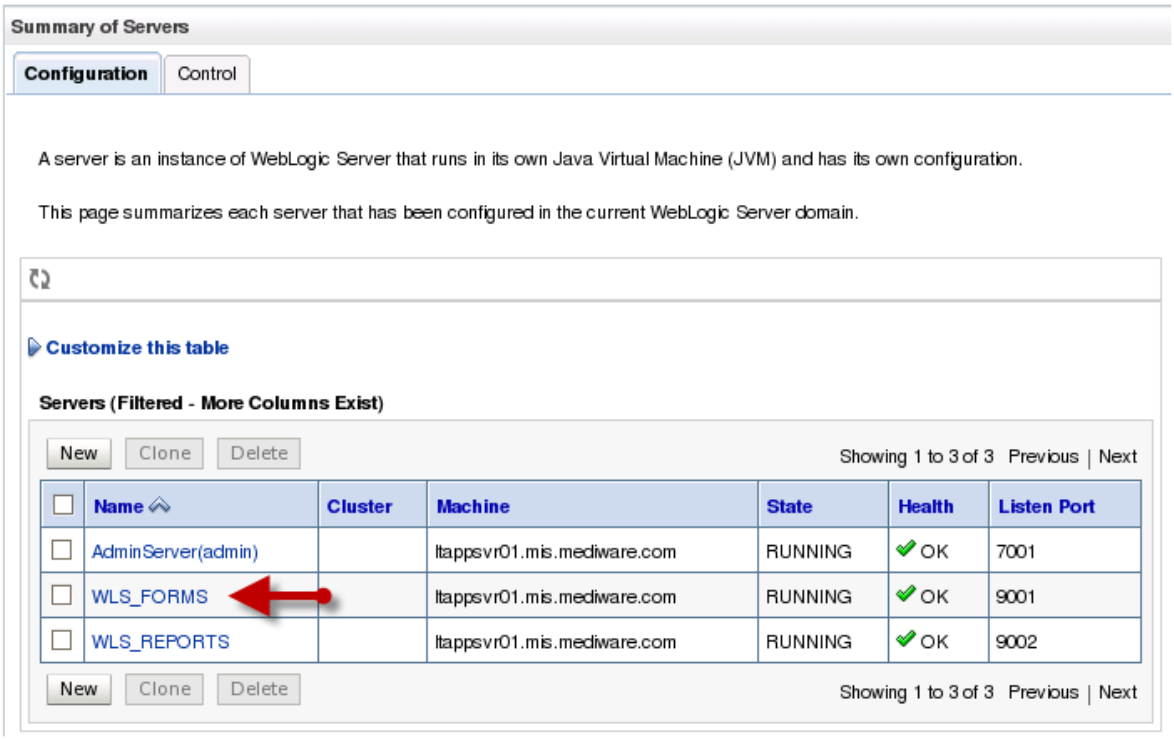
4. In the pane on the left under *Domain Structure*, expand **Environment** and click **Servers**.

Example:

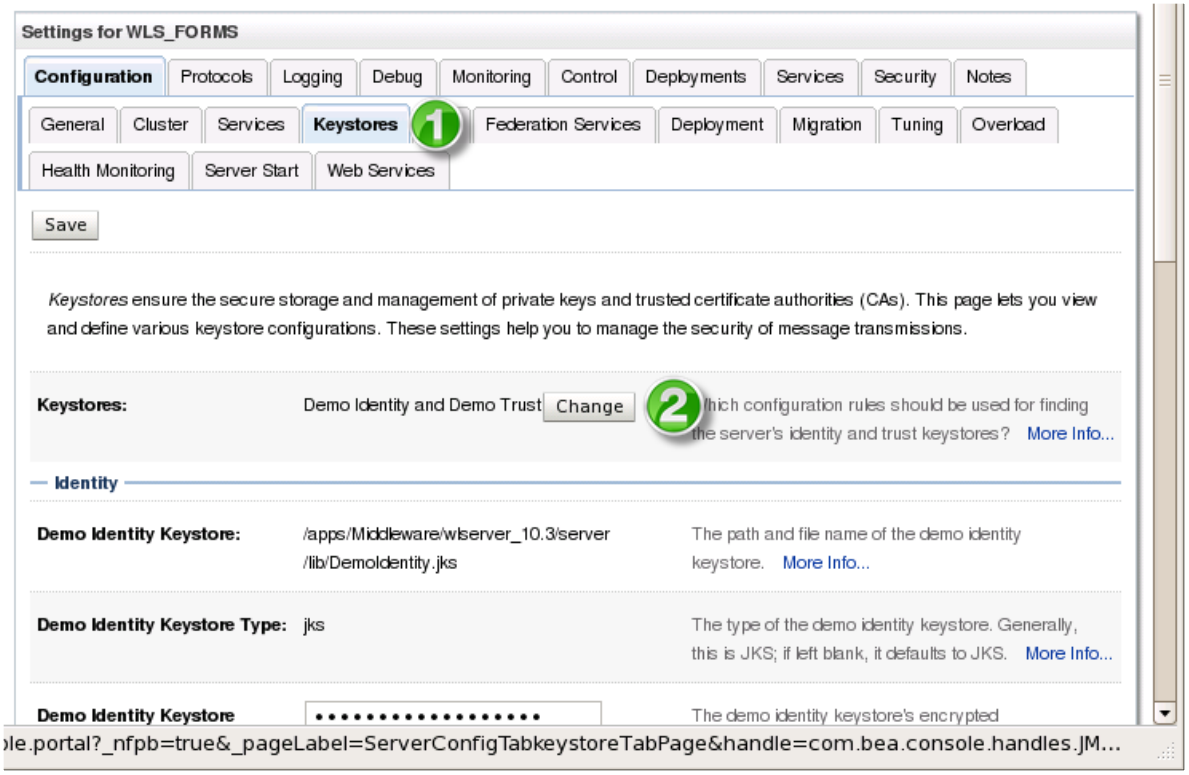


5. In the pane on the right under *Servers*, click WLS_FORMS.

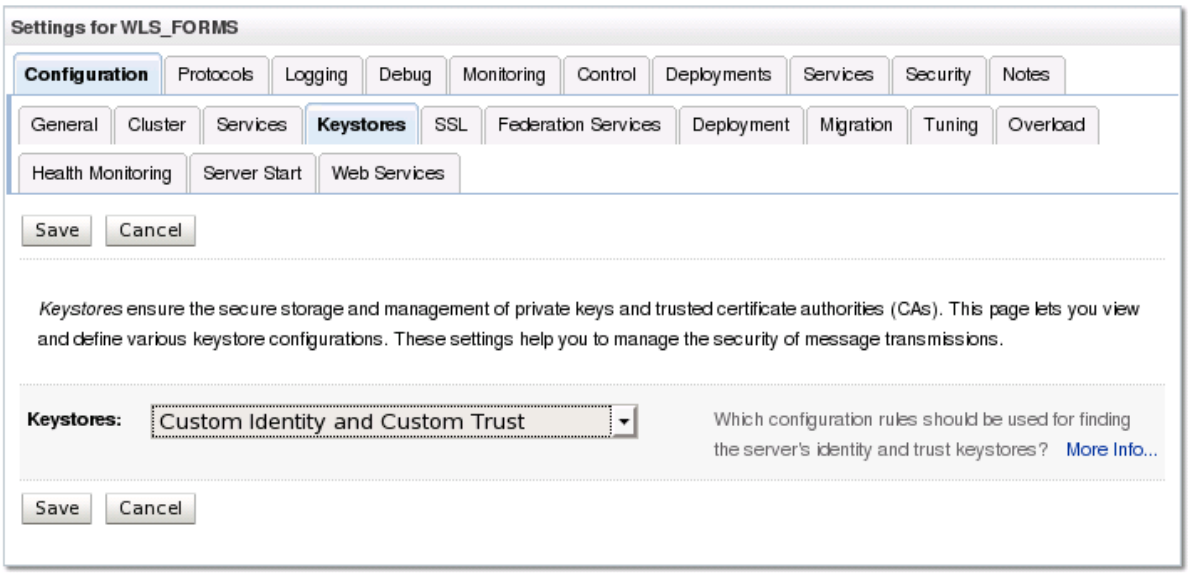
Example:



6. Select the **Keystores** tab, and then click **Change**.
   Example:



7. Select **Custom Identity and Custom Trust** from the *Keystores* drop-down and click **Save**.
   Example:

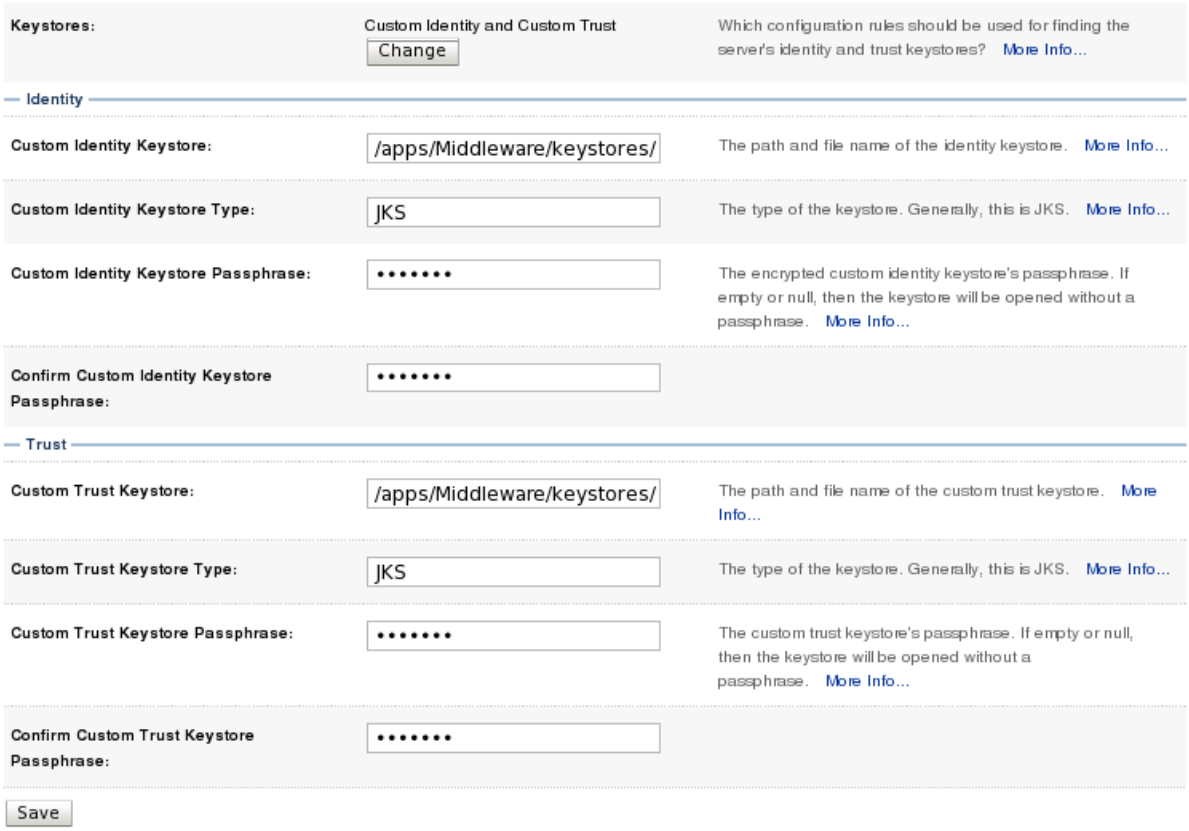


8. Fill out the Identity and Trust fields as outlined below.

   - *Custom Identity Keystore*: `/apps/Middleware/keystores/keystore.jks`
   - *Custom Identity Keystore Type*: `JKS`
   - *Custom Identity Keystore Passphrase*: `welcome`
   - *Confirm Custom Identity Keystore Passphrase*: `welcome`
   - *Custom Trust Keystore*: `/apps/Middleware/keystores/keystore.jks`
   - *Custom Trust Keystore Type*: `JKS`
   - *Custom Trust Keystore Passphrase*: `welcome`
   - *Confirm Custom Trust Keystore Passphrase*: `welcome`

   NOTE – The *passphrase* value used above (welcome) should be replaced with the *storepass* value from Part I, step 6.

Example:



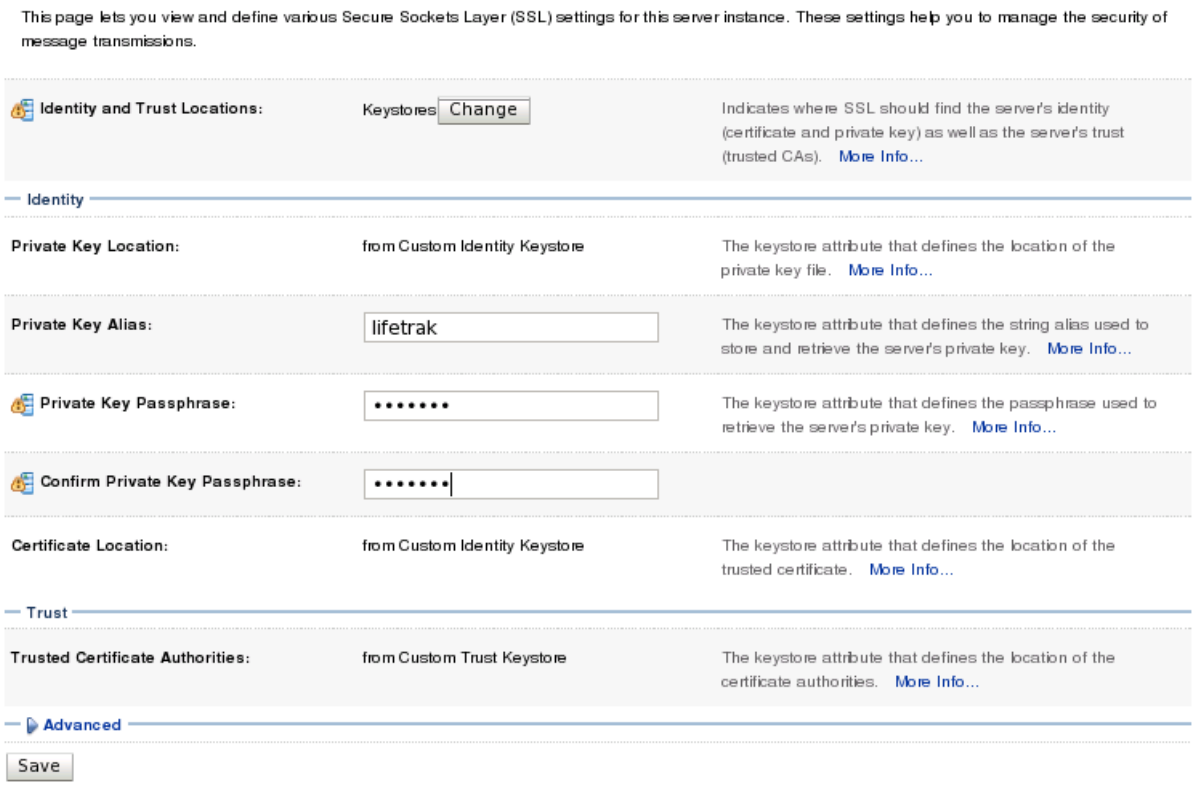
9. Click **Save**.
10. Select the **SSL** tab.

11. Complete the form as follows:

    - *Private Key Alias*: `lifetrak`
    - *Private Key Passphrase*: `welcome`
    - *Confirm Private Key Passphrase*: `welcome`

> **NOTE –** The *Private Key Alias* value (shown above as lifetrak) should be the same value entered as the *alias* in Part I, step 6.
>
> The *Private Key Passpharase* value is the *keypass* value as entered in Part I, step 6.

Example:

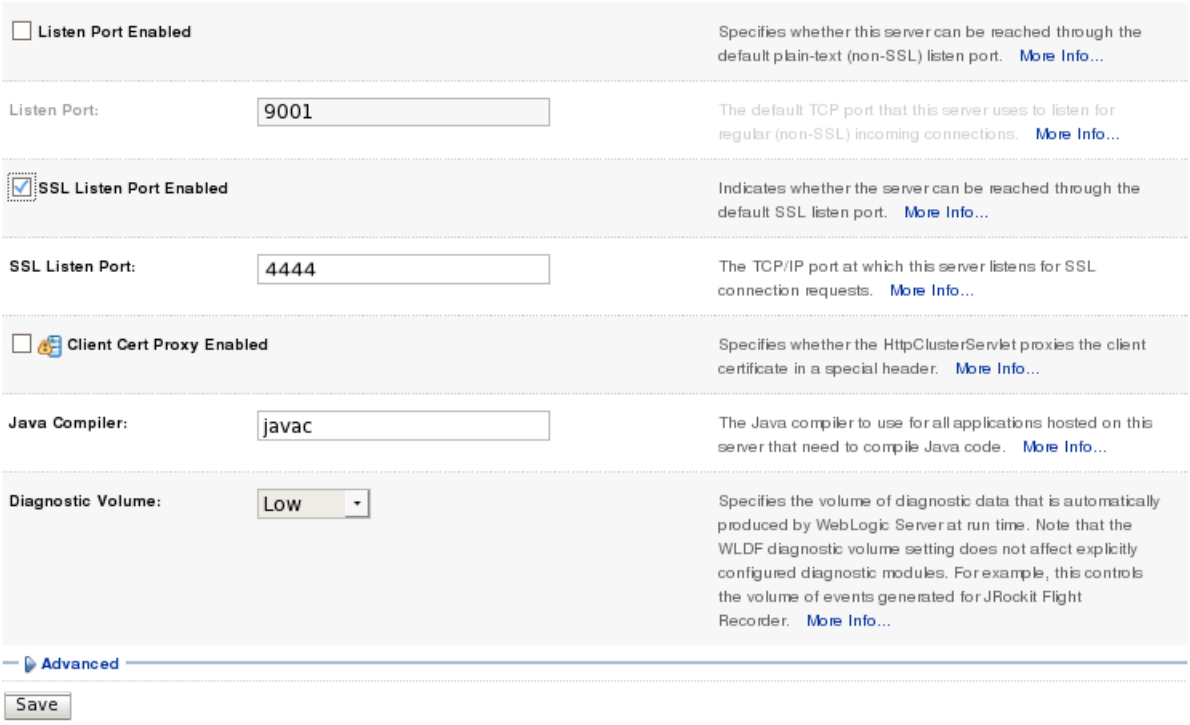


12. Click **Save**.
13. Select the **General** tab.
14. Clear the **Listen Port Enabled** check box.
15. Select the **SSL Listen Port Enabled** check box.

16. For *SSL Listen Port*, enter `4444`.

> **NOTE –** Make sure the port number entered is not used by any other process.

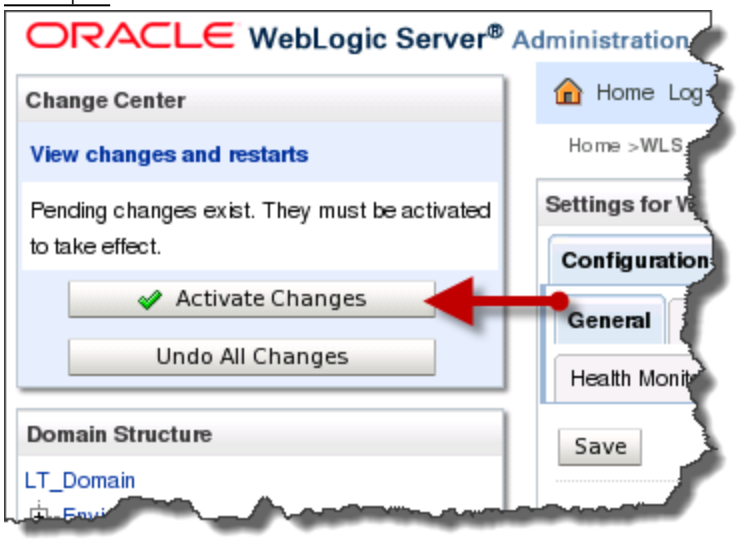
Example:



17. Click **Save**.

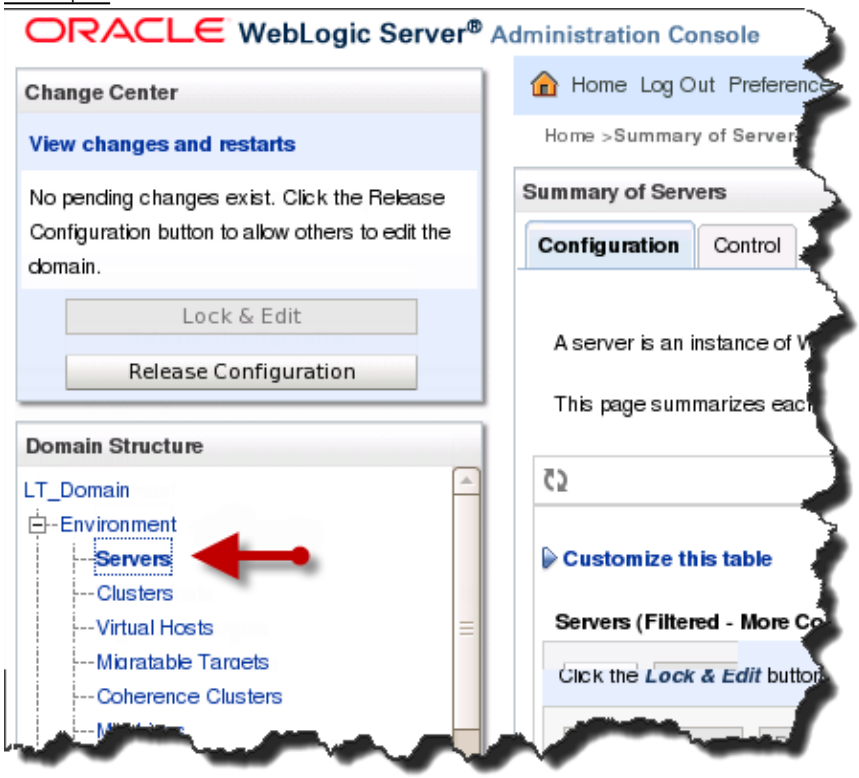
18. In the pane on the left under *Change Center*, click **Activate Changes**.

Example:



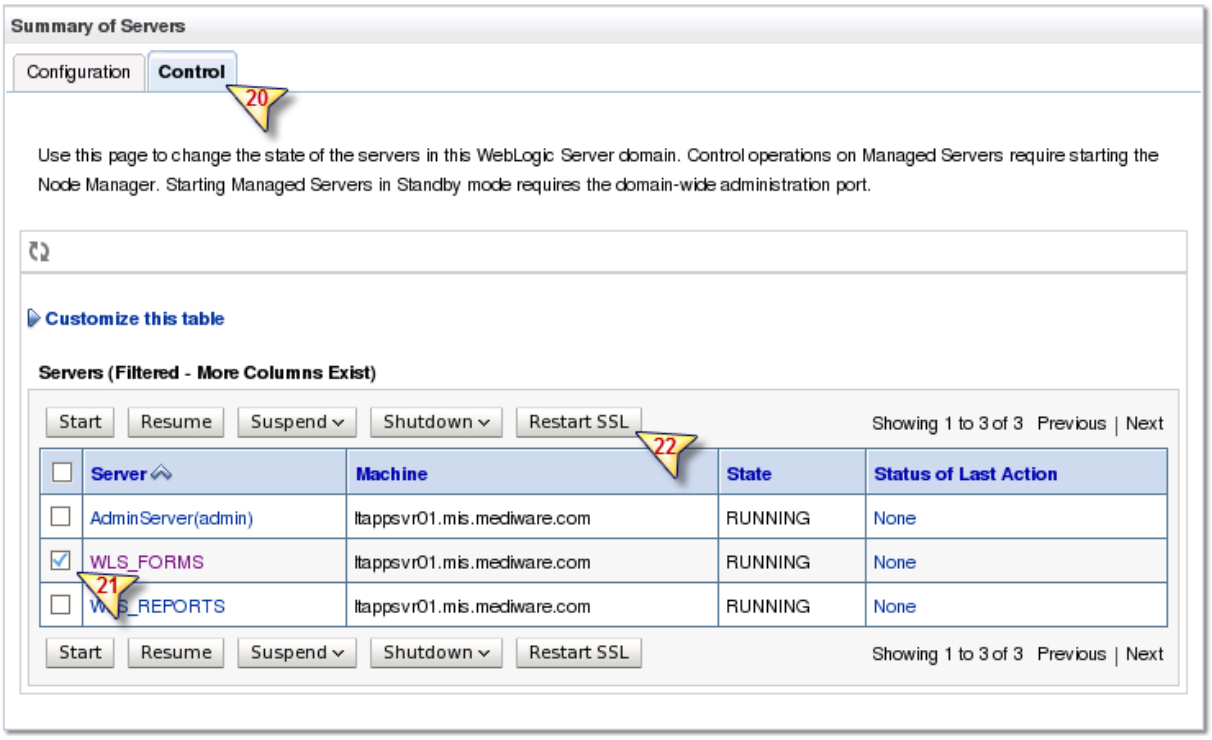
19. In the pane on the left under *Domain Structure*, click **Servers**.

Example:



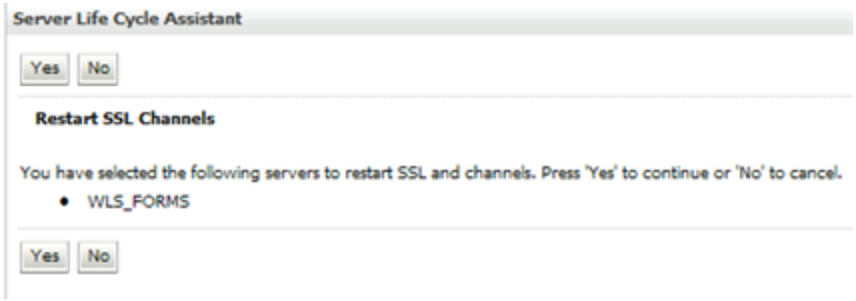
20. In the pane on the right, select the **Control** tab.

Example:



21. Select the **WLS_FORMS** check box.
22. Click **Restart SSL**. A confirmation message appears.

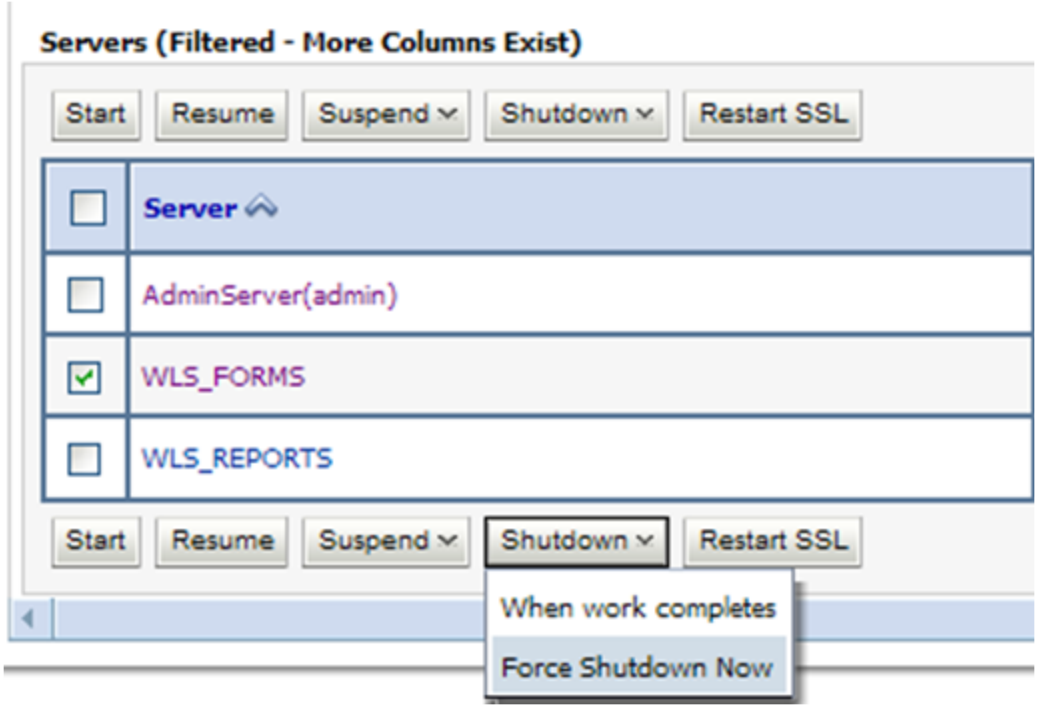
Example:



23. Click **Yes**.
24. Select the **WLS_FORMS** check box.

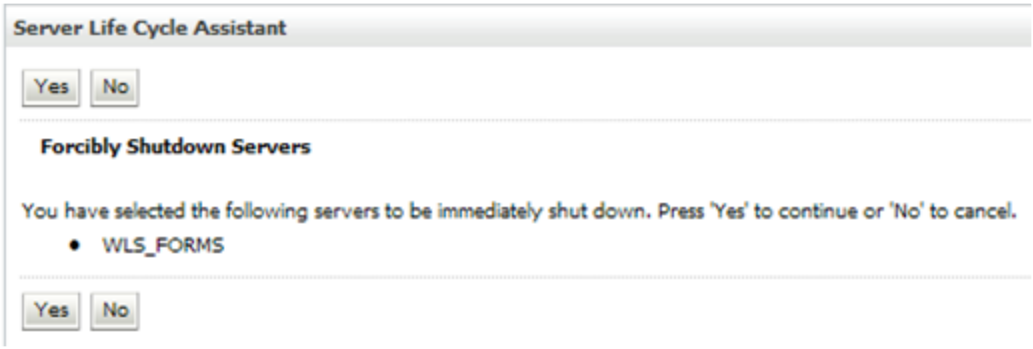
25. Click **Shutdown** > **Force Shutdown Now**.

Example:



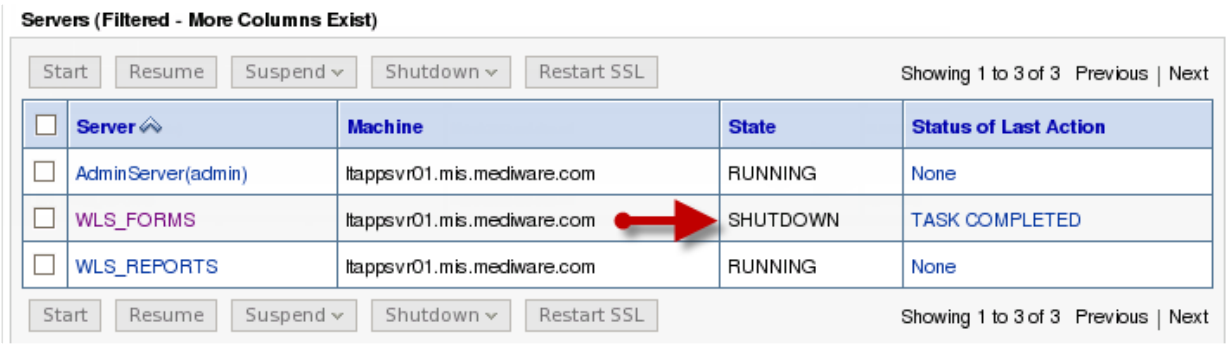
26. A confirmation message appears. Click **Yes**.

Example:



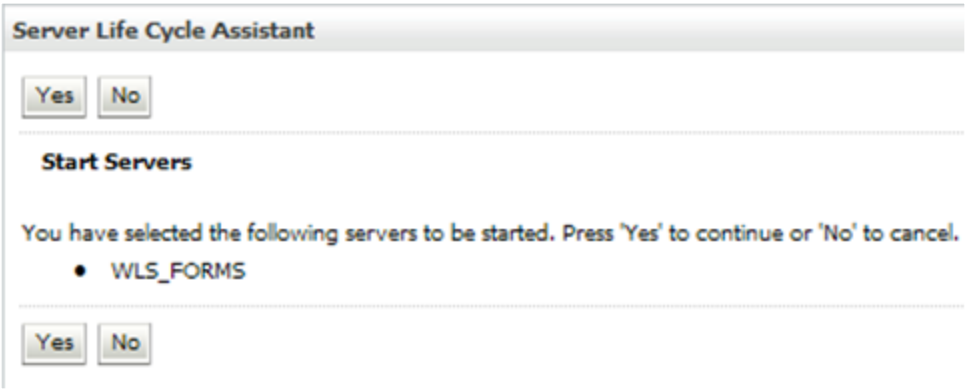
27. Verify the State for WLS_FORMS shows SHUTDOWN.

Example:



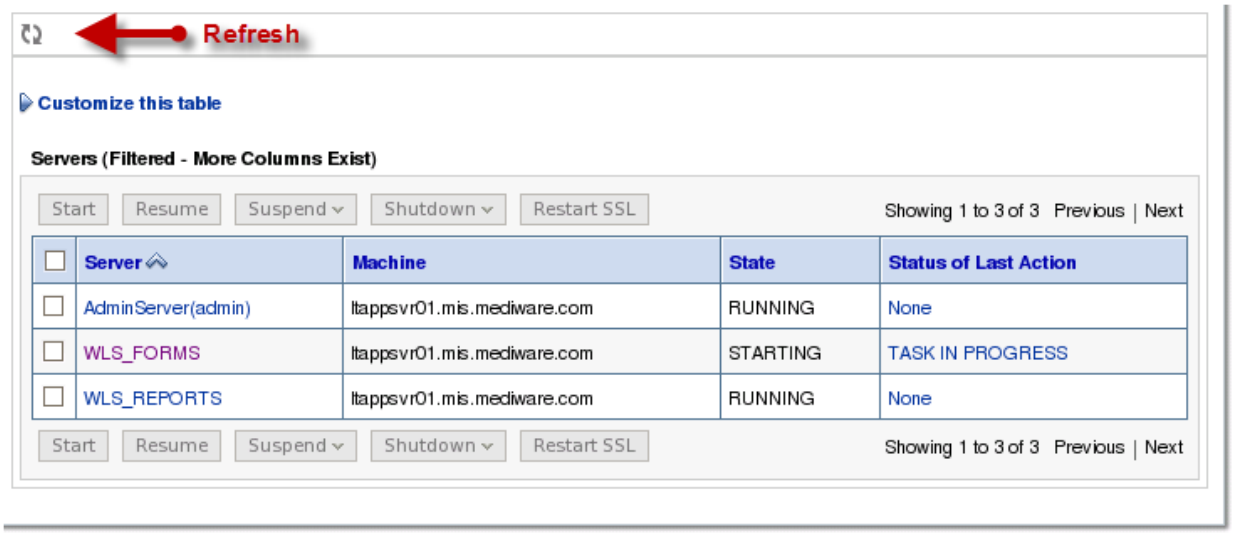
28. Select the **WLS_FORMS** check box.
29. Click **Start**. A confirmation message appears.

Example:



30. Click **Yes**. The startup process begins and the *State* for WLS_FORMS displays STARTING.

Example:



31. Click the **Refresh** icon. If needed, click the Refresh icon at regular intervals until the *State* of WLS_FORMS displays RUNNING.

Example:



A.1.3 Part III: Configure the Admin Server Trust Keystore

1. If a terminal window is not already open, right-click on the desktop and select **Open Terminal**.
2. Set the environment for Java Keytool by entering the following command:
   `$ . /apps/Middleware/user_projects/domains/LT_Domain/bin/setDomainEnv.sh`

   NOTE – Even though the command above displays on multiple lines, it should be entered as one line.

3. Change to the keystores directory for storing the keys by entering the following command:
   `$ cd /apps/Middleware/keystores`
4. Enter the following command to export the trusted Certificate(s) that signed the certificate in the Managed Server's keystore.
   `$ keytool -export -alias lifetrak –file rootCA.cer –keystore keystore.jks –storepass welcome`

   NOTE – Even though the command above displays on multiple lines, it should be entered as one line.

   NOTE – The values for *alias*, *keystore*, and *storepass* should be the same as entered in Part 1, step 6.

   NOTE – Running the command will create a file labeled rootCA.cer in the /apps/Middelware/keystores directory.

   NOTE – A "Certificate stored in file <rootCA.cer> message displays.

5. Enter the following command to import the trusted certificate exported in step 4 into this Trust Keystore.
   `$ keytool -import -trustcacerts -alias lifetrak -file rootCA.cer -keystore /apps/Middleware/jdk1.6.0_91/jre/lib/security/cacerts -storepass changeit`

   NOTE – Even though the command above displays on multiple lines, it should be entered as one line.

   NOTE – The values for alias and file should be the same as those used for step 4. The storepass value is the default for Java Standard.

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NOTE – This command will import the rootCA.cer values into the cacerts file, which is a Java Standard Trust Keystore file for the Admin Server.

6. It will prompt to trust this certificate. Type **yes** and press **Enter**. The configuration is complete.
7. Restart both the Admin Server and Managed Server. Make sure no errors occur relating to SSL or certificates in the logs and the AdminServer console.
8. If a terminal Right-click on the desktop and select **Open Terminal**.
9. Stop the application services by entering the following command:
   `$ ltapstop`
10. Search for any running process related to Middleware by entering the following command:
    `$ ps -ef|grep Middleware`
11. If you find any running process related to Middleware, manually kill them by entering the following command:
    `$ kill -9 <process Number>`
12. Start the application services by entering the following command:
    `$ ltapstart`

A.2 LT_APPLICATION_CONFIG Setup Guide

Column	Value	Description
APPL_ID	LIFETRAK	Default Value
TIME_ZONE_ENABLED	Y	If Time Zone is enabled, set value to Y; otherwise, set value to N
MULTI_CHAPTER	Y	Set value to Y for both single- and multi-chapter facilities
SINGLE_USER	Y	If Single User is enabled, set value to Y; otherwise, set value to N
STRONG_PWD_ENABLED	N	If Strong Password is enabled, set value to Y; otherwise, set value to N
SILENT_LOGIN	Y	If Silent Login is enabled, set value to Y; otherwise, set value to N
LOGIN_WARNING	N	If Login Warning is enabled, set value to Y; otherwise, set value to N
DEFAULT_DBASE	Y	If Default Database is enabled, set value to Y; otherwise, set value to N
EXTERNAL_AUTH	N	If External Authentication(for example, CAC Card) is enabled, set value to Y; otherwise, set value to N
SPECIAL_SCANNING	N	If Special Scanning is enabled, set value to Y; otherwise, set value to N
INT_ADDRESS_SUPPORT	N	If International Address Support is enabled, set value to Y; otherwise, set value to N
KTRAK_INTEGRATED	Y	If KTRAK (Quality & Compliance) is integrated, set value to Y; otherwise, set value to N
AD_ENABLED	N	If Active Directory is enabled, set value to Y; otherwise, set value to N

Column	Value	Description
AD_SERVER	mis.wellsky.com (This is only an example. Enter your own value.)	Active Directory's Server Value
AD_PORT	389 (This is only an example. Enter your own value.)	Active Directory's Port Value
AD_DOMAIN	mis (This is only an example. Enter your own value.)	Active Directory's Domain Value
TOUT_IDLE	0	Determines the amount of time (in minutes) an end user can stay idle before the application displays an alert warning. Default value is 0. Values of 0-4 are accepted.
TOUT_WARN	N/A	Determines the amount of time (in minutes) the warning alert is displayed before the system automatically logs the user out. Leave field blank if not using. Values of 0-3 are accepted.

A.3 CAC Setup and Configuration

This section applies to those clients who are using a Common Access Card (CAC) security configuration for the On-Site and oracle forms application. All others may skip to the next section.

A.3.1 Setup and Configuration

1. Install the CAC Agent Software for Apache using **web-agent-2025.6-Apache_v24_Linux_64bit.zip** software in the Onsite Linux Server, using the agent software installation notes.
    a. Copy the **web-agent-2025.6-Apache_v24_Linux_64bit.zip** in the **/apps/Software/CAC** directory as root user.
    b. Unzip **web-agent-2025.6-Apache_v24_Linux_64bit.zip** under **/apps directory**.
    c. Change the directory to **/apps/web_agents/apache24_agent/bin**.
    d. Run **agentadmin –i to** install the agent.

        ./agentadmin –i

    e. When prompted for the apache httpd.conf file, enter /etc/httpd/conf/httpd.conf if it is default Linux based Apache.
    f. Update /etc/httpd/conf.d/ssl.conf if it is set up with https.
    g. Change ownership of created directories using User and Group settings in **httpd.conf**. Enter **yes**.
    h. Take the **bc2026/BC_SERVER_INSTALL/CAC_Software/config_files/ssl.conf** and update required changes according to the server requirements.

2. Download the below oracle instant client zip files into the **/apps/Software/CAC** directory from Oracle website **https://www.oracle.com/database/technologies/instant-client/linux-x86-64-downloads.html**.

    ```
    instantclient-basic-linux.x64-19.28.0.0.0dbru.zip
    instantclient-sdk-linux.x64-19.28.0.0.0dbru.zip
    instantclient-sqlplus-linux.x64-19.28.0.0.0dbru.zip
    ```

3. As the root user, extract the above three packages into the /opt/instantclient_19_28 directory using the below commands:

    ```
    # cd /opt
    # unzip /apps/Software/CAC/instantclient-basic-linux.x64-
    19.28.0.0.0dbru.zip
    # unzip /apps/Software/CAC/instantclient-sdk-linux.x64-19.28.0.0.0dbru.zip
    # unzip /apps/Software/CAC/instantclient-sqlplus-linux.x64-
    19.28.0.0.0dbru.zip
    ```


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4. Install PHP-related Linux OS packages using the commands below as the root user.

```
# dnf -y install systemtap-sdt-devel

# dnf -y install https://dl.fedoraproject.org/pub/epel/epel-releaselatest-9.noarch.rpm

# dnf -y install http://rpms.remirepo.net/enterprise/remi-release-9.rpm

# dnf module reset php -y

# dnf module enable php:remi-8.4 -y

# dnf install php php-cli php-fpm php-mysqlnd php-gd php-xml php-mbstring php-devel -y

# php -v
```

5. Below is the procedure for PHP OCI software installation.

```
# export PHP_DTRACE=yes

# pecl install oci8
```

When prompted for the instantclient and its location, enter as below:

instantclient,/opt/instantclient_19_28

6. Edit /etc/php.ini file using vi editor
   a. Search variables_order parameter and update value from 'GPCS' to 'EGPCS'

      variables_order = "EGPCS"

   b. Update oci8.so extension at the end of the file like below and save the file.

      extension=/usr/lib64/php/modules/oci8.so

7. Copy bc2026/BC_SERVER_INSTALL/CAC_Software/config_files/www.conf file in the /etc/php-fpm.d directory. Be sure to backup the existing file.
8. Edit the /etc/sysconfig/httpd file and update below parameter:

```
### Below is the location for instant client software

LD_LIBRARY_PATH=/opt/instantclient_19_28:$LD_LIBRARY_PATH

export LD_LIBRARY_PATH

#### Below is the wallet related file's location

TNS_ADMIN=/var/www/bcapps/config/wallet

export TNS_ADMIN
```

9. Copy the bc2026/BC_SERVER_INSTALL/CAC_Software/cac directory in /var/www/html directory

10. Edit the /var/www/html/cac/.env.prod file and update required parameters for database connection, application server url
11. Restart the apache & php-fpm service by running below command:

```
# service php-fpm restart
# service httpd restart
```

12. Launch the onsite application using below CAC URL:

    https://cac2026.health.mil/cac/bconsiteagentapp/index.php

13. Launch the Oracle Application using below CACURL:

    https://cac2026.health.mil/cac/bcoracleagentapp/index.php

Important! For launching the Oracle-based application, you must set the registry editor using C:\BC2026\ regedit-bcapp2026R1-prddb.bat file. Run this file as administrator.

A.4 Blood Centers Application Links

A.4.1 .NET-Based Blood Centers Applications

NOTE – The .NET-based application is qualified on Google Chrome 64-bit version 84.0.4147.125.

OnSite Application

`http://bc2026.wellsky.com/BCOnsiteUI/index.html`

`https://cac2026.health.mil/cac/bconsiteagentapp/index.php (CAC-Enabled Onsite Application)`

`https://cac2026.health.mil/cac/bcoracleagentapp/index.php (CAC-Enabled Oracle Application)`

`https://bc2026.wellsky.com/okta/bconsiteapp (OKTA-Enabled)`

Fastrak Application

`http://bc2026.wellsky.com/fasttrak/fasttrak/sign-up`

`https://bc2026.wellsky.com/fasttrak/fasttrak/sign-up (SSL-Enabled)>`

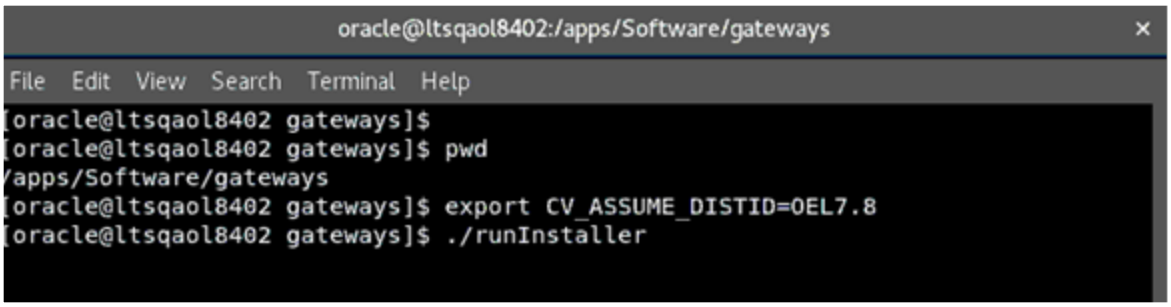
A.5 AGMGR Setup Instructions

NOTE – This is optional software. Check with WellSky for purchasing this software separately and installation.

A.5.1 Set Up the Oracle Gateway Software

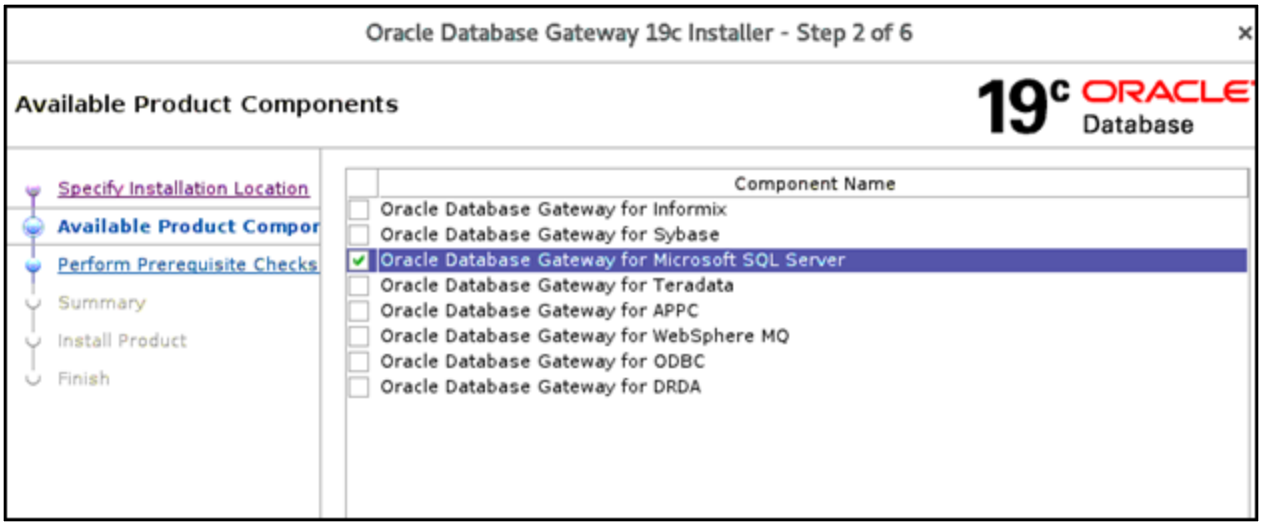
NOTE – Please perform the following procedure if you have Transfusion server (SQL Server's database). Otherwise, skip to Section A.6.2.

1. Go to https://www.oracle.com/database/technologies/oracle19c-linux-downloads.html.
2. Download Database Gateway 19c Software LINUX.X64_193000_gateways.zip.
3. Unzip the file in the /apps/Software directory.
4. Open VNC and log in as `oracle user`.
5. Change to the /apps/Software/gateways directory.
6. Run the following command on the terminal:
   `export CV_ASSUME_DISTID=OEL7.8`
7. Run the installer for installing the Gateway 19c software:
   `$ ./runInstaller`





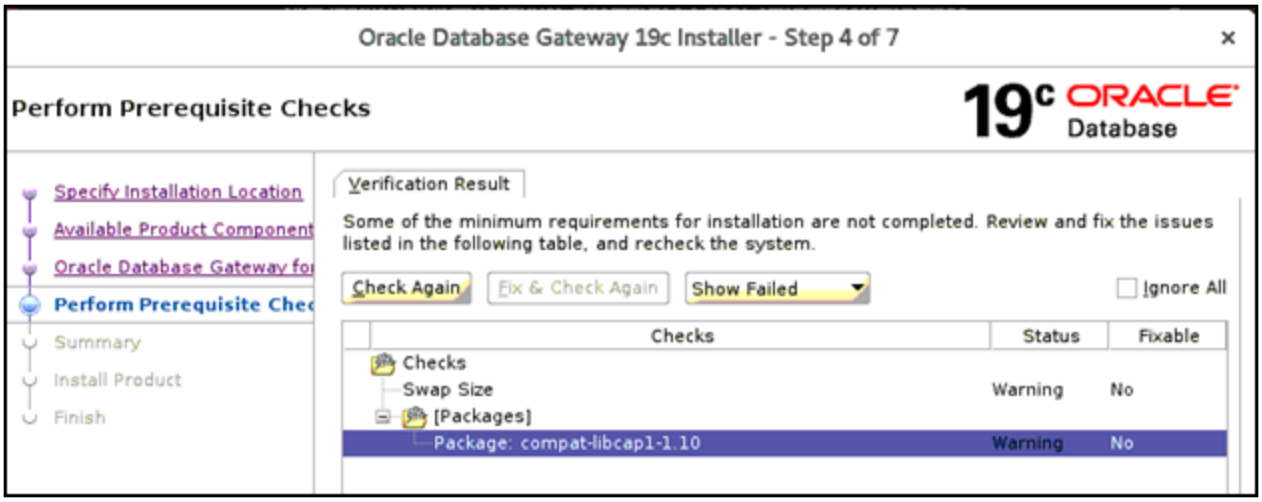
8. Set the Oracle base to **/apps/product** and the Software location as **/apps/product/19.0.0/tghome_1**.
9. Click **Next**.



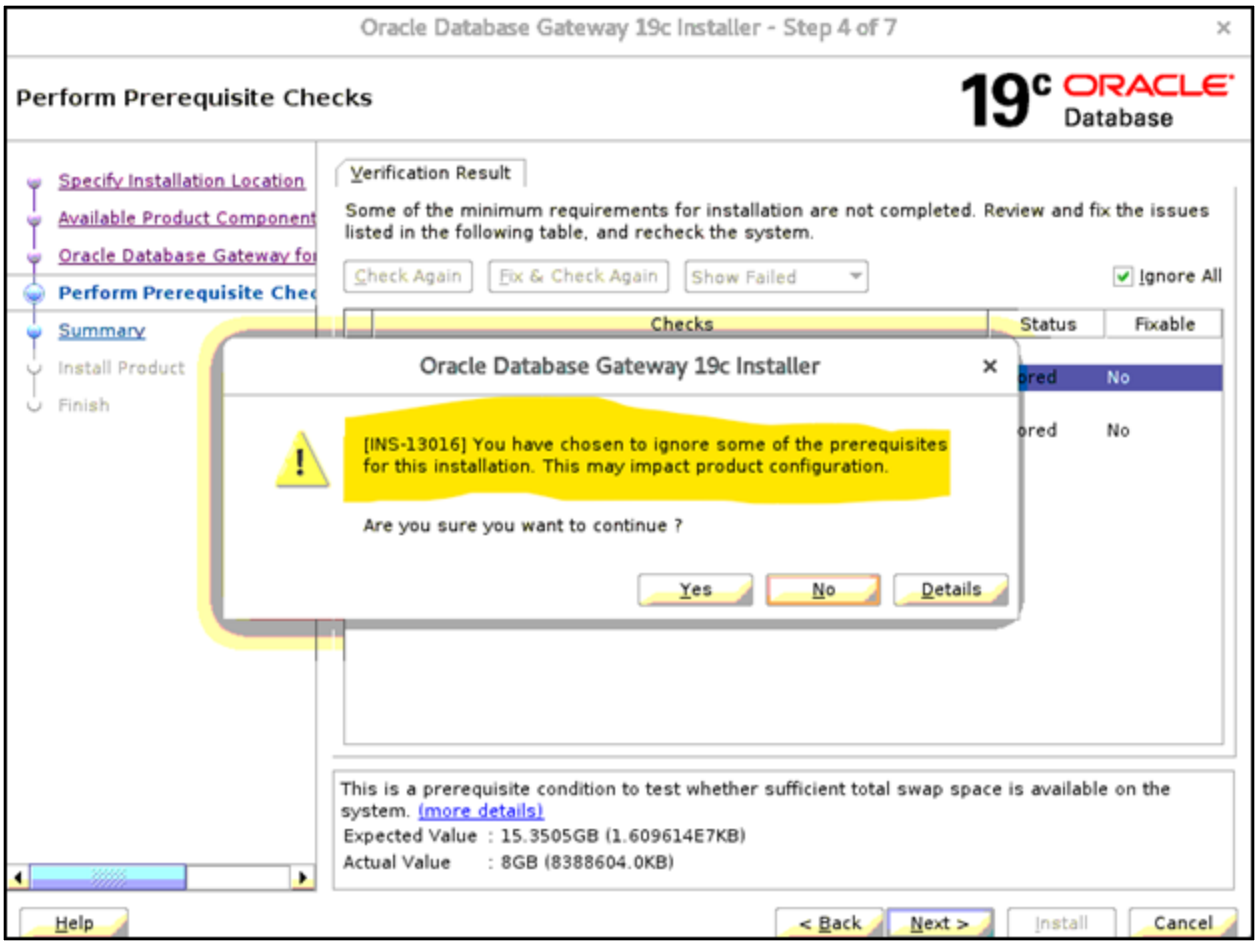
10. Select the **Oracle Database Gateway for Microsoft SQL Server** component, and click **Next**.



11. You will be prompted for the details for the Microsoft SQL Server. Enter the following:
    - Microsoft SQL Server Database Server Hostname
    - Microsoft SQL Server Database Server Port Number
    - Microsoft SQL Server Name
    - Microsoft SQL Server Database Name



12. Verify if any OS checks need to be addressed. It will look for a minimum of 16 GB of swap space, but you can install with a minimum of 8 GB. You can ignore the warning and click **Next**.



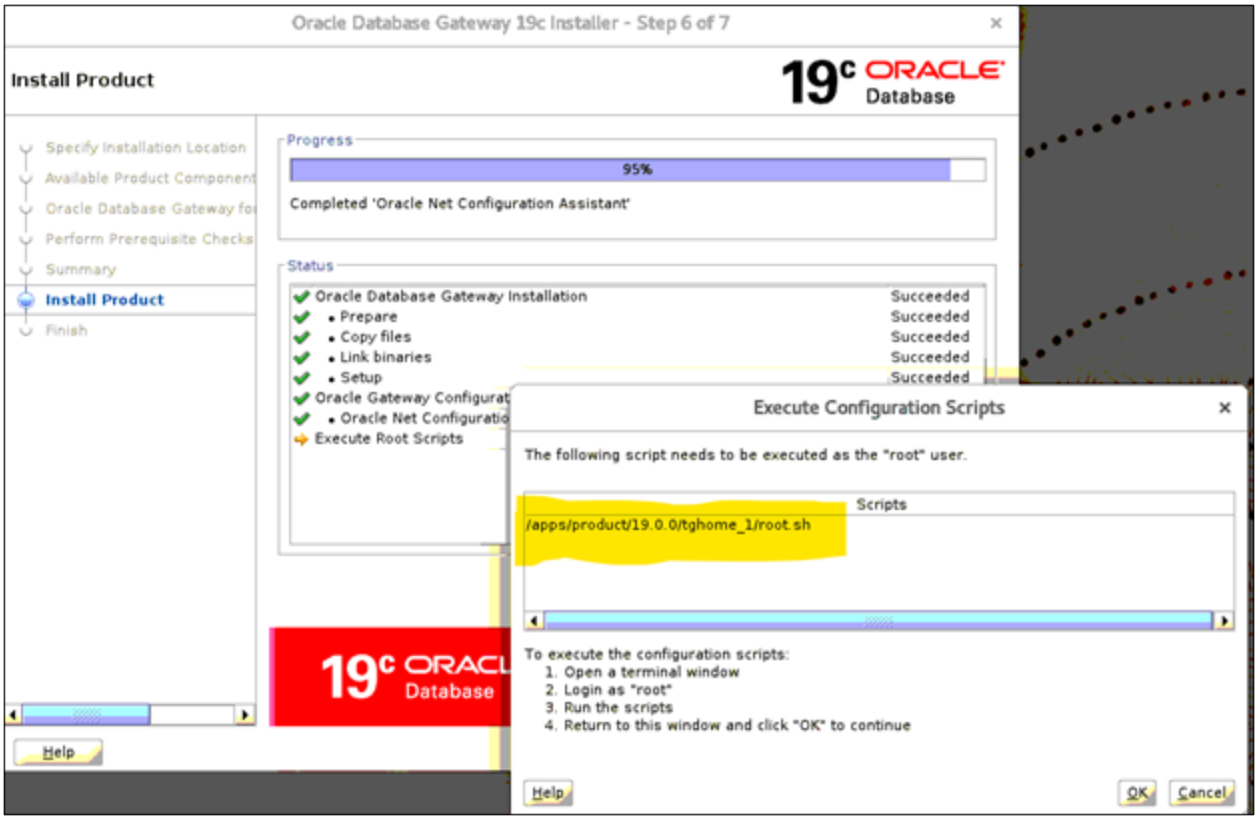
13. Click **Yes**.
14. Click **Install**.

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15. Select the **Perform typical configuration** option, and click **Next**.



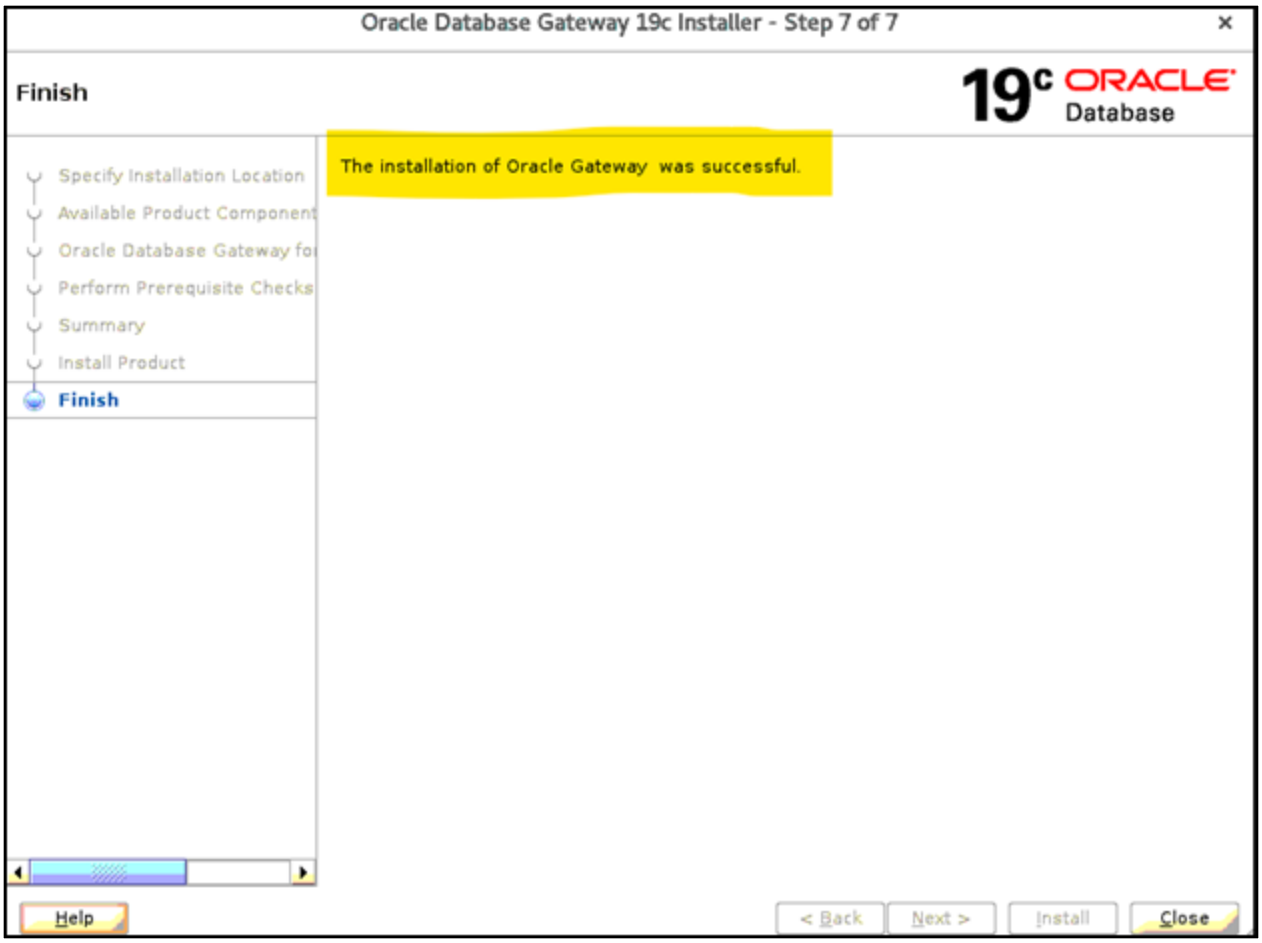
16. Log in as `root`.
17. Run the **/apps/product/19.0.0/tghome_1/root.sh** script.
18. Click **OK**.

```
[root@ltrndol8401 ~]#
[root@ltrndol8401 ~]# /apps/product/19.0.0/tghome_1/root.sh
Performing root user operation.

The following environment variables are set as:
    ORACLE_OWNER= oracle
    ORACLE_HOME=  /apps/product/19.0.0/tghome_1

Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.

Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
[root@ltrndol8401 ~]#
```



19. Click **Close**.
20. Change the directory to **/apps/product/19.0.0/tghome_1/dg4msql/admin**:
    `$ cd /apps/product/19.0.0/tghome_1/dg4msql/admin`
21. Verify the **initdg4msql.ora** file, and confirm the following content:
    `HS_FDS_CONNECT_INFO (variable values)`

22. Set up the listener.ora file in /apps/product/19.0.0/tghome_1/network/admin. The following is the format for the listener.ora file:

```
DGLISTENER =
  (DESCRIPTION_LIST =
    (DESCRIPTION =
       (ADDRESS = (PROTOCOL = TCP)(HOST =
ltsqaol8402.mis.mediware.com)(PORT = 1522))
       (ADDRESS = (PROTOCOL = IPC)(KEY = EXTPROC1522))
     )
   )

  SID_LIST_DGLISTENER=
    (SID_LIST=
       (SID_DESC=
          (SID_NAME=dg4msql)
          (ORACLE_HOME=/apps/product/19.0.0/tghome_1)

  (ENV="LD_LIBRARY_PATH=/apps/product/19.0.0/tghome_1/dg4msql
  /driver/lib:/apps/product/19.0.0/tghome_1/lib")
          (PROGRAM=dg4msql)
       )
    )
```

23. Start the listener after setting the following environment variables:

```
$ export ORACLE_HOME=/apps/product/19.0.0/tghome_1
$ export PATH=$PATH:$ORACLE_HOME/bin
$ export TNS_ADMIN=$ORACLE_HOME/network/admin
# lsnrctl start DGLISTENER
```

24. Set up the tns alias in Oracle Database Home (/apps/product/19.0.0/dbhome_1/network/admin/tnsnames.ora) using the following content:

```
dg4msql =
  (DESCRIPTION=
  ADDRESS=(PROTOCOL=tcp)(HOST=10.155.194.56)(PORT=1522))
    (CONNECT_DATA=(SID=dg4msql))
    (HS=OK)
  )
```


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25. Set up the Oracle database environment variable and test the tns alias **dg4msql**.

```
[oracle@ltsqaol8402 ~]$
[oracle@ltsqaol8402 ~]$ env|grep ORA
ORACLE_SID=bcqas
ORACLE_BASE=/apps/product/19.0.0
ORACLE_HOME=/apps/product/19.0.0/dbhome_1
[oracle@ltsqaol8402 ~]$ tnsping dg4msql

TNS Ping Utility for Linux: Version 19.0.0.0.0 - Production on 23-JUL-2024 20:03:29

Copyright (c) 1997, 2024, Oracle.  All rights reserved.

Used parameter files:
/apps/product/19.0.0/dbhome_1/network/admin/sqlnet.ora


Used TNSNAMES adapter to resolve the alias
Attempting to contact (DESCRIPTION= (ADDRESS=(PROTOCOL=tcp)(HOST=10.155.194.56)(PORT=1522)) (CONNECT_DATA=(SID=dg4msql)) (HS=OK))
OK (0 msec)
[oracle@ltsqaol8402 ~]$
```

26. Connect to the Oracle database and set up the DB link to the MS SQL database.

> **NOTE –** You need the MS SQL Server's **sa** user password.

```
[oracle@ltsqaol8402 ~]$
[oracle@ltsqaol8402 ~]$ sqlplus "/as sysdba"

SQL*Plus: Release 19.0.0.0.0 - Production on Tue Jul 23 20:06:01 2024
Version 19.23.0.0.0

Copyright (c) 1982, 2023, Oracle.  All rights reserved.


Connected to:
Oracle Database 19c Standard Edition 2 Release 19.0.0.0.0 - Production
Version 19.23.0.0.0

SQL> CREATE PUBLIC DATABASE LINK dg4msql CONNECT TO "sa" IDENTIFIED BY '[redacted]' USING 'dg4msql';

Database link created.
```

27. Test the SQL Server connection by running the following query on the Oracle database:
`SQL> select * from dual@dg4msql;`

```
SQL> select * from dual@dg4msql;

DU
--
X
```

28. Run the SQL Query for antigen values using the following query:
`SQL> SELECT count(*) FROM AntigenTestHistoryforBloodCentersPre2023@dg4msql;`

```
SQL>
SQL> SELECT count(*) FROM AntigenTestHistoryforBloodCentersPre2023@dg4msql;

  COUNT(*)
----------
         8

SQL>
```

29. Create View in SQL Server Database for one year of data using the columns below:

```
SELECT
        TRIM("DonationID")    AS unit_id,
    TRIM("Antigen")      AS antigen_code,
    TRIM("Interpretation") AS result_code,
    "AddDateTime"        AS insert_date,
    "AgCancelDT"         AS cancel_date,
    "AgTestInvalDT"      AS invalidate_date,
    TRIM("Active_Status")  AS tf_status
    FROM
    antigentesthistoryforbloodcenterspre2023@dg4msql
```

A.5.2 Install and Configure the AGMGR Schema

1. Create the /home/sbcsys/bcprd/AGFILES/archive directory on the Oracle Database Server as Linux user sbcsys and grant group permissions using the following commands:
   ```
   $ mkdir -p /home/sbcsys/bcprd/AGFILES/archive
   $ chmod 775 /home/sbcsys/bcprd/AGFILES
   $ chmod 777 /home/sbcsys/bcprd/AGFILES/archive
   ```
2. Extract AGMGR_BC2024.1.tar.gz under /oradata/Exports directory as Oracle user.
   ```
   $ tar -xvzf AGMGR_BC2024.1.tar.gz
   ```
   This will create the AGMGR_BC2024.1 directory under /oradata/Exports.
3. Change the directory to /oradata/Exports/AGMGR_BC2024.1 using the following command:
   ```
   $ cd /oradata/Exports/AGMGR_BC2024.1
   ```

Set up the AGMGR Database Schema

> **NOTE –** You can ignore this section if the database has already been upgraded to BC2024 R1.

4. Run the following shell script to configure the AGMGR schema:
   ```
   $ sh agmgr_config.sh
   ```
5. You will be prompted to provide the database name. Enter the production database name: **bcprd**.
   The AGMGR database user and related grants will be created.
6. You will be prompted for the database user's password. Enter the util database user's password. The default password is **util**.
   The AGMGR schema will be imported.

Set up the (Donor and Lab) Triggers and AGMGR Configuration Data

7. Change the directory /oradata/Exports/AGMGR_BC2024.1/DON_LAB_TRG:
   ```
   $ cd /oradata/Exports/AGMGR_BC2024.1/DON_LAB_TRG
   ```


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8. Run the following shell script:
   `# sh Config_DON_LAB_TRG.sh`
9. You will be prompted to provide the database name. Enter the production database name: **bcprd**.
   Donor and Lab's AGMGR-related triggers will be created.
10. Change the directory to /oradata/Exports/AGMGR_BC2024.1/AGDATA
    `$ cd /oradata/Exports/AGMGR_BC2024.1/AGDATA`
11. Run the following shell script to configure the AGMGR configuration data:
    `$ sh agdata_config.sh`
12. You will be prompted to provide the database name. Enter the production database name: **bcprd**.

Set up the AGMGR CRON Jobs

13. Change the directory to /oradata/Exports/AGMGR_BC2024.1/CRON_AG:
    `$ cd /oradata/Exports/AGMGR_BC2024.1/CRON_AG`
14. Copy the following scripts to the /home/sbcsys/bcprd/labexec directory as sbcsys user:
    - AGCORE.sh
    - AGCORE.sql
    - TF_Data_Extract.sh
    - TF_Data_Extract.sql
15. Copy the following directory and scripts to the **/home/sbcsys/bcprd** directory as **sbcsys** user:
    `AGLBL/BC_AGLBL_PRINT.sh`
    `AGLBL/BC_AGLBL_PRINT.sql`
16. Update the sbcsys OS crontab for the following content:

```
############ AGMGR CRON JOB  ################# BC2024R1

0,5,10,15,20,25,30,35,40,45,50,55 * * * * /home/sbcsys/bcprd/labexec/AGCORE.sh 1>> /tmp/bcprd_util/AGCORE.log 2>&1


0,5,10,15,20,25,30,35,40,45,50,55 * * * * /home/sbcsys/bcprd/labexec/TF_Data_Extract.sh 1>> /tmp/bcprd_util/TF_Data_Extract.log 2>&1

###### TieTag Label CRON Job for printing, need to run in nohup like below manually

cd home/sbcsys/bcprd/AGLBL/
nohup /home/sbcsys/bcprd/AGLBL/BC_AGLBL_PRINT.sh &
```

> **NOTE –** The TF_Data_Extract.sh and TF_Data_Extract.sql files are for extracting data from the Transfusion server. If you are not using the Transfusion server, ignore these files and the cron job as well.

Update the AGMGR Forms and Reports in Blood Centers Application Location

17. Copy the AGMGR-related forms and reports to the application server.

18. Change the directory to /oradata/Exports/AGMGR_BC2024.1/APPS_AG/bcorapp:

```
$ cd /oradata/Exports/AGMGR_BC2024.1/APPS_AG/bcorapp
$ cp -rf donor/frmx/* /home/bcorapp/donor/frmx
$ cp -rf donor/reports/* /home/bcorapp/donor/reports
$ cp -rf lab/frmx/* /home/bcorapp/lab/frmx
```

A.5.3 Setting Up the Zebra Label Printer

NOTE – Please verify the printer's setup and handling instructions. The instructions may vary for each printer model.

Network Printer Setup

1. Connect the printer to the network switch.
2. In Internet Explorer, type the IP address for the printer to log in to the printer using the printer IP address.
3. In the following example, 192.168.0.3 is the printer IP address. Please change it to match the printer's IP address.



4. Click View and Modify Printer Settings.

5. You will be prompted for a password. Enter **1234** (default password).



6. Click Submit Changes.



7. Click Access Granted.



8. Click General Setup.

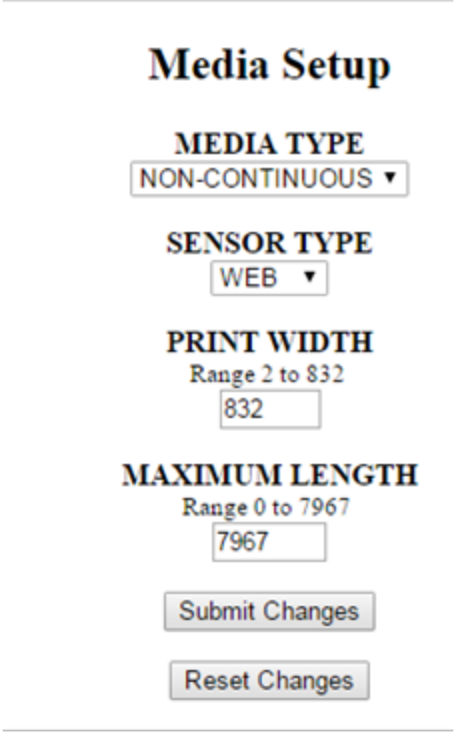


9. Compare the values in the above screenshot to what you see on the screen, and make sure they are the same. If they are not the same, update them.
10. Click Submit Changes.
11. Click View and Modify Printer Settings.
12. Click Save Current Configuration.
13. Click Serial Communications Setup.
14. Repeat steps 9-12 with the following screenshot.



15. Click Media Setup.

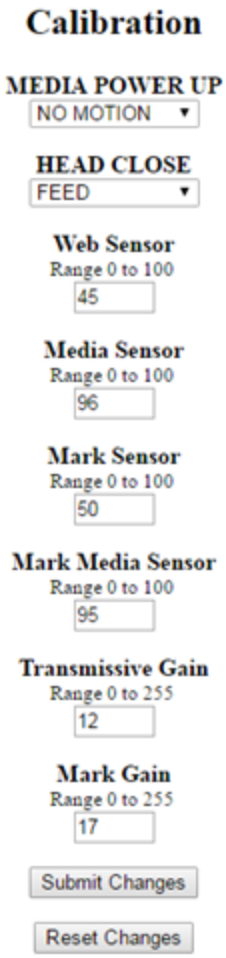
16. Repeat steps 9-12 with the following screenshot.



17. Click **Calibration**.

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18. Repeat steps 9-12 with the following screenshot.



19. Click ZPL Control.

20. Repeat steps 9-12 with the following screenshot.



21. Click **Advanced Setup**.
22. Repeat steps 9-12 with the following screenshot.



Linux Printer Setup

1. Go to **Applications** > **Sundry**, and click **Printer Settings**.
2. Click **Add** under Printer Settings.
3. Click **Network Printer** > **Find Network Printer**.
4. Enter the IP address and click **Find**.
5. You will be prompted to provide user credentials. Enter the user name, password, and domain name used on the laptop.
6. Click **Set authentication details now**, and enter the user name and password.

> **NOTE –** Enter the user name as domain name\user name.

7. Click **Forward** to search for drivers.
8. Select **Zebra** and click **Forward**.
9. Click **ZPL Label** and click **Forward**.
10. Enter the printer name (**SGX420D**) and click **Apply**.
11. Click **Print Test Page**.

A.6 Oracle Wallet Setup and Configuration

A.6.1 Oracle Wallet Setup in Database Server Environment

1. Create the Oracle Wallet Directory, log in as a root OS user to database server using Putty, create a **/opt/BCJOB/bcpstore** directory, then grant ownership to oracle:dba.
   - `# mkdir -p /opt/BCJOB/bcpstore`
   - `# chown -R oracle:dba /opt/BCJOB`

```
[root@cac2026 ~]# mkdir -p /opt/BCJOB/bcpstore
[root@cac2026 ~]# chown -R oracle:dba /opt/BCJOB
```

2. Log in as an Oracle OS user using Putty and create a Wallet using **mkstore**.
   a. Set the Oracle environment variable and select the SID as **prddb**.
   b. Create the Wallet using mkstore. It will prompt for a password.
      - `$ mkstore -wrl /opt/BCJOB/bcpstore -create`
   c. Enter a password. For example, **welcome1** and confirm the same.

```
[oracle@cac2026 ~]$
[oracle@cac2026 ~]$ . oraenv
ORACLE_SID = [oracle] ? prddb
The Oracle base has been set to /apps/product/19.0.0
[oracle@cac2026 ~]$
[oracle@cac2026 ~]$ mkstore -wrl /opt/BCJOB/bcpstore -create
Oracle Secret Store Tool Release 19.0.0.0.0 - Production
Version 19.4.0.0.0
Copyright (c) 2004, 2025, Oracle and/or its affiliates. All rights reserved.

Enter password:
Enter password again:
[oracle@cac2026 ~]$
```

   d. List the created Wallet using the command below:
      - `$ mkstore -wrl /opt/BCJOB/bcpstore -listCredential`
   e. When prompted for a password, enter the same password that you created above. For example, **welcome1**.
      - It's a blank Wallet without any database user's credentials.

```
[oracle@cac2026 ~]$ mkstore -wrl /opt/BCJOB/bcpstore -listCredential
Oracle Secret Store Tool Release 19.0.0.0.0 - Production
Version 19.4.0.0.0
Copyright (c) 2004, 2025, Oracle and/or its affiliates. All rights reserved.

Enter wallet password:
List credential (index: connect_string username)
[oracle@cac2026 ~]$
```

f. List the Wallet files under **/opt/BCJOB/bcpstore** using the command below:

- `$ ls -l /opt/BCJOB/bcpstore`

```
[oracle@cac2026 ~]$
[oracle@cac2026 ~]$ ls -l /opt/BCJOB/bcpstore
total 8
-rw-------. 1 oracle dba 270 Oct 15 15:55 cwallet.sso
-rw-------. 1 oracle dba   0 Oct 15 15:55 cwallet.sso.lck
-rw-------. 1 oracle dba 225 Oct 15 15:55 ewallet.p12
-rw-------. 1 oracle dba   0 Oct 15 15:55 ewallet.p12.lck
[oracle@cac2026 ~]$
```

3. Setup the Oracle network TNS files **sqlnet.ora** and **tnsnames.ora** under the **/apps/product/19.0.0/dbhome_1/network/admin** directory.

- Below is the Wallet location format for **sqlnet.ora**.

```
############################################################

WALLET_LOCATION =

  (SOURCE =

    (METHOD = FILE)

    (METHOD_DATA = (DIRECTORY = /opt/BCJOB/bcpstore))

)

SQLNET.WALLET_OVERRIDE = TRUE

############################################################
```

```
[oracle@cac2026 admin]$ cat sqlnet.ora
############################################################
WALLET_LOCATION =
    (SOURCE =
       (METHOD = FILE)
       (METHOD_DATA = (DIRECTORY = /opt/BCJOB/bcpstore))
)

SQLNET.WALLET_OVERRIDE = TRUE
############################################################
```

- Below is the Wallet TNS alias format for each database user.
- **Example: For UTIL DB user TNS alias**

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```
#########################################################
UTIL_PRDDB =
 (DESCRIPTION =
  (ADDRESS = (PROTOCOL = TCP)(HOST = 10.16.0.104)(PORT = 1521))
  (CONNECT_DATA =
   (SERVER = DEDICATED)
    (SERVICE_NAME = PRDDB)
  )
 )
#########################################################
```

- Example: For DONSYS DB user TNS alias

```
#########################################################
DONSYS_PRDDB =
 (DESCRIPTION =
(ADDRESS = (PROTOCOL = TCP)(HOST = 10.16.0.104)(PORT = 1521))
(CONNECT_DATA =
(SERVER = DEDICATED)
(SERVICE_NAME = PRDDB)
  )
 )
#########################################################
```

```
#############################################################################
#############################################################################
### WALLET Based TNS Aiades for BC DB Users
#############################################################################

UTIL_PRDDB =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = 10.16.0.104)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = PRDDB)
    )
  )

DONSYS_PRDDB =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = 10.16.0.104)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = PRDDB)
    )
  )
```

- This tnsnames.ora file needs all the Blood Center database users with the above format for each user separately.

- Below are most of the TNS aliases for Blood Center Database:
  - UTIL_PRDDB
  - DONSYS_PRDDB
  - SBCSYS_PRDDB
  - SRVSYS_PRDDB
  - LTAPI_PRDDB
  - DBADMN_PRDDB
  - VISTA_PRDDB
  - SFAPI_PRDDB
  - AGMGR_PRDDB

4. Add the Blood Center Database user credentials to the Wallet for each user separately, using the following commands:
   - Below is for adding *UTIL DB* user credentials in Wallet.
     - `$ mkstore -wrl /opt/BCJOB/bcpstore -createCredential UTIL_PRDDB UTIL util`
       - Here, the Wallet directory is: **/opt/BCJOB/bcpstore**.
       - TNS Alias: **UTIL_PRDDB**
       - DB user: **UTIL**
       - Password: **util**
   - When prompted for the Wallet password, enter **welcome1**, which you created in Step 2.

```
[oracle@cac2026 ~]$ . oraenv
ORACLE_SID = [oracle] ? prddb
The Oracle base has been set to /apps/product/19.0.0
[oracle@cac2026 ~]$ mkstore -wrl /opt/BCJOB/bcpstore -createCredential UTIL_PRDDB UTIL util
Oracle Secret Store Tool Release 19.0.0.0.0 - Production
Version 19.4.0.0.0
Copyright (c) 2004, 2025, Oracle and/or its affiliates. All rights reserved.

Enter wallet password:
[oracle@cac2026 ~]$
```

   - Below is for adding *DONSYS DB* user credentials in Wallet.
     - `$ mkstore -wrl /opt/BCJOB/bcpstore -createCredential DONSYS_PRDDB DONSYS donsys`
       - Here, the Wallet directory is: **/opt/BCJOB/bcpstore**.
       - TNS Alias: **DONSYS_PRDDB**
       - DB user: **DONSYS**
       - Password: **donsys**

- When prompted for the Wallet password, enter **welcome1**, which you created in Step 2.

```
[oracle@cac2026 ~]$
[oracle@cac2026 ~]$ mkstore -wrl /opt/BCJOB/bcpstore -createCredential DONSYS_PRDDB DONSYS donsys
Oracle Secret Store Tool Release 19.0.0.0.0 - Production
Version 19.4.0.0.0
Copyright (c) 2004, 2025, Oracle and/or its affiliates. All rights reserved.

Enter wallet password:
[oracle@cac2026 ~]$
```

- You must add each Blood Center database user credentials like the above format.

5. Verify all the added Blood Center DB users in Wallet using the command below. When prompted for the Wallet password, enter **welcome1**.
   - `$ mkstore -wrl /opt/BCJOB/bcpstore -listCredential`

```
[oracle@cac2026 ~]$ mkstore -wrl /opt/BCJOB/bcpstore -listCredential
Oracle Secret Store Tool Release 19.0.0.0.0 - Production
Version 19.4.0.0.0
Copyright (c) 2004, 2025, Oracle and/or its affiliates. All rights reserved.

Enter wallet password:
List credential (index: connect_string username)
6: LTAPI_PRDDB LTAPI
5: DBADMN_PRDDB DBADMN
4: SRVSYS_PRDDB SRVSYS
3: SBCSYS_PRDDB SBCSYS
2: DONSYS_PRDDB DONSYS
1: UTIL_PRDDB UTIL
9: VISTA_PRDDB VISTA
8: AGMGR_PRDDB AGMGR
7: SFAPI_PRDDB SFAPI
[oracle@cac2026 ~]$
```

   - The above screenshot has all the DB users that we normally use in the Blood Center application.

6. If a Database user's password is modified, you must update it in the Wallet file as well using the command below.
   - `$ mkstore -wrl /opt/BCJOB/bcpstore -modifyCredential UTIL_PRDDB UTIL utpass`
   - In the above command, the database user's UTIL password was modified to utpass.
   - When prompted for a Wallet password, enter **welcome1**.

```
[oracle@cac2026 ~]$ mkstore -wrl /opt/BCJOB/bcpstore -modifyCredential UTIL_PRDDB UTIL utpass
Oracle Secret Store Tool Release 19.0.0.0.0 - Production
Version 19.4.0.0.0
Copyright (c) 2004, 2025, Oracle and/or its affiliates. All rights reserved.

Enter wallet password:
[oracle@cac2026 ~]$
```

7. All the database CRON job scripts under the Oracle, DONSYS, and SBCSYS OS users are set up with Wallet user configuration. Please contact the WellSky Technical Teamif there are any specific issues.

A.6.2 Oracle Wallet Setup in BC Forms and Reports Application Server

1. If the Oracle Application Server is in a separate Linux Server, create an Oracle Wallet Directory, log in as a root OS user using Putty and create an **/opt/BCJOB/bcpstore** directory, then grant ownership to oracle:dba.
   - `# mkdir -p /opt/BCJOB/bcpstore`
   - `# chown -R oracle:dba /opt/BCJOB`

```
[root@cac2026 ~]# mkdir -p /opt/BCJOB/bcpstore
[root@cac2026 ~]# chown -R oracle:dba /opt/BCJOB
```

2. Copy the **cWallet.sso** file from BC database server's **/opt/BCJOB/bcpstore** directory to the BC Application Server's **/opt/BCJOB/bcpstore directory**.
3. Copy **sqlnet.ora** and **tnsnames.ora** files from the BC database server's **/apps/product/19.0.0/dbhome_1/network/admin** directory to the BC application server's **/apps/Middleware/Oracle_Home/user_projects/domains/BC_Domain/config/fmwconfig** directory.
4. Verify and update (if needed) the Oracle WebLogic forms server's **/apps/Middleware/Oracle_Home/user_projects/domains/BC_Domain/config/fmwconfig/servers/WLS_FORMS/applications/formsapp_14.1.2/config/formsweb.cfg formsweb.cfg** for userid parameters like below.
   - userid=/@DBADMN_PRDDB

```
[bcorapp]
form=ltilogon
pageTitle=Blood Center 2026 PRDDB Application
envFile=bcorapp.env
imageBase=codeBase
archive=frmall.jar,frmwebutil.jar,jacob.jar,sdonor.jar,shyperlink.jar,slaf.jar
baseHTMLjinitiator=basejpi.htm
splashScreen=no
width=800
height=600
separateframe=false
HTMLbodyAttrs=scroll="no" leftmargin="0" topmargin="0" marginwdith="0" marginheight="0"
otherparams=term=/apps/Middleware/Oracle_Home/forms/bcorapp.res  record=names p_uid=%p_uid_key%

userid=/@DBADMN_PRDDB
workingDirectory=/home/bcorapp/donor/frmx
allowJVMControllerAutoStart=true
baseSAAfile=webutilsaa.txt
fsalcheck=true
```

5. By default, it will **encrypt** this line once the forms server is restarted.

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```
[bcorapp]
form=ltilogon
pageTitle=Blood Center 2026 QASDB Application
envFile=bcorapp.env
imageBase=codeBase
archive=frmall.jar,frmwebutil.jar,jacob.jar,sdonor.jar,shyperlink.jar,slaf.jar
baseHTMLjinitiator=basejpi.htm
splashScreen=no
width=800
height=600
separateframe=false
HTMLbodyAttrs=scroll="no" leftmargin="0" topmargin="0" marginwdith="0" marginheight="0"
otherparams=term=/apps/Middleware/Oracle_Home/forms/bcorapp.res  record=names p_uid=%p_uid_key%

userid=(AES)zjk1IL70F8kDE9zDUIPydA==
workingDirectory=/home/bcorapp/donor/frmx
allowJVMControllerAutoStart=true
baseSAAfile=webutilsaa.txt
fsalcheck=true
```

A.6.3 Oracle Wallet Setup in BC Onsite Application Server

1. If the OnSite Application Server is in a separate Linux Server, Create an Oracle Wallet Directory, log in as a root OS user using Putty and create a **/var/www/bcapps/config/Wallet** directory.
   - `# mkdir -p /var/www/bcapps/config/Wallet`
2. Copy the **cWallet.sso** file from the BC database server's **/opt/BCJOB/bcpstore** directory to the BC Application Server's **/var/www/bcapps/config/Wallet** directory.
3. Copy the **sqlnet.ora** and **tnsnames.ora** files from the BC database server's **/apps/product/19.0.0/dbhome_1/network/admin** directory to the BC OnSite application server's **/var/www/bcapps/config/Wallet** directory.
4. Verify the **appsettings.json** file for each of the below APIs under their respective folders and make sure that (***Data Source, DIRECTORY*** and ***WalletTnsAdmin***) parameters reflect the correct values for a specific database.

   - BC_Onsite.API /var/www/html/BC_Onsite.API
   - BC_OnsiteAuth.API /var/www/html/BC_OnsiteAuth.API
   - BC_DonorCards.API /var/www/html/BC_DonorCards.API
   - BC_DonorSurvey.API /var/www/html/BC_DonorSurvey.API
   - fasttrakapi /var/www/html/fasttrakapi

A.6.4 Oracle Wallet files modification and updates

If a Blood Center database user's passwords are modified, they must be updated in the Wallet configuration in the database server and those changes must reflect in the application server's Wallet locations as well. Please follow the procedure below.

The command below is an example for the UTIL DB user's Wallet update. Tun the command below in the database server after setting the Oracle environment variable. For example: PRDDB.

```
$ mkstore -wrl /opt/BCJOB/bcpstore -modifyCredential UTIL_PRDDB UTIL
utpass
```

- In the above command, the Database user UTIL's password is modified as utpass.
- When it prompts for a Wallet password, enter **welcome1**

```
[oracle@cac2026 ~]$ mkstore -wrl /opt/BCJOB/bcpstore -modifyCredential UTIL_PRDDB UTIL utpass
Oracle Secret Store Tool Release 19.0.0.0.0 - Production
Version 19.4.0.0.0
Copyright (c) 2004, 2025, Oracle and/or its affiliates. All rights reserved.

Enter wallet password:
[oracle@cac2026 ~]$
```

The DBA must now copy the cWallet.sso file from the Database server's **/opt/BCJOB/bcpstore** directory to BC forms and report servers **/opt/BCJOB/bcpstore** and the same file must update BC Onsite Server's **/var/www/bcapps/config/Wallet** directory.

Otherwise, the updated password will *not* synchronize with the application server's Wallet configurations and the application will not run properly.

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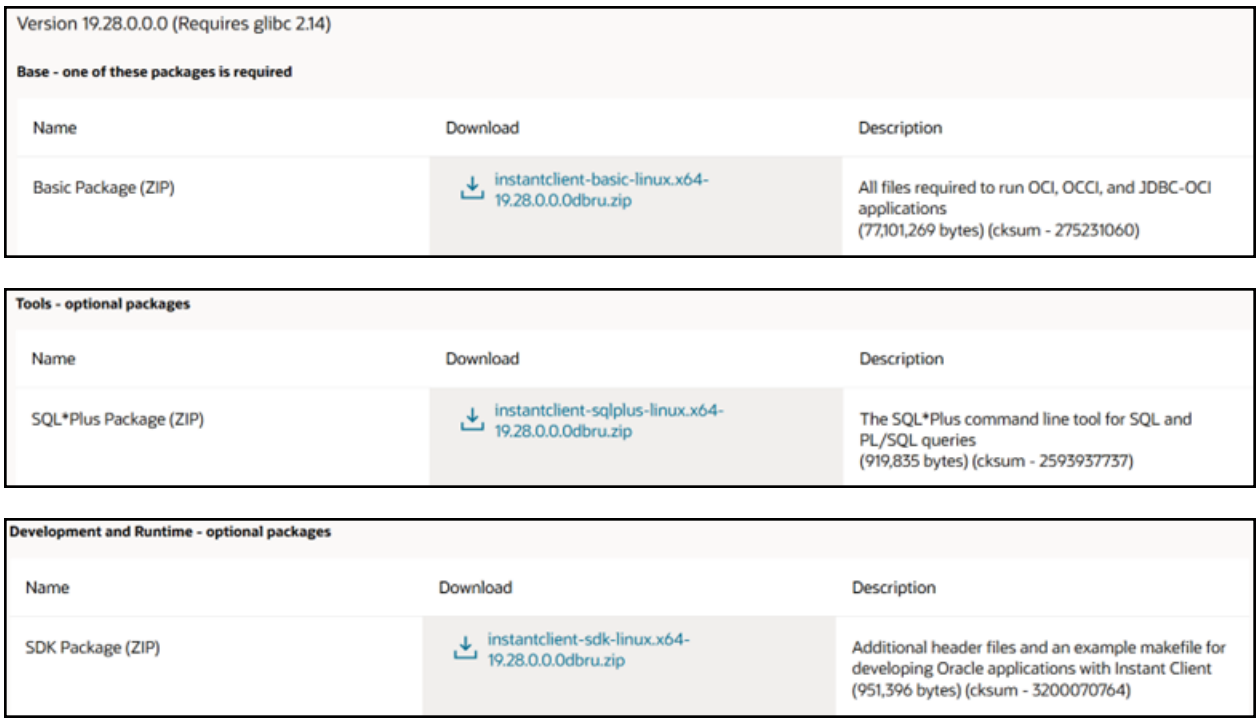
A.7 OKTA Setup and Configuration

NOTE – OKTA authentication setup in the application server requires Linux Apache httpd service and the default BC OnSite Application Server has the httpd service running.

NOTE – Please work with WellSky IT for OKTA authentication after following the installation steps.

1. Download the below Oracle instantclient packages into the **/apps/Software/OKTA** directory from the Oracle website.

   https://www.oracle.com/database/technologies/instant-client/linux-x86-64-downloads.html.



Version 19.28.0.0.0 (Requires glibc 2.14)

Base - one of these packages is required

Name	Download	Description
Basic Package (ZIP)	instantclient-basic-linux.x64-19.28.0.0.0dbru.zip	All files required to run OCI, OCCI, and JDBC-OCI applications (77,101,269 bytes) (cksum - 275231060)

Tools - optional packages

Name	Download	Description
SQL*Plus Package (ZIP)	instantclient-sqlplus-linux.x64-19.28.0.0.0dbru.zip	The SQL*Plus command line tool for SQL and PL/SQL queries (919,835 bytes) (cksum - 2593937737)

Development and Runtime - optional packages

Name	Download	Description
SDK Package (ZIP)	instantclient-sdk-linux.x64-19.28.0.0.0dbru.zip	Additional header files and an example makefile for developing Oracle applications with Instant Client (951,396 bytes) (cksum - 3200070764)

2. As the root user, extract the above three packages into the **/opt/instantclient_19_28** directory using the commands below:

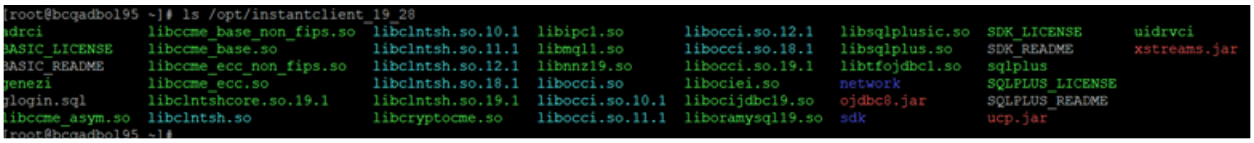
```
# cd /opt

# unzip /apps/Software/OKTA/instantclient-basic-linux.x64-
19.28.0.0.0dbru.zip

# unzip /apps/Software/OKTA/instantclient-sdk-linux.x64-
19.28.0.0.0dbru.zip

# unzip /apps/Software/OKTA/instantclient-sqlplus-linux.x64-
19.28.0.0.0dbru.zip
```

3. Verify the **/opt/instantclient_19_28** directory after extracting them.



4. Install PHP-related Linux OS packages using the commands below as the root user.

```
# dnf -y install systemtap-sdt-devel

# dnf -y install https://dl.fedoraproject.org/pub/epel/epel-release-latest-9.noarch.rpm

# dnf -y install http://rpms.remirepo.net/enterprise/remi-release-9.rpm

# dnf module reset php -y

# dnf module enable php:remi-8.2 -y

# dnf install php php-cli php-fpm php-mysqlnd php-gd php-xml php-mbstring php-devel -y

# php -v
```

```
[root@bcqadbol95 ~]# php -v
PHP 8.2.29 (cli) (built: Jul  1 2025 16:29:21) (NTS gcc x86_64)
Copyright (c) The PHP Group
Zend Engine v4.2.29, Copyright (c) Zend Technologies
    with Zend OPcache v8.2.29, Copyright (c), by Zend Technologies
[root@bcqadbol95 ~]#
```

5. Install **oci8** using the pecl command.

```
# pecl install oci8
```

   - When prompted for the instantclient and its location, enter like below:

```
instantclient,/opt/instantclient_19_28
```

6. Edit the **/etc/php.ini** file and add the below line at the end of the file and save. Make sure this line is *not* commented.

```
extension=/usr/lib64/php/modules/oci8.so
```

```
[root@bcqadbol95 ~]# tail /etc/php.ini
; certificate. This value must be a correctly hashed certificate directory.
; Most users should not specify a value for this directive as PHP will
; attempt to use the OS-managed cert stores in its absence. If specified,
; this value may still be overridden on a per-stream basis via the "capath"
; SSL stream context option.
;openssl.capath=

[ffi]
; see /etc/php.d/20-ffi.ini
extension=/usr/lib64/php/modules/oci8.so
```


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7. Edit the /etc/sysconfig/httpd file and update the parameter below

```
LD_LIBRARY_PATH=/opt/instantclient_19_28:$LD_LIBRARY_PATH

export LD_LIBRARY_PATH

TNS_ADMIN=/var/www/bcapps/config/wallet

export TNS_ADMIN
```

```
[root@bcqadbol95 ~]# cat /etc/sysconfig/httpd
#### Below is the location for instant client software
LD_LIBRARY_PATH=/opt/instantclient_19_28:$LD_LIBRARY_PATH
export LD_LIBRARY_PATH

#### Below is the wallet related files location
TNS_ADMIN=/var/www/bcapps/config/wallet
export TNS_ADMIN
[root@bcqadbol95 ~]#
```

8. Copy the okta directory from the Software CD -> BC2026 –> BC_SERVER_INSTALL directory to the /var/www/html directory.

9. Restart the Linux Apache service using the command below:

```
# service httpd restart
```

```
[root@bcqadbol95 ~]# service httpd restart
Redirecting to /bin/systemctl restart httpd.service
[root@bcqadbol95 ~]#
```

10. Please contact the WellSky IT team for OKTA authentication.

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Glossary

– # –

2D Barcode
A matrix code that provides a two-dimensional way of representing information in a scannable form.

– A –

Access Class
A numeric or alpha code that defines which kind of interdictions the user can view, place, or remove or if a user has special permissions within the application such as overrides.

Access Level
Defines the specific level of permission granted to the user to access system functions. Access Levels rank as follows: 0 = lowest; 9 = highest.

Action Interdiction
An Interdiction that can be placed on the Donor record that can be the source, existing, or a new interdiction.

Active Template
The status of the selected Health History template. The current template's status is Active. A terminated template's status is Inactive.

Alert Message
A message created by a user for other system users. Messages can be created and sent to a single recipient or multiple recipients or messages from a user can be forwarded to other designated recipients.

ALL
A Static Access Class that provides the user access to all interdictions (1-9 level) if all the interdictions are assigned to that class.

Allowable Results
A table defining the testing and qualification tests needed for a donor to become qualified.

Alpha Character
Letter(s) from the alphabet (as opposed to numeric character).

Alternate ID
A form of identification presented by a donor, such as a driver's license or passport.

ALTS
A Static Access Class that enables the user to update the alternate sponsor in the Registration program.

Answer
Defines the text that qualifies as a valid answer to a question in the Health History Question Bank.

API
Application Program Interface

Application ID
An alpha code that has been assigned to describe a particular Module.

Attribute
Coded information to identify special characteristics of a donor or unit; an informational type of interdiction. For example: Donor prefers non latex gloves for phlebotomy.

Attribute Code
An alpha/numeric code naming convention, associated to a specifric attribute to identify special characteristics of a donor or unit.

Audit Screen
Donation or deferral final review for acceptability.

Automatic Printing
An option available when generating a report; the report automatically prints to the default printer determined by your user login.

– B –

Band Code
Used as the header when resulting cellular level results for ABORH or other tests as configured.

Band Name
Used as the header when resulting cellular level results for ABORH or other tests as configured.

Banner Block
The very top of any Blood Centers screen that displays the name of the application, user ID, name of the screen/function that you are accessing, warehouse ID if needed, date and time of screen access, keyword, and program ID.

BDMS
Blood Donor Management System

Bill of Lading
The carrier's tracking identifier for the shipment.

BIO
A label that is placed on a product to indicate a Biohazard. Products with a BIO interdiction are labeled as Biohazard and the label will be printed with the universal Biohazard symbol.

BO Status
Designates that the status of an order is "back ordered."

Box Type
The type of box that you are packing products for shipment in, such as 'igloo cooler' or carton.

Box/Carton/Case
Interchangeable terms used to describe the container into which samples or products are packed.

– C –

CAC
Common Access Card A standard identification smart card issued by the Department of Defense that identifies active duty members of the Uniformed Services and other related personnel and contractors.

Campaign Theme
The theme that you are using for your campaign to market your mobile blood drive, such as pint-for-pint (pint of ice cream for pint of blood), football challenge (win football tickets), or free gift wrapping during the holidays when you donate blood.

Campaign Type
The type of campaign that you are using to market your mobile blood drive, such as media campaign or billboard campaign.

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Cell Type
Identifies groupings of products collected for use in eligibility determination. There are two kinds of cell types: RBC – Red Blood Cell and PLASM – Plasma

Center
One of many physical locations of a blood donor center. A Center may be referred to as a Chapter.

cGMP
Current Good Manufacturing Practice.

Chapter
One of many physical locations of a blood donor center. A Chapter may be referred to as a Center.

Charge Code
Code defined by each facility to indicate a specific charge for services and or products.

checksum
A character that is used to validate a string of numbers.

CMV
Cytomegalovirus

Codabar ABORh Values
Numeric values that are associated to specified ABORHs used in barcoding and resulting ABORH tests in Blood Centers.

Codabar Unit Scheme
Indicates if the unit number is Codabar alpha or Codabar numeric.

Common Access Card (CAC)
A standard identification smart card issued by the Department of Defense that identifies active duty members of the Uniformed Services and other related personnel and contractors.

Communication Code
A code for the specific type of communication sent to a donor. For example, customer service communication may have several sub codes to further idtentify the specific communication such as T-Shirt mailed, request for donation history, or donor apology.

Communication Letter
Letter that could be sent to a donor that is not related to interdictions. These communications mainly relate to infomration that the donor may have requested or customer service.

Communication Type
A code to indicate the type or category of communication that you would send to a donor, such as; Notification, Customer Service, Scheduling, Nursing Service, Medical Director or General.

Communications Log
A log of cummnications that are associated to a donor, these could include notification communications, donor relationship codes, request for information codes, medical director communication, and donor appointment scheduling as a few examples.

Compensation Card
A card number issued to the donor to maintain a record of the funds received by the donor for current and previous donations.

Component
Individual part of a pheresis or whole blood donation.

Consignee
Identifies to whom the product was shipped.

Consignee ID
A delivery address or site of the customer.

Consumable
Items that are used during the donation and manufacturing processes, that must be inventoried and replenished as needed.

Contact Type
Indication of the type of contact you are associating to a contract, such as Designated or Financial.

Contract Code
Two character code that indicates the type of contract that is being defined, such as Internal, Testing, Proficiency Testing, or Imported Blood.

CORE Routine
A system routine that reads uploaded test results and schedules additional tests, as needed, and places indicated interdictions.

CPU
Cost per unit

cron job
Cron is a time-based job scheduler. It enables you to schedule jobs (scripts) to run periodically at certain times or dates.

CUE
Confidential Unit Exclusion. A barcode applied by the donor to the donor record to indicate 'USE' or 'Do NOT Use' my blood. The system will still display the CUE on specific screens, even if the CUE is not in use by the facility.

Customer
The purchaser of the product.

Customer Type
A code categorizing the customer as one of the following: o IN for Internal o DI for Discard o EX for Export o MG for Manufacturing o RD for Research and Development o RG for Regular Customers

– D –

Data access
Authorized user access that is limited to information specifically required to perform a user's assigned tasks.

DBA Privileges
A user with DBA privileges has the authority to write to the database and create users.

Deferral
A code indicating if an interdiction is associated to a deferral period. For example: Y = Yes, donor or unit is deferred for this location. N = this interdiction is not associated to a deferral period.

Deferral Period
An amount of time associated to an interdiction that defines the duration of the interdiction. A Deferral Period can be defined as a fixed or variable amount of time.

Deferral Type
Defines whether a deferral is or is not for a fixed amount of time. An interdiction may have a fixed deferral period or various deferral periods associated with it. For example: A generic interdiction, HDT; Health History Deferral – Temporary, can have different deferral periods associated with the answer.

Device Type
Type of storage device for example Refrigerator, Freezer, Platelet Rotator, or Room Temperature Table.

Differential Code
Cellular differential code.

Discard Product
Product from a unit that has been discarded.

Discrepancy Check
A review of test results to check for discrepancies and validate the findings.

DNRSG
Donor signature.

Donation Class
The classification of a donor specific to a particular collection. Valid Donation Classes are: AP = Applicant; QL = Qualified; AN = Annual Donation; RG = Regular Donation. Donation Classes are static values and are defined in the Donation Class table. A user is not allowed to choose a donor's donation class. System logic determines a donor's donation class at registration.

Donation Credit
The number of credits the donor will receive for a particular procedure.

Donation Group
A group of donation types. Associating certain donation types together under a group such as Special group would include Autologous, Directed, or Therapeutic. Grouping donation types together aids in appointment scheduling.

Donation Type
The classification of a donor specific to a particular collection.

Donor Grade Match
Donors are given a numeric grade ranging from 0 to 5 according to the Donor Grading Matrix (GRADMAT) controlled under WellSky Restricted Validation Tables (MEDRVT). A valid entry is required to retrieve the donor grade matches. For example: An entry of 3 would return donor grade matches 3, 2 and 1. Refer to the Grading Matrix Query (QGRADMAT) screen.

Donor Grading Matrix
Defines critical donor data elements and assigns a grade based on a comparison with other donor data elements using the Find Donor block and other processes.

Double Blind Entry
Manual entry of a value twice prior to the system allowing the cursor to advance to the next field.

Download
To move data that was captured in a Cron job export file from the Blood Centers application server to the master mobile laptop.

DR Type Interdiction
An Informational type of interdiction related to donor recruitment, such as call after 5:00 p.m. or do not send e-mail.

DRIVE
A Static Access Class that enables the user to create and modify drives other than their own.

Drive Category
A category of drives that aids in the ability to schedule staff or limit the number of drives within that category from being booked on the same day. Example would be High School Drives, you may want extra staff as well as to limit that only 2 High School drives could be booked on the same day.

DRVBK
A Static Access Class that enables the user to enter a drive date in the past in the onsite registration module only.

DUFFP
Dummy FFP. This is a product type that allows further manufacturing into a cryo precipitate or a cryo precipitate and cryo poor plasma products.

Duplicate Donor
When a donor has two or more separate records in the database.

Dynamic Access Class

Those access classes that can be defined internally for classes of interdictions or properties (attributes).

Dynamic Master Code

An existing code type that may be added to internally or removed from (terminated) a code type. Compare to Static Master Code. Refer to the Master Codes appendix.

– E –

EDIPI

Electronic Data Interchange Person Identifier A unique number assigned to a record in a database that identifies a person and a personnel category.

EDN

Electronic Delivery Note

EIE

Enterprise Integration Engine - This is WellSky's enterprise level interface sub-system for the automatic transmission of data between a transmitting and target system.

Employee ID

The code that has been associated to the User ID or the employee ID associated to the User ID.

Enlistment Type

Types of enlistment lists to group enlistment criteria under the enlistment type.

EPASS

A Static Access Class that enables the user to change laptop passwords for mobile staff.

Equipment ID

The location where the product is stored.

Equipment Master Code

Master Codes that are related to the Equipment management module in Lab.

Equipment Type

The type of location where the product is stored or a type of laboratory/screening equipment.

Error Message

There are four types of Messages: System Error Messages must be acknowledged to continue. User Error Messages appear when the user makes an entry that cannot be accepted. Warning Messages display when a user attempts to perform an operation that should be considered carefully before committing data. Informational Messages provide information only and require no action to continue.

Existing Interdiction

The interdiction currently existing on the Donor; the interdiction is either a SURVE or INTER type, as defined by the Interdiction Codes table.

Export

To transfer data from the Blood Centers primary database server to a Blood Centers application server.

Extract

To move donor data that was captured offline during a mobile drive from the donor database on the mobile master laptop to the designated location on the mobile laptop hard drive.

– F –

Facility Code

Code defined in WellSky Transfusion for physical locations for transfusion service.

False Duplicates

Two donor records that appear to be the same donor but are, in fact, different donors.

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FFP
Fresh Frozen Plasma

Find Donor (vs. Search for)
Used to search for, identify, and retrieve donor records in critical programs. A window that displays automatically in the ONILK-REG (Lookup and Registration) screen. The functionality of the window can be customized and configured independently for each program that uses it.

Find Donor Block
Screen that allows a user to find a graded result for donors with the input demographic data.

Flat File
A file that is generated to send test results into the system.

Frequency Code
A code that has been assigned to identify how often a particular donor task is performed, as defined in the Master Code FRQCD.

Functional/Working Group
The user's group that defines their function within the organization. For example: accounting, registration, collections. Each user is assigned a group when created.

Future Template
The status of the selected Health History template. A Future Template will become Active on a specified effective date in the future and replace the current Active template. When the Future template replaces the current template, the current Active template is terminated and becomes Inactive.

– G –

GMT Offset
Greenwich Mean Time plus or minus a specified number of hours and minutes

Goal Type
Collection Goals used in Donor, Types of goals could be institution level, sponsor, recruiter, or even collection site.

Group ID
The working/functional group code that has been assigned to a user.

– H –

Half Address
When the number portion of an address contains one half; for example, 123 1/2.

HLA/RBC searches
Searches that will locate RBC and/or HLA antigen matched donors and/or units for recruitment.

HOLD Type Interdiction
A type of interdiction, such as a data entry hold or a hold on physical findings.

– I –

INFOR Type Interdiction
A type of interdiction, such as donor attributes, that should display during donor look up.

Informational Message
A message that provides information only and requires no action to continue; the user is only required to close the pop-up.

Institution
The code that represents the name of the institution.

Intended Use Code
Letter or number code that is in the 6th position of the product code that specify the usage of the product.

Inter Classes (Interdiction Classes)
Classes set up to group interdictions for ease of security assignments.

INTER Type Interdiction
An Interdiction category that is associated to a deferral or a process control in the system, such as test result related interdictions and donor history related interdictions.

Interdiction
A type of identifier for a unit or donor that can be a deferral type identifier or used to aid in management of a process.

Interdiction Owner
A code assigned to designate the owner of the interdiction. For example, D = Donor, U = Unit, B = Both.

Interdiction Type
A sub-type of interdictions such as informational, deferral, hold.

Interdiction Upgrade/Downgrade Logic
An interdiction that is not allowed multiple placements on the unit/donor and must be set to N to go through. See Interdiction Upgrade Matrix (INTUP) for more information.

INTUP Function
Interdiction Upgrade Matrix function.

Inventory Class
A code catezorizing an inventoried product as one of the following: - RE -- Regular - AD -- Autologous/directed - SP -- Special

Inventory Product Type
A category to place common products together. Valid inventory product types are associated in Master Codes, code type PRTYP in Distribution.

ISBT
International Society Blood Transfusion

ISBT Code 128
ISBT 128 is the global standard for the identification, labeling, and information processing of human blood, cell, tissue, and organ products across international borders and disparate health care systems. The standard has been designed and perfected over a period of almost two decades to ensure the highest levels of accuracy, safety, and efficiency for the benefit of donors, patients, and health care professionals around the world.

– J –

Job Code
Specific code that indicates what job the staff member can perform.

– K –

Keyword
An abbreviation for a program name, a menu command, a master code, a system parameter or a validation table.

– L –

Lab Code
Lab user affiliation, which should be RT for all users.

Label Code
Code associated to a specific testing and storage statement for labels.

Line-by-line Basis
Where data is committed to the system as each field is completed by pressing ENTER.

LOB
Line of Business

Long Audit Process
When you use the DONAUDIT or the DEFAUDIT screens to audit the donation back office. This will record questions and answers as well as physical findings codes.

Lot Category
A code that identifies a valid lot category each consumable bag is associated to, as defined in Master Code BAGID.

Lot Number
A required entry code that identifies each consumable bag using a maximum of 30 characters.

LOV
The List of Values contains all valid data for a particular field.

– M –

Manual Hold
A hold that is placed manually on the Donor during the registration process. If a suspect duplicate donor is found during registration, two or more donors can be placed on hold and not allowed to continue.

Manufacturing Process
The steps used to transform the units collected into products that can be shipped.

Mapsco
A company that provides printed street level maps.

MAS90 Data
Financial or Billing interface file format.

Master Code Type
A classification under which Master Codes are grouped. A table of underlying codes is associated to each code type. Master Code types are defined and stored by WellSky.

Master Codes
A set of code types used by individual modules to provide flexibility to the application. Some Master Codes define valid data/values for other tables in the system (for example, Street Types: CIR, AVE, ST, BLVD). WellSky may be required to add new or modify existing Master Codes.

Maximum Cell Loss
The maximum loss of red cell or plasma a donor may have over a specified number of days and still maintain eligibility. If the donor goes over the maximum amount in the specified number of days, he becomes ineligible.

Merge Donors
To join two donor records that belong to the same donor into a single donor record.

Message ID
The code that identifies an Alert Message.

Mul Flag
Multiple Flag in the interdiction table for donor.

– N –

New Donor
A first time donor.

NFT
Not for Transfusion. A label that is placed on a product. Not for Transfusion for ISBT the system will print Discard, for codabar the system will print 'NFT' on the label.

Nickname
The donor's familiar name that is associated to a first name for use in the donor grading matrix for possible donor matches when searching for a donor record.

Notification Letter Template
A template that defines the information included in a letter sent to a donor. A Notification Letter is

a Controlled Document.

Notification Mode
One of two modes, Standard and Notification, in which the Donor Detail Query screen can operate. If this screen is called from the Donor Notification functions of the application, the Donor Detail Query screen will run in Notification mode.

NT Results
Results that are not tested.

Number Value (for a Master Code)
An optional entry made when defining a Master Code; entering Number Values establishes the valid entries for a specific parameter defined by a particular Master Code. Refer to the Master Codes appendix before making an entry.

– O –

Offline Data Entry
Entry of data when recovering from downtime that bypasses certain system controls to allow proper documentation of activities that occurred while system was down.

OR Status
Designates that the status of an order is "ordered."

Order Class
Defines the urgency of the order- stat, asap, emergency, routine.

Order Entry Module
Provides computer aids to take, record, and track orders for the blood center.

Order Mode
A code that indicates how the order was placed (e-mail, phone, interface).

Order Number
A number that identifies an order; it is generated by the system after the first line item is entered.

Order Type
Special, Auto Directed, Regular, Export.

Ordered Tier
Freshness indicator for the products orderd.

OSUPR
A Static Access Class that enables Supervisor override in the onsite registration module. It allows the user the ability to re-enter data for health history, physical findings and to modify registration information.

Overlapping Interdictions
Interdictions associated to more than one class.

– P –

Parent Question
A question in the Health History Question Bank that can trigger a related question to display. The related question usually requires more detailed information from the donor.

Payee
The billing address of the Customer. Use the Payee screen to define new payees in the system or to revise details on previously entered payees. The payee can be different from the customer. This is the second screen required in entry of new customer accounts.

PDF 417 Barcode format
A stacked, linear barcode symbol format. This is a 2D barcode used by a CAC, State driver's license, and other State-issued IDs.

Physical Code
A type of physical finding, such as temperature.

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Physical Findings Template

A template for the donor physical. You may define a specific template for each of the different donation cycles, as well as a template for future use. This table is institutional and can only be defined once for all chapters. The chapter should remain at the default value.

Physical Findings Values

A screen used to set up physical entries that the donor must pass in order to donate. The allowable range or the allowable character value can be documented and/or associated to an interdiction for placement if the physical does not match the allowed. This table is institutional.

PLASM Loss

Defines the acceptable amount of plasma loss for the specified procedure to be considered valid by the system. The standard loss includes the amount of blood that is routinely taken for the procedure and all test tubes collected. If the blood center desires, the amount of blood circulating in an apheresis machine (extra-corporeal volume) can be added into the eligibility calculations. The value entered in Define Procedure Codes (PROCS) is used if the PLASMA Loss field is left blank during collection.

Plate Sequence

Determines which type of result it is. I = The result code is for initial results; R = The result code is for repeat results; If the result can be both initial and repeat entry it must entered twice.

Point of data responsibility

The person who has authority to enter or change data in any database within Life-Trak/Distribution.

PRINT

A print mode that enables all non-supervisory users to view, print, or delete reports that only they have generated. The print mode for supervisors is SPRINT.

Print Header

The name that displays on the header for reports run at the institution level.

Procedure Code

The type of draw the donor will have.

Product Type

A code categorizing the product as one of the following:

Program

Each screen or window is a separate functional program governed by a set of codes.

– Q –

QC

Quality Control.

Quarantine

A status assigned to suspect products that prevents them from being allocated, packed, and shipped; pending investigation into their viability.

Question Code

The five-letter code associated with a particular question that is available to be included on the Health History Questionnaire.

– R –

Randomize Flag

A code used to designate the order in which the questions are displayed on the Donor response screen. For example: Y = Display the questions in no particular order. N = Display the questions in numerical order according to the template.

RBC Loss

Defines the acceptable amount of red cell loss for the specified procedure to be considered valid by the system. The standard loss includes the amount of blood that is routinely taken for

the procedure and all test tubes collected. If the center desires, the amount of blood circulating in an apheresis machine (extra-corporeal volume) can be added into the eligibility calculations. The RBC value entered in Define Procedure Codes (PROCS) is used if the RBC Loss field is left blank during collection.

Recruiter Caretaker

Staff member that will have access to drives created by specific recruiter to allow modifications to those blood drives.

Reference Code

The data that is used in query or audit screens. This is the value that users see in other screens.

Registration Hold

A type of interdiction that is placed automatically on a donor record at a defined process in the system as a result of unexpected field level entries, or placed manually during the registration process. Deferral period and manual placement allowance must be verified. Registration Holds are defined in Interdiction Codes (INTER), type HOLDS. Holds placed on the donor record can be removed by using the Remove Donor Interdiction or Remove Unit Interdiction functionality by a System Administrator or WellSky Staff.

Registration Holds Mapping Table (PLIREGIN)

Configuration table that allows definition of various interdictions that are placed by certain outcomes in the application.

Registration Override

When a donor is not suitable for donation but is allowed to donate by a user with appropriate authority, who overrides the restriction.

Releasing an Interdiction

The action that should be taken if the interdiction should not have been placed on a donor or unit record.

Repeat Donor

A donor who has donated previously.

Response Code

Designates the code for the valid responses to each question in the question bank. Y = Yes, N = No, U = Unanswered.

Restricted Validation Table

Tables that are set up to run the system. Valid values define the functionality pieces of the system.

Return Slip

A report of products that were returned to a center.

RMF Read More Field

This field is used when a question code has a –D suffix, indicating that the question has additional information to be displayed. Valid entries are Y or N, which designates whether or not the Question Details window automatically pops up in the Donor Response screen. Entering a Y in the RMF field will require the donor to see additional information related to a question.
Example: Referring to the screen picture, Q Code H0010 is the basic question (Are you allergic to iodine and/or latex?). Q Code H0010-D displays more detailed information, which appears in the Question Details window. The RMF field is set to Y, so the window will pop up automatically. If the RMF field is set to N (or blank), the donor must click the Read More button to display the Question Details window.

RPFF

Recovered Plasma Fresh Frozen

RPL

Recovered Plasma Liquid

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– S –

Safeblood Format
Electronic resulting file format.

Salvage
Products that are used for further manufacturing.

Sample Pack List Report
A report of samples that are being sent to a specific laboratory for testing.

Scheduled Test
Tests that are automatically assigned to a unit. An expected test.

Sequence (For a Master Code)
An optional entry made when defining a Master Code; the Sequence is specific to each code type. WellSky will note if Sequence entry is required. Otherwise, this field may be left blank. Refer to the Master Codes appendix before making an entry.

SLD
System Label Decision--whether to label the product either NFT or BIO.

SOP
Standard operational procedures.

Source Interdiction
The interdiction being placed on the donor record; the interdiction being placed is either a SURVE or INTER type, as defined by the Interdiction Codes table.

Source Product Type
Product type that will be used to make another product type via modification or conversion.

SPKEY
A Static Access Class that enables the user to reprint a donor card in donor look-up.

Sponsor Category
Indicator for how successful the drives have been or should be for the sponsor.

Sponsor Rating
Indicator for how successful the drives have been or should be for the sponsor.

SPRINT
A print mode that enables supervisors to view, print, or delete reports that other users have generated. The print mode for non-supervisory users is PRINT.

SQL
Structured Query Language

STAFF
A Static Access Class that enables the user to have access to the drive sign off process.

Standard Mode
One of two modes, Standard and Notification, in which the Donor Detail Query screen can operate. If this screen is called from the Donor Profile or Shipment functions of the application, the Donor Detail Query screen will run in Standard mode.

STAT
Used to indicate urgent or rushed.

Static Access Class
Those access classes that cannot be defined internally due to their explicit functionality. These are defined only by WellSky.

Static Master Code
An existing code type that may not be added to internally or removed from a code type. Only

WellSky may add, modify, or remove a Static Master Code. Compare to Dynamic Master

Strong Password

A password that has more complex configuration than a traditional password. For example, a strong password could consist of at least 15 characters (and no more than 50 characters) with at least two upper case and two lower case alpha characters, two numbers, and two special characters.

SUPER Mode

One of two modes in which the Notification Letters screen can operate. SUPER mode enables you to authorize, print, and send the letter to the donor. USER mode allows you to generate a notification letter.

Super User

A user who is defined by a site as having additional privileges/security access than other users in the system.

Supervisor Mode

Allows the user to access reports generated by all users.

SURVE Type Interdiction

A surveillance-type interdiction related to test results and whether or not a donor deferral period should be upgraded depending on a donor incurring a second occurrence of the same interdiction code.

System Date

The current date and time to which the system is set using the set date screen.

System Error Message

A message that informs the user of any system errors that have occurred.

System Name

The name that displays on the Application Menu at the first entry screen at log on.

System news

A notification system to inform the user of occurrences that may affect the Blood Centers application; for example, when down time is scheduled for maintenance or when changes have been made to the system.

System Parameters

Configurable elements that define various components, types of values to enter. For example: The amount of time a password is valid for.

System User

A user who has the appropriate access level, has a login/password and can access the system, and is found in the ADD USER list.

– T –

Tax ID

The Federal Employment Identification Number (FEIN).

Test Kit

A table that lists the test kit associated to each screening test that was performed in order to print the testing table onto the COA report.

Test Lab

The laboratory that performs the designated tests on the samples. The specific tests conducted by a particular lab are defined in Lab Test Table (PLLABTES). The Test Lab is used to define the results that should post onto the donor's record.

Tier

Tiers define timeframes until product expiration. • Tier 1 = 0 to 7 (days to expiration, oldest) • Tier 2 = 8 to 37 (days to expiration) • Tier 3 = 38 to 42 (days to expiration, freshest) Minimum and

maximum values cannot overlap from one tier to another.

Tracking Number
Identifier that is attached to shipments and is used to track the shipment.

Traditional Password
A password that has a less complex configuration than a strong password. A traditional password must contain at least six characters (and no more than 50 characters) and must contain at least one alpha character and one numeric character.

Transfusion Implicated Interdiction
A type of interdiction that is mapped to the appropriate test result in the Define Test Results (RESULT) table and is manually placed on the donor, but automatically goes through this logic when appropriately mapped.

TVIEW
A Static Access Class that provides the user with the ability to view test results; View ability is associated to the view level, V Level field in Define Tests (TESTS).

– U –

Unit Interdiction
An interdiction placed on a unit for deferral purposes.

Unit Scheme
The one-digit Scheme currently in use at your institution, as defined in the Master Code BARCD. Valid Unit Schemes are C for Codabar or I for ISBT.

UNLDF
A Static Access Class that enables the user to access the Unlink Deferral button.

UNOTE
A Static Access Class that enables the user to add notes to donation in Donation window.

Unpack Slip
A list of items that the system generates if the entire order is cancelled, accounting for the items to be unpacked from the order.

Unscheduled Test
A test that is not expected on a unit but is interfaced into the system. Or an additional test result on an already-resulted, scheduled test.

Up Flag
Upgrade Flag, in the Donor Interdiction Code table that indicates if the interdiction should be upgraded or not via the INTUP table (upgrade matrix).

Upgrade Matrix
Configuration table that defines the action of two interdictions that when used in conjunction can elicit the placement or removal of another interdiction.

Upload
To move data that was captured offline from the hard drive of the mobile master laptop to the Blood Centers application server.

User Class
A group of valid users that the administrator can assign to an access class.- Not used

User Error Message
A message that appears when the user makes an entry that cannot be accepted.

User Mode
Restricts access to only reports generated by the user.

USER Mode

One of two modes in which the Notification Letters screen can operate. USER mode allows you to generate a notification letter. SUPER mode allows you to authorize, print, and send the letter to the donor.

UTB

Unit Test Base

– V –

Valid System User

A user who has the appropriate access level, has a login/password and can access the system, and is found in the ADD USER list.

Validation Table

A list of restricted functions that can only be accessed by Users with administrator-level access due to their impact and interaction with the Blood Centers application.

VBN

Vendor Batch Number

Venipuncture Attribute Code

A code that is associated to information about a particular venipuncture during the donation process. For example: No Blood Collected, Donor Walked Out, or Double Stick.

View Level

Numeric value attached to interdictions and test results that will limit the ability of end users to see specific test results and interdiction codes.

VWSSN

A Static Access Class that provides a user with access to view the SSN of the donor in the list browser.

– W –

Walkout

A donor who signed in but did not complete the donation process.

Warning Message

A message that displays when a user attempts to perform an operation that should be carefully considered before committing data to the database.

Working/Functional Group

The user's group that defines their function within the organization. For example: accounting, registration, collections. Each user is assigned a group when created.